A. OVERALL COVER PAGE

Project Title: Alzheimer's Disease Research Centers	
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Human Subjects: Yes **HS Exempt:** NA **Exemption Number:** **Phase III Clinical Trial:** NA	**Vertebrate Animals:** No
hESC: No	**Inventions/Patents:** No

B. OVERALL ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

The USC ADRC focuses on factors that modify the effect of APOE e4 on cognitive decline and ATN (vi) biomarkers, emphasizing vascular, lipid, inflammatory, genetic, and ethnic influences. In the next 5-years, the USC ADRC has three overarching Aims:

1) Recruit and follow a clinical cohort that reflects the population of Los Angeles and is enriched for APOEe4, focusing on factors that modify the effect of APOEe4 on cognitive decline and AD biomarkers. Our goal is to build a multi-sourced, participant registry with APOE genotype and Social and Environmental Determinants of Health (SEDoH).

2) Promote local ADRD research at USC: 1. Integrate research programs by holding annual events, pilot fundings, and harmonization of protocols. 2. Share data, tissues, biofluids, and a participant registry through a common USC ADRC database. 3. Train the next generation of ADRD researchers and educate participants and the public.

3) Further national collaborations: Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, comprehensive imaging to national initiatives.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

For this reporting period, is there one or more Revision/Supplement associated with this award for which reporting is required?

No

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

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B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

In addition, the USC ADRC sponsored or co-sponsored the following Educational Events:
• March 8, 2025: USC ADRC Latino Study Participant Appreciation Event (research participants)
o Organized by Aranda & Ringman (Delivered primarily in Spanish)
• May 23, 2025 : Southern California ADC Symposium (investigators)
o Organized by Grill, Yassine, Brewer, Tan
• November 1, 2025: 9th Verdugo Hills Brain Health Forum (lay community)
o Organized by Danner and Chui
• April 14, 2026: CPBH Symposium: Advances in Personalized Brain Health (Target audience: scientists)
o Organized by Yassine, Kolchins & Center for Personalized Brain Health (CPBH)

New USCADRC website Learning Hub has pages for 1) Education and Training,2) REC Invited Speaker Seminar, 3) Symposia, and 4) Neuropathology Teaching Set.

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

• Spring 2026: Strategic planning with Cores and Internal Advisory Committee (IAC) related to vascular and metabolic contributions of apolipoprotein E4 to Alzheimer pathology (ATN (v) among diverse communities.
• Fall 2026 External Advisory Board (EAB): meeting, including progress reports from Development Projects.
• Enhance ADRC database with ATN(v) biomarkers.
• Build APOE4 enriched cohorts for CLARITI (focused on Amyloid Targeting Treatment) and SCAN (focusing on communities that are representative of our catchment area.

B.2 – Administrative Accomplishments

This RPPR covers Year 6 accomplishments for the 6- month period from 7/1/2025 to 12/31/2025. period).

1) Major Activities

Administration Core [mPIs: Helena Chui (contact) and Arthur Toga; Associate Director: Hussein Yassine] foster our overarching scientific goals, coordinates administrative functions aligned with NIA-ADRC program, attracts early-stage and new investigators through development projects, promotes collaborative local and national research, and promotes training/education with REC and ORE Cores.

Calendar year 2025 was challenging. As we experienced a 4-month funding gap from March 1 to June 30, 2025, we used various strategies to continue UDS follow-up and mitigate loss of key staff and personnel. Almost simultaneously with receiving the NOA for the ADRC's competitive renewal on 7/30/2025, USC implemented University-wide budget cuts, resulting in layoffs and hiring freezes. These financial challenges led to the postponement of the start-up of new development projects and the convening of our External Advisory Board (EAB) until 2026. Nonetheless, as this RPPR shows, all ADRC Cores remained productive.

On 10/27/2025, we received a vigorous response to the ADRC Development Project RFA, receiving 18 applicants (8 from faculty and 10 from post-doctoral students). We decided to combine Year 6 and Year 7 DP funds, selected 2 faculty and 4 post-docs as finalists, and submit 6 proposals in this RPPR for approval to start in Year 7.

Clinical Core (Lon Schneider, Elizabeth Joe) recruits and retains participants enriched for *APOE* ε4, performs standardized evaluations/diagnoses using the NACC Uniform Data Set, characterizes participants by *APOE* genotype and ATN(v) biomarkers, collects biofluids, obtains informed consent for autopsy, and maintains a research participant registry with the ORE and Data cores.

We trained our clinicians and transitioned successfully from English versions of UDS 3.0 to UDS 4.0. We were approved as a participating site for CLARiTI and ALZ-NET; recruitment is actively underway.

Data Management and Statistical Core (Arthur Toga/Dan Nation and David Conti/Xinhui Wang) provides a centralized data management framework, ensures timely and accurate data entry and management, and provides biostatistical support.

Data coordination via local REDCap with NACC (UDS v4.0) and NCRAD (for plasma biomarkers) is in place. The Centralized Umbrella Project encompasses a growing number of ADRC and ADRC-affiliated clinical and imaging datasets. New developments are underway to build participant tracking/dashboard, cohort discovery tools (ARC Builder), and ATNv data consolidation across clinical phenotype, biomarkers, and neuroimaging findings. Significant additions have been made to the ADRC website, including a Data Usage Agreement to request data access and a Learning hub with linkages to REC seminars/symposia and a Neuropathology Teaching Set.

Neuropathology Core (Annie Hiniker, Deborah Hawes) performs standardized neuropathological diagnoses, distributes tissues, develops a digital pathology library, and interrogates the effect of APOEε4 on AD neuropathology and perivascular inflammation.

Steady progress is being made to upgrade the characterization of brain tissue by backfilling ApoE genotype and upgrading diagnoses to current NACC 11 standards for TDP-43 LATE/FTLD, alpha-synucleinopathies, and Abeta amyloidosis, and capturing resulting images in a digital library.

Outreach, Recruitment, and Engagement Core (Maria Aranda, John Ringman) focuses on participant recruitment enriched for APOE ε4, develops a research participant registry in concert with the Clinical and Data Cores, enhances recruitment from public safety-net hospitals, and engages in bi-directional education, information and resource sharing.

The online USC GeneScreen registry continues to enroll community members in the greater Los Angeles area who are interested in participating in ADRC research. To date, the registry has enrolled 668 members, of which 188 are APOE ε4 carriers.

Biomarker Core (Robert Rissman) coordinates collection, processing, storage, and sharing of biofluids, facilitates biomarker research through an integrated informatics toolkit, accelerates local, national, and international biomarker discovery efforts.

We have been working with Meso Scale Discovery (MSD) scientists to develop new and sensitive assays for a large number of novel biomarkers of interest. Refinements have been achieved in APOE4-specific ultra-sensitive MSD assays, which would allow for rapid screening of biofluids.

Imaging Core (Arthur Toga/Kirsten Lynch): Provides imaging for the ATN(v) framework for ADRC projects, performs high quality control and MRI analytics, develops technical and scientific approaches for ultra-high field 7T MRI, assists in PET protocol development, quality control, and analytics.

We are working towards a unified minimum standard acquisition protocol that can serve as guide for future ADRC-affiliated projects, to ensure that data collected across independent project remains interoperable and comparable, and transitioning to a model where all raw data regardless of origin, is reprocessed through the Core's harmonized pipelines to ensure that downstream metrics are methodologically consistent, maximizing the statistical power of the aggregate ADRC dataset. We have developed a noninvasive method using vascular space occupancy and arterial spin labeling for measuring "microvascular volumetric pulsatility."

Research Education Component (Hussein Yassine, Duke Han) supports ADRC scientists by monitoring the progress of ADRC scientists in their early-career milestones.

We sponsor weekly ADRC REC Webinar series, an annual research symposium titled "Advances in Personalized Brain Health," participate in the annual Southern California ADC Symposium, and sponsor applications by and mentor recipients of early-stage investigator's Development Projects.

2) Specific Objectives

- The Admin Core received a $3 Million Gift from the Pattiz Foundation to support the ADRC. This includes $1 million to support an endowed Professorship in Neuropathology. This generous gift will be used to accelerate digital pathology and to support recruitment in LA County Department of Health Service Hospitals.
- The Clinical Core collected 331 PBMC samples, 348 buffy coat samples, and 126 plasma samples. We received plasma biomarker results from NCRAD for 283 Aβ40, 272 Aβ42, 250 GFAP, 250 NfL, and 270 ptau217.
- The Data Core completed the harmonization of 14 ADRC and ADRC-affiliated datasets and made 47 total clinical and image analysis datasets available. From 7/1/25 to 12/31/25, MR scans from 2,489 participants and 1,068 PET scans from 1,073 participants were uploaded into the LONI IDA repository for USC ADRC affiliated studies.
- The Neuropathology Core completed 9 autopsies, backfilled APOE genotyping on > 600 cases lacking clear genotype, implemented a new REDCap tissue request form, and sent brain tissues to 8 outside investigators (3 new).
- In the ORE Core, the online GeneScreen registered 39 new enrollees. To date, the registry has enrolled 668 members; of these, 188 (28%) are APOE ε4 carriers.
- The Biomarker Core: Processed 143 plasma and 143 WBC, 18 sera, and 3 CSF. The Core analyzed over 450 CSF sample for AD biomarkers and 500 DNA samples for APOE genotype.
- The Imaging Core completed quality control for 114 amyloid-PET scans, 75 Tau-PET scans, and 917 structural MRI, and added the computation of Centiloids for amyloid-PET and CenTauR for tau-PET.
- The REC Core organized 13 REC webinar seminars, sent 25 scholars to the May 2025 Southern California ADC symposium, and invited 2 USC ADRC speakers (L Joe and A Hiniker). In coordination with the Admin Core, the REC selected 4 post-doctoral scholars for Year 7 ADRC development projects.

3) Significant Results

In 2025, NCBI lists approximately 100 publications attributed to the USC ADRC (approximately 50 published between 7/1/25 and 12/31/25). Of these 50 publications, we highlight 16 below significant results related to our central research themes.

Apolipoprotein E4: Metabolism, Lipids, and Inflammation

Nedelkov D, Tsokolas ZE, Rodrigues MS, Sible I, Han SD, Kerman BE, Renteln M, Mack WJ, Pascoal TA, Yassine HN, Alzheimer's Disease Neuroimaging I. Increased cerebrospinal fluid and plasma apoE glycosylation is associated with reduced levels of Alzheimer's disease biomarkers. Alzheimers Res Ther. 2025;17(1):151. Epub 20250710. doi: 10.1186/s13195-025-01795-7. PubMed PMID: 40635030; PMCID: PMC12243426.

The apolipoprotein E (APOE) epsilon4 allele is the strongest genetic risk factor for late-onset Alzheimer's disease (AD). ApoE is glycosylated with an O-linked Core-1 sialylated glycan at several sites. We examined apoE glycosylation (total and secondary) in a cohort of cerebrospinal fluid (CSF, n = 181) and plasma (n = 178) samples from the *Alzheimer's Disease Neuroimaging Initiative (ADNI)* stratified into 4 groups by cognitive trajectory over 4 years. RESULTS: **We observed decreasing glycosylation**

(reduced % of apoE being glycosylated) from apoE2 > apoE3 > apoE4 in the CSF and in plasma (apoE3 > apoE4), with stronger effect sizes for secondary glycosylation in CSF. In CSF, greater apoE4 glycosylation was associated with lower t-tau and p-tau181 levels. These results indicate strong associations between apoE glycosylation and biomarkers of AD pathology independent of apoE genotype, warranting a deeper understanding of the functional role of apoE glycosylation in AD tau pathology.

Li B, Wang S, Kerman B, Hugo C, Shwab EK, Shu C, Chiba-Falek O, Arvanitakis Z, H NY. Microglial States Are Susceptible to Senescence and Cholesterol Dysregulation in Alzheimer's Disease. Aging Cell. 2025:e70189. Epub 20250811. doi: 10.1111/acel.70189. PubMed PMID: 40790613.

We hypothesized that dysregulated cholesterol metabolism is central to cellular senescence in AD. We analyzed single-cell RNA-seq data from the *ROSMAP and SEA-AD* cohorts to uncover cell type-specific senescence pathologies. RESULTS: **Homeostatic, inflammatory, phagocytic, lipid-processing, and neuronal-surveillance microglial states were associated with AD-related senescence in both ROSMAP (152,459 microglia nuclei from six brain regions) and SEA-AD (82,486 microglia nuclei) via integrative analysis.** To validate these findings, we applied cholesterol-related senescence signatures to snRNA-seq data from iPSC-derived microglia (iMGs) exposed to myelin, Abeta, apoptotic neurons, and synaptosomes. **Treatment with AD-related substrates altered cholesterol-associated senescence signatures in iMGs**. DISCUSSION: This study provides the first human evidence that dysregulated cholesterol metabolism in microglia drives cellular senescence in AD. Targeting cholesterol pathways in senescent microglia is an attractive strategy to attenuate AD progression.

Wang S, Li B, Li J, Cai Z, Hugo C, Sun Y, Qian L, Tcw J, Chui HC, Dikeman D, Asante I, Louie SG, Bennett DA, Arvanitakis Z, Remaley AT, Kerman BE, Yassine HN. Cellular senescence induced by cholesterol accumulation is mediated by lysosomal ABCA1 in APOE4 and AD. Mol Neurodegener. 2025;20(1):15. Epub 20250204. doi: 10.1186/s13024-025-00802-7. PubMed PMID: 39901180; PMCID: PMC11792374.

We examined the association between cellular senescence and ABCA1 expression in human postmortem brain samples using transcriptomic, histological, and biochemical analyses. RESULTS: Transcriptomic profiling of the human dorsolateral prefrontal cortex from the *Religious Order Study/Memory Aging Project (ROSMAP)* cohort revealed the upregulation of cellular senescence transcriptome signatures in AD, which correlated with ABCA1 expression and oxysterol levels. **Greater caveolin-1 expression was observed in APOE4-TR mouse models and AD human brains.** Oxysterol induced mTORC1 activation and senescence were regulated by ABCA1 lysosomal trapping. **Treatment of APOE4-TR mice with cyclodextrin reduced brain oxysterol levels, ABCA1 lysosome trapping, mTORC1 activation, and attenuated senescence and neuroinflammation markers. In human iPSC-derived astrocytes, the reduction of cholesterol by cyclodextrin attenuated inflammatory responses**. CONCLUSIONS: Oxysterol accumulation in APOE4 and AD induced ABCA1 and caveolin-1 expression, contributing to lysosomal dysfunction and increased cellular senescence markers. This study provides novel insights into how cholesterol metabolism accelerates features of brain cellular senescence pathway and identifies therapeutic targets to mitigate these processes.

Weissberger GH, Yassine HN, Axelrod J, Fenton L, Noriega-Makarskyy DT, Molinare C, Williams JT, Oyen E, Kim P, Lai MHC, Han SD. Cognitive functioning predicts vulnerability to financial exploitation in APOE e4 carriers. Int Psychogeriatr. 2025:100162. Epub 20251107. doi: 10.1016/j.inpsyc.2025.100162. PubMed PMID: 41206326; PMCID: PMC12711538.

An emerging body of work suggests that **financial exploitation vulnerability (FEV)** may be an early marker of Alzheimer's disease (AD). We examined how APOE e4 genotype and cognition at baseline interact to predict subsequent FEV one year later. Participants were 95 older adults without dementia aged 50 or older (M age = 69.09, SD = 7.51; 72.63 % female; 72.63 % White non-Hispanic). Main effects of APOE and cognition were not found. However, **cognition and APOE status interacted to predict Year 2 FEV.** Specifically, worse cognitive functioning predicted higher FEV only for e4 carriers. Cognitive domain analyses revealed more nuanced relationships. **Findings suggest that cognition is a relevant risk factor of FEV for e4 carriers, and support the emerging idea that FEV may be an early marker of AD-related neuropathological changes.**

Vascular and Genetic Contributions to Cognitive Impairment and Dementia

Marmarelis V, Billinger S, Joe E, Shin D, Hashem S, Rizko J, Hazen E, Cardim D, Burns J, Zhang R, Chui HC. Dysregulation of cerebral perfusion dynamics is associated with Alzheimer's disease. Alzheimers Dement (Amst). 2025;17(3):e70134. Epub 20250718. doi: 10.1002/dad2.70134. PubMed PMID: 40689334; PMCID:

PMC12272300.

A novel physio-marker, termed "cerebrovascular dynamics index" (CDI), was developed to improve the diagnosis of MCI or dementia due to Alzheimer's disease (AD). The CDI quantifies the regulation dynamics of cerebral perfusion and oxygenation through **predictive dynamic modeling using relevant time-series data**. Participants were enrolled from *USC ADRC, U Kansas UDRC, and UT Southwestern* and RESULTS: Cross-sectional results demonstrated **excellent diagnostic performance of CDI in differentiating 90 MCI/AD patients from 77 controls (area under the curve (AUC) = 0.96)**, which surpassed the commonly used biomarker of amyloid positron emission tomography-standardized uptake value ratio (PET-SUVR) (AUC = 0.78) or cognitive screening tests of Mini-Mental State Examination (MMSE) and Montreal Cognitive Assessment (MoCA) (AUC = 0.91 and 0.92, respectively). **The CDI can also be used for disease staging because it differentiated MCI (n=56) from mild dementia (n=34) due to AD (AUC = 0.98).** These findings offer the promise of a high-performance diagnostic physio-marker for MCI and AD, which can be obtained in a comfortable, rapid, and automated manner in clinical settings.

Lohman T, Shenasa F, Sible I, Kapoor A, Engstrom AC, Dutt S, Head E, Sordo L, Alitin JPM, Gaubert A, Nguyen A, Pahlevan N, Nation DA. Blood pressure dynamic instability and neurodegeneration in older adults. J Alzheimers Dis. 2025;108(3):1276-86. Epub 20251017. doi: 10.1177/13872877251386443. PubMed PMID: 41105529.

We investigated whether systolic average real variability (ARV) and increased pulse pressure variability, a form of BPV associated with arterial stiffness (ASI), exacerbates markers of neurodegeneration among normal older adults (N = 105). Results: **We describe a significant interaction between higher ARV and higher ASI with left-sided medial temporal lobe atrophy in both the region-of-interest (left hippocampus beta = -252.79, p = 0.0002, right hippocampus beta = -193.56, p = 0.001, left entorhinal cortex beta = -0.13, p = 0.007**), using false discovery rate-corrected voxel-based morphometry analysis (p = 0.03). The interactive effect was **also significantly associated with increased plasma NfL (beta = 3.88, p = 0.01), but not GFAP**. Conclusions: The interaction between ARV and ASI is independently associated with neurodegenerative markers, including medial temporal lobe atrophy and plasma NfL, in older adults. These findings suggest that greater hemodynamic instability is associated with increased risk for neurodegenerative processes.

Engstrom AC, Dutt S, Alitin JPM, Kapoor A, Lohman T, Sible IJ, Marshall AJ, Shenasa F, Joyce JL, Gaubert A, Ferrer F, Nguyen A, Shao X, Wang DJJ, Nation DA. Age differences in spontaneous cerebrovascular reactivity at rest. Geroscience. 2025. Epub 20251030. doi: 10.1007/s11357-025-01935-9. PubMed PMID: 41166037.

Deficits in cerebral blood flow (CBF) and **cerebrovascular reactivity (CVR)** may indicate vulnerability to vascular dysfunction and cognitive impairment. Prior findings on age-related change in CVR are mixed, and no studies to date have investigated age group differences in spontaneous CVR (sCVR) at rest. The present study compared whole brain and regional sCVR at rest using capnography and pCASL-MRI between 24 healthy younger adults (ages 18-34) and 163 cognitively unimpaired older adults (ages 55-87). **Relative to younger adults, older adults had deficits in both whole brain and regional sCVR at rest. Among the regions examined, medial temporal lobe areas showed the largest age differences in sCVR at rest. Age differences in sCVR and CBF variability were larger than age differences in mean CBF.** Results show age-related deficits in sCVR are present even in cognitively unimpaired older adults, suggesting sCVR may be indicative of cerebrovascular aging. Further research should investigate sCVR as a means for early detection of age-related vascular dysfunction that may predate irreversible vascular brain injury.

Autosomal Dominant Alzheimer Disease (ADAD)

Leone R, Kobeleva X, Rowe B, Choupan J, Ringman JM, Barisano G. Alterations in MRI-visible perivascular spaces precede dementia diagnosis by 18 years in autosomal dominant Alzheimer's disease. Alzheimers Dement. 2025;21(8):e70588. doi: 10.1002/alz.70588. PubMed PMID: 40851076; PMCID: PMC12375433.

Perivascular space (PVS) alterations are traditionally linked to cardiovascular risk factors and aging but may also play a direct role in Alzheimer's disease (AD). In this cross-sectional study of 96 non-demented individuals (62 ADAD mutation carriers) from the *USC ADRC*, we quantified PVS count fraction and mean diameter in white matter and basal ganglia using automated magnetic resonance imaging analysis. RESULTS: **Compared to non-carriers, mutation carriers showed lower PVS count fraction in white matter and basal ganglia, and larger PVS diameter in basal ganglia and the temporal lobe.**

Changes were evident up to 18 years before expected dementia onset and followed trajectories similar to amyloid beta 42 and tau biomarkers. DISCUSSION: In ADAD mutations carriers, the temporal evolution of PVS alterations aligns with Abeta and tau changes in the cerebrospinal fluid.

Fernandez SG, Oria CG, Petit D, Annaert W, Ringman JM, Fox NC, Ryan NS, Chavez-Gutierrez L. Spectrum of gamma-Secretase dysfunction as a unifying predictor of ADAD age at onset across PSEN1, PSEN2 and APP causal genes. Mol Neurodegener. 2025;20(1):48. Epub 20250426. doi: 10.1186/s13024-025-00832-1. PubMed PMID: 40281586; PMCID: PMC12032737

In Autosomal Dominant Alzheimer's Disease (ADAD), Age At symptom Onset (AAO) is relatively consistent among carriers of the same PSEN1 mutation, but more variable for PSEN2 and APP variants, We performed biochemical assessment of gamma-secretase dysfunction on PSEN1, PSEN2 and APP mutations, including disease-associated, unclear and benign variants. RESULTS: **The integration of PSEN1, PSEN2 and APP correlation data shows parallel but shifted lines, suggesting a common pathogenic mechanism with gene-specific shifts in onset. We found overall "delays" in AAOs of 27 years for PSEN2 and 8 years for APP variants, compared to PSEN1. Notably, extremely inactivating PSEN1 variants delayed onset**. CONCLUSION: This study supports a **unified model of ADAD pathogenesis wherein gamma-secretase dysfunction and the resulting shifts in Abeta profiles are central to disease onset across all 3 of these causal genes.** Our findings also support the development of therapeutic strategies modulating gamma-secretase across different genetic ADAD forms and potentially more broadly in AD.

Biomarkers: Clinical, Blood, Brain & Retina

Jann K, Park G, Ringman JM, Kim H. Classifying mild cognitive impairment from normal cognition: fMRI complexity matches tau PET performance. Alzheimers Dement (Amst). 2025;17(3):e70159. Epub 20250812. doi: 10.1002/dad2.70159. PubMed PMID: 40799842; PMCID: PMC12340428.

In clinical settings, brain entropy mapping via resting-state functional magnetic resonance imaging (fMRI) has emerged as a marker of impaired brain function related to tauopathy. Convolutional neural networks, initially trained and evaluated via 5-fold cross-validation on ADNI data, were subsequently tested on an independent external cohort (EEAJ) using an ensemble approach. RESULTS: **The fMRI entropy classifier matched the tau PET model in accuracy and outperformed it in F1 score (0.64 vs. 0.61) and area under the curve (AUC; 0.73 vs. 0.67). On the independent external validation dataset (EEAJ), fMRI sample entropy showed a comparable F1 score (0.88) to tau PET (0.88) and achieved a notably higher AUC (0.94 vs. 0.92).** DISCUSSION: Our findings suggest that fMRI entropy could be a non-invasive imaging marker alternative to tau PET for detecting AD-related cognitive impairment.

Alishahi F, Beam CR, Gatz M, Schneider LS, Nation DA, Yassine HN, Kaplan H, Ganesan S, Pappas I, Davis DW, Zandi E. High precision and cost-effective multiplex quantification of amyloid-beta40, amyloid-beta42, p181Tau, p217Tau, neurofilament light chain, and glial fibrillary acidic protein from plasma and serum. J Alzheimers Dis. 2025;106(2):739-55. Epub 20250701. doi: 10.1177/13872877251340999. PubMed PMID: 40356374.

We aimed to develop a low-cost, sensitive, and accurate multiplex assay to quantify Abeta(40), Abeta(42), p181Tau, p217Tau, NfL, and GFAP biomarkers in plasma and serum based on a widely available technology. The utility of 5ADCSI was tested in matched cerebrospinal fluid (CSF) and plasma or serum of a cohort of cognitively normal (CN: n = 35), with mild cognitive impairment (MCI: n = 17), and with AD (n = 11) individuals. RESULTS: **The 5ADCSI demonstrated high specificity and sensitivity, with excellent precision. The AUC of receiver-operator characteristic curve for differentiating CN from AD for plasma/serum and CSF are 0.75, and 0.80 for Abeta(42/40), 0.95, 0.91 for p217Tau, 0.76, 0.81 for p181Tau, and 0.73 and 0.78for GFAP, respectively**. The wide availability of the base technology of 5ADCSI is an advantage over other similar methods and would allow cost-effective large-scale studies for validation of blood biomarkers for early diagnosis of AD.

Saca L, Gaggar R, Pappas I, Benzinger T, Reiman EM, Shiroishi MS, Joe EB, Ringman JM, Yassine HN, Schneider LS, Chui HC, Nation DA, Zlokovic BV, Toga AW, Chakhoyan A, Barnes S. Automatic detection of arterial input function for brain DCE-MRI in multi-site cohorts. Magn Reson Med. 2025. Epub 20250813. doi: 10.1002/mrm.70020. PubMed PMID: 40808286.

This work proposes a robust deep learning model that can precisely extract an **arterial input function (AIF)** from Dynamic Contrast Enhanced (DCE) MRI images. We conducted a retrospective analysis of DCE-MRI images from 289 participants, totaling 384 scans, from five institutions. A 3D UNet model was

implemented and trained on manually drawn AIF regions. This UNet was then applied to a separate dataset of 326 participants with a total of 421 DCE-MRI images with analyzed AIF quality and K(trans) values. RESULTS: **The resulting 3D UNet model achieved an average AIFitness score of 93.9 compared to 99.7 for manually selected AIFs in gray and white matter, and white matter K(trans) values were 0.45/min x 10(-3) and 0.45/min x 10(-3), respectively. The intraclass correlation between automated and manual K(trans) values was 0.89. The separate replication dataset yielded an AIFitness score of 97.0 and white matter K(trans) of 0.44/min x 10(-3)**. CONCLUSION: A 3D UNet model with additional convolutional neural network kernels and a modified Huber loss function derives AIFitness scores and DCE-MRI-derived metrics, such as K(trans) maps, with no significant differences in gray and white matter between manually drawn and automated AIFs.

Shou Q, Cen S, Chen NK, Ringman JM, Kim H, Jack CR, Jr., Borowski BJ, Senjem ML, Arani A, Wang DJJ, and for the Alzheimer's Disease Neuroimaging I. Generative diffusion model enables quantification of calibration-free arterial spin labeling perfusion magnetic resonance imaging data in an Alzheimer's disease cohort. Alzheimers Dement (Amst). 2025;17(4):e70214. Epub 20251030. doi: 10.1002/dad2.70214. PubMed PMID: 41181519; PMCID: PMC12572821.

M0 images were missing in Siemens ASL data in Alzheimer's Disease Neuroimaging Initiative (ADNI) dataset, prohibiting cerebral blood flow (CBF) quantification. METHODS: **A conditional latent diffusion model was trained and evaluated on in-house datasets, then applied to the Siemens missing data in ADNI-3**. RESULTS: The diffusion model generated M0 images with high fidelity (SSIM = 0.918 +/- 0.023, PSNR = 31.361 +/- 2.537) and minimal CBF bias (mean difference is 0.21 +/- 1.58 mL/100 g/min). Both generated and acquired CBF showed similar spatial patterns and decreasing trends with AD progression in specific AD-related regions. CONCLUSION: This study shows the **potential of diffusion models for imputing missing modalities in large-scale studies exploring the use of ASL as a biomarker of AD.** The performance of classification between Alzheimer's disease (AD) patients and cognitive normal people is better when using generated CBF maps than using non-quantitative perfusion images.

Lend Expertise to National and International Collaborations

Urich TJ, Tsiknia AA, Ali N, Park J, Mack WJ, Cortessis VK, Dinalo JE, Yassine HN. APOE epsilon4 and Dietary Patterns in Relation to Cognitive Function: An Umbrella Review of Systematic Reviews. Nutr Rev. 2025;83(7):e2113-e26. doi: 10.1093/nutrit/nuae156. PubMed PMID: 39499795; PMCID: PMC12166175.

Carrying the apolipoprotein epsilon4 allele (APOE epsilon4) is the strongest genetic risk factor for late-onset Alzheimer's disease. This umbrella review of systematic reviews evaluates the effect of **dietary interventions** on cognitive and brain-imaging outcomes by APOE status. RESULTS: *Six total reviews were included in the final analysis*. RESULTS: Both observational studies and clinical trials yielded inconclusive results attributed to both practical limitations associated with longitudinal follow-up and issues of methodological quality. **Except for the Mediterranean diet, dietary interventions, such as the ketogenic diet, nutraceuticals, and supplements, were generally not effective in older APOE epsilon4 carriers.** This review considers plausible biological mechanisms that might explain why older and cognitively impaired APOE epsilon4 carriers were less likely to benefit. CONCLUSION: This review identifies notable gaps in the literature, such as a **shortage of studies conducted in middle-aged and cognitively healthy APOE epsilon4 carriers assessing the impact of dietary interventions and provides suggestions for novel trial designs.**

Yassine HN, Hugo C, O'Donovan B, Stephens IO, Johnson LA, Cole G, Tcw J, Johansson J, Kisler K, Chiba-Falek O. APOE-Targeted Therapeutics for Alzheimer's Disease. J Neurosci. 2025;45(46). Epub 20251112. doi: 10.1523/JNEUROSCI.1388-25.2025. PubMed PMID: 41224653; PMCID: PMC12614056.

Apolipoprotein E (APOE) is the major lipid transport protein in the brain. Produced primarily by astrocytes and microglia, it delivers cholesterol and other lipids for membrane repair, synaptic maintenance, and immune regulation. Through interactions with specific lipid receptors, APOE maintains neuronal and vascular health. In this review, we organize the effects of **APOE4 into three interconnected "hits"** that modulate disease progression: **disrupted lipid handling, neurovascular inflammation, and neuronal dysfunction.** We review emerging **APOE-related therapeutic strategies** designed to address these mechanisms directly, **including structure correctors to restore normal protein folding, agents that enhance lipidation and receptor-mediated lipid transport, approaches that modulate downstream inflammatory responses, and gene therapies aimed at isoform switching or allele silencing.** By aligning therapeutic strategies with specific aspects of APOE4 biology, such approaches have the

potential to provide broader and safer benefits, complement existing disease-modifying therapies, and improve outcomes for individuals at heightened genetic risk of AD.

Ferdows NB, Aranda MP. Decomposing Racial and Ethnic Disparities in Risk and Protective Factors of Dementia in the U.S. Clin Gerontol. 2025:1-14. Epub 20250717. doi: 10.1080/07317115.2025.2534651. PubMed PMID: 40673372; PMCID: PMC12349561.

This study investigates racial/ethnic disparities in dementia risk and protective factors using data from the Health and Retirement Study (HRS) and the Harmonized Cognitive Assessment Protocol (HCAP). A retrospective analysis of 3,495 individuals aged 65+ from the 2016 HCAP linked to the HRS was conducted. RESULTS: **Black participants had 2.883 times higher odds of developing dementia compared to Whites, while Hispanic participants had 1.230 times higher odds (not statistically significant). Mid- and late-life risk and protective factors explained 32% of the cognitive gap between Black and White participants, and 70% between Hispanic and White participants.** Addressing disparities in education, wealth, cardiovascular risks, depression, and hearing loss can reduce cognitive dysfunction in older adults.

B.4 Training and Professional Development

- Core A – Administrative Core
  - February 25, 2025: CPBH Symposium: Advances in Personalized Brain
  - May 23, 2025: Southern California ADC Symposium (USC ADRC presentations: E Joe, A Hiniker)

- Core B - Clinical Core: Presentation (E Joe) at Southern California ADC Symposium; REC Seminars

- Core C - Data Core: supported: Weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.

- Core D - Neuropathology Core: Presentation (A Hiniker) at Southern California ADC Symposium. Two neuropathology fellows were trained during the 2024-2025 academic year.

- Core E- ORE Core: provided training and professional development to 8 PhD or MD professionals.

- Core F- Biomarker Core: 1 technician trained for processing biofluids

- Core G- Imaging Core: Weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.

- Core H- Research Education Component: We held an educational, multi-disciplinary symposium with 26 speakers attended by 80+ people in person (150+ total); planning next symposium for April 2026; regular REC seminars (n=13) with attendance averaging (n=29) for the current reporting period.

C. OVERALL PRODUCTS

C.1 PUBLICATIONS

Are there publications or manuscripts accepted for publication in a journal or other publication (e.g., book, one-time publication, monograph) during the reporting period resulting directly from this award?

Yes

Publications Reported for this Reporting Period

Public Access Compliance	Citation
Complete	Engstrom AC, Alitin JP, Kapoor A, Dutt S, Lohman T, Sible IJ, Marshall AJ, Shenasa F, Gaubert A, Ferrer F, Nguyen A, Bradford DR, Rodgers K, Sordo L, Head E, Shao X, Wang DJ, Nation DA. Spontaneous cerebrovascular reactivity at rest in older adults with and without mild cognitive impairment and memory deficits. medRxiv : the preprint server for health sciences. 2024 June 19. PubMed PMID: 38946941; PubMed Central PMCID: PMC11213117; DOI: 10.1101/2024.06.18.24309109.
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Complete	Digma LA, Young CB, Winer JR, Cody KA, Younes K, Sheng J, Insel PS, Rissman RA, Sperling R, Mormino EC. Continuum of Core 1 Biomarkers in Preclinical Alzheimer's Disease. medRxiv : the preprint server for health sciences. 2025 September 19. PubMed PMID: 41001450; PubMed Central PMCID: PMC12458534; DOI: 10.1101/2025.09.17.25336007.
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Complete	Ho PC, Peavy GM, Lee H, Gu Y, Kukull WA, Choi YB, Yu WH, Reyes-Dumeyer D, Henderson VW, Tee BL, Feldman HH, Li C, Yang HS, Lupo JL, Hsiung GR, Liu C, Wong EC, Mayeux R, Ta Park VM, Jun GR, Chui HC, Wang LS, Chow TW. Protocol for the Asian Cohort for Alzheimer's Disease (ACAD) Study. Alzheimer's & dementia (New York, N. Y.). 2025 October;11(4):e70189. PubMed PMID: 41439122; PubMed Central PMCID: PMC12720140; DOI: 10.1002/trc2.70189.
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Complete	Shou Q, Cen S, Chen NK, Ringman JM, Kim H, Jack CR Jr, Borowski BJ, Senjem ML, Arani A, Wang DJJ. Generative diffusion model enables quantification of calibration-free arterial spin labeling perfusion magnetic resonance imaging data in an Alzheimer's disease cohort. Alzheimer's & dementia (Amsterdam, Netherlands). 2025 October;17(4):e70214. PubMed PMID: 41181519; PubMed Central PMCID: PMC12572821; DOI: 10.1002/dad2.70214.
Complete	Shou Q, Cen S, Chen NK, Ringman JM, Kim H, Jack CR Jr, Borowski BJ, Senjem ML, Arani A, Wang DJJ. Generative diffusion model enables quantification of calibration-free arterial spin labeling perfusion magnetic resonance imaging data in an Alzheimer's disease cohort. Alzheimer's & dementia (Amsterdam, Netherlands). 2025 October;17(4):e70214. PubMed PMID: 41181519; PubMed Central PMCID: PMC12572821; DOI: 10.1002/dad2.70214.

Public Access Compliance	Citation
Complete	Sanjuan-Ruiz I, Serneels L, Craessaerts K, Goate A, Annaert W, Chávez-Gutiérrez L, Shi Y, Sheikh-Bahaei N, Jen JC, Ramos EM, Campan M, Ward PM, Magaki S, Bartlone K, Vinters HV, Craig DW, Ringman JM, De Strooper B. The curious case of a heterozygous loss-of-function PSEN1 variant associated with early-onset Alzheimer's disease. Molecular neurodegeneration advances. 2025 October 20;1(1):4. PubMed PMID: 41127441; PubMed Central PMCID: PMC12537781; DOI: 10.1186/s44477-025-00004-x.
Complete	Sanjuan-Ruiz I, Serneels L, Craessaerts K, Goate A, Annaert W, Chávez-Gutiérrez L, Shi Y, Sheikh-Bahaei N, Jen JC, Ramos EM, Campan M, Ward PM, Magaki S, Bartlone K, Vinters HV, Craig DW, Ringman JM, De Strooper B. The curious case of a heterozygous loss-of-function PSEN1 variant associated with early-onset Alzheimer's disease. Molecular neurodegeneration advances. 2025 October 20;1(1):4. PubMed PMID: 41127441; PubMed Central PMCID: PMC12537781; DOI: 10.1186/s44477-025-00004-x.
PMC Journal - In process	Engstrom AC, Dutt S, Alitin JPM, Kapoor A, Lohman T, Sible IJ, Marshall AJ, Shenasa F, Joyce JL, Gaubert A, Ferrer F, Nguyen A, Shao X, Wang DJJ, Nation DA. Age differences in spontaneous cerebrovascular reactivity at rest. GeroScience. 2025 October 30. PubMed PMID: 41166037; DOI: 10.1007/s11357-025-01935-9.
PMC Journal - In process	Engstrom AC, Dutt S, Alitin JPM, Kapoor A, Lohman T, Sible IJ, Marshall AJ, Shenasa F, Joyce JL, Gaubert A, Ferrer F, Nguyen A, Shao X, Wang DJJ, Nation DA. Age differences in spontaneous cerebrovascular reactivity at rest. GeroScience. 2025 October 30. PubMed PMID: 41166037; DOI: 10.1007/s11357-025-01935-9.
Complete	Edwards L, Gonzalez AI, Thomas KR, Smirnov DS, Brenner EK, Nation DA, Chang F, Blennow K, Salmon DP, Galasko D, Bangen KJ. Interactive effects of blood-brain barrier breakdown and Alzheimer's disease biomarker status on cognitive decline in older adults without dementia. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 November;21(11):e70910. PubMed PMID: 41216906; PubMed Central PMCID: PMC12603782; DOI: 10.1002/alz.70910.
Complete	Edwards L, Gonzalez AI, Thomas KR, Smirnov DS, Brenner EK, Nation DA, Chang F, Blennow K, Salmon DP, Galasko D, Bangen KJ. Interactive effects of blood-brain barrier breakdown and Alzheimer's disease biomarker status on cognitive decline in older adults without dementia. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 November;21(11):e70910. PubMed PMID: 41216906; PubMed Central PMCID: PMC12603782; DOI: 10.1002/alz.70910.
Complete	Oomens JE, Biber SA, Albert M, Alosco ML, Arfanakis K, Benzinger TLS, Brewer JB, Burns JM, Chin E, Chin NA, Clark LR, Cohen AD, Dage JL, Detre JA, Dickerson BC, DiFilippo FP, Donohue MC, van Dyck CH, Fan AP, Grabowski TJ, Habes M, Harrison TM, Hedden T, Hohman TJ, Jagust WJ, Jefferson AL, Jicha GA, Kecskemeti SR, Kantarci K, Keene DC, Knoefel JE, Kukull WA, Lin W, Lao P, Lepping RJ, Levendovszky SR, Masurkar AV, McConathy JE, Rivera-Mindt M, Morhardt DJ, Noll DC, Okonkwo OC, Parrish TB, Rabinovici GD, Rahman-Filipiak A, Reiman EM, Risacher SL, Rosen H, Rudolph MD, Schneider LS, Silbert LC, Song AW, Stark CEL, Swerdlow RH, Thompson PM, Vaillancourt D, Villemagne VL, Wolk DA, Qiu D, Foroud TM, Johnson SC, Mormino EC. The Consortium for Clarity in ADRD Research Through Imaging (CLARiTI): Overview of consortium sites and anticipated enrollment. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 November;21(11):e70880. PubMed PMID: 41208739; PubMed Central PMCID: PMC12598401; DOI: 10.1002/alz.70880.

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Complete	Oomens JE, Biber SA, Albert M, Alosco ML, Arfanakis K, Benzinger TLS, Brewer JB, Burns JM, Chin E, Chin NA, Clark LR, Cohen AD, Dage JL, Detre JA, Dickerson BC, DiFilippo FP, Donohue MC, van Dyck CH, Fan AP, Grabowski TJ, Habes M, Harrison TM, Hedden T, Hohman TJ, Jagust WJ, Jefferson AL, Jicha GA, Kecskemeti SR, Kantarci K, Keene DC, Knoefel JE, Kukull WA, Lin W, Lao P, Lepping RJ, Levendovszky SR, Masurkar AV, McConathy JE, Rivera-Mindt M, Morhardt DJ, Noll DC, Okonkwo OC, Parrish TB, Rabinovici GD, Rahman-Filipiak A, Reiman EM, Risacher SL, Rosen H, Rudolph MD, Schneider LS, Silbert LC, Song AW, Stark CEL, Swerdlow RH, Thompson PM, Vaillancourt D, Villemagne VL, Wolk DA, Qiu D, Foroud TM, Johnson SC, Mormino EC. The Consortium for Clarity in ADRD Research Through Imaging (CLARiTI): Overview of consortium sites and anticipated enrollment. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 November;21(11):e70880. PubMed PMID: 41208739; PubMed Central PMCID: PMC12598401; DOI: 10.1002/alz.70880.
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Complete	Spencer B, Quach B, Salehi S, Rissman RA. Systemic delivery of anti-sense oligonucleotide targeting a-synuclein for the treatment of multiple system atrophy. Research square. 2025 November 7. PubMed PMID: 41282154; PubMed Central PMCID: PMC12637822; DOI: 10.21203/rs.3.rs-7908950/v1.
Complete	Weissberger GH, Yassine HN, Axelrod J, Fenton L, Noriega-Makarskyy DT, Molinare C, Williams JT, Oyen E, Kim P, Lai MHC, Han SD. Cognitive functioning predicts vulnerability to financial exploitation in APOE e4 carriers. International psychogeriatrics. 2025 November 7:100162. PubMed PMID: 41206326; PubMed Central PMCID: PMC12711538; DOI: 10.1016/j.inpsyc.2025.100162.
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Complete	Seto M, Klinger HM, Clifton M, Janve VA, Brown JA, Birkenbihl C, Coughlan G, Townsend DL, Wang TC, Properzi M, Hanseeuw B, Chhatwal J, Yang HS, Rissman RA, Aisen P, Cuppels M, Donohue MC, Raman R, Johnson KA, Sperling RA, Dumitrescu L, Hohman TJ, Buckley RF. Blood-based Transcriptomics Reveal Sex- and Amyloid-Modulated Biology of Plasma pTau217 in Preclinical Alzheimer's Disease. bioRxiv : the preprint server for biology. 2025 November 24. PubMed PMID: 41394556; PubMed Central PMCID: PMC12697722; DOI: 10.1101/2025.11.21.689770.
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Complete	Lohman T, Kapoor A, Engstrom AC, Joyce J, Quiring M, Alitin JPM, Gaubert A, Nguyen A, Head E, Malhas R, Rodgers KE, Bradford D, Lashin B, Han SD, Mather M, Nation DA. Low frequency BOLD oscillations, APOE4, and plasma pTau(217). medRxiv : the preprint server for health sciences. 2025 November 27. PubMed PMID: 41358315; PubMed Central PMCID: PMC12676400; DOI: 10.1101/2025.11.25.25340991.
Complete	Janve VA, Seto M, Sperling RA, Aisen PS, Rissman RA, Koran MEI, Dumitrescu L, Buckley RF, Hohman TJ. Blood gene expression network expression strongly relates to brain amyloid burden. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 December;21(12):e70982. PubMed PMID: 41416491; PubMed Central PMCID: PMC12715697; DOI: 10.1002/alz.70982.
Complete	Janve VA, Seto M, Sperling RA, Aisen PS, Rissman RA, Koran MEI, Dumitrescu L, Buckley RF, Hohman TJ. Blood gene expression network expression strongly relates to brain amyloid burden. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 December;21(12):e70982. PubMed PMID: 41416491; PubMed Central PMCID: PMC12715697; DOI: 10.1002/alz.70982.
Complete	Lin PJ, Jiang Z, Liu Y, Li W, Qi Y, Sun J, Ni S, Shu L, Xia M, Toga AW. A vision-language foundation model for Alzheimer's disease diagnosis using MRI and clinical data. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 December;21(12):e71029. PubMed PMID: 41457461; PubMed Central PMCID: PMC12745493; DOI: 10.1002/alz.71029.
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Complete	Njamnshi WY, Chiotis K, Tsoy E, Blazhenets G, Maiti P, Zhang J, Allen IE, Rocha S, Shankar RP, Amuiri A, Lawson M, Aisen P, Beckett L, Eloyan A, Hammers DB, Kirby K, Koeppe R, Kukull WA, Toga AW, Atri A, Clark D, Day GS, Duara R, Graff-Radford NR, Grant I, Honig LS, Johnson ECB, Jones DT, Masdeu J, Mendez MF, Musiek E, Onyike CU, Riddle M, Rogalski E, Salloway S, Sha SJ, Turner RS, Wingo TS, Wolk DA, Womack K, Kramer JH, Touroutoglou A, Brickhouse MJ, Jack CR Jr, Vemuri P, Taurone A, Thangarajah M, Nudelman K, Carrillo MC, Dickerson BC, Apostolova LG, Possin KL, Rabinovici GD, La Joie R. Tablet-based Cognitive Assessment Tool (TabCAT) in a multisite study of early-onset Alzheimer's disease. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 December;21(12):e71012. PubMed PMID: 41395729; PubMed Central PMCID: PMC12703652; DOI: 10.1002/alz.71012.

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Complete	Njamnshi WY, Chiotis K, Tsoy E, Blazhenets G, Maiti P, Zhang J, Allen IE, Rocha S, Shankar RP, Amuiri A, Lawson M, Aisen P, Beckett L, Eloyan A, Hammers DB, Kirby K, Koeppe R, Kukull WA, Toga AW, Atri A, Clark D, Day GS, Duara R, Graff-Radford NR, Grant I, Honig LS, Johnson ECB, Jones DT, Masdeu J, Mendez MF, Musiek E, Onyike CU, Riddle M, Rogalski E, Salloway S, Sha SJ, Turner RS, Wingo TS, Wolk DA, Womack K, Kramer JH, Touroutoglou A, Brickhouse MJ, Jack CR Jr, Vemuri P, Taurone A, Thangarajah M, Nudelman K, Carrillo MC, Dickerson BC, Apostolova LG, Possin KL, Rabinovici GD, La Joie R. Tablet-based Cognitive Assessment Tool (TabCAT) in a multisite study of early-onset Alzheimer's disease. Alzheimer's & dementia : the journal of the Alzheimer's Association. 2025 December;21(12):e71012. PubMed PMID: 41395729; PubMed Central PMCID: PMC12703652; DOI: 10.1002/alz.71012.
Complete	Kelly ML, Lipior S, Schaefer SY, Han SD, Leech KA. Standardizing the Measurement and Definition of Post-Stroke Cognitive Impairment: Implications for Stroke Recovery and Rehabilitation. Neurorehabilitation and neural repair. 2025 December 30:15459683251399133. PubMed PMID: 41467585; PubMed Central PMCID: PMC12755724; DOI: 10.1177/15459683251399133.
Complete	Kelly ML, Lipior S, Schaefer SY, Han SD, Leech KA. Standardizing the Measurement and Definition of Post-Stroke Cognitive Impairment: Implications for Stroke Recovery and Rehabilitation. Neurorehabilitation and neural repair. 2025 December 30:15459683251399133. PubMed PMID: 41467585; PubMed Central PMCID: PMC12755724; DOI: 10.1177/15459683251399133.
In Process at NIHMS	GPU Accelerated Modeling of Cortical Radial and Tangential Connectivity Changes in Neurodegeneration. Medical image computing and computer-assisted intervention : MICCAI ... International Conference on Medical Image Computing and Computer-Assisted Intervention.
In Process at NIHMS	Robust Topographical Representation for Longitudinal Propagation of Tau Pathology. Medical image computing and computer-assisted intervention : MICCAI ... International Conference on Medical Image Computing and Computer-Assisted Intervention.
In Process at NIHMS	A call to address structural barriers to the representation of Hispanics/Latinos in clinical trials on Alzheimer’s disease and related dementias: A micro-meso-macro perspective. Alzheimer's & dementia (New York, N. Y.).
In Process at NIHMS	Age differences in spontaneous cerebrovascular reactivity at rest. GeroScience.
In Process at NIHMS	Impaired parasympathetic network function confers susceptibility to neurodegeneration and memory impairment in older adults with elevated beat-to-beat blood pressure variability. Journal of Alzheimer's disease : JAD.

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Category	Explanation
Other	https://adrc.usc.edu/
Other	https://brainhealth.usc.edu
Other	https://keck.usc.edu/cpbh/

C.3 TECHNOLOGIES OR TECHNIQUES

Category	Explanation
Other	We developed a sensitive assay to measure total APOE and APOE4 protein levels in CSF and plasma.
Other	We developed a sensitive assay to detect a brain microvascular pericyte injury biomarker sPDGFRβ in the cerebrospinal fluid (CSF) that indicates BBB disruption (See section C.5).

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Have inventions, patent applications and/or licenses resulted from the award during the reporting period? No

If yes, has this information been previously provided to the PHS or to the official responsible for patent matters at the grantee organization? No

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C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

Category	Explanation
Other	We have previously validated sPDGFRβ as a sensitive and novel biomarker of pericyte injury (Sagare et al., Neuroscience Letters, 2015; 607:97-101) in the brain that reflects BBB disruption in MCI individuals (Montagne et al., Neuron, 2015; 85, 296–302), independent of Aβ and tau (Nation et al., Nature Medicine, 2019; 25(2):270-276).
Other	We have identified a combination of capture and detecting antibodies for the development of the Meso Scale Discovery (MSD) assay to detect soluble sPDGFRβ and CypA (Sweeney et al., Alzheimer's & Dementia, 2020; 16(6):821–830).
Other	We recently developed a sensitive assay to measure total APOE and APOE4 protein levels in CSF and plasma.
Protocols	Protocols: Additionally, Biomarkers Core has developed new assays to measure CSF and plasma levels of soluble platelet-derived growth factor-β (sPDGFRβ), CSF cyclophilin-A (CypA), and MMP9 levels.
Other	We have found that high baseline levels of sPDGFRβ predicted future cognitive decline in APOE4 carriers (Montagne et al., Nature, 2020; 581:71-76).
Protocols	Protocols: The Biomarkers Core has developed a novel panel of molecular biomarkers in CSF/plasma that integrate neurovascular unit (NVU) cell- and system-specific biomarkers with established AD biomarkers such as Aβ, total tau, and pTau, to simultaneously analyze changes in vascular/BBB injury markers (Qalb, fibrinogen, Cyclophilin-A, active MMP9 and ferritin) in relation to inflammatory, neuronal injury, Aβ and tau biomarkers.

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

See the ORE and neuropathology cores for details on clinical core resource sharing of clinical materials and research specimens. See Administrative Tables and overall component for description of resource sharing.

C.5.c Data Management and Sharing

Data Type	Has data been generated to date?	Has the data been shared?	Status of Data Sharing	Repository	Unique Identifiers/Digital Object Identifiers
To better understand the ATNv biomarker data for our ADRC clinical participants, the Data Core began compiling and consolidating A, T, N, and vascular biomarker data from multiple sources. Specifically, we extract data from NACC UDS (demographics), NCRAD (plasma and APOE genotype), IDA (MRI/PET scans), and additional data from project-specific databases (e.g., VCS/PPG, DVR, etc.) stored in the IDA system. This aims to track the availability of ATNv data and characterize ADRC clinical participants who are actively being followed.	Y	N	The data has been accessed by USC faculty and staff but there are no DOIs from the usage of that data yet.	Not Applicable	Not Applicable

Description:
The data has been accessed by USC faculty and staff but there are no DOIs from the usage of that data yet. 64 individuals have downloaded clinical data from one or more of the studies ('vcd','ppg','vcsgt','dvr'). All but eight of them have a USC email address. Investigators can apply for data access by completing the ADRC Data Use Agreement (DUA), which is available on the USC ADRC website

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Are significant prospective changes to the Data Management and Sharing Plan being requested for the coming year (e.g., change in repository, change in timeline, or change in scientific direction)?
No Change

D. OVERALL PARTICIPANTS

D.1 WHAT INDIVIDUALS HAVE WORKED ON THE PROJECT?

Commons ID	Sr/Key	Name	ORCID	Degree(s)	Role	Cal	Aca	Sum	Foreign Org	Component(s)	Country	SS	Common Forms Documentation
CHANGCHUI	Y	CHUI, HELENA Chang		MD	PD/PI	1.6	0.0	0.0		Admin-Core-001 (Core A: Administrative Core)		NA	Chui OS.pdf
TOGA22	Y	TOGA, ARTHUR W		PHD,BS,MS	PD/PI	3.1	0.0	0.0		Additional-Cores-001 (Core H: Imaging Core), Admin-Core-001 (Core A: Administrative Core), Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	AToga_OS_commonsform_1.9.26.pdf
ARANDA	Y	Aranda, Maria P.		PHD,BS,MOTH,MPA	Co-Investigator	1.5	0.0	0.0		Recruitment-Core-001 (Core E: Outreach, Recruitm…nt (ORE) Core)		NA	ARANDA,Maria_CurrentPendingSupport_
BIENKOWSKI2	Y	Bienkowski, Michael	0000-0002-8231-8893	BS,PHD	Co-Investigator	0.5	0.0	0.0		Research-Education-001 (Core G: Research Education…mponent (REC))		NA	BienkowskiOS_cpos-1922122.pdf
MBRASKIE	Y	Braskie, Meredith Nicole	0000-0002-6439-4243	PHD,BOTH	Co-Investigator	0.6	0.0	0.0		Additional-Cores-001 (Core H: Imaging Core)		NA	Braskie_Commons_OS_2026_01_12_CER
LINADOR	Y	D'Orazio, Lina			Co-Investigator	1.7	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	D'Orazio_OS_01192026.pdf
DHAN06	Y	HAN, DUKE	0000-0003-2453-9230	MA,BS,PHD	Co-Investigator	0.7	0.0	0.0		Research-Education-001 (Core G: Research Education…mponent (REC))		NA	Han cpos-1872450.pdf
MIKETHESCOT	Y	Harrington, Michael Gordon	0000-0002-7923-8032	MB	Co-Investigator	1.2	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	Harrington cpos-1947628.pdf
DHAWES	Y	Hawes, Debra		MD	Co-Investigator	1.2	0.0	0.0		Neuropathology-Core-001 (Core D: Neuropathology Core)		NA	

Commons ID	Sr/Key	Name	ORCID	Degree(s)	Role	Cal	Aca	Sum	Foreign Org	Component(s)	Country	SS	Common Forms Documentation
HINIKER	Y	Hiniker, Annie E	0000-0002-0055-6610	AB,MD,PHD	Co-Investigator	1.8	0.0	0.0		Neuropathology-Core-001 (Core D: Neuropathology Core)		NA	HIniker cpos-2030378.pdf
EJOE01	Y	Joe, Elizabeth Bartelt	0000-0002-5222-0772	MD	Co-Investigator	1.9	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	JoeE OS Jan.pdf
KIRSTENML	Y	Lynch, Kirsten Mary	0000-0003-3976-5783	BS,PHD	Co-Investigator	2.7	0.0	0.0		Additional-Cores-001 (Core H: Imaging Core)		NA	KLynch_OS.pdf
DANIEL_NATION108	Y	Nation, Daniel A	0000-0002-9123-0658	BA,MS,PHD	Co-Investigator	0.8	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	Nation.pdf
PAWLUCZYK	Y	Pawluczyk, Sonia	0009-0005-4723-6031		Co-Investigator	1.1	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	SPawluczyk ScienCV OS_Jan 2026_2026
RINGMAN2	Y	RINGMAN, JOHN M	0000-0001-5429-0777	MD,MS,BA	Co-Investigator	2.0	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core), Recruitment-Core-001 (Core E: Outreach, Recruitm…nt (ORE) Core)		NA	Ringman OS_1.27.2026.pdf
RRISSMAN	Y	Rissman, Robert	0000-0001-9245-8278	PHD,BA	Co-Investigator	1.0	0.0	0.0		Biomarker-Core-001 (Core F: Biomarker Core)		NA	Rissman cpos-2025746.pdf
LSCHNEID	Y	SCHNEIDER, LON S	0000-0002-4636-6150	MS,MD	Co-Investigator	1.0	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	Schneider OS 2026 Jan29 cpos-2009363
SEGALGI	Y	SEGAL-GIDAN, FREDDI I	0000-0002-4776-9840	BS,PHD,BA,OTH	Co-Investigator	0.5	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	Segal_Gidan OS 012126.pdf

Commons ID	Sr/Key	Name	ORCID	Degree(s)	Role	Cal	Aca	Sum	Foreign Org	Component(s)	Country	SS	Common Forms Documentation
NASIM.SHEIKH-BAHAEI	Y	Sheikh-Bahaei, Nasim	0000-0001-7029-7215		Co-Investigator	1.5	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	Nasim Sheikh-Bahaei Other support 2026
YONGGANGSHI	Y	Shi, Yonggang	0000-0002-3466-4302	PHD	Co-Investigator	0.5	0.0	0.0		Additional-Cores-001 (Core H: Imaging Core)		NA	Shi_OS Jan26.pdf
ATOWFIGHI	Y	Towfighi, Amytis	0000-0002-9565-6787	MD,BS	Co-Investigator	0.3	0.0	0.0		Recruitment-Core-001 (Core E: Outreach, Recruitm…nt (ORE) Core)		NA	Towfighi_Other Support ADRC 2026.pdf
XINHUI_WANG	Y	Wang, Xinhui	0000-0002-1180-4624		Co-Investigator	3.2	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	Xinhui Wang OS 011626.pdf
HYASSINE	Y	Yassine, Hussein N	0000-0002-2483-649X	MD	Co-Investigator	1.4	0.0	0.0		Admin-Core-001 (Core A: Administrative Core), Research-Education-001 (Core G: Research Education…mponent (REC))		NA	cpos-1989741 Yassine 012926.pdf
ZISS01	Y	ZISSIMOPOULOS, JULIE M	0009-0006-7008-132X	PHD	Co-Investigator	0.3	0.0	0.0		Research-Education-001 (Core G: Research Education…mponent (REC))		NA	Zissimopoulos_OS2026.pdf
	N	Ard, Tyler			Technician	1.0	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	
	N	Avina, Miguel			Technician	8.5	0.0	0.0		Neuropathology-Core-001 (Core D: Neuropathology Core)		NA	
	N	Coleman, Nikaya			Technician	2.6	0.0	0.0		Biomarker-Core-001 (Core F: Biomarker Core)		NA	
	N	Ngayan, Pedro			Technician	1.1	0.0	0.0		Neuropathology-Core-001 (Core D: Neuropathology Core)		NA	
	N	Aksman, Leon			Staff scientist (Doctoral level)	2.7	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	

Commons ID	Sr/Key	Name	ORCID	Degree(s)	Role	Cal	Aca	Sum	Foreign Org	Component(s)	Country	SS	Common Forms Documentation
	N	Sagare, abhay			Staff scientist (Doctoral level)	0.6	0.0	0.0		Biomarker-Core-001 (Core F: Biomarker Core)		NA	
	N	Ali, Nada			Administrator	5.2	0.0	0.0		Research-Education-001 (Core G: Research Education…mponent (REC))		NA	
	N	Aranda, Michelle			Administrator	0.6	0.0	0.0		Admin-Core-001 (Core A: Administrative Core)		NA	
	N	Kolchins, Tiffany			Co-Administrator	2.5	0.0	0.0		Admin-Core-001 (Core A: Administrative Core)		NA	
	N	Boudreau, Julia			Data Manager	1.9	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
RIJWAN.RIJU01	Y	Ahammad, Rijwan Uddin	0000-0002-1387-1943	PHD	Development Project Awardee	0.0	0.0	0.0		Project-8229 (7.3_APOE-Dependent Impact …y_Ahammad_new)		NA	Rijwan (NIH BIO_cv-2034093.pdf Rijwan_cpos-2003528.pdf
MARYAMFOTOUHI	Y	Fotouhi, Maryam	0000-0003-4569-7094	MD	Development Project Awardee	0.0	0.0	0.0		Project-8231 (7.5_Platelet-derived growt…_ Fotouhi_new)		NA	Biosketch_Jan2025_MFotouhi.pdf OS_M.Fotouhi.pdf
HOSUNGKIM	Y	Kim, Hosung	0000-0002-2269-8644		Development Project Awardee	0.0	0.0	0.0		Project-8212 (7.1_Glymphatic Dysfunction…sease_Kim_new)		NA	NIH biosketch_H Kim_cv-2032257.pdf cpos-1999821.pdf
SILVERBOW	Y	Kisler Elliott, Kassandra	0000-0002-4161-2498	PHD,BOTH,BS	Development Project Awardee	0.0	0.0	0.0		Project-8228 (7.2_Synergistic Impact of …gy_Kisler_new)		NA	Kisler NIH Biosketch New Format 01.21.2 Kisler Other Support New Format 01.30.2
CIORANA	Y	Roman Ortiz, Ciorana	0000-0002-3554-4943	PHD,BS	Development Project Awardee	0.0	0.0	0.0		Project-8230 (7.4_APOE ε4-linked taste p…man-Ortiz_new)		NA	NIH Biographical Sketch Common Form_ADRC_CRO.pdf cpos-2000207 (1).pdf
THORWALD	Y	Thorwald, Max Andrew	0000-0003-0095-5344	PHD,BS	Development Project Awardee	0.0	0.0	0.0		Project-8232 (7.6_Iron Chelation as a St…_Thorwald_new)		NA	Thorwald SciENcv 2026 ADRC DP.pdf OS_M.Thorwald_1.23.26.pdf

Commons ID	Sr/Key	Name	ORCID	Degree(s)	Role	Cal	Aca	Sum	Foreign Org	Component(s)	Country	SS	Common Forms Documentation
	N	Mullen, Kimberley			Lab Assistant	1.7	0.0	0.0		Biomarker-Core-001 (Core F: Biomarker Core)		NA	
	N	Barlow, James			Lab Manager	2.4	0.0	0.0		Biomarker-Core-001 (Core F: Biomarker Core)		NA	
	N	Pedroza, Adriana			NP Tester	6.8	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
	N	Montoya, Lucy			Program Manager	0.6	0.0	0.0		Admin-Core-001 (Core A: Administrative Core)		NA	
	N	Church, Carol			Project Assistant	9.0	0.0	0.0		Neuropathology-Core-001 (Core D: Neuropathology Core)		NA	
	N	Estrada, Monica			Project Assistant	1.1	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	
	N	Fernandez, Emily			Project Assistant	2.0	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
	N	Ferman, Diana			Research Associate	2.7	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
	N	Sibilia, Francesca			Research Associate	2.6	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	
	N	Ayala, Cynthia			Research Coordinator	3.4	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
	N	Teodoro, Liberty			Research Nurse	1.6	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	
	N	Luo, James			Systems Analyst	5.2	0.0	0.0		Data-Management-Core-001 (Core C: Data Management an…tistical Core)		NA	
	N	Slowinska, Maya			UDS Coordinator	1.4	0.0	0.0		Clinical-Core-001 (Core B: Clinical Core)		NA	

Glossary of acronyms: Sr/Key - Senior/Key Cal - Person Months (Calendar) Aca - Person Months (Academic) Sum - Person Months (Summer)	Foreign Org - Foreign Organization Affiliation SS - Supplement Support RS - Reentry Supplement DS - Diversity Supplement OT - Other NA - Not Applicable

D.2 PERSONNEL UPDATES

D.2.a Level of Effort

Will there be, in the next budget period, either (1) a reduction of 25% or more in the level of effort from what was approved by the agency for the PD/PI(s) or other senior/key personnel designated in the Notice of Award, or (2) a reduction in the level of effort below the minimum amount of effort required by the Notice of Award?

Yes

David Conti, MS, PhD, has stepped down from his role as Co-Leader, Data Management and Statistics Core at USC to assume the position of Chair of Biostatistics & Informatics at ColoradoSPH, starting January 1, 2026. Leadership at the USC ADRC is currently reviewing potential candidates from the School of Population and Public Health to fill the vacancy.

Caleb Finch PhD (Founding Director of the USC ADRC 1984-2003) retired in January 2025. We are in the process of assembling a new IAC, which will be chaired by Carolyn Meltzer (Dean Keck School of Medicine.

D.2.e Multi-PI (MPI) Leadership Plan

Will there be a change in the MPI Leadership Plan for the next budget period?

No

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Chui, Helena Chang

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-1348-0931

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: USC Department of Neurology, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Effect of Diabetes and AD Pathology on Brain Imaging and Cognition in Latino Adults
***Status of Support:**	Current
Proposal/Award Number:	4R01AG078362
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	08/2027
***Total Anticipated Proposal/Project Amount:**	$2,682,813

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6

***Overall Objectives:** The overall goal is to gain an understanding of how metabolic and Alzheimer's risk relate to brain measures and cognition in 200 Northern American Latino middle-aged adults, an understudied group that is at higher risk for both T2DM and AD compared with non-Latino Whites.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Multi-Organ Chart of Personalized Susceptibility of Alzheimer's Disease and Aging

***Status of Support:** Current

Proposal/Award Number: RF1AG092412

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** $3,462,615

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.36
2027	0.36
2028	0.36
2029	0.36

***Overall Objectives:** The proposed Multi-organ Chart has a substantial impact on public health by potentially charting susceptibility to Alzheimer's disease and aging at an individual level. It employs advanced artificial intelligence and machine learning (AI/ML) methodologies, coupled with multi-omics (MRI, genetics, and proteomics) and multi-organ approaches (brain, eye, and heart), to comprehensively model the neurobiological and genetic underpinnings involved in the disease. The derived individualized susceptibility scores can improve precision in diagnosis and prognosis compared to previous single-organ and single-omics approaches, ultimately improving healthcare outcomes

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: P30AG066530

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,693,508

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.6
2027	1.6
2028	1.6
2029	1.6
2030	0.4

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Asian Cohort for Alzheimer Disease (ACAD)

***Status of Support:** Current

Proposal/Award Number: U19AG079774

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2023

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $8,325,999

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.45
2027	0.45
2028	0.22

***Overall Objectives:** The ACAD project will build the first major AD genetics study for Asians in US and Canada. Successful completion will lead to new genetic and lifestyle screening markers for Asian Americans and insights about novel therapeutic targets for AD. ACAD will be a first network for recruiting and studying AD in Asian Americans, serving the unmet needs of Alzheimer's disease research for Asian Americans.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Neurovascular calcification, Alzheimer's Disease and Related Dementias in Two Native South American Populations
***Status of Support:**	Current
Proposal/Award Number:	RF1AG082201
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2023
***Proposal/Active Project End Date: (MM/YYYY):**	05/2026
***Total Anticipated Proposal/Project Amount:**	$2,452,559

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3

***Overall Objectives:** To clarify how intracranial internal carotid artery calcification (iICAC) can affect risk for Alzheimer's disease and related dementias (ADRD), we propose to study the relationship between regional brain volume and iICAC in two native South American populations with high iICAC prevalence: the Tsimane and Moseten people of Bolivia.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The Role of Sex in GABAergic-mediated, Alzheimer's disease-related episodic memory impairments from mid to late life
***Status of Support:**	Current
Proposal/Award Number:	R01AG079512
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2022
***Proposal/Active Project End Date: (MM/YYYY):**	06/2026
***Total Anticipated Proposal/Project Amount:**	$810,935

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.36

***Overall Objectives:** This project proposes and investigates a GABAergic model in humans, testing the impact of sex on GABA, brain activity, and episodic memory in middle-aged and older adults at genetic risk for developing AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Large-scale harmonization and integration of multi-modal ADNI data for the early detection of Alzheimer's disease and related dementias
***Status of Support:**	Current
Proposal/Award Number:	R01NS128486
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2022
***Proposal/Active Project End Date: (MM/YYYY):**	05/2027
***Total Anticipated Proposal/Project Amount:**	$794,031

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12
2027	0.12

***Overall Objectives:** The goal is to identify specific AD and ADRD disease pathology markers and how they evolve. This project aims to improve the early detection of AD and ADRD so that future disease-modifying therapy can be allocated more efficiently to patients.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Validation of Imaging and Blood-based Small Vessel VCID Biomarkers in Multi-ethnic Population
***Status of Support:**	Current
Proposal/Award Number:	UFINS100614 – Phase II
***Source of Support:**	NINDS
***Primary Place of Performance:**	University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2022

***Proposal/Active Project End Date: (MM/YYYY):** 05/2027

***Total Anticipated Proposal/Project Amount:** $2,623,751

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48

***Overall Objectives:** Cerebral small vessel disease (SVD) is the most common vascular cause of dementia, a major contributor to mixed Alzheimer disease and vascular dementia, and the cause of about one fifth of all strokes worldwide. The primary objective of the next 5 years is to carry out comprehensive multi-site clinical validation of candidate VCID biomarkers in longitudinal studies of diverse populations that are typical in clinical setting in the US.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer disease

***Status of Support:** Current

Proposal/Award Number: 5P01AG052350-08

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6
2026	0.54
2027	0.54

***Overall Objectives:** This program project focuses on alterations in the neurovascular unit (e.g., blood-brain barrier, pericytes) in persons carrying the apolipoprotein E4 and autosomal dominant AD mutations to clarify vascular contributions in AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Testing Hypothesized Pathways Linking Infection, Physical Activity, ApoE Genotype, and Biological Sex To Low Dementia Prevalence And Reduced Brain Atrophy In Two Native American Populations
***Status of Support:**	Current
Proposal/Award Number:	R01AG054442
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2022
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$14,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	0.3

***Overall Objectives:** This research will investigate the risk of cognitive impairment, dementia and brain atrophy in relation to physical activity, infection and immune pathways, biological sex, and APOE genotype among two cohorts of Native Americans whose lifestyle and environment require high levels of physical activity and expose them to high pathogen burdens, most like the human preindustrial past.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Quantitative Blood Vessel Biomarkers in AD and VCID
***Status of Support:**	Current
Proposal/Award Number:	R01AG072490
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2021
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$2,193,562

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.15

***Overall Objectives:** The goal of this application is to develop a class of cerebral blood vessel hemodynamic and morphological metrics and to characterize cerebral blood vessel dysfunction in AD and VCID.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** The Clinical Significance of Incidental White Matter Lesions on MRI Amongst a Diverse Population with Cognitive Complaints (INDEED)

***Status of Support:** Current

Proposal/Award Number: 5U19NS120384

***Source of Support:** NINDS

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $965,334

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48
2027	0.12

***Overall Objectives:** This study proposes to comprehensively examine the impact of individual and combined MRI measures of white matter injury on cognitive performance among a diverse, non-demented, stroke free population with cognitive complaints over an extended period of observation.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Deep neural networks to study mild traumatic brain injury as a risk factor for Alzheimer's disease

***Status of Support:** Pending

Proposal/Award Number: N/A

***Source of Support:** Department of Defense

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2026

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** $988,878

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.24
2028	0.24
2029	0.24

***Overall Objectives:** We hypothesize that TBI alters neuroanatomical and biomarker-defined aging trajectories toward ADRD in ways that can be quantified from structural MRI and from demographics, education, and genetics (e.g., APOE) to forecast near- to mid-term adverse cognitive change.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Mechanisms of Cerebral Amyloid Angiopathy and Neurovascular Inflammation in Alzheimer's Disease

***Status of Support:** Pending

Proposal/Award Number: pending

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2030

***Total Anticipated Proposal/Project Amount:** $4,166,435

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6

Year	Person Months
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** This research addresses a critical public health challenge by exploring novel therapeutic strategies to reduce neurovascular inflammation and cerebral amyloid angiopathy (CAA), key contributors to Alzheimer's disease (AD) progression.

***Statement of Potential Overlap:** None

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Chui, Helena in SciENcv on 2026-01-27 16:09:12

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: TOGA, ARTHUR

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-7902-3755

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: University of Southern California

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Global Alzheimer's Association Interactive Network (GAAIN)
***Status of Support:**	Current
Proposal/Award Number:	GAAIN-2-20-678486
***Source of Support:**	Alzheimer's Association
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2026
***Proposal/Active Project End Date: (MM/YYYY):**	12/2026

***Total Anticipated Proposal/Project Amount:** $999,731

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.01

***Overall Objectives:** The Global Alzheimer's Association Interactive Network (GAAIN) unites a diverse and geographically distributed network of data partners within a federated data platform designed to foster cohort discovery, collaboration and sharing. GAAIN represents the first open access, federated Alzheimer's disease data discovery platform of its kind. GAAIN is advancing research into the causes, prevention and treatment of Alzheimer's and other neurodegenerative diseases through a global cooperative of sharing, investigation and discovery. Using GAAIN, researchers can discover clinical, genetic, imaging and other data collected across many independent studies; build cohorts and connect with GAAIN data partners.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** MJFF Bioinformatics: Infrastructure, data aggregation, distribution, and compliance activities in PPMI and complementary research efforts

***Status of Support:** Current

Proposal/Award Number: MJFF-017025/MJFF-028768

***Source of Support:** Michael J. Fox Foundation

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2026

***Proposal/Active Project End Date: (MM/YYYY):** 12/2026

***Total Anticipated Proposal/Project Amount:** $1,524,559

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

***Overall Objectives:** LONI's overarching objective is to provide an informatics infrastructure wherein data from multiple MJFF initiatives are safely and securely housed, integrated, harmonized, described and made available for researchers to explore and download.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Multi-Organ Chart of Personalized Susceptibility to Alzheimer's Disease and Aging

***Status of Support:** Current

Proposal/Award Number: 1 RF1AG092412-01

***Source of Support:** NIH

***Primary Place of Performance:** Columbia University

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** $75,797

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06
2027	0.06
2028	0.06
2029	0.06

***Overall Objectives:** The major goal of this project is to distill AD-related information from high-dimensional imaging data into lower-dimensional intermediate phenotypes, called Multiorgan AI-derived Endophenotypes (MAE). The overarching hypothesis is that the MAEs holistically model cross-organ pathological and neurobiological effects by linking the multi-scale causal pathway of ADRD: genetics –›proteomics –›imaging (MAE) –›cognitive decline disease endpoint (diagnosis), thus emerging as robust instruments for personalized AI for precision medicine.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 2 P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,693,508

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3.2
2027	3.2
2028	3.2
2029	3.2
2030	3.2

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Risk and Resilience, Clinical Presentation, and Biomarker Profiles of Chronic Traumatic Encephalopathy and Related Dementias: The DIAGNOSE CTE Research Project II

***Status of Support:** Current

Proposal/Award Number: 5 R01NS139383-02

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2024

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** $15,538,237

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06
2027	0.06
2028	0.06
2029	0.06

***Overall Objectives:** The goals of this R01 are to retain and grown DIAGNOSE and examine the clinical and biomarker course and profiles of traumatic encephalopathy syndrome (TES) (Aim 1); investigate risk/resilience factors of TES (Aim 2); and compare biomarkers of amyloid, p-tau, neurodegeneration, neuroinflammation, and white matter injury between TES and AD syndromes (Aims 3-4). Our hypotheses are that TES has unique clinical and biomarker profiles, and RHI and non-RHI risk factors influence the development of TES.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	ADRC Consortium for Clarity in ADRD Research Through Imaging (CLARiTi)
***Status of Support:**	Current
Proposal/Award Number:	5 U01AG08230-02
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2023
***Proposal/Active Project End Date: (MM/YYYY):**	08/2029
***Total Anticipated Proposal/Project Amount:**	$467,581

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6

***Overall Objectives:** Dr. Toga's USC team will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, providing assistance the site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy to use, web-based data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The Wisconsin Registry for Alzheimer's Prevention
***Status of Support:**	Current
Proposal/Award Number:	5 R01AG027161-14
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2023

***Proposal/Active Project End Date: (MM/YYYY):** 02/2028

***Total Anticipated Proposal/Project Amount:** $108,628

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06
2027	0.06
2028	0.06

***Overall Objectives:** Examine biomarker and cognitive longitudinal rate of change in people at risk for Alzheimer's Disease, and identify factors that increase risk and resilience to Alzheimer's Disease.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Education and Cognitive Functioning in Later Life: The Nation's High School Class of 1972

***Status of Support:** Current

Proposal/Award Number: 5 U01AG078533-02

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 07/2027

***Total Anticipated Proposal/Project Amount:** $598,950

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06
2027	0.06

***Overall Objectives:** Using new survey, biological, and MRI data collected from the National Longitudinal Study of the High School Class of 1972, this project will transform our understanding of how, why, and for whom education shapes risk of Alzheimer's Disease and Related Dementias (AD/ADRD).

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** The Health & Aging Brain Study – Health Disparities (HABS-HD)

***Status of Support:** Current

Proposal/Award Number: 5 U19AG078109-04

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $3,656,510

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.69
2027	0.69

***Overall Objectives:** Will provide a secure data repository and data sharing infrastructure for neuroimaging data acquired for this study and uploaded to the LONI Image and Data Archive (IDA). Will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, aiding the site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy to use, web-based data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Association LatAM FINGERS

***Status of Support:** Current

Proposal/Award Number: SG-22-971689

***Source of Support:** Alzheimer's Association

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $119,273

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	0.06

***Overall Objectives:** Dr. Toga's USC team will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, providing assistance the site(s) archiving neuroimaging data, securely storing MRI data and providing an easy to use, web-based data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked. Additionally, Dr. Toga's team will integrate clinical data collected in an external REDCap electronic data capture system, supporting data transfer activities.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium

***Status of Support:** Current

Proposal/Award Number: 5 U24AG074855-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $4,046,287

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.35

***Overall Objectives:** LONI's primary objective is to provide a system for housing, searching and interrogating harmonized AD data and linkage with access to NIAGADS.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Longitudinal multicenter head-to-head harmonization of tau PET tracers

***Status of Support:** Current

Proposal/Award Number: 5 R01AG073267-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $551,980

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06

***Overall Objectives:** The USC group will provide a secure data repository and data sharing infrastructure for neuroimaging data acquired for this study and uploaded to the LONI Image and Data Archive (IDA). Dr. Toga's USC team will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, providing assistance the site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy-to-use data archiving and sharing infrastructure. All upload, download, edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** KBASE2: Korean Brain Aging Study, Longitudinal Endophenotypes and Systems Biology

***Status of Support:** Current

Proposal/Award Number: 5 U01AG072177-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2021

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $678,150

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:** LONI will provide a secure, stable platform for de-identifying, uploading and storing MRI and PET scans and clinical data acquired for the study. Data de-identification methods will be applied to protect participant privacy. LONI will provide support to sites and investigators and

access to user management components whereby investigators can grant or revoke investigator accesses to their study data.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** North American Prodromal Syncleinopathy (NAPS2)

***Status of Support:** Current

Proposal/Award Number: 5 U19AG071754-04

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2021

***Proposal/Active Project End Date: (MM/YYYY):** 04/2027

***Total Anticipated Proposal/Project Amount:** $322,245

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1
2027	0.1

***Overall Objectives:** The North American Prodromal Synucleinopathy Consortium for RBD, Stage 2 (NAPS2) will evaluate individuals with RBD prospectively to more fully develop and analyze biomarkers of synucleinopathy and perform other key tasks required for planning neuroprotective clinical trials. The overarching goal of NAPS is to enable optimal development of neuroprotective clinical trials to prevent or delay overt synucleinopathies.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Understanding the long-term impact of COVID-19 on the brain through advanced MR imaging and spectroscopy

***Status of Support:** Current

Proposal/Award Number: 5 R01NS124065-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2021

***Proposal/Active Project End Date: (MM/YYYY):** 06/2026

***Total Anticipated Proposal/Project Amount:** $85,394

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.02

***Overall Objectives:** The USC group will provide a secure data repository and data sharing infrastructure for neuroimaging and clinical data acquired for this study and uploaded to the LONI Image and Data Archive (IDA).

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** National Alzheimer's Coordinating Center

***Status of Support:** Current

Proposal/Award Number: 5 U24AG072122-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2021

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $3,291,750

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.21

***Overall Objectives:** LONI's primary objective is to provide a secure data repository and data sharing infrastructure for neuroimaging data collected by participating ADRCs and to provide data search, visualization and download interfaces that help meet the data management and data sharing objectives of NACC and the wider Alzheimer's disease research community. LONI will provide a customized Data Use application request/review system to manage data use request and review/approval processes.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Neurocognitive Aging & Analytics Research Education (NAARE)

***Status of Support:** Current

Proposal/Award Number: 5 R25AG069711-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2021

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $124,090

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:** The USC site will help to develop appropriate curriculum for this project and will coordinate the ongoing execution of this program, including amongst USC faculty summer mentors.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Biomarker Consortium – Down Syndrome (ABC-DS)

***Status of Support:** Current

Proposal/Award Number: 5 U19AG068054-05

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $1,355,841

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.16

***Overall Objectives:** To test hypotheses related to how Alzheimer's Disease in Down syndrome may parallel sporadic Alzheimer's Disease within an amyloid, tau, neurodegeneration AT(N) framework and to identify modifiers of risk of conversion/progression.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The Clinical Significance of Incidental White Matter Lesions on MRI Amongst a Diverse Population with Cognitive Complaints (INDEED)
***Status of Support:**	Current
Proposal/Award Number:	5 U19NS120384-05
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2020
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$617,526

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24

***Overall Objectives:** The goal of this project is to support a large prospective clinical research study in the U.S. of patients engaged with the health care system because of incidental white matter (WM) lesions found on neuroimaging, who present with cognitive complaints, and are at risk for cognitive decline.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	ARTFL LEFFTDS Longitudinal Frontotemporal Lobar Degeneration, Cycle 2 (ALLFTD2)
***Status of Support:**	Current
Proposal/Award Number:	2 U19AG063911-06
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2019
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$648,166

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1
2027	0.1
2028	0.1
2029	0.1
2030	0.1

***Overall Objectives:** The major goals of this project is to formalize the merger of the ARTFL and LEFFTDS studies to create an integrated North American research consortium to study FTLD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Data Analysis and Coordination Center for the MACS-WIHS Combined Cohort Study

***Status of Support:** Current

Proposal/Award Number: 5 U01HL146193-07

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2019

***Proposal/Active Project End Date: (MM/YYYY):** 03/2026

***Total Anticipated Proposal/Project Amount:** $26,275

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.04

***Overall Objectives:** The MACS-WIHS Data Analysis and Coordination Center (DACC) brings together two datacenters that have served successfully as the scientific partner and operations center for the MACS and WIHS cohort studies, respectively. The merged datacenter will utilize best practices from both groups to advance knowledge of the basic science, clinical science, and epidemiology of HIV infection in the US, with a focus on HIV-related comorbidities. We will provide epidemiological and statistical expertise to scientific initiatives, as well as continue our quality assurance program in partnership with the clinical research sites that integrates expertise in data management, study coordination, statistical methodology, and scientific disciplines.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Clinical Trials Consortium (ACTC)

***Status of Support:** Current

Proposal/Award Number: 5 U24AG057437-08

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 12/2017

***Proposal/Active Project End Date: (MM/YYYY):** 08/2028

***Total Anticipated Proposal/Project Amount:** $409,490

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.36
2027	0.36
2028	0.36

***Overall Objectives:** The ACTC-IP will serve as an extensible, secure, and scalable information, collaboration, and process management infrastructure to facilitate the effective execution of program- and study-related research activities by ACTC Units, investigators, study teams, and performance sites. The ACTC-IP will provide facilities to collect, store, process, analyze, visualize, audit and share scientific, operational, and administrative data and metadata using standardized data models, formats, and taxonomies. The Informatics Unit will facilitate sharing of software, archived study datasets (including images), assessment instruments, forms and procedures, and sharing of specimens (as directed by the BARC). All ACTC archival datasets will be included on the Global Alzheimer's Association Interactive Network (GAAIN); ACTC data can be accessed via the GAAIN platform; full or partial archived datasets can be shared via the dedicated ACTC Dataset-Sharing Portal.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: 5 P01AG052350-08

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2016

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.65
2027	1.65
2028	1.65

***Overall Objectives:** Program project to study imaging and molecular biomarkers of neurovascular dysfunction in individuals at genetic risk for AD both familial and sporadic.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Characterizing Cognitive Decline in Late Life Depression: The ADNI-D Project

***Status of Support:** Current

Proposal/Award Number: 5 R01MH098062-09

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2013

***Proposal/Active Project End Date: (MM/YYYY):** 05/2027

***Total Anticipated Proposal/Project Amount:** $163,335

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.06

***Overall Objectives:** The major goals of this project are to identify the role of cortical atrophy, amyloid deposition, and tau accumulation as factors contributing to accelerated cognitive decline in late life depression.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Disease Neuroimaging Initiative (ADNI4)

***Status of Support:** Current

Proposal/Award Number: 2 U19AG024904-18

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2004

***Proposal/Active Project End Date: (MM/YYYY):** 07/2028

***Total Anticipated Proposal/Project Amount:** $4,971,080

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3
2027	0.3
2028	0.3

***Overall Objectives:** The overarching goal of the Informatics Core (IC) is to secure, link and disseminate all ADNI data coming from the participating sites and the other ADNI cores providing a resource for validating biomarkers for clinical trials. This involves de-identifying and receiving imaging data from the sites, integrating clinical data from the Clinical Core, incorporating analysis results from other cores, managing the data access systems utilized by the Data Publication and Sharing Committee, providing user interfaces to explore, visualize, interpret and download ADNI data and providing comprehensive information about the ADNI study through the public website. The Informatics Core has provided ongoing systems and support for achieving this goal and for sharing ADNI data with thousands of investigators worldwide. The Informatics Core has provided leadership in the design and deployment of the ADNI data-sharing model and has been instrumental in the data sharing effort.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Biomarker Evaluation in Young Onset NeuroDegeneration for Early Detection (BEYONDD)

***Status of Support:** Pending

Proposal/Award Number: R56AG075744

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 12/2026

***Proposal/Active Project End Date: (MM/YYYY):** 11/2031

***Total Anticipated Proposal/Project Amount:** $223,290

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.06
2028	0.06
2029	0.06
2030	0.06
2031	0.06

***Overall Objectives:** Dr. Toga's USC team will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, providing assistance the site(s) archiving neuroimaging data, securely storing MRI data and providing an easy to use, data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked. Additionally, Dr. Toga's team will create relabeled image files, with additional de-identification methods applied, at intervals or at study conclusion.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** National Alzheimer's Coordinating Center

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $415,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.1
2028	0.1
2029	0.1
2030	0.1
2031	0.1

***Overall Objectives:** LONI's primary objective is to provide a secure data repository and data

sharing infrastructure for neuroimaging data collected by participating ADRCs and to provide data search, visualization and download interfaces that help meet the data management and data sharing objectives of NACC and the wider Alzheimer's disease research community. LONI will provide a customized Data Use application request/review system to manage data use request and review/approval processes.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Resilience and Environmental Stressors Integrated with Sensors and Tracking in Alzheimer's Disease (RESIST-AD)
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$259,375

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.1
2028	0.1
2029	0.1
2030	0.1
2031	0.1

***Overall Objectives:** The USC group will provide a secure data repository and data sharing infrastructure for de-identified research data acquired for this study and uploaded to one configured project in the LONI Image and Data Archive (IDA).

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	US POINTER Imaging Ancillary Study
***Status of Support:**	Pending

Proposal/Award Number: R01AG062689

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $584,785

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.2
2028	0.2
2029	0.2
2030	0.2
2031	0.2

***Overall Objectives:** The POINTER imaging ancillary study leverages the rare opportunity provided by the ongoing POINTER multidomain lifestyle intervention trial to examine intervention effects on changes in Alzheimer's disease and cerebrovascular pathophysiology, measured with MRI and PET neuroimaging. This study also aims to determine whether these biomarkers at baseline predict cognitive response to the intervention, which could inform precision medicine approaches to healthcare.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Data Archive for the Brain Initiative (DABI)

***Status of Support:** Pending

Proposal/Award Number: R24MH140088

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2031

***Total Anticipated Proposal/Project Amount:** $5,670,640

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2027	1.2
2028	1.2
2029	1.2
2030	1.2
2031	1.2

***Overall Objectives:** The purpose of this proposal for a Data Archive for the Brain Initiative (DABI) is to develop web-accessible data archives to capture, store, and curate data related to the BRAIN Initiative proposals that collect invasive human neurophysiological data and make them broadly available and accessible to the research community for furthering research.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Head-to-head characterization of MAO-B PET and TSPO PET tracers in Alzheimer's disease

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2031

***Total Anticipated Proposal/Project Amount:** $163,550

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.1
2028	0.1
2029	0.1
2030	0.1
2031	0.1

***Overall Objectives:** The USC group will provide a secure data repository and data sharing infrastructure for neuroimaging data acquired for this study and uploaded to the LONI Image and Data Archive (IDA). Dr. Toga's USC team will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, providing assistance the

site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy-to-use data archiving and sharing infrastructure. All upload, download, edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The Health & Aging Brain Study – Health Disparities (HABS-HD)
***Status of Support:**	Pending
Proposal/Award Number:	U19AG078109
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2031
***Total Anticipated Proposal/Project Amount:**	$264,046

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.1
2028	0.1
2029	0.1
2030	0.1
2031	0.1

***Overall Objectives:** Will provide a secure data repository and data sharing infrastructure for neuroimaging data acquired for this study and uploaded to the LONI Image and Data Archive (IDA). Will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, aiding the site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy to use, web-based data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	SCAN: Standardized Centralized Alzheimer's and Related Dementias Neuroimaging

***Status of Support:** Pending

Proposal/Award Number: U24AG067418

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2021

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $3,025,505

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.6
2028	0.6
2029	0.6
2030	0.6
2031	0.6

***Overall Objectives:** The USC group will provide a secure data repository and data sharing infrastructure for PET and T1 MRI scans acquired by participating NACC centers along with support to center personnel for uploading these to the LONI Image and Data Archive (IDA). Development efforts developed for the main SCAN study (e.g. custom de-identification methods, data validations, query API) will be leveraged for this sub-project.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Early Onset AD Consortium - the LEAD Study (LEADS)

***Status of Support:** Pending

Proposal/Award Number: U01AG057195

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2018

***Proposal/Active Project End Date: (MM/YYYY):** 11/2031

***Total Anticipated Proposal/Project Amount:** $631,250

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12
2027	0.12
2028	0.12
2029	0.12
2030	0.12

***Overall Objectives:** We will serve as the central repository for the de-identification, storage and dissemination of neuroimaging and other data. The Institute will develop the procedures for the collection of uniform, compatible neuroimaging, neuropsychological and associated data. We will assist with the establishment of the protocols, Q/C and secure data transfer for efficient and accurate ingestion into a robust, comprehensive database system.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by TOGA, ARTHUR in SciENcv on 2026-01-09 14:08:29

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Aranda, Maria

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-5229-4993

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

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*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Alzheimer Disease Research Center (ADRC)
***Status of Support:**	Current
Proposal/Award Number:	P30AG066530-06
***Source of Support:**	National Institute on Aging (NIA)
***Primary Place of Performance:**	University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.5
2027	1.5
2028	1.5
2029	1.5
2030	1.5

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through it administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Demonstrating the Efficacy of Unidos en el Ciudadado: A Spanish-Language Program for Latino Caregivers of Persons Living with Dementia

***Status of Support:** Current

Proposal/Award Number: 24-10126

***Source of Support:** California Department of Public Health (CDPH)

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2024

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $2,000,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	3
2026	2.25
2027	2.25
2028	3

***Overall Objectives:** We will test the efficacy of Unidos en el Cuidado® (United in Caring, herein Unidos), a 3-session, culturally-adapted, group psychoeducational and skill-building intervention for 226 Spanish-speaking family caregivers in California

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The United States Cost of Dementia Model: Quantifying Current and Future Impacts of Alzheimer's Disease and Related Dementias
***Status of Support:**	Current
Proposal/Award Number:	1U01AG086827-01
***Source of Support:**	National Institute on Aging (NIA)
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2024
***Proposal/Active Project End Date: (MM/YYYY):**	03/2029
***Total Anticipated Proposal/Project Amount:**	$8,032,512

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6
2026	0.6

***Overall Objectives:** The overarching goal of this project is to provide national, annual, comprehensive cost estimates of dementia, utilizing the best available data sources. Furthermore, the aim is to develop research infrastructure that will significantly increase the national research capacity for generating cost estimates. Innovative simulation methods will be employed to analyze the factors influencing these costs and their impact on various demographic groups.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	USC Alzheimer's Disease Resource Centers for Minority Aging Research (USC AD-RCMAR)
***Status of Support:**	Current
Proposal/Award Number:	5P30AG043073-09
***Source of Support:**	National Institute on Aging (NIA)

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2023

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $3,602,796

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	1

***Overall Objectives:** The overarching goal is to identify racial and ethnic differences in risk, diagnosis and treatment and, once identified, elucidate opportunities for potential AD prevention and treatment regimen across diverse populations through mentoring of pilot- funded RCMAR scientists. A key goal of the Research and Education Component is to lead the research and training component for scientific fellows through mentored pilot projects.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Aranda, Maria in SciENcv on 2026-01-30 09:00:45

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Bienkowski, Michael S

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-8231-8893

*POSITION TITLE: Assistant Professor of Physiology and Neuroscience and Biomedical Engineering

*ORGANIZATION AND LOCATION: University of Southern California, LOS ANGELES, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Multiscale Modeling of Hippocampus Vulnerability to Alzheimer's Disease Neurodegeneration
***Status of Support:**	Current
Proposal/Award Number:	R01AG092662-01
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2026
***Proposal/Active Project End Date: (MM/YYYY):**	12/2030
***Total Anticipated Proposal/Project Amount:**	$3,478,945

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** The accumulation of Alzheimer's disease (AD) pathology and neurodegenerative atrophy of the hippocampus are believed to be strongly associated with cognitive decline, but how cell type-specific neurodegeneration leads to disconnection of hippocampal networks and cognitive impairment is unknown. Building on our previous research creating the Hippocampus Gene Expression Atlas (HGEA), our team will develop a multiscale computational model of HGEA-defined cell types (HGEA-NET) that will simulate the structural and functional disconnection of

hippocampal networks at varying stages of cognitive decline with new cell type-specific transcriptomic and connectomic datasets from human AD patients and mouse models. The development of the HGEA-NET will impact AD research as a translational ‘Rosetta Stone’ through which new strategies to prevent or reverse AD cognitive impairment could be tested

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer’s Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2029

***Total Anticipated Proposal/Project Amount:** $67,592

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through it administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** There is no overlap

***Proposal/Active Project Title:** Mesoscale correlative light-electron microscopy (CLEM) computational pathoconnectomes of degenerated retinas

***Status of Support:** Current

Proposal/Award Number: R01EY035527-01

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 05/2028

***Total Anticipated Proposal/Project Amount:** $675,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6

***Overall Objectives:** To this date, we are still uncovering the exact morphological and functional changes that retina cells undergo throughout retinal degenerative disease timecourse. Furthermore, as common to many neurodegenerative diseases, our knowledge is incomplete when it comes to understanding how these morphological changes to cells affects their role in neural networks, as well as the factors that impact these changes in connectivity. With this proposal, we will take what we have learned from multiscale computational modeling of extracted early data from patho-connectomes, or connectomics volumes constructed from early degeneration stages pathological or neurally degenerating tissues, and pursue large scale network creation and modeling for all four stages of retinal degeneration. These models will be integrated in our parallel multiscale Admittance Method (AM)-NEURON computational platform, which integrates modeling of exogenous electric field application with neural activity of complex networks to provide insights into the physiological consequences of morphological changes on retinal signaling.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** THEIA: Total Human Eye-allotransplantation Innovation Advancement (THEIA)

***Status of Support:** Current

Proposal/Award Number: 1AY2AX000042-01

***Source of Support:** ARPA-H

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** $267,968

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2025	0.84

***Overall Objectives:** Whole eye transplantation with optic nerve regeneration for restoration of brain circuits and visual perception.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Morphologic and Functional Assessment of Field-Potentiated Optic Nerve Regeneration

***Status of Support:** Current

Proposal/Award Number: R01EY035715-01

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 03/2024

***Proposal/Active Project End Date: (MM/YYYY):** 02/2028

***Total Anticipated Proposal/Project Amount:** $286,164

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.46
2027	0.46

***Overall Objectives:** In this project, we propose to develop electric field application into a technology that will synergize that with current strategies for optic nerve regeneration by providing directional cues for regenerating axons, leading to more targeted growth and increased recovery of visual function.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Restoration of optic nerve function driven by in vivo multimodal electrical stimulation

***Status of Support:** Current

Proposal/Award Number: 1 R01EY035375-02

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 02/2028

***Total Anticipated Proposal/Project Amount:** $276,558

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.46

***Overall Objectives:** The proposed research builds upon promising results showing that electric fields direct regeneration of retinal ganglion cell axons in vitro and in vivo, even when applied after degeneration has started to set in. This project will lay the groundwork for the future development of a breakthrough electric field-based therapy to regenerate the optic nerve using novel waveforms and systems.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: 5 P01AG052350-07

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $412,500

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.75
2026	0.75
2027	0.75

***Overall Objectives:** Program project to study imaging and molecular biomarkers of neurovascular dysfunction in individuals at genetic risk for AD both familial and sporadic.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	GCR: Reprogramming Biological Neural Networks with Field-Based Engineered Systems
***Status of Support:**	Current
Proposal/Award Number:	NSF OIA-2121164
***Source of Support:**	NSF
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	10/2021
***Proposal/Active Project End Date: (MM/YYYY):**	09/2026
***Total Anticipated Proposal/Project Amount:**	$196,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.62

***Overall Objectives:** Using Royal College of Surgeons (RCS) rats, a strain of rats affected by genetic blindness, we propose to slow and control the progression of retinal degeneration using bioengineered devices. The questions that we intend to ask in the proposed effort are: What are the morphological changes experienced by the neural network in neurodegenerative conditions? What are the underlying mechanisms for these changes? Can external or implantable bioengineered devices be exploited to alter these changes? If so, what are the mechanisms for which this change is effected and do the morphologic changes induced by electrical stimulation lead to positive functional outcomes?

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	A Translational 3D Map of Hippocampal Cell Types to Drive Investigations of Alzheimer's Disease
***Status of Support:**	Current
Proposal/Award Number:	5 K01AG066847-04
***Source of Support:**	NIH
***Primary Place of Performance:**	5 K01AG066847-04
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2021

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $125,766

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	6

***Overall Objectives:** Alzheimer's disease (AD) drug development relies on both mouse and clinical research, but translational resources to bridge the gap between mouse and human AD are limited. Building upon my previous research creating the mouse Hippocampus Gene Expression Atlas (HGEA), this proposal will investigate translational gene expression patterns across hippocampal neuronal cell types in mouse AD models and humans (with and without AD). The completion of this research will produce a comprehensive human HGEA that, together with the mouse HGEA, can help guide AD drug development and reveal the hippocampal neuronal cell types that are primarily affected by AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Subiculum Cell Type Susceptibility to Alzheimer's Disease Neurodegeneration

***Status of Support:** Pending

Proposal/Award Number: R01AG103317

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $4,149,990

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3
2027	3
2028	3
2029	3
2030	3

***Overall Objectives:** Atrophy of the subiculum (SUB) is an early clinical marker of Alzheimer's

disease (AD), but the cellular mechanisms underlying its selective vulnerability remain unknown. The proposed project will leverage spatial transcriptomics, high-resolution morphological reconstruction, and circuit mapping in human post-mortem tissue samples and across aging timepoints in male and female 5xFAD/PS19 mice to identify cell type-specific cellular and molecular signatures of neurodegeneration in the SUB. The experiments proposed here will determine how specific SUB cell types are susceptible to AD and identify molecular pathways associated with morphological neurodegeneration that can be targeted for future treatment strategies that slow or prevent cell death.

***Statement of Potential Overlap:** There is no overlap.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Bienkowski, Michael in SciENcv on 2026-01-23 20:28:07

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Braskie, Meredith

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-6439-4243

*POSITION TITLE: Assistant Professor of Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 5 P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,693,508

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Charting the female brain during midlife: assessing the interplay between menopause and modifiable factors in individual aging trajectories
***Status of Support:**	Current
Proposal/Award Number:	149622098_2
***Source of Support:**	Wellcome Leap
***Primary Place of Performance:**	University of Southern Californa
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2025
***Proposal/Active Project End Date: (MM/YYYY):**	05/2028
***Total Anticipated Proposal/Project Amount:**	$1,715,953

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48
2027	0.48
2028	0.48

***Overall Objectives:** Major Goals: Launch an international initiative to pool neuroimaging data from multiple datasets with women scanned during the perimenopausal transition period and chart changes associated with factors specific to women's health. Map brain structural and microstructural changes throughout midlife for females while determining the set of genetic and environmental factors such as diet and initiation of hormone replacement therapy that may influence ADRD risk at different stages of brain aging.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Causal and Event Based Modeling of Brain Alterations in ADRD
***Status of Support:**	Current
Proposal/Award Number:	1 R01AG087513-02
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2025
***Proposal/Active Project End Date: (MM/YYYY):**	12/2029
***Total Anticipated Proposal/Project Amount:**	$3,213,541

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6

***Overall Objectives:** Large-scale aging and epidemiological datasets can be harmonized alongside studies specific to Alzheimer's disease and related dementias to generate computational models that can map normal brain aging and identify the first non-invasive indicators of abnormal brain aging and ADRD pathology and the modifiable lifestyle factors that can potentially alter the course of the abnormalities. Here, we propose advanced modeling techniques of harmonized high dimensional data from existing and public datasets to 1) create normative aging models of the entire brain at the individual level to identify the first signs of deviations at local brain regions; 2) use high dimensional event based modeling and subtyping approaches to identify the order in which different microstructural and structural properties of brain tissue start to deviate from normality in the progression of disease; and 3) use marginal sensitivity analysis to model the extent to which modifications in lifestyle factors may causally relate to changes in trajectories of brain abnormalities in the progression of dementia.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Associations between vascular risk factors and functional connectivity

***Status of Support:** Current

Proposal/Award Number: AWD-00009090

***Source of Support:** Sammons Foundation

***Primary Place of Performance:** University of Texas, Dallas

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2024

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $82,500

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

***Overall Objectives:** To relate fMRI lag to vascular risk factors

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults

***Status of Support:** Current

Proposal/Award Number: 1 RF1AG078362-01

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $3,755,733

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8

***Overall Objectives:** The goal of this grant is to evaluate the relationship between type II diabetes status and the brain's response to glucose ingestion in non-demented older adults.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** The Health & Aging Brain Study – Health Disparities (HABS-HD)

***Status of Support:** Current

Proposal/Award Number: 5 U19AG078109-04

***Source of Support:** NIH

***Primary Place of Performance:** University of North Texas Health Science Center

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $2,358,185

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4

***Overall Objectives:** Neuroimaging Core: Analyze and store neuroimaging data Project 2: Associate vascular, metabolic, and inflammation risk to cognition and Alzheimer's disease-related neuroimaging biomarkers.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Neurocognitive Aging & Analytics Research Education (NAARE)
***Status of Support:**	Current
Proposal/Award Number:	5 R25AG069711-05
***Source of Support:**	NIH
***Primary Place of Performance:**	California State University of Fullerton
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2021
***Proposal/Active Project End Date: (MM/YYYY):**	05/2026
***Total Anticipated Proposal/Project Amount:**	$37,442

***** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48

***Overall Objectives:** This project will develop an exhaustive research training experience in neurocognitive aging ranging from basic neurobiology of neurodegenerative conditions, modifiable risk factors and health disparities related to neurocognitive aging, which will subsequently establish a talented, committed, and sustained pipeline of underrepresented students reflective of community demographics, who have unique perspectives, capable of addressing critical neurocognitive aging challenges in our changing world. The USC site will help to develop appropriate curriculum for this project and will coordinate the ongoing execution of this program, including amongst USC faculty summer mentors.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study
***Status of Support:**	Current
Proposal/Award Number:	5 R01 AG058533-05
***Source of Support:**	NIH

***Primary Place of Performance:** University of North Texas Health Science Center

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2020

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $1,035,235

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2

***Overall Objectives:** Dr. Braskie's lab will assess global and regional amyloid and tau burden in regions selected a priori for their prior differential binding in AD patients versus non-demented participants. We will perform statistical analyses on these measures evaluate the relationship between amyloid and tau positivity and age of dementia onset in Mexican-Americans, and to relate amyloid burden to cognitive measures as outlined in the Aims and Hypotheses.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Brain Changes During Perimenopause

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** Steven and Alexandra Cohen Foundation

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2027

***Proposal/Active Project End Date: (MM/YYYY):** 12/2031

***Total Anticipated Proposal/Project Amount:** $7,638,919

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	2.4
2028	2.4
2029	2.4
2030	2.4
2031	2.4

***Overall Objectives:** This innovative, mechanism-driven 3-year longitudinal project will follow 300 women aged 50 and older through a critical time in women's health – perimenopause. Using detailed

hormone profiles, cognitive assessments, and state-of-the-art multimodal MRI measures, we will assess how the brain changes in response to a rapidly changing endogenous hormonal environment. By linking hormonal fluctuations and menopausal symptoms to changes in cerebrovascular health, brain structure, neuroinflammation, and plasma ptau231, we seek to clarify how perimenopause influences Alzheimer's disease risk. Our results will help identify important mechanisms underlying brain changes during this time, improve the ability to predict risk for cognitive decline, and lay the groundwork for personalized interventions to protect women's brains during midlife.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Alzheimer's Risk Intervention Study with Estradiol (ARISE)
***Status of Support:**	Pending
Proposal/Award Number:	1R01AG094445-01
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$2,140,709

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.21
2027	2.21
2028	2.21
2029	2.21
2030	2.21

***Overall Objectives:** Determining whether menopausal hormone therapy (MHT) reduces subclinical atherosclerosis and improves brain and cognitive markers of Alzheimer's disease and related dementias (ADRD) risk has immense public health implications for reducing morbidity and mortality, particularly given that cardiovascular disease (CVD) and ADRD disproportionately affect postmenopausal women and have enormous health and economic implications for society. Simultaneous beneficial effect of MHT on atherosclerosis progression for CVD and brain imaging markers for ADRD risk will impact both clinical practice and future clinical studies investigating CVD and ADRD interventions. Our large ~3 year randomized controlled MHT clinical trial aims to demonstrate a treatment effect on brain imaging markers of ADRD risk and brain health and also uses cutting edge neuroimaging measures of brain structure, connectivity, and cerebrovascular health that will allow us to investigate the mechanisms underlying any MHT effect.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Braskie, Meredith in SciENcv on 2026-01-12 19:04:18

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: D'Orazio, Lina Mercedes

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-6190-2935

*POSITION TITLE: Associate Professor of Clinical Neurology

*ORGANIZATION AND LOCATION: USC Keck School of Medicine, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: P30AG066530-06

***Source of Support:** NIH/NIA

***Primary Place of Performance:** USC Keck School of Medicine

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8
2030	1.8

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Quantitative cerebral blood vessel imaging biomarkers for AD and VCID
***Status of Support:**	Current
Proposal/Award Number:	RF1AG072490
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	USC Keck School of Medicine (subaward)
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2025
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$527,940

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6

***Overall Objectives:** The goal is to develop a class of cerebral blood vessel hemodynamic and morphological metrics and to characterize cerebral blood vessel dysfunction in AD and VCID.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID)
***Status of Support:**	Current
Proposal/Award Number:	RF1AG084072
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	USC Keck School of Medicine
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2023
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$2,487,668

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

***Overall Objectives:** Vascular contributions to cognitive impairment and dementia (VCID) have been recognized as an important cause of clinical dementia. This project aims to systematically evaluate

novel metrics of small vessel morphology/density and perfusion as imaging biomarkers of VCID in a multiethnic longitudinal cohort of 200 subjects (including 50 Asian Americans) that are enriched for small vessel VCID. This project is expected to result in a powerful suite of imaging tools for comprehensive characterization of the morphology and function of cerebral small vessels, as well as to fill in the important gap in health disparities of Asian Americans in ADRD research.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Validation of Imaging and Blood-based Small Vessel Biomarkers in Multi-ethnic Population
***Status of Support:**	Current
Proposal/Award Number:	UF1NS100614 – Phase II
***Source of Support:**	NIH
***Primary Place of Performance:**	USC Keck School of Medicine
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2023
***Proposal/Active Project End Date: (MM/YYYY):**	07/2026
***Total Anticipated Proposal/Project Amount:**	$2,623,751

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.9

***Overall Objectives:** Cerebral small vessel disease (SVD) is the most common vascular cause of dementia, a major contributor to mixed Alzheimer's disease and vascular dementia, and the cause of about one fifth of all strokes worldwide. The primary objective of the next 5 years is to carry out comprehensive multi-site clinical validation of candidate VCID biomarkers in longitudinal studies of diverse populations that are typical in clinical settings in the US.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Cyclical deficits in emotion regulation as a risk factor for alcohol misuse in premenopausal females
***Status of Support:**	Current
Proposal/Award Number:	R01AA030065
***Source of Support:**	NIH

***Primary Place of Performance:** USC Keck School of Medicine

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2023

***Proposal/Active Project End Date: (MM/YYYY):** 02/2028

***Total Anticipated Proposal/Project Amount:** $1,095,088

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6

***Overall Objectives:** The proposed project will build a better understanding of cyclical changes in objective (heart rate variability) and subjective (self-reported) emotion regulation as a risk factor for cyclical alcohol misuse in females.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Phenotype and Genotype of Autosomal Dominant Alzheimer's Disease in Jalisco, Mexico

***Status of Support:** Current

Proposal/Award Number: R01AG69013

***Source of Support:** NIH/NIA

***Primary Place of Performance:** USC Keck School of Medicine

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $3,255,386

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** To phenotypically and genetically characterize persons with genetic neurodegenerative disease at the Centro de Investigación Biomédica de Occidente (CIBO) Genetics clinic in Guadalajara and the University of Guadalajara (U de G) Polyclinic in Tepatitlan, Jalisco, Mexico (Centro Universitario de los Altos, or CUAltos).

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by D'Orazio, Lina in SciENcv on 2026-01-20 00:10:39

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Han, Duke

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-2453-9230

*POSITION TITLE: Professor of Psychology, Family Medicine, Neurology, and Gerontology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	THE NIH PRECISION BRAIN HEALTH NETWORK: Open and Inclusive Measurement of Neuropsychological
***Status of Support:**	Current
Proposal/Award Number:	1U24AG092760-01
***Source of Support:**	The Pennsylvania State University/NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2025
***Proposal/Active Project End Date: (MM/YYYY):**	07/2030
***Total Anticipated Proposal/Project Amount:**	$2,163,200

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8
2030	1.8

***Overall Objectives:** The goal of this U24 grant is to develop, evaluate, and disseminate scientifically rigorous cognitive assessment tools that improve the precision and reliability of brain health research, with a focus on measuring cognitive change over time rather than just detecting impairment. This work will ensure that cognitive tests are valid, scalable, and applicable across real-world settings, enabling earlier detection of cognitive decline and more effective intervention strategies. Through a

coordinated research network, this project will leverage existing NIH investments, systematically identify gaps in current cognitive assessments, and implement open-access digital tools that improve research efficiency and accelerate discoveries in Alzheimer's disease and Alzheimer's disease-related dementias (AD/ADRD) prevention.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Examining mobility restrictions through the lens of post-stroke apathy
***Status of Support:**	Current
Proposal/Award Number:	1R01HD115526 - 01A1
***Source of Support:**	National Institute of Health
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$106,672

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** This project seeks to understand the association between apathy and restricted mobility post-stroke and identify the neural correlates of post-stroke apathy.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Alzheimer Disease Research Center (ADRC)
***Status of Support:**	Current
Proposal/Award Number:	P30AG066530-06

***Source of Support:** National Institute of Health

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 06/2030

***Total Anticipated Proposal/Project Amount:** $97,901

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.8
2028	0.8
2029	0.8
2030	0.8

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** ARMCADA-Advancing Reliable Measurement in Cognitive Aging and Decision-making Ability

***Status of Support:** Current

Proposal/Award Number: U24AG082022

***Source of Support:** Northwestern University /NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2023

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $834,264

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.42
2027	2.4
2028	2.4

***Overall Objectives:** The goal of this project is to create a network of decision making scientists to develop and test a measure of decision making that can be used in research studies of aging.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Mid-Career Development and Mentoring on Financial Vulnerability and Alzheimer's disease
***Status of Support:**	Current
Proposal/Award Number:	K24AG081325
***Source of Support:**	National Institute of Health
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2023
***Proposal/Active Project End Date: (MM/YYYY):**	03/2028
***Total Anticipated Proposal/Project Amount:**	$834,264

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3
2027	3
2028	3

***Overall Objectives:** The overall goal of this study is to mentor trainees and conduct studies of financial vulnerability and Alzheimer's Disease in large cohort studies of aging.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Research Network on decision Neuroscience and Aging
***Status of Support:**	Current
Proposal/Award Number:	R24AG076847
***Source of Support:**	The University of Texas at Dallas/NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $47,025

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24
2027	0.24

***Overall Objectives:** The overall goal of this study is to continue supporting the multidisciplinary Scientific Research Network on Decision Neuroscience and Aging (SRNDNA)

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Finance, Cognition, and Default Network in Aging (FCDNA)

***Status of Support:** Current

Proposal/Award Number: RF1AG068166/R01AG068166

***Source of Support:** National Institute of Health

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2021

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $3,608,795

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2

***Overall Objectives:** The overall goal of this study is to examine the association of financial exploitation in old age with markers of Alzheimer's Disease.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Cerebrovascular Resistance in Cognitive Aging and Alzheimer's Disease Risk (NCE)

***Status of Support:** Current

Proposal/Award Number: R01AG060049

***Source of Support:** National Institute of Health

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2019

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $79,260

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:** The Major goal of this project is to investigate markers of artery stiffness and resistance in the brains of older adults who may be at risk for Alzheimer's disease

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Finance, Cognition, and Default Network in Aging (FCDNA) – Renewal

***Status of Support:** Pending

Proposal/Award Number: R01AG068166

***Source of Support:** National Institute of Health

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $3,946,906

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	1.2
2028	1.2
2029	1.2
2030	1.2
2031	1.2

***Overall Objectives:** The overall goal of this study is to examine the association of financial exploitation vulnerability in older age with markers of Alzheimer's Disease.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Cerebrovascular resistance in cognitive aging and Alzheimer's disease risk-Renewal
***Status of Support:**	Pending
Proposal/Award Number:	R01AG60049
***Source of Support:**	National Institute of Health
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2031
***Total Anticipated Proposal/Project Amount:**	$123,357

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.6
2028	0.6
2029	0.6
2030	0.6
2031	0.6

***Overall Objectives:** This project is focused on furthering our understanding of cerebrovascular resistance as a novel vascular risk factor for cognitive decline and Alzheimer's dementia

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Advancing Mobile Monitoring of Intraindividual Change in Alzheimer's Disease and Related Dementias: Building an Open and Accessible Infrastructure
***Status of Support:**	Pending
Proposal/Award Number:	U19AG087932

***Source of Support:** The Pennsylvania State University/ NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 12/2025

***Proposal/Active Project End Date: (MM/YYYY):** 11/2030

***Total Anticipated Proposal/Project Amount:** $493,645

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.8
2027	0.8
2028	0.8
2029	0.8
2030	0.8

***Overall Objectives:** The overall goal of this project is to build the platform, M2 CogKit, for the mobile monitoring of cognitive change that is designed to promote innovation with an open-source model.

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Han, Duke in SciENcv on 2026-01-29 22:46:50

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Harrington, Michael

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-7923-8032

*POSITION TITLE: Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID)
***Status of Support:**	Current
Proposal/Award Number:	1RF1AG084072
***Source of Support:**	NIH
***Primary Place of Performance:**	USC
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2023
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$2,487,668

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3

***Overall Objectives:** Vascular contributions to cognitive impairment and dementia (VCID) have been recognized as an important cause of clinical dementia. This project aims to systematically evaluate novel metrics of small vessel morphology/density and perfusion as imaging biomarkers of VCID in a multiethnic longitudinal cohort of 200 subjects (including 50 Asian Americans) that are enriched for small vessel VCID. This project is expected to result in a powerful suite of imaging tools for comprehensive characterization of the morphology and function of cerebral small vessels, as well as to fill in the important gap in health disparities of Asian Americans in ADRD research.

***Statement of Potential Overlap:** No overlap

***Proposal/Active Project Title:** Validation of Imaging and Blood-based Small Vessel Biomarkers in Multi-ethnic Population

***Status of Support:** Current

Proposal/Award Number: UO1NS100614 - Phase II

***Source of Support:** NIH

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2023

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $2,623,751

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3

***Overall Objectives:** Cerebral small vessel disease (SVD) is the most common vascular cause of dementia, a major contributor to mixed Alzheimer's disease and vascular dementia, and the cause of about one fifth of all strokes worldwide. The primary objective of the next 5 years is to carry out comprehensive multi-site clinical validation of candidate VCID biomarkers in longitudinal studies of diverse populations that are typical in clinical settings in the US.

***Statement of Potential Overlap:** No overlap

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: 2PO1AG052350-06

***Source of Support:** NIH

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2016

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $8,538,416

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2

***Overall Objectives:** This program project focuses on alterations in the neurovascular unit (e.g., blood-brain barrier, pericytes) in persons carrying the apolipoprotein E4 and autosomal dominant AD mutations to clarify vascular contributions in AD.

***Statement of Potential Overlap:** No overlap

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Pending

Proposal/Award Number: P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2
2028	1.2
2029	1.2
2030	1.2

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through it administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** No overlap

In-Kind Contributions

***Status of Support:** Current

***Source of Support:** NIH

***Receipt (or Anticipated Receipt) Date of In-Kind Contribution: (MM/YYYY):** 01/2026

***Summary of In-Kind Contribution:** Serving as Co-Investigator for the study VCS-II

***U.S. Dollar Value of In-Kind Contribution:** $7,100

Person-Month(s) (or Partial Person-Months) Per Year Devoted to the In-Kind Contribution:

Year	Person Months
2026	0.6

***Overall Objectives:** I perform clinical neurology assessments (UDS).

***Statement of Potential Overlap:** No overlap

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Harrington, Michael in SciENcv on 2026-01-26 17:21:41

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Hiniker, Annie

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-0055-6610

*POSITION TITLE: Associate Professor of Pathology

*POSITION TITLE: Associate Professor of Pathology

*ORGANIZATION AND LOCATION: University of Southern California Keck School of Medicine, Los Angeles 90033, California, United States

*ORGANIZATION AND LOCATION: University of Southern California Keck School of Medicine, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Discovery and Trial Ready Cohort for Limbic Age Related TDP-43 Encephalopathy (LATE)
***Status of Support:**	Current
Proposal/Award Number:	1R01AG095017-01
***Source of Support:**	NIA
***Primary Place of Performance:**	University of California Irvine (USC subaward)
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$17,948,254

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.36
2027	0.36
2028	0.36
2029	0.36
2030	0.36

***Overall Objectives:** This study will recruit and characterize 420 participants aged 85 and older across five Alzheimer's disease research centers to validate clinical markers, discover new

biomarkers, and develop recruitment strategies for early-phase LATE clinical trials.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Alzheimer Disease Research Center (ADRC)--Neuropathology Core (director)
***Status of Support:**	Current
Proposal/Award Number:	P30AG066530-06
***Source of Support:**	NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	03/2030
***Total Anticipated Proposal/Project Amount:**	$23,693,514

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education, and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Targeting proteostatic mechanisms to inhibit LRRK2-mediated neurodegeneration and neuropathology
***Status of Support:**	Current
Proposal/Award Number:	7R01NS135607-02
***Source of Support:**	NINDS
***Primary Place of Performance:**	University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2028

***Total Anticipated Proposal/Project Amount:** $2,764,802

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3.6
2027	3.6
2028	3.6

***Overall Objectives:** To characterize the proteostatic mechanisms upstream and downstream of LRRK2 using patient-derived neurons and rare postmortem LRRK2 patient brain tissue by (1) defining the role of LRRK2's subcellular localization in its cellular functions and degradation, (2) testing if newly identified LRRK2 protein degradation pathways rescue neurotoxicity in patient-derived neurons and (3) defining the mechanisms by which LRRK2 mutations can drive both synucleinopathy and tauopathy using RT-QuIC, detailed neuropathology, and cell-type specific transcriptomics on LRRK2 patient brain tissue.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer's Disease Hallmark Pathology and Associated Inflammation in the Retina

***Status of Support:** Current

Proposal/Award Number: 5R01AG055865-07

***Source of Support:** NIA/NIH

***Primary Place of Performance:** Cedars-Sinai (University of Southern California subaward)

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2018

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $165,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24
2027	0.24
2028	0.24

***Overall Objectives:** The major goals are to provide human eyes and cortical tissues obtained at autopsy from normal and age matched AD patients to compare AD changes in each site

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Early signatures of a-synuclein seeds in human brain
***Status of Support:**	Pending
Proposal/Award Number:	RAG104621A
***Source of Support:**	NIH
***Primary Place of Performance:**	Case Western Reserve University, (Subaward USC)
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$3,500,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8

***Overall Objectives:** The overall goals of this proposal are to determine the cellular microenvironments, lipid alterations and cellular responses that accompany unique alpha-synuclein and tau protein assemblies identified in mixed pathologies of Lewy body and Alzheimer's disease.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The role of microglial lipid metabolism, cellular senescence and neuroinflammation in the pathogenesis of AD
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2031
***Total Anticipated Proposal/Project Amount:**	$4,141,154

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.36

***Overall Objectives:** This study explores how lipid buildup in microglia contribute to Alzheimer's disease by causing these cells—and nearby neurons—to age and develop features of senescence.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Mechanisms of Cerebral Amyloid Angiopathy and Neurovascular Inflammation in Alzheimer's Disease

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2030

***Total Anticipated Proposal/Project Amount:** $4,166,435

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** This research addresses a critical public health challenge by exploring novel therapeutic strategies to reduce neurovascular inflammation and cerebral amyloid angiopathy (CAA), key contributors to Alzheimer's disease (AD) progression.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Modifiers of tau seeds in human brain
***Status of Support:**	Pending
Proposal/Award Number:	1R01AG096873-01
***Source of Support:**	NIA
***Primary Place of Performance:**	Case Western Reserve University, U of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$4,096,721

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4

***Overall Objectives:** The overall goals of this proposal are to map the microenvironments associated with tau seeds in the human brain and to determine how common co-existing protein pathologies and related cellular responses influence abnormal tau seeds and their ability to accumulate.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Hiniker, Annie in SciENcv on 2026-01-29 13:07:58

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Joe, Elizabeth

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-5222-0772

*POSITION TITLE: Assistant Professor of Clinical Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center

***Status of Support:** Current

Proposal/Award Number: P30AG066530-06

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Barriers to early identification of dementia in a safety net healthcare system

***Status of Support:** Current

Proposal/Award Number: 1R03AG082997

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $331,458

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4

***Overall Objectives:** This study will identify barriers to early identification of dementia for older adults who receive healthcare in a safety net health system, to identify specific barriers that can be targeted for future interventions to improve early identification of dementia in the safety net population.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: P01AG052350

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $12,320,954

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4

Year	Person Months

***Overall Objectives:** This program project focuses on alterations in the neurovascular unit (e.g., blood- brain barrier, pericytes) in persons carrying the apolipoprotein E4 and autosomal dominant AD mutations to clarify vascular contributions in AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	The Clinical Significance of Incidental White Matter Lesions on MRI Amongst a Diverse Population with Cognitive Complaints (INDEED)
***Status of Support:**	Current
Proposal/Award Number:	5U19NS120384
***Source of Support:**	NINDS
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2020
***Proposal/Active Project End Date: (MM/YYYY):**	08/2027
***Total Anticipated Proposal/Project Amount:**	$965,334

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2

***Overall Objectives:** This study proposes to comprehensively examine the impact of individual and combined MRI measures of white matter injury on cognitive performance among a diverse, non-demented, stroke free population with cognitive complaints over an extended period of observation.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Joe, Elizabeth in SciENcv on 2026-01-16 17:04:53

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Lynch, Kirsten

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-3976-5783

*POSITION TITLE: Assistant Professor of Research - Neurology

*ORGANIZATION AND LOCATION: Keck School of Medicine of USC, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 5 P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $4,436,949

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.06
2027	2.06
2028	2.06
2029	2.06
2030	2.06

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:** Vascular Contributions to Dementia and Genetic Risk Factors for Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: 5 P01AG052350-08

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.35
2027	1.35
2028	1.35

***Overall Objectives:** This is a continuing multi-disciplinary program on vascular contributions to dementia and genetic risk factors for Alzheimer's disease with multiple projects, cores, institutions and investigators. The overall goal of the program is to test the 'neurovascular hypothesis' of AD and establish whether the neurovasculature has a major role in the pathogenesis of early cognitive decline, and therefore could be a key new therapeutic target to treat early cognitive impairment, dementia and early AD. The results of this work may lead to new ways of thinking about pathogenesis, treatment and early prevention of cognitive impairment, dementia and AD for which we still do not have effective therapies.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:** Biomarkers of neurodegeneration in the US and India: A cross-national evaluation of the biological underpinnings of ADRD

***Status of Support:** Current

Proposal/Award Number: RF1AG088003-01

***Source of Support:** NIH

***Primary Place of Performance:** Johns Hopkins University

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2029

***Total Anticipated Proposal/Project Amount:** $2,969,745

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.71
2027	0.71
2028	0.71
2029	0.71

***Overall Objectives:** There is a critical need to identify connections between biological factors – not only of neurodegeneration but of vascular, nutritional, and inflammatory systems – underlying ADRD pathology and cognitive outcomes across socioeconomically diverse contexts. We will use cross-national data in the US and India to assess the interplay between context, social factors, and ADRD biology using recently collected cross-nationally harmonized neurodegenerative biomarkers as well as information on cognitive outcomes and other markers of vascular, nutritional, and inflammatory systems. Our results will set the stage to leverage biology to understand social differences in cognition that would not be clear from research in just one country, with the ultimate goal of preventing and/or delaying the onset of ADRD from a global perspective.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Noneducational sources of later-life cognitive reserve and resilience among older adults with and without formal education in India
***Status of Support:**	Current
Proposal/Award Number:	1RF1AG087965-01
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2024
***Proposal/Active Project End Date: (MM/YYYY):**	06/2029
***Total Anticipated Proposal/Project Amount:**	$4,048,263

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.71
2027	0.71
2028	0.71

Year	Person Months
2029	0.71

***Overall Objectives:** Using longitudinal data from a nationally representative study of older adults in India, we will identify the roles of life course employment, social relationships, and social engagement in cognitive reserve, cognitive change, and ADRD risk, overall and according to access to early-life formal education. Our results will advance scientific theory on resilience to aging-related brain pathology in the absence of education and support the identification of intervention targets to reduce the global ADRD burden for diverse older populations.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:** Multilingualism as a factor of resilience to Alzheimer's disease and related dementias in India

***Status of Support:** Current

Proposal/Award Number: 5 R01AG080473-03

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 12/2022

***Proposal/Active Project End Date: (MM/YYYY):** 11/2027

***Total Anticipated Proposal/Project Amount:** $4,048,263

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.75
2027	0.75

***Overall Objectives:** Determine whether bilingualism modifies the association between Alzheimer's biomarkers and cognitive outcomes, 2) evaluate whether the protective effect of bilingualism differs across diverse life-course environmental determinants of health, and 3) deconstruct language use within bilinguals in India to understand how bilingualism protects against Alzheimer's.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Large-scale harmonization and integration of multi-modal ADNI data for the early detection of Alzheimer's disease and related dementias
***Status of Support:**	Current
Proposal/Award Number:	5 R01NS128486-04
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2022
***Proposal/Active Project End Date: (MM/YYYY):**	05/2027
***Total Anticipated Proposal/Project Amount:**	$1,189,381

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.26
2027	2.26

***Overall Objectives:** The goal of this application is to harmonize trans-national cohorts of the Alzheimer's Disease Neuroimaging Initiative (ADNI) to improve the diagnostic classification of AD and ADRD. The hypothesis is that by harmonizing the multi-modal ADNI datasets and building predictive models from each modality integrated into comprehensive ensembles, novel classifiers and features for early AD diagnosis and differentiation from ADRD can be identified.

***Statement of Potential Overlap:** There is commitment overlap for Dr. Lynch between 5 R01NS128486-04 and the application under consideration. If this application is funded Dr. Lynch will request approval to reduce her effort on 5 R01NS128486-04 to 1 calendar month. If other pending applications are funded, Dr. Lynch will reduce her effort appropriately.

***Proposal/Active Project Title:**	Structural and diffusion changes of perivascular space in aging, cognitive decline and Alzheimer's disease
***Status of Support:**	Current
Proposal/Award Number:	5 R01AG070825-05
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2021
***Proposal/Active Project End Date: (MM/YYYY):**	06/2026
***Total Anticipated Proposal/Project Amount:**	$4,124,992

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.26

***Overall Objectives:** This study will identify pathological alterations to the brain glia- lymphatic pathway (i.e. perivascular spaces) is expected to aid in Alzheimer's disease diagnosis and disease monitoring. It is also expected to fill the gap in knowledge regarding the role of the perivascular spaces in Alzheimer's disease and its pathological neuroimaging signature.

***Statement of Potential Overlap:** There is commitment overlap for Dr. Lynch between 5 R01AG070825-05 and the application under consideration. If this application is funded, Dr. Lynch will request approval to reduce her effort on 5 R01AG070825-05 to 0 calendar months for the remaining 6 months of the project. If other pending applications are funded, Dr. Lynch will reduce her effort appropriately.

***Proposal/Active Project Title:**	The role of modifiable ADRD risk factors in cerebrovascular pathology, neurodegeneration and cognitive decline in India
***Status of Support:**	Pending
Proposal/Award Number:	1 R01AG096398-01A1
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2031
***Total Anticipated Proposal/Project Amount:**	$4,145,714

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	3
2028	3
2029	3
2030	3
2031	3

***Overall Objectives:** The overarching goal of this project is to understand how neuroimaging-based brain pathologies vary in India and are related to modifiable ADRD risk factors and cognitive decline. The project aims to: (1) Identify how cerebrovascular pathologies are associated with ADRD-related modifiable risk factors; (2) Determine how modifiable ADRD risk factors influence micro- and

macrostructural neurodegenerative patterns; (3) Determine how brain pathologies mediate the relationship between modifiable ADRD risk factors and prospective cognitive decline.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Harmonized Diagnostic Assessment of Dementia (DAD) for Longitudinal Aging Surveys in India (LASI)
***Status of Support:**	Pending
Proposal/Award Number:	2R01AG051125-11
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	12/2025
***Proposal/Active Project End Date: (MM/YYYY):**	11/2030
***Total Anticipated Proposal/Project Amount:**	$33,729,292

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** Our overarching goal is to develop a deeply phenotyped, nationally representative, longitudinal study of late-life cognition and AD/ADRD in India and provide the highest quality, cutting-edge data to the larger research community to advance the understanding of risk factors of late-life cognitive decline and AD/ADRD.

***Statement of Potential Overlap:** none

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Lynch, Kirsten in SciENcv on 2026-01-15 15:27:19

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Nation, Daniel A

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-9123-0658

*POSITION TITLE: Professor

*POSITION TITLE: Professor of Gerontology and Medicine

*POSITION TITLE: Professor

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	The Los Angeles Older Americans Independence Center (LA OAIC)
***Status of Support:**	Current
Proposal/Award Number:	P30AG094848
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$7,331,459

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	0.5
2027	0.5
2028	0.5
2029	0.5
2030	0.5

***Overall Objectives:** The LA OAIC is a three-institution collaboration in the LA area will catalyze the conduct of highly innovative human trials of gerotherapeutics, including nutraceuticals, drugs, EVs, and regenerative therapies to promote interventions to modulate aging biology for healthspan extension.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:** University of Southern California Alzheimer's Disease Research Center

***Status of Support:** Current

Proposal/Award Number: P30AG066530

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.8
2027	0.8
2028	0.8
2029	0.8
2030	0.8

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through it administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Locus Coeruleus imaging markers in preclinical Alzheimer's disease, cerebrovascular disease and cognitive decline
***Status of Support:**	Current
Proposal/Award Number:	R01AG082073
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	02/2024
***Proposal/Active Project End Date: (MM/YYYY):**	03/2028
***Total Anticipated Proposal/Project Amount:**	$7,473,104

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4

***Overall Objectives:** The overall goal is to examine brain MRI markers of locus coeruleus integrity as biomarkers of preclinical Alzheimer's disease and in relation to brain MRI markers of cerebrovascular disease, and as predictors of longitudinal cognitive decline in older adults.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Vascular Contributions to Dementia and Genetic Risk Factors for Alzheimer's Disease
***Status of Support:**	Current
Proposal/Award Number:	P01AG052350
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2022
***Proposal/Active Project End Date: (MM/YYYY):**	08/2028
***Total Anticipated Proposal/Project Amount:**	$16,196,003

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	3.6
2027	3.6
2028	3.6

***Overall Objectives:** The overall goal of this continuing program project is to test the 'neurovascular hypothesis' of AD and establish whether the neurovasculature has a major role in the pathogenesis of early cognitive decline, and therefore could be a key new therapeutic target to treat cognitive impairment, dementia and early AD.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:** Endothelial progenitor cells and neurovascular injury in the aging brain

***Status of Support:** Current

Proposal/Award Number: R01AG064228

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2019

***Proposal/Active Project End Date: (MM/YYYY):** 06/2030

***Total Anticipated Proposal/Project Amount:** $2,487,155

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2
2027	2
2028	2
2029	2
2030	2

***Overall Objectives:** This project is focused on furthering our understanding of potentially protective cellular factors in age-related neurovascular dysfunction and cognitive decline. Specifically, we are studying how endothelial progenitor cell dysfunction contributes to blood-brain barrier permeability both in vitro using vascular microbrains and in vivo using brain MRI.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Cerebral blood flow regulation and blood pressure in Fabry disease
***Status of Support:**	Pending
Proposal/Award Number:	R21AG103457
***Source of Support:**	NIH/NINDS
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2028
***Total Anticipated Proposal/Project Amount:**	$510,644

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6

***Overall Objectives:** To investigate 24-hour ambulatory blood pressure and cerebrovascular mechanisms of brain injury in both monogenic (Fabry disease) and sporadic (age-related) forms of cerebral small vessel disease.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Hemodynamic Mechanisms of Aortic Arch Stiffness and Aorta-Brain Coupling Underlying Brain Injury and Cognitive Decline in Older Adults
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$4,023,413

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	0.25
2027	0.25
2028	0.25
2029	0.25
2030	0.25

***Overall Objectives:** This project aims to investigate the role of age-related changes in aortic biomechanics and their interactions with cerebral blood vessels in contributing to dementia risk. By focusing on the unique physiology of aorta-brain vascular coupling, the study aims to clarify how these age-related changes affect brain health.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:** Circulating mitochondrial-derived microproteins and risks for cognitive impairment

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2031

***Total Anticipated Proposal/Project Amount:** $1,937,411

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5
2027	0.5

***Overall Objectives:** Study mitochondrial-derived microproteins in cognitive impairment.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Hemodynamic risk factors in cognitive decline and Alzheimer's disease.
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2031
***Total Anticipated Proposal/Project Amount:**	$3,870,635

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2
2027	2
2028	2
2029	2
2030	2
2031	2

***Overall Objectives:** Study blood pressure variability in cognitive decline and Alzheimer's disease.

***Statement of Potential Overlap:** There is no overlap.

***Proposal/Active Project Title:**	Geroscience of sex differences SCORE (GeroSCORE)
***Status of Support:**	Pending
Proposal/Award Number:	U54AG099000
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	02/2026
***Proposal/Active Project End Date: (MM/YYYY):**	01/2031
***Total Anticipated Proposal/Project Amount:**	$7,495,704

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** This center aims to advance our understanding of the role of biological sex across biological, behavioral, and population levels, which will accelerate discoveries in sex-based differences in key aging outcomes including biological aging, physical function, disability and disease, and cognitive and brain function.

***Statement of Potential Overlap:** There is no overlap.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Nation, Daniel in SciENcv on 2026-01-08 16:33:11

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Pawluczyk, Sonia

PERSISTENT IDENTIFIER (PID): https://orcid.org/0009-0005-4723-6031

*POSITION TITLE: Assistant Professor

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer's Disease Neuroimaging Initiative (ADNI4)

***Status of Support:** Current

Proposal/Award Number: 5U19AG024904-18

***Source of Support:** Northern California Institute for Research and Education

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $36,840

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5

***Overall Objectives:** To conduct the neuroimaging initiative at USC.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Alzheimer's Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 2P30AG066530-06

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $2,270,226

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.4
2027	1.4
2028	1.4
2029	1.4
2030	1.4

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** A Phase II/III Randomized, Double-Blind, Placebo-Controlled, Cognitive Endpoint, Multicenter Study of Potential Disease Modifying Therapies in Individuals at Risk for and with Dominantly Inherited Alzheimer's Disease (DIAN-TU-001)

***Status of Support:** Current

Proposal/Award Number: DIAN-TU-001

***Source of Support:** Esai Incorporated

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2025

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $58,170

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1

Year	Person Months
2027	0.1

***Overall Objectives:** The purpose of this study is to assess the safety, tolerability, biomarker, cognitive and clinical efficacy of investigational products in participants with an Alzheimer's disease-causing mutation by determining if treatment with the study drug slows the rate of progression of cognitive/clinical impairment or improves disease-related biomarkers.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Alzheimer's Clinical Trials Consortium

***Status of Support:** Current

Proposal/Award Number: 5U24AG057437-08

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $996,229

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4

***Overall Objectives:** The proposal aims to create an optimized infrastructure to optimally design, efficiently launch, rapidly recruit with improved diversity, and successfully complete NIH supported clinical trials in AD and related dementias.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** USC California Alzheimer Disease Center

***Status of Support:** Current

Proposal/Award Number: 23-10416

***Source of Support:** State of California

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2023

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $1,409,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8

***Overall Objectives:** This center is established to support the clinical care of and follow-up on patients with dementia and provide educational outreach to the community. Goals are to address the needs of our understudied and underserved communities and provide a model of statewide collaboration.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Phase II randomized controlled trial of benfotiamine in early Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: R01AG086634-01

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2022

***Proposal/Active Project End Date: (MM/YYYY):** 06/2026

***Total Anticipated Proposal/Project Amount:** $837,100

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8

***Overall Objectives:** A clinical trial of benfotiamine for Alzheimer's disease.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	AHEAD 3-45 Study: A Placebo-Controlled, Double-Blind, Parallel-Treatment Arm, 216 Week Study to Evaluate Efficacy and Safety of Treatment With BAN2401 in Subjects With Preclinical Alzheimer's Disease-5451840005
***Status of Support:**	Current
Proposal/Award Number:	011970-00001
***Source of Support:**	Eisai Incorporated
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2020
***Proposal/Active Project End Date: (MM/YYYY):**	12/2027
***Total Anticipated Proposal/Project Amount:**	$2,638,930

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3
2027	0.3

***Overall Objectives:** A clinical trial of lecanemab for preclinical Alzheimer's Disease

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Pawluczyk, Sonia in SciENcv on 2026-01-29 20:00:10

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Ringman, John Matthew

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-5429-0777

*POSITION TITLE: Professor of Clinical Neurology

*ORGANIZATION AND LOCATION: Department of Neurology, Keck School of Medicine at USC, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Discovery and Trial Ready Cohort for Limbic predominant Age related TDP-43 Encephalopathy (D-TRC LATE)
***Status of Support:**	Current
Proposal/Award Number:	R01AG095017
***Source of Support:**	NIH
***Primary Place of Performance:**	University of California Irvine, University of Southern Cali
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$3,128,129

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2
2028	1.2
2029	1.2
2030	1.2

***Overall Objectives:** The overall aim of this proposal is to assess the utility of and refine the current diagnostic recommendations for LATE, develop and test novel biomarkers for LATE, and identify effective strategies for recruitment and sustaining of diverse participants for early phase clinical trials of LATE.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)
***Status of Support:** Current
Proposal/Award Number: P30AG066530
***Source of Support:** NIA
***Primary Place of Performance:** University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025
***Proposal/Active Project End Date: (MM/YYYY):** 03/2030
***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2
2027	2
2028	2
2029	2
2030	2

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Effects of Acute Stress Exposure on Plasma beta-amyloid Levels
***Status of Support:** Current
Proposal/Award Number: R21AG080414
***Source of Support:** NIA
***Primary Place of Performance:** University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):** 01/2024
***Proposal/Active Project End Date: (MM/YYYY):** 12/2025

***Total Anticipated Proposal/Project Amount:** $468,308

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6

***Overall Objectives:** Stress has been linked to increased risk for Alzheimer's disease in observational studies in human populations. However, the mechanism by which stress increases Alzheimer's disease risk in unclear. Rodent models suggest that acute stress exposure increases beta-amyloid levels, which may be one possible mechanism linking stress exposure to increased Alzheimer's disease risk. With the advent of reliable plasma markers of beta-amyloid, we can now test whether acute stress exposure also upregulates beta-amyloid levels in humans. This study will test whether a brief exposure to an acute laboratory stressor increases plasma levels of beta-amyloid-40 and beta-amyloid-42 and whether any observed increases are related to the magnitude of the cortisol response to the stressor. 38 healthy adult participants will be enrolled in this study and complete stress and no-stress control sessions (within-subject) to determine whether acute stress exposure increases bet-amyloid levels in plasma in the 5 hours following stress onset.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID)

***Status of Support:** Current

Proposal/Award Number: R01AG084072

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2023

***Proposal/Active Project End Date: (MM/YYYY):** 07/2026

***Total Anticipated Proposal/Project Amount:** $2,487,668

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.8

***Overall Objectives:** Vascular contributions to cognitive impairment and dementia (VCID) have been recognized as an important cause of clinical dementia. This project aims to systematically evaluate novel metrics of small vessel morphology/density and perfusion as imaging biomarkers of VCID in a multiethnic longitudinal cohort of 200 subjects (including 50 Asian Americans) that are enriched for small vessel VCID. This project is expected to result in a powerful suite of imaging tools for comprehensive characterization of the morphology and function of cerebral small vessels, as well as

to fill in the important gap in health disparities of Asian Americans in ADRD research.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Validation of Imaging and Blood-based Small Vessel Biomarkers in Multi-ethnic Population
***Status of Support:**	Current
Proposal/Award Number:	U01NS100614
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2023
***Proposal/Active Project End Date: (MM/YYYY):**	07/2026
***Total Anticipated Proposal/Project Amount:**	$2,623,751

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.7

***Overall Objectives:** Cerebral small vessel disease (SVD) is the most common vascular cause of dementia, a major contributor to mixed Alzheimer's disease and vascular dementia, and the cause of about one fifth of all strokes worldwide. The primary objective of the next 5 years is to carry out comprehensive multi-site clinical validation of candidate VCID biomarkers in longitudinal studies of diverse populations that are typical in clinical settings in the US.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Shape-based personalized AT(N) imaging markers of Alzheimer's disease
***Status of Support:**	Current
Proposal/Award Number:	RF1AG077578
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	06/2023
***Proposal/Active Project End Date: (MM/YYYY):**	05/2026

***Total Anticipated Proposal/Project Amount:** $2,169,981

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1

***Overall Objectives:** We will develop novel imaging markers to allow more personalized characterization of the AT(N) status of individual subjects. Building upon our extensive experience in brain shape analysis, we will develop tools to quantify the topographic pattern of cortical tau and Aβ pathology and build a personalized analysis framework for the early detection of localized brain atrophy.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Effects of Resonance-Frequency Breathing on Preclinical Alzheimer's Disease Biomarkers and Cognition

***Status of Support:** Current

Proposal/Award Number: R01AG080652

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2023

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $3,888,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.6

***Overall Objectives:** In this clinical trial, we compare the effects of 10 weeks of resonance-frequency breathing and random-paced breathing on plasma AD biomarkers and on learning rates in African-American and European-American adults age 50-70. We will also test whether paced breathing effects relate to magnetic resonance imaging biomarkers of brain waste clearance (cerebrospinal flow and perivascular space volume).

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Vascular contributions to dementia and genetic risk factors for Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: 5P01AG052350-08

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2025	0.6

***Overall Objectives:** This program project focuses on alterations in the neurovascular unit (e.g., blood-brain barrier, pericytes) in persons carrying the apolipoprotein E4 and autosomal dominant AD mutations to clarify vascular contributions in AD.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Neurodevelopment in children from families with genetic frontotemporal dementia and Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: R01AG071756

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of California San Francisco

***Proposal/Active Project Start Date: (MM/YYYY):** 02/2022

***Proposal/Active Project End Date: (MM/YYYY):** 01/2027

***Total Anticipated Proposal/Project Amount:** $802,137

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	0.2

***Overall Objectives:** This project will compare cognition and neuroanatomy in children with FTD mutations and children with AD mutations to typically developing children with the hypothesis that cognitive profiles and neural circuits associated with each disease will show abnormalities in mutation carriers.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Structural and diffusion changes of perivascular space in aging, cognitive decline and Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: R01AG070825

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 06/2026

***Total Anticipated Proposal/Project Amount:** $4,124,992

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3

***Overall Objectives:** This study will identify pathological alterations to the brain glia- lymphatic pathway (i.e. perivascular spaces) is expected to aid in Alzheimer’s disease diagnosis and disease monitoring. It is also expected to fill the gap in knowledge regarding the role of the perivascular spaces in Alzheimer’s disease and its pathological neuroimaging signature.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Phenotype and Genotype of Autosomal Dominant Alzheimer’s Disease in Jalisco, Mexico

***Status of Support:** Current

Proposal/Award Number: 1R01AG069013

***Source of Support:** NIH Fogarty

***Primary Place of Performance:** University of Southern California, University of Guadalajara

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2020

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $3,255,386

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1

***Overall Objectives:** To facilitate research in autosomal dominant AD in Jalisco state by providing the technology for research and clinical genetics as well as funding and training to develop a cohort that will prospectively be followed using standardized clinical and cognitive measures, enabling additional research studies.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Surface-Based Fiber Tracking and Modeling Techniques for Mapping the Superficial White Matter Connectome with Diffusion MRI

***Status of Support:** Current

Proposal/Award Number: 2R01EB022744

***Source of Support:** NIH/NIBIB

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2016

***Proposal/Active Project End Date: (MM/YYYY):** 01/2027

***Total Anticipated Proposal/Project Amount:** $562,847

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.2
2027	0.2

***Overall Objectives:** Over the last funding period from 2016 to 2022, we have developed various cutting-edge tools for connectome modeling that include the reconstruction of fiber orientation distributions (FODs) from multi-shell dMRI, fiber tracking via the propagation of Frenet-Serret and parallel transport frames (PTFs) tract filtering with topographic regularity, and groupwise consistency,

and deep learning algorithms for distortion correction. We have also conducted rigorous validations of fiber tractography against tracer injections in mouse brains and released multiple probabilistic atlases of important fiber pathways in human brain. Building upon our extensive experiences in developing these cutting-edge computational tools for connectome modeling and our long track record in developing advanced surface mapping and shape analysis tools, we will develop a series of surface-based modeling techniques to greatly advance SWM connectivity research.

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Ringman, John in SciENcv on 2026-01-27 12:21:48

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Rissman, Robert

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-9245-8278

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: UNIVERSITY OF SOUTHERN CALIFORNIA, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	The role of TREM2 in Alzheimer's disease related neuropathogenesis in people with HIV
***Status of Support:**	Current
Proposal/Award Number:	1R01AG092255-01
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	UNIVERSITY OF CALIFORNIA, SAN DIEGO
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$3,776,237

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.48
2026	0.48
2027	0.48
2028	0.48
2029	0.48

***Overall Objectives:** Major Goals: To investigate human immunodeficiency virus infection of the brain as a stimulus to promote Alzheimer's disease neuropathogenesis through dysregulation of inflammatory signaling.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:**	Advancing Diagnosis of Dementia in Primary Care Practice(ADDPCP)
***Status of Support:**	Current
Proposal/Award Number:	R01AG085689
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	UNIVERSITY OF SOUTHERN CALIFORNIA
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2024
***Proposal/Active Project End Date: (MM/YYYY):**	05/2029
***Total Anticipated Proposal/Project Amount:**	$993,648

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.48
2026	0.48
2027	0.48
2028	0.48

***Overall Objectives:** This project will expand collaborations with primary care practices using the cognitive complaints toolkit for Alzheimer's disease (ACCT-AD), including evaluating the accuracy of primary care diagnosis using expert second-opinion and biomarkers.

***Statement of Potential Overlap:** Nothing to report.

***Proposal/Active Project Title:**	Impact of Comorbidities on the Utility of Plasma Amyloid and Tau to Predict Alzheimer's Disease
***Status of Support:**	Current
Proposal/Award Number:	5R01AG088623-02
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	UNIVERSITY OF SOUTHERN CALIFORNIA
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2024
***Proposal/Active Project End Date: (MM/YYYY):**	05/2029
***Total Anticipated Proposal/Project Amount:**	$2,973,790

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2025	0.6
2026	0.6
2027	0.6
2028	0.6

***Overall Objectives:** Major Goals: This proposal aims aim to characterize plasma p-tau217 and Aβ42/40 in the HABS-HD study.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Impact of TBI and Cognitive Decline on Alzheimer's Disease Brain-Derived Exosome Cargo

***Status of Support:** Current

Proposal/Award Number: 1RF1AG079303-01A1

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF SOUTHERN CALIFORNIA

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2023

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $2,614,250

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.9

***Overall Objectives:** Major Goals: The objective of this project is to determine if circulating proteins carried in neuronal exosomes have utility as biomarkers for neurodegeneration and how this is impacted by TBI, PTSD in twins from the VETSA Study.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** The Health & Aging Brain Study – Health Disparities (HABS-HD)

***Status of Support:** Current

Proposal/Award Number: 5U19AG078109-03

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF NORTH TEXAS HLTH SCI CTR

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $149,880,426

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.2
2026	1.2

***Overall Objectives:** Major Goals: (1) to collect imaging, clinical, biological and genetic data resulting in the largest longitudinal cohort of diverse populations examining AT(N) biomarkers across the three most prevalent racial/ethnic groups in the U.S.; (2) collect life-course exosome, as well as, sociocultural and behavioral data in alignment with the NIA Framework for Health Disparities Research, and examine how these factors affect the timing, sequence and trajectories of AT(N) biomarkers among diverse populations; (3) to disseminate HABS- HD data and samples to the global scientific community in a timely manner.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Vascular Contributions to Dementia and Genetic Risk Factors for Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: 2P01AG052350-08

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6

Year	Person Months
2026	0.6
2027	0.6

***Overall Objectives:** Major Goals: The overall goal of this continuing program project is to test the 'neurovascular hypothesis' of AD and establish whether the neurovasculature has a major role in the pathogenesis of early cognitive decline, and therefore could be a key new therapeutic target to treat cognitive impairment, dementia and early AD.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Precision Medicine for Inflammatory Treatment for Alzheimer's Disease in Down Syndrome

***Status of Support:** Current

Proposal/Award Number: 4R01AG073979-02

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF SOUTHERN CALIFORNIA

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $3,896,985

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.2
2026	1.2

***Overall Objectives:** Major Goals: The major goal of this project is to explore the biological makeup of DS-AD is as heterogeneous as the sporadic AD population and that identifying specific subsets of DS- AD for particular treatments will yield meaningful therapeutic responses.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Novel Systemic Delivery of Peptide-Mediated Anti-Sense Oligonucleotides for Dementia with Lewy Bodies

***Status of Support:** Current

Proposal/Award Number: 7R01AG072053-03

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF SOUTHERN CALIFORNIA

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2021

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $3,059,589

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6

***Overall Objectives:** Major Goals: The major goal of this project is to examine the pharmacokinetics and toxicology of systemic ApoB11:2'-MOe si a-syn following intra-peritoneal delivery and to examine the ability of the ApoB11:2'-MOe si a-syn to reduce a-syn and improve survival of neurons and improve cognitive ability and motor coordination in an a-syn tg in an a-syn tg mouse model of DLB.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: P30AG066530-06

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $4,770,384

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.2
2026	1.2
2027	1.2
2028	1.2

Year	Person Months
2029	1.2

***Overall Objectives:** Major Goals: (1) to collect imaging, clinical, biological and genetic data resulting in the largest longitudinal cohort of diverse populations examining AT(N) biomarkers across the three most prevalent racial/ethnic groups in the U.S.; (2) collect life-course exosome, as well as, sociocultural and behavioral data in alignment with the NIA Framework for Health Disparities Research, and examine how these factors affect the timing, sequence and trajectories of AT(N) biomarkers among diverse populations; (3) to disseminate HABS- HD data and samples to the global scientific community in a timely manner.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF NORTH TEXAS HLTH SCI CTR

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2020

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $45,656,674

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.6

***Overall Objectives:** Major Goals: The major goal of this project is to address an important health disparity faced by Mexican Americans suffering from AD and MCI.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Alzheimer's Clinical Trials Consortium (ACTC)

***Status of Support:** Current

Proposal/Award Number: 5U24AG057437-07

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF SOUTHERN CALIFORNIA

***Proposal/Active Project Start Date: (MM/YYYY):** 12/2017

***Proposal/Active Project End Date: (MM/YYYY):** 08/2028

***Total Anticipated Proposal/Project Amount:** $28,436,169

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.4
2027	1.4
2025	1.2

***Overall Objectives:** Major Goals: The timeframe, complexity, and expense of the recruitment process and site activation for Alzheimer's disease (AD) trials are extremely challenging, and indeed trial enrollment and site network development represent a bottleneck for drug development for the AD field. This proposal aims aim to create an optimized infrastructure to optimally design, efficiently launch, rapidly recruit with improved diversity, and successfully complete NIH supported clinical trials in AD and related dementias.

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** Alzheimer's Disease Neuroimaging Initiative (ADNI)

***Status of Support:** Current

Proposal/Award Number: 2U19AG024904-16

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** NORTHERN CALIFORNIA INSTITUTE/RES/EDU

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2004

***Proposal/Active Project End Date: (MM/YYYY):** 07/2028

***Total Anticipated Proposal/Project Amount:** $59,412,691

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.12
2026	0.12

Year	Person Months
2027	0.12

***Overall Objectives:** Major Goals: The overall goal of the (ADNI) is to discover, standardize, and validate biomarkers for AD treatment trials. Validation is accomplished by comparing and correlating clinical/cognitive with biomarker data. Impact of ADNI has been to optimize, standardize and validate biomarkers, especially brain amyloid by PET and CSF measurements of Aβ peptides, termed “Aβ amyloid- phenotyping."

***Statement of Potential Overlap:** Nothing to report

***Proposal/Active Project Title:** The Neighborhoods Study: Contextual Disadvantage and Alzheimer's Disease and Related Dementias (ADRD)

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** UNIVERSITY OF SOUTHERN CALIFORNIA

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2026

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** $2,372,915

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48
2027	0.48
2028	0.48
2029	0.48
2030	0.48

***Overall Objectives:** This project leverages data within 22 Alzheimer’s Disease Research Centers across the US to conduct the largest study of its kind on social determinants of health in the context of AD and to develop a novel collaborative infrastructure of contextual exposure for future social-biological phenotypic evaluation-- a potential pathway to new therapeutics directly responsive to the NIA mission.

***Statement of Potential Overlap:** Adjustments to avoid overlap will be made to my other support prior to funding of this application.

***Proposal/Active Project Title:**	a-Synuclein vulnerability mechanisms and therapeutics in Alzheimer's disease
***Status of Support:**	Pending
Proposal/Award Number:	1R01AG103122-01
***Source of Support:**	National Institute on Aging
***Primary Place of Performance:**	UNIVERSITY OF SOUTHERN CALIFORNIA
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$2,253,046

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** Nothing to report.

***Statement of Potential Overlap:** Adjustments to avoid overlap will be made to my other support prior to funding of this application.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Rissman, Robert in SciENcv on 2026-01-26 18:10:20

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Schneider, Lon S.

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-4636-6150

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: The University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Alzheimer's Disease Neuroimaging Initiative (ADNI4)
***Status of Support:**	Current
Proposal/Award Number:	5U19AG024904-18
***Source of Support:**	Northern California Institute for Research and Education
***Primary Place of Performance:**	The University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2025
***Proposal/Active Project End Date: (MM/YYYY):**	07/2026
***Total Anticipated Proposal/Project Amount:**	$36,840

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:** To conduct the neuroimaging initiative at USC.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Improving Equitable Access to Neurology Care to Achieve Pharmacoequity for Older Adults with ADRD
***Status of Support:**	Current
Proposal/Award Number:	1R01NS143735-01

***Source of Support:** National Institute of Neurological Disorders and Stroke

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 07/2029

***Total Anticipated Proposal/Project Amount:** $97,024

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24
2027	0.24
2028	0.24
2029	0.24

***Overall Objectives:** The goal of this study is to understand the multilevel barriers to and facilitators of equitable access to neurology care and develop a community-informed, evidence-based patient navigation intervention to optimize neurology access from primary care for Black and Hispanic older adults, while improving access to future novel ADRD therapies.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Alzheimer's Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 2P30AG066530-06

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $2,270,226

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2

Year	Person Months
2028	1.2
2029	1.2
2030	1.2

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Allopregnanolone as Regenerative Therapeutic for Alzheimer's: Phase 2 Clinical Trial

***Status of Support:** Current

Proposal/Award Number: R01AG063826-04

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2025

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $48,687

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** Within this trial, efficacy of Allo to modify rate of change on ADAS-Cog14 and chronic safety will be investigated. Exploratory analyses will determine validity of MRI-based measures to function as a surrogate biomarker of regenerative target engagement and iPSC-based strategy to function as predictive biomarker to identify regenerative responders and non-responders.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** ADRC Consortium for Clarity in ADRD Research Through Imaging

***Status of Support:** Current

Proposal/Award Number: 5U01AG082350-02

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $510,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3

***Overall Objectives:** The primary goals of this project are to collect longitudinal ATN imaging including PET and MRI, and blood on ADRC Clinical Core participants to gain insights on the multiple intersecting causes of cognitive impairment, and to link (via NACC) the imaging and plasma collection to all available cognitive and neurobehavioral data collection, genetics and eventually neuropathology on these same participants.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** DIAN-TU E2814 Trial in Dominantly Inherited Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: NGSA-REQ-00009716

***Source of Support:** Washington University

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2024

***Proposal/Active Project End Date: (MM/YYYY):** 05/2029

***Total Anticipated Proposal/Project Amount:** $489,274

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8

***Overall Objectives:** Dr. Schneider serves as the project arm leader for the E2814 arm of the DIAN-TU platform trial.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Alzheimer's Clinical Trials Consortium
***Status of Support:**	Current
Proposal/Award Number:	5U24AG057437-08
***Source of Support:**	National Institutes of Aging
***Primary Place of Performance:**	The University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2023
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$996,229

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** The proposal aims to create an optimized infrastructure to optimally design, efficiently launch, rapidly recruit with improved diversity, and successfully complete NIH supported clinical trials in AD and related dementias.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	USC California Alzheimer Disease Center
***Status of Support:**	Current
Proposal/Award Number:	23-10416
***Source of Support:**	State of California
***Primary Place of Performance:**	The University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2023
***Proposal/Active Project End Date: (MM/YYYY):**	06/2028
***Total Anticipated Proposal/Project Amount:**	$1,409,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2
2028	1.2

***Overall Objectives:** This center is established to support the clinical care of and follow-up on patients with dementia and provide educational outreach to the community. Goals are to address the needs of our understudied and underserved communities and provide a model of statewide collaboration.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Phase II randomized controlled trial of benfotiamine in early Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: R01AG086634-01

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2022

***Proposal/Active Project End Date: (MM/YYYY):** 06/2026

***Total Anticipated Proposal/Project Amount:** $837,100

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.24

***Overall Objectives:** A clinical trial of benfotiamine for Alzheimer's disease.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Safety/Tolerability/Immunogenicity of first-in-human Aβ DNA vaccine, AV-1959D Phase 1 trials AD subjects: based on IND18953 cleared by FDA

***Status of Support:** Current

Proposal/Award Number: 5R01AG074983-04

***Source of Support:** National Institutes of Aging

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 02/2022

***Proposal/Active Project End Date: (MM/YYYY):** 01/2027

***Total Anticipated Proposal/Project Amount:** $828,596

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8

***Overall Objectives:** To assess the safety and immunogenicity of Aβ DNA vaccine AV-1959D and to develop it through early clinical trials.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** AHEAD 3-45 Study: A Placebo-Controlled, Double-Blind, Parallel-Treatment Arm, 216 Week Study to Evaluate Efficacy and Safety of Treatment With BAN2401 in Subjects With Preclinical Alzheimer's Disease-5451840005

***Status of Support:** Current

Proposal/Award Number: 011970-00001

***Source of Support:** Eisai Incorporated

***Primary Place of Performance:** The University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2020

***Proposal/Active Project End Date: (MM/YYYY):** 12/2027

***Total Anticipated Proposal/Project Amount:** $2,638,930

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.84
2027	0.84

***Overall Objectives:** A clinical trial of lecanemab for preclinical Alzheimer's Disease

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	FDA approved IND 29644: Multicenter Phase 1 Trial to Evaluate Safety and Immunogenicity of Preventive Tau Vaccine, AV-1980R/A, in cognitively unimpaired preclinical Alzheimer's Disease participants
***Status of Support:**	Pending
Proposal/Award Number:	1R01AG092949-01
***Source of Support:**	National Institutes of Aging
***Primary Place of Performance:**	The University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2025
***Proposal/Active Project End Date: (MM/YYYY):**	07/2030
***Total Anticipated Proposal/Project Amount:**	$934,990

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8
2030	1.8

***Overall Objectives:** The primary objective of this project is to assess the safety and tolerability of the tau vaccine, AV-1980R/A, and the secondary objective is to analyze its immunogenicity in a Phase 1 study involving preclinical Alzheimer's disease (AD) participants. This Phase 1 trial is based on FDA-cleared IND 29644.

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Schneider, Lon in SciENcv on 2026-01-29 20:22:43

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Segal-Gidan, FREDDI I

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-4776-9840

*POSITION TITLE: Associate Professor

*POSITION TITLE: Associate Professor

*ORGANIZATION AND LOCATION: University of Southern California, Keck School of Medicine, Los Angeles, California, United States

*ORGANIZATION AND LOCATION: University of Southern California, Keck School of Medicine, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimers Disease Research Center

***Status of Support:** Current

Proposal/Award Number: 2 P30AG066539-06

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,693,508

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.48

***Overall Objectives:** NIH-funded research center that supports basic and clinical research focused on Alzheimers disease and related disorders through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Advancing Diagnosis of Dementia in Primary Care Practice (ADDPCP)

***Status of Support:** Current

Proposal/Award Number: R01G085689

***Source of Support:** NIH/NIA

***Primary Place of Performance:** Rancho los Amigos National Rehabilitation Center

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2024

***Proposal/Active Project End Date: (MM/YYYY):** 05/2029

***Total Anticipated Proposal/Project Amount:** $13,167,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2

***Overall Objectives:** Assessing the accuracy of dementia diagnosis by primary care community providers

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Evaluation of a new cognitive screening approach in primary care among underserved populations

***Status of Support:** Current

Proposal/Award Number: CDPH 24-10130

***Source of Support:** California Dept. of Public Health

***Primary Place of Performance:** Los Angeles County Dept. of Health

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2024

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $4,787,847

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2

***Overall Objectives:** Evaluate the impact of the implementation of the cognitive health assessment (CHA), a screening approach for dementia, in primary care community based clinics in San Francisco and Los Angeles counties.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** California Alzheimers Disease Center

***Status of Support:** Current

Proposal/Award Number: CDPH 23-10643

***Source of Support:** California Dept. of Public Health (CDPH)

***Primary Place of Performance:** Rancho los Amigos National Rehabilitation Center

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2023

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $1,409,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3.6

***Overall Objectives:** One of 10 state-funded centers that provides comprehensive clinical evaluation, diagnosis and ongoing treatment for persons with dementia.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Segal-Gidan, FREDDI in SciENcv on 2026-01-21 12:50:54

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Sheikh-Bahaei, Nasim

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0001-7029-7215

*POSITION TITLE: Associate Professor of Radiology and Neurology, Chief of Neuroradiology, MD, MRCP, FRCR

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles , California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Discovery and Trial Ready Cohort for Limbic Predominant Age Related TDP-43 Encephalopathy (TRC LATE)
***Status of Support:**	Current
Proposal/Award Number:	R01 AG095017
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California, Los Angeles
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$17,948,254

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** Limbic-predominant age-related TDP-43 encephalopathy neuropathologic change (LATE-NC) is a common pathological condition that significantly contributes to cognitive decline in older adults and plays a critical role in Alzheimer's-type dementia in older age groups. This study will recruit and characterize 420 participants aged 85 and older across five Alzheimer's disease

research centers to validate clinical markers, discover new biomarkers, and develop recruitment strategies for early-phase LATE clinical trials.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)
***Status of Support:** Current
Proposal/Award Number: P30AG066530-06
***Source of Support:** NIA
***Primary Place of Performance:** University of Southern California, Los Angeles
***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025
***Proposal/Active Project End Date: (MM/YYYY):** 03/2030
***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2
2028	1.2
2029	1.2
2030	1.2

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** ADRD risk factors in Middle Eastern and North African immigrants in the US
***Status of Support:** Current
Proposal/Award Number: AARG-23-1150420
***Source of Support:** Alzheimer's Association
***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 01/2023

***Proposal/Active Project End Date: (MM/YYYY):** 01/2026

***Total Anticipated Proposal/Project Amount:** $200,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3

***Overall Objectives:** This study proposes to conduct a preliminary recruitment of aging MENA individuals to assess the prevalence of ADRD risk factors and their relationship with immigration factors and perceived stress, along with impacts on brain health

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Large-scale harmonization and integration of multi-modal ADNI data for the early detection of Alzheimer's disease and related dementias

***Status of Support:** Current

Proposal/Award Number: R01NS128486-01

***Source of Support:** NINDS

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2022

***Proposal/Active Project End Date: (MM/YYYY):** 05/2027

***Total Anticipated Proposal/Project Amount:** $3,952,966

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	1

***Overall Objectives:** The aim of this study is to harmonize ADNI dataset including all clinical, imaging, CSF and other biomarkers and develop an algorithm for improving the diagnosis of Alzheimer’s disease and related dementia utilizing AI and ML.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Structural and diffusion changes of perivascular space in aging, cognitive decline and Alzheimer's disease
***Status of Support:**	Current
Proposal/Award Number:	R01AG070825
***Source of Support:**	NIA
***Primary Place of Performance:**	University of Southern California, Los Angeles
***Proposal/Active Project Start Date: (MM/YYYY):**	10/2021
***Proposal/Active Project End Date: (MM/YYYY):**	06/2026
***Total Anticipated Proposal/Project Amount:**	$4,118,155

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.5

***Overall Objectives:** This grant aims to investigate the role of PVS in cognitive decline and dementia using ADNI, USC ADRC, HCP and VCID datasets. We quantify and score the PVS in different stages of dementia and evaluate its association with cognitive decline and other biomarkers of AD.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Sheikh-Bahaei, Nasim in SciENcv on 2026-01-30 13:15:20

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Shi, Yonggang

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-3466-4302

*POSITION TITLE: Professor of Neurology, Biomedical Engineering, and Electrical and Computer Engineering

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 5 P30AG066530-06

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $1,925,563

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6
2030	0.6

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID)

***Status of Support:** Current

Proposal/Award Number: 1 RF1AG084072-01

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2023

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $2,487,668

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** Vascular contributions to cognitive impairment and dementia (VCID) have been recognized as an important cause of clinical dementia. This project aims to systematically evaluate novel metrics of small vessel morphology/density and perfusion as imaging biomarkers of VCID in a multiethnic longitudinal cohort of 200 subjects (including 50 Asian Americans) that are enriched for small vessel VCID. This project is expected to result in a powerful suite of imaging tools for comprehensive characterization of the morphology and function of cerebral small vessels, as well as to fill in the important gap in health disparities of Asian Americans in ADRD research.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Shape-based personalized AT(N) imaging markers of Alzheimer's disease

***Status of Support:** Current

Proposal/Award Number: 1 RF1AG077578-01A1

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2023

***Proposal/Active Project End Date: (MM/YYYY):** 05/2026

***Total Anticipated Proposal/Project Amount:** $3,788,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2026	3

***Overall Objectives:** We will develop novel imaging markers to allow more personalized characterization of the AT(N) status of individual subjects. 55Building upon our extensive experience in brain shape analysis, we will develop tools to quantify the topographic pattern of cortical tau and Aß pathology and build a personalized analysis framework for the early detection of localized brain atrophy.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** The Health & Aging Brain Study - Health Disparities (HABS-HD)

***Status of Support:** Current

Proposal/Award Number: 5 U19AG078109-03

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2022

***Proposal/Active Project End Date: (MM/YYYY):** 08/2027

***Total Anticipated Proposal/Project Amount:** $916,269

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2
2027	1.2

***Overall Objectives:** Will provide a secure data repository and data sharing infrastructure for neuroimaging data acquired for this study and uploaded to the LONI Image and Data Archive (IDA). Will oversee the neuroimaging archiving activities including custom configuration for the study in the repository, access management, aiding the site(s) archiving neuroimaging data, securely storing MRI and PET data and providing an easy to use, web-based data archiving and sharing infrastructure. All upload/download/edit and deletion activities will be tracked.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Quantitative cerebral blood vessel imaging biomarkers for AD and VCID

***Status of Support:** Current

Proposal/Award Number: 7 RF1AG072490-02

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2021

***Proposal/Active Project End Date: (MM/YYYY):** 03/2026

***Total Anticipated Proposal/Project Amount:** $2,408,375

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.46

***Overall Objectives:** Specific mechanisms linking cerebrovascular dysfunction and AD/VCID pathology remain unclear. This application will develop a class of cerebral blood vessel imaging biomarkers to characterize and differentiate the vascular pathology in AD and VCID.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 5 P30AG066530-05

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2020

***Proposal/Active Project End Date: (MM/YYYY):** 02/2026

***Total Anticipated Proposal/Project Amount:** $16,428,655

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.46

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Tau-induced connectome imaging markers of Alzheimer's disease
***Status of Support:**	Current
Proposal/Award Number:	4 R01AG064584-02
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2020
***Proposal/Active Project End Date: (MM/YYYY):**	03/2026
***Total Anticipated Proposal/Project Amount:**	$2,130,568

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.75

***Overall Objectives:** Our project will for the first time provide the comprehensive and in vivo characterization of the fiber pathways affected by tau pathology in Alzheimer's disease (AD). This systematic investigation is important for improving our understanding of the biological mechanism of AD and provide more targeted imaging makers for the early prediction of AD at the prodromal stage.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Surface-Based Fiber Tracking and Modeling Techniques for Mapping the Superficial White Matter Connectome with Diffusion MRI
***Status of Support:**	Current
Proposal/Award Number:	5 R01EB022744-07
***Source of Support:**	NIH/NIBIB
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2016
***Proposal/Active Project End Date: (MM/YYYY):**	01/2028
***Total Anticipated Proposal/Project Amount:**	$562,847

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.3
2027	2.3
2028	2.3

***Overall Objectives:** Over the last funding period from 2016 to 2022, we have developed various cutting-edge tools for connectome modeling that include the reconstruction of fiber orientation distributions (FODs) from multi-shell dMRI, fiber tracking via the propagation of Frenet-Serret and parallel transport frames (PTFs) tract filtering with topographic regularity, and groupwise consistency, and deep learning algorithms for distortion correction. We have also conducted rigorous validations of fiber tractography against tracer injections in mouse brains and released multiple probabilistic atlases of important fiber pathways in human brain. Building upon our extensive experiences in developing these cutting-edge computational tools for connectome modeling and our long track record in developing advanced surface mapping and shape analysis tools, we will develop a series of surface-based modeling techniques to greatly advance SWM connectivity research.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Multi-tracer tau PET harmonization for multimodal modeling of Alzheimer's disease complexity
***Status of Support:**	Pending
Proposal/Award Number:	1R01AG103481-01
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2026
***Proposal/Active Project End Date: (MM/YYYY):**	06/2031
***Total Anticipated Proposal/Project Amount:**	$4,104,965

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	3.6
2028	3.6
2029	3.6
2030	3.6
2031	3.6

***Overall Objectives:** Our project will provide fundamentally novel solutions for the harmonization of tau PET imaging data from different tracers while preserving biological variability and diagnostic information. We will develop novel tools for personalized tau pathology analysis and longitudinal prediction of tau topography. We will also develop foundation models for early diagnosis, longitudinal prediction, and survival analysis of Alzheimer's disease.

***Statement of Potential Overlap:** None

Certification:

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I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Shi, Yonggang in SciENcv on 2026-01-12 20:15:35

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Towfighi, Amytis

*POSITION TITLE: Professor of Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** REACH-EpiVCID

***Status of Support:** Current

Proposal/Award Number: 1RF1NS139183-01

***Source of Support:** NINDS

***Primary Place of Performance:** Rancho Los Amigos National Rehabilitation Center

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2024

***Proposal/Active Project End Date: (MM/YYYY):** 07/2027

***Total Anticipated Proposal/Project Amount:** $2,403,545

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	2
2026	2
2027	2

***Overall Objectives:** DNA and micro RNA sequencing data and derived data under the REACH-EpiVCID will be registered and submitted to dbGaP. Phenotypic and DNA common variant genotype data for 700 White, Black, Hispanic and Asian ICH survivors with extensive SDOH, BP control and outcome data is being collected under the parent REACH-ICH award.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Race/Ethnicity, Hypertension and Prevention of VCID and Stroke after Intracerebral Hemorrhage

***Status of Support:** Current

Proposal/Award Number: 2R01NS093870-06A1

***Source of Support:** NINDS

***Primary Place of Performance:** Rancho Los Amigos National Rehabilitation Center

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2022

***Proposal/Active Project End Date: (MM/YYYY):** 07/2027

***Total Anticipated Proposal/Project Amount:** $1,855,014

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.2
2026	1.2
2027	1.2

***Overall Objectives:** REACH-ICH aims to: 1. Determine whether modifiable social determinants of health (SDOH) in association with established biological factors, predict increased risk for Vascular contributions to Cognitive Impairment and Dementia (VCID) after intracerebral hemorrhage (ICH); 2) determine whether modifiable SDOH predict risk of poor hypertension control after ICH; 3) determine whether modifiable SDOH are associated with decreased engagement with our standardized BP management program. The data generated will be used to develop novel strategies for BP management and prevention of VCID and recurrent stroke after ICH, and are likely to have implications for preventing CSVD-related VCID and stroke, even in those who have not suffered an ICH. We will enroll and follow longitudinally 700 survivors of intracerebral hemorrhage divided among Black, Hispanic, White, Asian individuals, all of whom will receive a standardized state-of-the-art evidence-based program of blood pressure management.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Southern California Clinical and Translational Science Institute

***Status of Support:** Current

Proposal/Award Number: UL1TR001855

***Source of Support:** NIH/NCATS

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $45,831,091

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	3
2026	3
2027	3

***Overall Objectives:** The goal of SC CTSI is to be a leader in engaging diverse communities and special populations in clinical and translational research to improve their health in partnership with the Los Angeles County Department of Health Services and broad-based community health organizations throughout Los Angeles.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** UCLA-UCI Center for Eliminating Cardio-Metabolic Disparities in Multi-Ethnic Populations (UC END-DISPARITIES)

***Status of Support:** Current

Proposal/Award Number: 1 P50 MD017366-01

***Source of Support:** NIH- National Institutes of Health

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2021

***Proposal/Active Project End Date: (MM/YYYY):** 08/2026

***Total Anticipated Proposal/Project Amount:** $17,012,820

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	2.4
2026	2.4

***Overall Objectives:** UC END-DISPARITIES will function through three highly integrated cores and three center research projects across southern California, all supporting Pilot Awards and Community Catalyst Awards and partnering with key community, health system, and academic partners to prevent cardiometabolic disease and reduce its harmful impact in minority communities.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

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Certified by Towfighi, Amytis in SciENcv on 2026-01-21 19:09:50

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Wang, Xinhui

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-1180-4624

*POSITION TITLE: Assistant Professor of Research Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: 2P30AG066530-06

***Source of Support:** NIH

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,693,508

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3
2027	3
2028	3
2029	3
2030	3

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:**	Worldwide Mapping of Air Pollution Exposure Patterns on Aging Brain Health
***Status of Support:**	Current
Proposal/Award Number:	R01ES033961
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2022
***Proposal/Active Project End Date: (MM/YYYY):**	06/2028
***Total Anticipated Proposal/Project Amount:**	$1,955,199

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	4.8
2027	4.8
2028	4.8

***Overall Objectives:** This proposal will address this gap through a new environmental working group within the ENIGMA consortium. This will be the first international study to assess how AP 1) affects the structural integrity of the CAN in older adults across geographically diverse datasets, 2) interacts with biopsychosocial factors to influence heterogeneity in AD risk within the aging brain, and 3) influences brain integrity across the life course.

***Statement of Potential Overlap:** none

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Wang, Xinhui in SciENcv on 2026-01-16 19:50:11

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Yassine, Hussein N.

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-2483-649X

*POSITION TITLE: Professor (with tenure)

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults
***Status of Support:**	Current
Proposal/Award Number:	4R01AG078362-02
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	08/2027
***Total Anticipated Proposal/Project Amount:**	$2,307,632

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8

***Overall Objectives:** To relate blood measures of diabetes and AD risk to cognitive testing and scans that evaluate brain structure, function, and health of the brain's blood vessels.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:**	Drug Development for Alzheimer's Neuroinflammation
***Status of Support:**	Current
Proposal/Award Number:	1U01AG094622-01
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2025
***Proposal/Active Project End Date: (MM/YYYY):**	07/2030
***Total Anticipated Proposal/Project Amount:**	$8,128,559

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4
2027	2.4
2028	2.4
2029	2.4
2030	2.4

***Overall Objectives:** Alzheimer's disease (AD) is a debilitating aging disease with no effective therapies. Brain inflammation is one of the key mechanisms present in the Alzheimer's brain. We propose the development of AD therapies based on reducing the activity of calcium-dependent phospholipase A2, a key driver of brain inflammation.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:**	Alzheimer Disease Research Center (ADRC)
***Status of Support:**	Current
Proposal/Award Number:	2P30AG066530-06
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	03/2030
***Total Anticipated Proposal/Project Amount:**	$23,696,676

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.4
2027	1.4
2028	1.7
2029	1.7
2030	1.7

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through its administrative, clinical, imaging, pathology, education, and information transfer cores.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:**	Impact of circulating levels of endocrine disrupting toxicants on cognition, brain activity and Alzheimer's disease biomarkers across the menopausal transition
***Status of Support:**	Current
Proposal/Award Number:	
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	07/2025
***Proposal/Active Project End Date: (MM/YYYY):**	06/2030
***Total Anticipated Proposal/Project Amount:**	$4,055,366

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2028	0.6
2029	0.6

***Overall Objectives:** The overarching goal of this project is to test whether exposure to endocrine disrupting persistent organic pollutants (POPs) have differential impacts on cognitive and brain health in midlife women and men, and represent a potentially modifiable risk factor for AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Brain cPLA2 as a mechanism for neuroinflammation in AD/ADRD with and without APOE4
***Status of Support:**	Current
Proposal/Award Number:	4R01AG076124-02
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	05/2025
***Proposal/Active Project End Date: (MM/YYYY):**	04/2027
***Total Anticipated Proposal/Project Amount:**	$1,568,908

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.8
2026	1.8

***Overall Objectives:** The elucidation of potentially modifiable molecular pathways involved in Alzheimer's disease (AD)/Alzheimer's Disease Related Dementia (ADRD) is of great scientific interest and offers hope for improved public health.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:**	Finance, Cognition, and Default Network in Aging (FCDNA)
***Status of Support:**	Current
Proposal/Award Number:	RF1AG068166/R01AG068166
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	05/2021
***Proposal/Active Project End Date: (MM/YYYY):**	04/2026

***Total Anticipated Proposal/Project Amount:** $3,608,795

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** The overall goal of this study is to examine the association of financial exploitation in old age with markers of Alzheimer's Disease.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** ApoE, ABCA1 and endosomal dysregulation in AD

***Status of Support:** Current

Proposal/Award Number: 1R01AG067063

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2020

***Proposal/Active Project End Date: (MM/YYYY):** 02/2026

***Total Anticipated Proposal/Project Amount:** $3,421,918

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.12

***Overall Objectives:** Activity regulate endosomal trafficking in human and animal tissues.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:** Alzheimer's Abeta, Tau, Apolipoproteins and Blood-brain barrier

***Status of Support:** Current

Proposal/Award Number: 5R01NS034467

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 03/2020

***Proposal/Active Project End Date: (MM/YYYY):** 02/2026

***Total Anticipated Proposal/Project Amount:** $3,947,137

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** Understanding at the cellular, molecular and systems level, and in allele-specific and gender-specific manner apolipoprotein E (apoE) effects on the cerebrovascular system, specifically, 1) how they contribute to Alzheimer's disease (AD) neurovascular dysfunction, neurodegenerative, amyloid-β (Aβ) and tau disorders; and 2) how targeting apoE-low density lipoprotein receptor-related protein 1 (LRP1) interactions on different cerebrovascular cell types can mitigate and/or block cerebrovascular and brain damage causing dementia in AD.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:** Mechanisms of ApoE glycosylation and AD Tau Pathology

***Status of Support:** Pending

Proposal/Award Number: 1R21AG098756-01A1

***Source of Support:** NIH/NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $444,350

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.6
2028	0.6

***Overall Objectives:** Mechanisms of ApoE glycosylation and AD Tau Pathology will determine how

sugar molecules attached to the Alzheimer's risk protein apolipoprotein E (ApoE) change its interactions with the toxic protein tau and will identify the enzymes that add or remove these sugars in human brain cells. By revealing whether ApoE glycosylation can be targeted to reduce harmful tau buildup, this work could open new avenues for therapies that delay or prevent Alzheimer's disease.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** SMART-APOE4: Adaptive Physical Activity Intervention Trial

***Status of Support:** Pending

Proposal/Award Number: 1R01AG104578-01

***Source of Support:** National Institute on Aging

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $4,136,736

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.96
2028	0.96
2029	0.96
2030	0.96
2031	0.96

***Overall Objectives:** SMART-APOE4: Adaptive Physical Activity Intervention Trial tests whether goal-oriented physical activity interventions can help adults increase regular activity levels and support brain health, thereby reducing Alzheimer's disease risk. It also explores whether personalized genetic counseling influences motivation or adherence to physical activity among individuals with and without genetic risk for Alzheimer's disease.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:** Precision Vascular Risk Management in APOE E4 Carriers

***Status of Support:** Pending

Proposal/Award Number: 1R01AG104255-01

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2031

***Total Anticipated Proposal/Project Amount:** $4,131,800

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.6
2028	0.6
2029	0.6
2030	0.6
2031	0.6

***Overall Objectives:** Precision Vascular Risk Management in APOE E4 Carriers is a 6-month clinical trial for the management of high blood pressure and high cholesterol in individuals carrying genetic risk for Alzheimer's disease. This proposal is relevant to public health as it works to understand how controlling vascular risk factors affects the risk of developing Alzheimer's disease dementia when started in mid life before the onset of dementia.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

***Proposal/Active Project Title:** The role of microglial lipid metabolism, cellular senescence and neuroinflammation in the pathogenesis of AD

***Status of Support:** Pending

Proposal/Award Number: 1R01AG100287-01

***Source of Support:** NIH/NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2031

***Total Anticipated Proposal/Project Amount:** $4,141,154

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8
2030	1.8

***Overall Objectives:** The role of microglial lipid metabolism, cellular senescence and neuroinflammation in the pathogenesis of AD study explores how lipid buildup in microglia contribute to Alzheimer's disease by causing these cells—and nearby neurons—to age and develop features of senescence. The research focuses on a protein called ABCA1, which usually helps remove cholesterol from cells but may instead be causing harmful cholesterol accumulation in Alzheimer's if trapped in lysosomes. Using human brain tissue and iPSC cell models, the team aims to map brain cellular senescence and test whether lowering ABCA1 levels can reduce measures of senescence in neurons and microglia which could lead to new treatments targeting lipid metabolism for AD.

***Statement of Potential Overlap:** None

***Proposal/Active Project Title:**	Proactive Health Precision Health Dashboard Harnessing the Power of Molecular Data, Digital Recordings, AI, and Individual Data Sharing
***Status of Support:**	Pending
Proposal/Award Number:	N/A
***Source of Support:**	Metabosensor
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	12/2025
***Proposal/Active Project End Date: (MM/YYYY):**	11/2030
***Total Anticipated Proposal/Project Amount:**	$593,073

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6
2027	0.6
2028	0.6
2029	0.6

Year	Person Months
2030	0.6

***Overall Objectives:** We aim to revolutionize an individual's ability to monitor their health, enabling personalized and proactive approaches to delay disease onset. We aim to find better ways to prevent disease, pursue wellness, and age in a healthy manner. Our metabolomics/exposomics/digital data with AI technologies can expand biomarker measurement to thousands of biomolecules, capturing the influences of environmental chemicals, gut microbiome, diet, exercise, and socioeconomic factors. It complements genomics data by providing a dynamic longitudinal assessment of health changes over time. Our 'digital twin' ecosystem will employ FAIR data principles to allow AI-based analysis to inform about trajectories of changes in health within each individual and define individuals who are similar empowering individuals to actively manage their health while sharing information with others in a protected way.

***Statement of Potential Overlap:** Overall, there is no scientific conflict between funded, pending, and the present application; they are complementary studies. If pending applications are awarded, effort on active projects will be adjusted within allowable levels.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Yassine, Hussein in SciENcv on 2026-01-29 18:53:53

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Zissimopoulos, Julie

PERSISTENT IDENTIFIER (PID): https://orcid.org/0009-0006-7008-132X

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Health Economics Investigation of the U.S. POINTER Trial Lifestyle Intervention
***Status of Support:**	Current
Proposal/Award Number:	
***Source of Support:**	Alzheimer's Association
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	01/2026
***Proposal/Active Project End Date: (MM/YYYY):**	12/2027
***Total Anticipated Proposal/Project Amount:**	$316,674

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1

***Overall Objectives:** The U.S. POINTER trial evaluated whether a two-year lifestyle intervention could protect cognitive function among older adults at risk for dementia, finding meaningful cognitive benefits in both structured and self-directed programs, even with relatively low resource intensity. However, economic evaluation is needed to assess the value of these interventions, identify which components drive health and cost outcomes, and estimate their longer-term individual and societal impacts. This project will address these gaps by estimating short- and medium-term economic effects and laying the groundwork for lifetime impact analysis using dynamic microsimulation.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** Improving Dementia Detection and Diagnosis in Diverse Populations

***Status of Support:** Current

Proposal/Award Number: 25HSR-ADRD-1487712

***Source of Support:** Alzheimer's Association

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 09/2027

***Total Anticipated Proposal/Project Amount:** $1,190,186

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.6

***Overall Objectives:** To identify the Medicare benefits, healthcare encounters, and diagnostic factors that most effectively lead to timely and accurate dementia diagnoses across populations. Using advanced machine learning and statistical methods, the study will uncover how patient, provider, plan, and policy characteristics influence diagnostic equity, informing advocacy and policy efforts to improve detection for all.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** USC/UCLA Center on Biodemography and Population Health

***Status of Support:** Current

Proposal/Award Number: P30AG017265

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** $3,959,438

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.5
2027	0.5

Year	Person Months
2028	0.5
2029	0.5
2030	0.5

***Overall Objectives:** The major goal of this grant is to understand biological pathways through which socio-demographic factors affect health risks across the life-course.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** Dynamic Microsimulation of Parkinsons

***Status of Support:** Current

Proposal/Award Number: MJFF-027212

***Source of Support:** Michael J. Fox Foundation

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $1,190,186

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1

***Overall Objectives:** To develop a comprehensive dynamic microsimulation model, VALUE-PD, to quantify the long-term, life-course value of disease-modifying therapies (DMTs) for Parkinson's disease. This model will evaluate the health, quality of life, and economic impacts of DMTs on patients, families, and broader stakeholders to inform treatment decisions, policy, and future innovation.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** Alzheimer Disease Research Center (ADRC)

***Status of Support:** Current

Proposal/Award Number: P30AG066530

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2030

***Total Anticipated Proposal/Project Amount:** $23,694,145

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.3
2027	0.3
2028	0.3
2029	0.3

***Overall Objectives:** An NIH funded research center that supports basic and clinical AD research through it administrative, clinical, imaging, pathology, education and information transfer cores.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** Use and impact of novel and repurposed therapeutics for Alzheimer's disease and related dementia in diverse populations

***Status of Support:** Current

Proposal/Award Number: R33AG081811

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2025

***Proposal/Active Project End Date: (MM/YYYY):** 03/2028

***Total Anticipated Proposal/Project Amount:** $1,945,521

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1.2

***Overall Objectives:** The goal of this project is to address the significant disparities in access to and utilization of Alzheimer's Disease and Related Dementias (AD/ADRD) therapeutics, particularly in underserved non-Hispanic Black communities. The overarching aim is to promote equitable access to future AD/ADRD treatments by identifying and quantifying both the demand for these therapies and the barriers preventing their use from the perspective of non-Hispanic Black stakeholders.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:**	Network on Life Course Health Dynamics and Disparities in 21st Century America Renewal
***Status of Support:**	Current
Proposal/Award Number:	R24AG045061
***Source of Support:**	NIA
***Primary Place of Performance:**	USC
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2024
***Proposal/Active Project End Date: (MM/YYYY):**	05/2029
***Total Anticipated Proposal/Project Amount:**	$2,042,579

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25
2028	0.25

***Overall Objectives:** The major goals are to grow an existing Network on life course disparities in health and aging; stimulate research to address key knowledge gaps in life course and geographic disparities in health; forge multidisciplinary collaborations between researchers; provide development and training opportunities; and develop and disseminate data and analytic resources related to life course health research.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:**	The United States Cost of Dementia Model: Quantifying Current and Future Impacts of Alzheimer’s Disease and Related Dementias
***Status of Support:**	Current

Proposal/Award Number: U01AG086827

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2024

***Proposal/Active Project End Date: (MM/YYYY):** 03/2029

***Total Anticipated Proposal/Project Amount:** $8,245,873

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3
2027	3
2028	3

***Overall Objectives:** The overarching goal of this project is to provide national, annual, and comprehensive cost estimates for dementia, utilizing the best available data sources. Furthermore, the aim is to develop research infrastructure that will significantly increase the national research capacity for generating cost estimates. Innovative simulation methods will be employed to analyze the factors influencing these costs and their impact on various demographic groups.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:** USC Alzheimer's Disease/Alzheimer's Disease Related Dementias Resource Center for Minority Aging Research (USC AD/ADRD RCMAR)

***Status of Support:** Current

Proposal/Award Number: P30AG043073

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2023

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** $3,602,796

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.4

Year	Person Months
2027	1.4

***Overall Objectives:** The goal is to increase the number and success of researchers focusing on the health and economic wellbeing of the elderly. Special topic area is Alzheimer's disease and other dementias.

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:**	Identifying and Addressing Social Determinants of Health to Reduce the National Burden of and Inequities in Dementia
***Status of Support:**	Current
Proposal/Award Number:	R01AG075916-01A1
***Source of Support:**	NIA
***Primary Place of Performance:**	USC
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2023
***Proposal/Active Project End Date: (MM/YYYY):**	12/2028
***Total Anticipated Proposal/Project Amount:**	$134,224

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	1

***Overall Objectives:** Our long-term goal is to reduce the burden of and disparities in ADRD. To use multiple approaches to provide a comprehensive evidence base on the impacts of social determinants of health (SDoH).

***Statement of Potential Overlap:** n/a

***Proposal/Active Project Title:**	Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (CeASES ADRD).

***Status of Support:** Pending

Proposal/Award Number: P30AG066589

***Source of Support:** NIA

***Primary Place of Performance:** USC

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2020

***Proposal/Active Project End Date: (MM/YYYY):** 08/2031

***Total Anticipated Proposal/Project Amount:** $5,536,057

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8
2028	1.8
2029	1.8
2030	1.8

***Overall Objectives:** The goal is to advance innovative social science ADRD research, to increase knowledge and technical capability of social science researchers, to broaden the set of researchers working in the field and, to disseminate resources and communicate findings for impact.

***Statement of Potential Overlap:** n/a

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Ahammad, Rijwan Uddin

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-1387-1943

Position Title: Postdoc Research Associate

Organization and Location:University of Southern California, San Diego, California, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Nagoya University, Nagoya, Not Applicable, N/A, Japan	Doctor of Philosophy (PHD)	04/2014	04/2022	Neuroscience
University of Dhaka, Dhaka, Not Applicable, N/A, Bangladesh	Master of Science (MS)	10/2008	04/2010	Biochemistry and Molecular Biology
University of Dhaka, Dhaka, Not Applicable, N/A, Bangladesh	Bachelor of Science (BS)	04/2004	04/2008	Biochemistry and Molecular Biology

Appointments and Positions

2024 - present	Postdoc Research Associate, University of Southern California, San Diego, California, United States
2022 - 2024	Postdoctoral Researcher, University of California, San Diego (UCSD), San Diego, California, United States
2020 - 2022	Research Assistant, Dept. of Cell Pharmacology, Nagoya University, Nagoya, Not Applicable, N/A, Japan
2014 - 2022	Teaching Assistant, Dept. of Cell Pharmacology, Nagoya University, Nagoya, Not Applicable, N/A, Japan
2014 - 2016	Research Student, Dept. of Cell Pharmacology, Nagoya University, Nagoya, Not Applicable, N/A, Japan
2011 - 2014	Research Officer, Emerging Diseases and Immunobiology, International Centre for Diarrheal Diseases Research, (icddr,b), Dhaka, Not Applicable, N/A, Bangladesh
2009 - 2010	Research Fellow, Molecular Genetics, International Centre for Diarrheal Diseases Research, (icddr,b), Dhaka, Not Applicable, N/A, Bangladesh

Products

Products Closely Related to the Proposed Project

1. Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. Anti-Sense Oligonucleotide as a Therapeutic for Synucleinopathies: Pharmacokinetic, Safety and Efficacy Evaluation. bioRxiv. 2025 May 7; PubMed Central PMCID: PMC12247940.
2. Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. A splice-switching antisense oligonucleotide targeting APP reduces accumulation of α-synuclein in a mouse model of Parkinson's disease. Alzheimers Dement (N Y). 2025 Apr-Jun;11(2):e70117. PubMed Central PMCID: PMC12185244.
3. Leitão ADG, Ahammad RU, Spencer B, Wu C, Masliah E, Rissman RA. Novel systemic delivery of a peptide-conjugated antisense oligonucleotide to reduce α-synuclein in a mouse model of Alzheimer's disease. Neurobiol Dis. 2023 Oct 1;186:106285. PubMed Central PMCID: PMC10584037.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Funahashi Y, Ahammad RU, Zhang X, Hossen E, Kawatani M, Nakamuta S, Yoshimi A, Wu M, Wang H, Wu M, Li X, Faruk MO, Shohag MH, Lin YH, Tsuboi D, Nishioka T, Kuroda K, Amano M, Noda Y, Yamada K, Sakimura K, Nagai T, Yamashita T, Uchino S, Kaibuchi K. Signal flow in the NMDA receptor-dependent phosphoproteome regulates postsynaptic plasticity for aversive learning. Sci Signal. 2024 Sep 10;17(853):eado9852. PubMed PMID: 39255336.
2. Ahammad RU, Nishioka T, Yoshimoto J, Kannon T, Amano M, Funahashi Y, Tsuboi D, Faruk MO, Yamahashi Y, Yamada K, Nagai T, Kaibuchi K. KANPHOS: A Database of Kinase-Associated Neural Protein Phosphorylation in the Brain. Cells. 2021 Dec 24;11(1) PubMed Central PMCID: PMC8750479.
3. Funahashi Y, Ariza A, Emi R, Xu Y, Shan W, Suzuki K, Kozawa S, Ahammad RU, Wu M, Takano T, Yura Y, Kuroda K, Nagai T, Amano M, Yamada K, Kaibuchi K. Phosphorylation of Npas4 by MAPK Regulates Reward-Related Gene

Expression and Behaviors. Cell Rep. 2019 Dec 3;29(10):3235-3252.e9. PubMed PMID: 31801086.

4. Jahan I, Ahammad RU, Khalid MM, Rahman MI, Hayat S, Islam B, Mohammad QD, Islam Z. Toll-like receptor-4 299Gly allele is associated with Guillain-Barré syndrome in Bangladesh. Ann Clin Transl Neurol. 2019 Apr;6(4):708-715. PubMed Central PMCID: PMC6469239.
5. Jahan I, Ahammad RU, Farzana KS, Khalid MM, Islam MB, Rahman MI, Nahar S, Kabir Y, Mohmmad QD, Islam Z. Tumor necrosis factor-alpha -863C/A polymorphism is associated with Guillain-Barré syndrome in Bangladesh. J Neuroimmunol. 2017 Sep 15;310:46-50. PubMed PMID: 28778444.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Ahammad, Rijwan Uddin in SciENcv on 2026-01-27 15:18:28

OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Ahammad, Rijwan Uddin

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-1387-1943

Position Title: Postdoc Research Associate

Organization and Location: University of Southern California, San Diego, California, United States

Personal Statement

As an undergraduate in Bangladesh, I co-launched one of the first neuroimmune disorder projects in the country, studying genetic and molecular risk factors for Guillain-Barré syndrome. This experience provided early training in scientific rigor, project management, and problem-solving, as I independently designed and executed experiments, collected and analyzed patient samples, and integrated complex datasets. During my PhD in Professor Kozo Kaibuchi's lab in Japan, I investigated molecular mechanisms of memory and synaptic plasticity. I demonstrated how NMDA receptor-Rho signaling regulates Shank3 phosphorylation and aversive learning (first-author, Science Signaling) and developed phosphoproteomic tools, including the Kinase Associated Neuronal Phospho Signalling (KANPHOS) database (first-author, Cells). This work required mastery of neuronal culture, viral gene manipulation, in vivo behavioral assays, calcium imaging, patch-clamp electrophysiology, super-resolution imaging, and large-scale data integration, strengthening my ability to connect molecular mechanisms to systems-level function. In my postdoctoral research at UCSD and USC ATRI, I have focused on α-synuclein pathology in Alzheimer's disease and synucleinopathies, with translational emphasis on antisense oligonucleotide (ASO) therapeutics. I have led preclinical studies on ASO pharmacokinetics, safety, and efficacy in mouse models, resulting in first-author publications in Alzheimer's & Dementia: Translational Research & Clinical Interventions, with additional manuscripts under revision in Molecular Pharmacology and in preparation for submission to Neurotherapeutics. I have also contributed to collaborative projects examining neuroinflammation and APP/α-synuclein interactions. These experiences have equipped me with the technical expertise and project management skills to successfully execute the proposed ADRC Development Project. The proposed project will investigate APOE-dependent cognitive and molecular consequences of α-syn/Aβ co-pathology, bridging mechanistic insights with disease-relevant models. I am particularly focused on understanding how APOE genotype modulates vulnerability to mixed proteinopathies, a critical unresolved question in Alzheimer's disease and synucleinopathies. This project is designed to generate high-quality preliminary data for future independent funding while expanding my expertise in multi-omics approaches, including proteomics, phosphoproteomics, and transcriptomics. By integrating behavioral, cellular, and molecular readouts, I will establish mechanistic links between protein aggregation, immune response, and functional deficits, advancing both fundamental knowledge and translational opportunities. Beyond the bench, I am committed to mentoring future scientists and fostering collaborative research communities. I have mentored undergraduate and graduate students, served as a guest editor, and organized postdoctoral seminar series. To date, I have 17 peer-reviewed publications, including five first-author papers in the past five years. This Development Project builds directly upon my training in molecular mechanisms, preclinical therapeutics, and multi-omics profiling, positioning me to establish an independent research program focused on APOE-dependent mechanisms of co-pathology in Alzheimer's disease and related dementias.

Honors

2026 - 2027	ADRC Development Project (DP) award, USC ADRC
2020 - 2020	Best Research Presentation Award, Nagoya University Graduate School of Medicine, Japan
2016 - 2019	PhD scholarship, Otsuka Toshimi Scholarship Foundation, Japan
2009 - 2010	NISCT Fellowship, Ministry of Science and Technology, Bangladesh

Contribution to Science

1. 1. Antisense Oligonucleotide (ASO) Therapeutics for Synucleinopathies: During my postdoctoral fellowship, I have focused on translational neuroscience by developing antisense oligonucleotides (ASOs) as disease modifying therapies for α-synuclein driven neurodegeneration. Building on my PhD training in basic neuroscience, I aimed to bridge mechanistic understanding with preclinical therapeutics. To overcome the major challenge of CNS delivery, I employed a peptide (ApoB11) mediated systemic delivery via intraperitoneal injection, achieving robust brain penetration, prolonged CNS retention, and efficient target engagement. My work spanned the full translational pipeline; from ASO design and dose optimization to pharmacokinetics, pharmacodynamics, and toxicity assessment, demonstrating a safe and effective strategy for α-synuclein suppression. Treatment with ApoB11 conjugated ASOs reduced α-synuclein pathology, preserved neurons, mitigated

neuroinflammation, and improved functional outcomes in disease relevant models. This integrated approach represents an innovative and comprehensive preclinical characterization, establishing systemic peptide conjugated ASOs as a promising disease modifying strategy with strong potential for clinical translation. Publications: a) Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. Anti-Sense Oligonucleotide as a Therapeutic for Synucleinopathies: Pharmacokinetic, Safety and Efficacy Evaluation. bioRxiv. 2025 May 7; PMID: 40654713 b) Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. Systemic antisense oligonucleotide treatment ameliorates α-synuclein pathology in DLB model mice. (under review) Presentations: I. Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. Systemic delivery of ApoB11-conjugated ASO in DLB mice. AAIC 2025, (Oral Presentation) II. Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. Anti-sense oligonucleotide as a therapeutic for synucleinopathies. SfN 2025 Nanosymposium. (Oral presentation)

2. 2. Mechanistic and Translational Insights into α-Synuclein Pathology: In parallel, I investigated modulation of amyloid precursor protein (APP) processing as a means to modulate α-synuclein accumulation indirectly. Using a splice-switching oligonucleotide targeting APP, I demonstrated that APP modulation can also lower α-synuclein accumulation and rescues hippocampal neurodegeneration in vivo, establishing a mechanistic link between APP and α-synuclein pathology. Additionally, in collaboration, systemic delivery of peptide-conjugated ASOs reduced α-synuclein pathology and improved cognition in Alzheimer's disease mouse models, highlighting the broader translational potential of this approach. Together, these studies represent an innovative, complete translational pipeline, from molecular design and delivery platform development to functional rescue in vivo. Which provides the conceptual and technical foundation for my propose research proposal on α-synuclein and Aβ co-pathology and therapeutic intervention. Publications: a) Ahammad RU, Spencer B, Quach B, Salehi S, Rissman RA. A splice-switching antisense oligonucleotide targeting APP reduces accumulation of α-synuclein in a mouse model of Parkinson's disease. Alzheimers Dement (N Y). 2025 Apr-Jun;11(2):e70117. doi: 10.1002/trc2.70117. eCollection 2025 Apr-Jun. PubMed PMID: 40556661; PubMed Central PMCID: PMC12185244. b) Leitão ADG, Ahammad RU, Spencer B, Wu C, Masliah E, Rissman RA. Novel systemic delivery of a peptide-conjugated antisense oligonucleotide to reduce α-synuclein in a mouse model of Alzheimer's disease. Neurobiol Dis. 2023 Oct 1;186:106285. doi: 10.1016/j.nbd.2023.106285. Epub 2023 Sep 9. PubMed PMID: 37690676; PubMed Central PMCID: PMC10584037.
3. 3. Phosphoproteomic Tools and Resources for Neuronal Signaling: During the first part of my doctoral training, I focused on understanding how phosphorylation regulates neuronal signaling and synaptic function. To systematically map kinase-associated phosphorylation events in the brain, I developed KANPHOS (Kinase-Associated Neural Phosphorylation Database), the first publicly accessible, brain-specific phosphoproteomics database cataloging kinase-substrate relationships. I personally performed phosphopeptide enrichment, liquid chromatography tandem mass spectrometry (LC-MS/MS) analysis, data curation, and validation with phosphorylation-specific antibodies, creating a resource that enables researchers to explore signaling networks in both normal and diseased states. In parallel, I actively involved in the developed of KISS (Kinase Interacting Substrate Screening), a novel phosphoproteomics methods to capture kinase-oriented phosphorylation events with higher specificity and coverage, advancing the ability to systematically interrogate signaling cascades in neurons. Together, these tools and methods provide a foundational platform for mechanistic and systems level studies of brain signaling pathways, synaptic physiology, and neurobiology, and continue to support the broader neuroscience community. Publications: a) Ahammad RU, Nishioka T, Yoshimoto J, Kannon T, Amano M, Funahashi Y, Tsuboi D, Faruk MO, Yamahashi Y, Yamada K, Nagai T, Kaibuchi K. KANPHOS: A Database of Kinase-Associated Neural Protein Phosphorylation in the Brain. Cells. 2021 Dec 24;11(1). doi: 10.3390/cells11010047. PubMed PMID: 35011609; PubMed Central PMCID: PMC8750479. b) Shohag MH, Nishioka T, Ahammad RU, Nakamuta S, Yura Y, Hamaguchi T, Kaibuchi K, Amano M. Phosphoproteomic Analysis Using the WW and FHA Domains as Biological Filters. Cell Struct Funct. 2015;40(2):95-104. doi: 10.1247/csf.15004. Epub 2015 Jun 27. PubMed PMID: 26119529.
4. 4. Synaptic Plasticity and Phosphorylation Dependent Signaling in Learning and Memory: During my doctoral training, I investigated phosphorylation-dependent regulation of postsynaptic density (PSD) proteins and their role in synaptic plasticity and learning. Building on the phosphoproteomic tools and methods developed in the first part of my PhD (KANPHOS, KISS, and PIKISS), I applied these approaches alongside kinase manipulation and behavioral models to identify novel phosphorylation substrates and pathways. I provided the first direct evidence that NMDA receptor–driven signaling modulates the Rho/Rho-kinase pathway, establishing a mechanistic link between excitatory neurotransmission and cytoskeletal regulation. Furthermore, I uncovered a previously unrecognized role of Rho-kinase in regulating synaptic plasticity beyond its classical function in actin cytoskeleton modulation, connecting kinase activity to dendritic spine morphology, PSD protein phosphorylation, and functional plasticity underlying learning and memory. Collectively, these studies yielded new mechanistic insights into how phosphorylation signaling shapes synaptic structure, neuronal function, and behavior, providing foundational tools and methodologies that continue to support mechanistic and systems-level studies in neurobiology and synaptic physiology. Publications: a) Funahashi Y, Ahammad RU (Co-First Author), Zhang X, Hossen E, Kawatani M, Nakamuta S, Yoshimi A, Wu M, Wang H, Wu M, Li X, Faruk MO, Shohag MH, Lin YH, Tsuboi D, Nishioka T, Kuroda K, Amano M, Noda Y, Yamada K, Sakimura K, Nagai T, Yamashita T, Uchino S, Kaibuchi K. Signal flow in the NMDA receptor-dependent phosphoproteome regulates postsynaptic plasticity for aversive learning. Sci Signal. 2024 Sep 10;17(853):eado9852. doi: 10.1126/scisignal.ado9852. Epub 2024 Sep 10. PubMed PMID: 39255336.
5. 5. Genetic and Molecular Risk Factors in Neuroimmune Disorders: After my MSc, I worked at icddr,b investigating genetic determinants of Guillain-Barré syndrome (GBS) in the Bangladeshi population. I identified risk alleles in TLR4 and FAS and

clarified the lack of association for certain CD1 variants, providing novel insights into immune signaling pathways that influence susceptibility and nerve damage. This work strengthened my expertise in human genetics, immunobiology, and translational research, laying a foundation for my later studies in neurodegeneration. Publications: a) Jahan I, Ahammad RU, Khalid MM, Rahman MI, Hayat S, Islam B, Mohammad QD, Islam Z. Toll-like receptor-4 299Gly allele is associated with Guillain-Barré syndrome in Bangladesh. Ann Clin Transl Neurol. 2019 Apr;6(4):708-715. doi: 10.1002/acn3.744. eCollection 2019 Apr. PubMed PMID: 31019995; PubMed Central PMCID: PMC6469239. b) Jahan I, Ahammad RU, Farzana KS, Khalid MM, Islam MB, Rahman MI, Nahar S, Kabir Y, Mohmmad QD, Islam Z. Tumor necrosis factor-alpha -863C/A polymorphism is associated with Guillain-Barré syndrome in Bangladesh. J Neuroimmunol. 2017 Sep 15;310:46-50. doi: 10.1016/j.jneuroim.2017.06.005. Epub 2017 Jun 20. PubMed PMID: 28778444.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

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NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Ahammad, Rijwan Uddin

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-1387-1943

*POSITION TITLE: Postdoc Research Associate

*ORGANIZATION AND LOCATION: University of Southern California, San Diego, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	APOE-Dependent Impact on α-Synuclein and Aβ Cross-Seeding: Behavioral and Molecular Signatures of Co-Pathology
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USC Alzheimer's Disease Research Center
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$40,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3

***Overall Objectives:** The overall objective of this project is to define how APOE genotype particularly APOE4 modulates the interaction between α-synuclein and amyloid-β to drive mixed proteinopathies and accelerated neurodegeneration in Alzheimer's disease and related dementias. Although α-syn/Aβ co-pathology is highly prevalent and strongly associated with faster cognitive decline and therapeutic resistance, the APOE-dependent mechanisms governing cross-protein aggregation, clearance, and neuroinflammatory responses remain undefined. This project will integrate behavioral, molecular, and cell-type–specific analyses in APOE3 and APOE4 knock-in mouse models to determine how APOE4 alters cognitive vulnerability, protein aggregation dynamics, and glial-mediated clearance in the context of α-syn/Aβ co-expression. By linking APOE-dependent behavioral impairments with underlying aggregation, microglial dysfunction, and synaptic injury, this work will establish a mechanistic framework explaining why APOE4 carriers exhibit more aggressive and heterogeneous dementia phenotypes. The resulting dataset will provide a critical foundation for future R01-level,

multi-omics studies aimed at identifying APOE-specific therapeutic targets and informing precision treatment strategies for mixed Alzheimer's and Lewy body dementias.

***Statement of Potential Overlap:** None

Certification:

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OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Fotouhi, Maryam

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4569-7094

Position Title: Postdoctoral research associate

Organization and Location:Department of Radiology, University of Southern California, Los Angeles, California, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
University of Southern California, Los Angeles, CA, United States	Postdoctoral Fellow	07/2023	07/2026	Radiological Sciences
Iran University of Medical Sciences, Tehran, Not Applicable, N/A, Iran	DOCTOR OF MEDICINE	01/2014	01/2022	Medicine

Appointments and Positions

2023 - present	Postdoctoral research associate, Department of Radiology, University of Southern California, Los Angeles, California, United States
2021 - 2023	Research scientist, Cancer Research Center, Tehran University of Medical Sciences, Tehran, Not Applicable, N/A, Iran
2021 - 2022	Physician Doctor, Emergency Department, Shahid Beheshti Hospital, Maragheh University of Medical Sciences, Maragheh, Not Applicable, N/A, Iran
2018 - 2020	Undergraduate Project Manager, Quantitative MRI and Spectroscopy Group (QMISG), Tehran University of Medical Sciences, Tehran, Not Applicable, N/A, Iran
2018 - 2020	Medical Intern, Iran University of Medical Sciences, Tehran, Not Applicable, N/A, Iran
2018 - 2019	Undergraduate Researcher, Universal Scientific Education and Research Network (USERN), Tehran University of Medical Sciences, Tehran, Not Applicable, N/A, Iran

Certification:

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OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Fotouhi, Maryam

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-4569-7094

Position Title: Postdoctoral research associate

Organization and Location: Department of Radiology, University of Southern California, Los Angeles, California, United States

Personal Statement

I have dedicated my research to understanding brain clearance pathways and their role in neurodegeneration, focusing on perivascular spaces and their contribution to cerebrovascular and neurodegenerative disorders. My work examines how perivascular space dysfunction affects glymphatic clearance, amyloid accumulation, and white matter degeneration, ultimately influencing cognitive decline and dementia. By integrating multi-modal MRI analysis with large-scale neuroimaging harmonization, I aim to develop noninvasive imaging biomarkers to improve early disease detection, risk assessment, and therapeutic monitoring in cerebral amyloid angiopathy, Alzheimer's disease, and cerebral small vessel disease. A key aspect of my research involves leading the harmonization of the Alzheimer's Disease Neuroimaging Initiative dataset, addressing scanner variability and technical inconsistencies that obscure disease signatures. By standardizing multi-modal MRI, clinical, and biomarker data across transnational cohorts and integrating them into predictive models, my goal is to enhance biomarker reliability and improve early diagnosis of Alzheimer's disease and related dementias. My recent findings show that specific regional perivascular space volume fractions are strong predictors of postmortem cerebral amyloid angiopathy, highlighting the potential of MRI based perivascular space quantification as a noninvasive biomarker for cerebrovascular amyloid burden and a feasible alternative to amyloid PET scans. Additionally, my work investigates the interplay between perivascular space enlargement and white matter hyperintensities, key markers of small vessel disease, and vascular contributions to cognitive impairment. Evidence suggests that perivascular space dysfunction accelerates white matter hyperintensity formation, and my research aims to clarify its relationship with neuroinflammation and blood-brain barrier integrity. Understanding these mechanisms will refine our knowledge of cerebrovascular contributions to dementia and inform new intervention strategies to preserve cognitive function in aging populations. As a researcher dedicated to advancing perivascular space biomarkers, I have the expertise and technical foundation to ensure the success of this project. My goal is to establish perivascular space as an imaging biomarker for cerebrovascular dysfunction, refine its clinical utility for early disease detection, and develop standardized MRI based protocols for widespread adoption. This funding will support multimodal neuroimaging analyses, integration of harmonized datasets into predictive models, and exploration of perivascular space dysfunction's impact on cognitive decline and vascular pathology. With the support of this grant, I will further develop computational approaches to enhance biomarker discovery, improve diagnostic precision, and ultimately contribute to more effective treatment strategies for neurodegenerative diseases.

Honors

2025 - 2025	Conference fellowship, Alzheimer's Association, the Rainwater Charitable Foundation, and CurePSP.
2025 - 2025	Tavel grant for American Association of Neuroradiology (ASNR) Annual Meeting , University of southern California Postdoctoral Affairs
2025 - 2025	Trainee (Educational) Stipend award, International Society for Magnetic Resonance in Medicine
2025 - 2025	Full membership, Sigma Xi Scientific Research Honor Society

Contribution to Science

1. Developing harmonization technique for the ADNI dataset and validating the PVS quantification technique on the large-scale dataset: my research focused on developing harmonization techniques for the ADNI dataset and validating perivascular space (PVS) quantification on large-scale neuroimaging data to improve biomarker reliability in neurodegenerative diseases. PVS plays a critical role in Alzheimer's disease, cerebral amyloid angiopathy, and small vessel disease, yet inconsistencies in imaging protocols across research sites have hindered its clinical utility. To address this, we developed a harmonization framework for ADNI, ensuring standardized PVS quantification and enhancing its role as a noninvasive MRI-based biomarker for early disease detection and risk stratification. My work extends beyond harmonization by validating PVS quantification in large datasets, revealing strong associations between PVS enlargement, white matter hyperintensities, and small vessel disease, reinforcing the mechanistic link between glymphatic dysfunction and neurodegeneration. These efforts lay the foundation for integrating multi-modal biomarkers into precision medicine, improving early diagnosis, patient stratification for clinical trials, and treatment monitoring. a. Fotouhi, M., Sheikh-Bahaei, N., Choupan, J. (In Process). Classifying small WMH lesions and

PVS on T1-MRI using radiomics features and machine learning. Radiology: Artificial Intelligence. b. Jannat, R., Lynch, K., Fotouhi, M., Cen, S., Choupan, J., Sheikh-Bahaei, N., Pandey, G., Varghese, B. A. (2025). Advancing 1.5T MR Imaging: Achieving 3T Quality through Deep Learning Super-Resolution Techniques. Frontiers in Human Neuroscience. c. Samadi Khoshe Mehr, F., Fotouhi, M., Choupan, J. (2025). Perivascular Space Enlargement: A Pathway to Iron Accumulation in the Aging Brain, International Society for MRI in Medicine (ISMRM).

2. Developing AI-Based Algorithms for Early Detection of Liver Cancer Using Diffusion and Perfusion MRI: My previous research focus integrated AI-driven imaging biomarkers with MR analysis techniques to enhance early detection and classification of hepatocellular carcinoma, addressing a critical need for more accurate, noninvasive diagnostic tools in liver oncology. Traditional methods often struggled to differentiate benign from malignant liver nodules, particularly in cirrhotic patients, leading to diagnostic uncertainty and the over-reliance on biopsies. By leveraging radiomics and machine learning, my work aims to enhance liver cancer diagnosis by identifying subtle imaging patterns beyond human perception, enabling more precise risk stratification and clinical decision-making. Through this work, I aim to bridge computational advancements with real-world clinical practice, ultimately improving early detection, treatment planning, and patient outcomes in liver cancer care. a. Arian, A., Fotouhi, M., Samadi Khoshe Mehr, F., Setayeshpour, B., Delazar, S., Nahvijou, A., & Nasiri-Toosi, M. (2025). Differentiating benign and hepatocellular carcinoma cirrhotic nodules: radiomics analysis of water restriction patterns with diffusion MRI. British Journal of Radiology, 98(1171), 1155-1164. b. Fotouhi, M., Samadi Khoshe Mehr, F., Delazar, S., Shahidi, R., Setayeshpour, B., Nassiri Toosi, M., Arian, A. (2023). Assessment of LI-RADS efficacy in classification of hepatocellular carcinoma and benign liver nodules using DCE-MRI features and machine learning.European Journal of Radiology Open, 11, 100535. c. M. Fotouhi, F. Samadi Khoshe Mehr, B. Setayeshpour, A. Arian. Impact of data harmonization on the performance of cirrhotic nodule classification models with diffusion radionics, European conference of Radiology (ECR), Vienna, 2025.
3. Investigating the Impact of COVID-19 on Brain Structure and Function: The global health crisis caused by COVID-19 brought attention to the widespread effects of viral infections on the brain and the complex interplay between systemic diseases and neurological health. In my efforts to understand these relationships, I have investigated how pre-existing conditions such as cancer influence the progression of coronavirus infections, shedding light on the role of immunosuppression and systemic vulnerabilities in disease outcomes. Beyond systemic interactions, I have contributed to research examining the neurological manifestations of COVID-19, particularly its impact on brain structure and function. By synthesizing findings from neuroimaging studies and clinical data, I have explored how viral infections may disrupt neural networks and contribute to long-term cognitive and neurological complications. These efforts are part of a larger endeavor to bridge the fields of infectious disease, neurobiology, and neuroimaging, ultimately contributing to a deeper understanding of how systemic illnesses shape brain health and influence recovery trajectories. a. Fotouhi, M., Samami, E., Mohseni, S., Nasrollahizadeh, A., Haddadi, M., Mirbeyk, M., Saghazadeh, A., Rezaei, N. (2021). How Prevalent Is Cancer in Confirmed Cases with Coronaviruses and Severe Acute Respiratory Syndromes? Advances in Experimental Medicine and Biology, Springer. b. Afsahi, A. M., Norbash, A. M., Syed, S. F., Sedaghat, M., Afsahi, G., Shahidi, R., Tajabadi, Z., Bagherzadeh-Fard, M., Karami, S., Yarahmadi, P., Shirdel, S., Asgarzadeh, A., Baradaran, M., Khalaj, F., Sadeghsalehi, H., Fotouhi, M., Habibi, M. A., Jang, H., Alavi, A., Sedaghat, S. (2023). Brain MRI findings in neurologically symptomatic COVID-19 patients: a systematic review and meta-analysis. Journal of Neurology, 270(11), 5131-5154.

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CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Fotouhhi, Maryam

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-4569-7094

*POSITION TITLE: Postdoctoral Scholar - Research Associate

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Platelet-derived growth factor receptor-β in Plasma and Cerebrospinal Fluid as a Marker of Pericyte Injury: Association Analysis with White Matter Changes in Alzheimer's Disease
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USC Alzheimer's Disease Research Center
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	03/2026
***Proposal/Active Project End Date: (MM/YYYY):**	04/2027
***Total Anticipated Proposal/Project Amount:**	$40,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.34

***Overall Objectives:** The primary aim of this research is to determine whether soluble platelet-derived growth factor receptor-β levels in cerebrospinal fluid and plasma are associated with quantitative perivascular space burden on magnetic resonance imaging and with cognitive performance in individuals across the Alzheimer's disease spectrum. In Alzheimer's disease and vascular cognitive impairment, neurovascular unit integrity and perivascular clearance efficiency are critical determinants of protein accumulation and neurodegeneration. Pericyte degeneration, reflected by elevated soluble platelet-derived growth factor receptor-β, is hypothesized to relate to greater perivascular space volume and altered diffusion-based clearance indices. By linking these fluid and imaging measures to baseline and longitudinal cognition, this project will test a pericyte-driven vascular-clearance axis in vivo with potential implications for future risk stratification and trial enrichment.

***Statement of Potential Overlap:** None

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OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Kim, Hosung

Position Title: Associate Research Professor of Neurology

Organization and Location:University of Southern California, Los Angeles, California, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
McGill University, Montreal, Quebec, Canada	Doctor of Philosophy (PHD)	09/2005	12/2011	Biomedical Engineering
Hanyang University, Seoul, Not Applicable, N/A, Korea South	Master of Engineering (MENG)	03/2001	02/2003	Biomedical Engineering
Hanyang University, Seoul, Not Applicable, N/A, Korea South	Bachelor of Engineering (BENG)	03/1993	02/2001	Mechanical Engineering

Appointments and Positions

2025 - present	Associate Research Professor of Neurology, University of Southern California, Los Angeles, California, United States
2016 - 2025	Assistant Professor of Neurology, University of Southern California, Los Angeles, California, United States
2014 - 2016	Postdoctoral scholar, University of California, San Francisco, San Francisco, California, United States
2012 - 2014	Postdoctoral scholar, Montreal Neurological Institute, Montreal, Quebec, Canada

Products

Products Closely Related to the Proposed Project

1. Chung J, Park G, Ryu WS, Schellingerhout D, Kim HR, Gwak DS, Haddad E, Jahanshad N, Kim BJ, Hong KS, Oh H, Jeong SW, Kim JT, Park MS, Choi KH, Lee K, Park TH, Park SS, Park JM, Kang K, Yu KH, Oh MS, Lee SJ, Kim JG, Cha JK, Kim DH, Lee J, Han MK, Cho YJ, Lee BC, Bath PM, Wardlaw JM, Bae HJ, Kim H, Kim DE. Distinct spatiotemporal patterns of white matter hyperintensity progression. Nat Commun. 2025 Oct 24;16(1):9360. PubMed Central PMCID: PMC12552450.
2. Jann K, Park G, Ringman JM, Kim H. Classifying mild cognitive impairment from normal cognition: fMRI complexity matches tau PET performance. Alzheimers Dement (Amst). 2025 Jul-Sep;17(3):e70159. PubMed Central PMCID: PMC12340428.
3. Oh Y, Kim JS, Park G, Yoo SW, Ryu DW, Kim H. Clinical correlates of data-driven subtypes of deep gray matter atrophy and dopamine availability in early Parkinson's disease. NPJ Parkinsons Dis. 2025 Jun 12;11(1):165. PubMed Central PMCID: PMC12162818.
4. Chai Y, Zhang H, Robles C, Kim AS, Janhanshad N, Thompson PM, van der Werf Y, van Heese EM, Kim J, Joo EY, Aksman L, Kang KW, Shin JW, Trang A, Ha J, Lee E, Moon Y, Kim H. Precise perivascular space segmentation on magnetic resonance imaging from Human Connectome Project-Aging. medRxiv. 2025 Mar 20; PubMed Central PMCID: PMC11957161.
5. Jann K, Park G, Kim H. Classifying Mild Cognitive Impairment from Normal Cognition: fMRI Complexity Matches Tau PET Performance. bioRxiv. 2025 Jan 17; PubMed Central PMCID: PMC11761101.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Park G, Khan MH, Andrushko JW, Banaj N, Borich MR, Boyd LA, Brodtmann A, Brown TR, Buetefisch CM, Conforto AB, Cramer SC, Dimyan M, Domin M, Donnelly MR, Egorova-Brumley N, Ermer ER, Feng W, Geranmayeh F, Hanlon CA, Hordacre B, Jahanshad N, Kautz SA, Khlif MS, Liu J, Lotze M, MacIntosh BJ, Mohamed FB, Nordvik JE, Piras F, Revill KP, Robertson AD, Schranz C, Schweighofer N, Seo NJ, Soekadar SR, Srivastava S, Tavenner BP, Thielman GT, Thomopoulos SI, Vecchio D, Werden E, Westlye LT, Winstein CJ, Wittenberg GF, Ferris JK, Yu C, Thompson PM, Liew SL, Kim H. Associations between contralesional neuroplasticity and motor impairment through deep learning-derived MRI regional brain

age in chronic stroke (ENIGMA): a multicohort, retrospective, observational study. Lancet Digit Health. 2026 Jan 22; PubMed PMID: 41577565.

2. Yook S, Park HR, Seo D, Joo EY, Kim H. Obstructive sleep apnea subtyping based on apnea and hypopnea specific hypoxic burden is associated with brain aging and cardiometabolic syndrome. Comput Biol Med. 2025 Feb;185:109604. PubMed PMID: 39721413.
3. Ryu WS, Schellingerhout D, Park J, Chung J, Jeong SW, Gwak DS, Kim BJ, Kim JT, Hong KS, Lee KB, Park TH, Park SS, Park JM, Kang K, Cho YJ, Park HK, Lee BC, Yu KH, Oh MS, Lee SJ, Kim JG, Cha JK, Kim DH, Lee J, Park MS, Kim D, Bang OY, Kim EY, Sohn CH, Kim H, Bae HJ, Kim DE. Deep learning-based automatic segmentation of cerebral infarcts on diffusion MRI. Sci Rep. 2025 Apr 16;15(1):13214. PubMed Central PMCID: PMC12003832.
4. Weaver O, Gano D, Zhou Y, Kim H, Tognatta R, Yan Z, Ryu JK, Brandt C, Basu T, Grana M, Cabriga B, Alzamora MDPS, Barkovich AJ, Akassoglou K, Petersen MA. Fibrinogen inhibits sonic hedgehog signaling and impairs neonatal cerebellar development after blood-brain barrier disruption. Proc Natl Acad Sci U S A. 2024 Jul 30;121(31):e2323050121. PubMed Central PMCID: PMC11295022.
5. Liu M, Zhu AH, Maiti P, Thomopoulos SI, Gadewar S, Chai Y, Kim H, Jahanshad N. Style transfer generative adversarial networks to harmonize multisite MRI to a single reference image to avoid overcorrection. Hum Brain Mapp. 2023 Oct 1;44(14):4875-4892. PubMed Central PMCID: PMC10472922.

Certification:

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I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Kim, Hosung

Position Title: Associate Research Professor of Neurology

Organization and Location: University of Southern California, Los Angeles, California, United States

Personal Statement

I am an Associate Professor of Research Neurology. My research focuses on developing an MRI data processing tools and analytic platform that assesses neurodegeneration of brain structure. To further understand glymphatic function and brain aging associated with neurodegenerative disorders, my group develops methods to learn the pattern of various aspects of brain anatomical and networking variability using multimodal neuroimaging data including structural/functional MRI, diffusion tensor imaging and EEG. Using advanced pattern analytic approaches with machine learning and deep learning on multicontrast MRI features, my current research seeks to measure the brain age using multimodal MRI including structural, diffusion, and perfusion imaging, and predict disease progression and effects of treatment in various neurological diseases including Alzheimer's disease, Parkinson's disease, vascular dementia, sleep disorders such as REM-sleep behavior disorder, insomnia, and sleep apnea, and epilepsy. I use the Artificial Intelligence (AI) to ultimately understand the underlying pathomechanisms in these diseases. In this proposal, As a PI, he will be responsible for overseeing the project, disseminating the data, findings by publications, and supervising the postdoc to ensure the fulfillment of the analysis.

Honors

2025	Epstein Alzheimer's Research Award, Epsten Foundation
2025	Certificate of Appreeciation, Recognition of Achievement, Korean Epilepsy Society
2017	Faculty Fellowship Award, Baxter Foundation
2016	Trainee Stipends Award, ISMRM 24th Annual Meeting & Exhibition
2013	Sleep Research Society Abstract Excellence Award, SLEEP 2013
2011	Student Traveling Awards, International Conference in Medical Image Computing and Computer Assisted Intervention
2010	Young Investigator Awards, Annual Meeting of American Epilepsy Society
2015 - 2017	Banting Postdoctoral Fellowships, Canadian Institutes of Health Research
2007 - 2009	FRSQ Doctoral Training Awards, Fonds de la Recherche en Santé Quebec/ Health Research Funds in Quebec

Contribution to Science

1. I have developed machine learning techniques to predict brain aging as well as various clinical outcome in various neurological diseases. I have also developed brain MRI analysis approaches to better understand disease progression and the role of different biomarkers in Parkinson's disease. Finally, my on-going research focuses on the role of brain glymphatic clearance function in relation to sleep, cardiovascular disease factors, and neurodegenerative diseases. These studies are fundamental for the proposed research project.
2. I have assessed imaging evidence of pathophysiology and provided a better understanding of progression in various sleep disorders including sleep apnea, insomnia, and narcolepsy using various quantitative methods on multimodal brain MRI. In my previous studies, I provided MRI analytic techniques, by which disruption in brain network, decreases in cortical volume, and changes in medial temporal lobe morphology were found in relation to sleep deprivation and OSA-induced oxygen desaturation. We also found such brain structural and functional changes are reversible with treatment. These findings demonstrated that there are structural and functional damages, which may further lead to neurodegeneration and aging.
3. We have developed deep learning algorithms and AI approaches to improve MRI QC and removal of image artifacts as well as harmonizing MRI data that are acquired from different scanners and protocols. We also developed a pipeline to segment neonatal brain and cortex and measure cortical morphology under head motion.
4. I have shown that extraction of features on multi-contrast MRI can capture the variations related to brain abnormalities related to brain development, degeneration and aging. I have showed that extraction of features on multi-contrast MRI can capture the variations related to microstructural pathology. These features are shown to be very useful for the analysis of brain MRI data in various aging groups with and without dementia.
5. Neuroimaging Biomarkers of Neurofluid Clearance and Neurodegeneration I have developed deep learning–based

neuroimaging methods to quantify neurofluid clearance function, with a focus on the glymphatic system. Using a semi-supervised neural network, I established an automated approach to segment perivascular spaces from MRI, enabling scalable in vivo assessment of glymphatic pathway integrity. In parallel, I developed convolutional neural network models integrating resting-state fMRI–derived cortical complexity measures to classify mild cognitive impairment and Alzheimer's disease. These models achieved diagnostic accuracy comparable to tau PET, establishing noninvasive, cost-effective imaging biomarkers for early neurodegeneration.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

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NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Kim, Hosung

*POSITION TITLE: Associate Research Professor of Neurology

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Brain Stroke Detection with 360 Degree Multi-angular 2.5D X-ray Synthesis from Orthogonal planar 2D X-ray Images Using Convolutional Transformer and Diffusion Models
***Status of Support:**	Current
Proposal/Award Number:	
***Source of Support:**	OTOM
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	11/2024
***Proposal/Active Project End Date: (MM/YYYY):**	08/2025
***Total Anticipated Proposal/Project Amount:**	$110,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.5

***Overall Objectives:** The primary aim of this research is to develop a novel methodology to construct a 3D CT brain image from multiplanar 2D X-ray head images and apply it to stroke detection. In the field of stroke detection, the 3D anatomical structure of the brain is important. It not only enables rapid diagnosis and surgery, but also supports surgical planning and patient follow-up. Several medical imaging techniques, including computed tomography (CT) and magnetic resonance imaging (MRI), play an important role in the detection of stroke. However, they have drawbacks such as radiation exposure, limited availability, and high cost. On the other hand, X-ray imaging, which is widely available and inexpensive, excels in visualizing internal anatomical structures, but poses challenges in stroke detection. Therefore, the model for 3D CT reconstruction from 2D X-ray images is an innovative approach that offers significantly reduced radiation exposure, wide availability, and low cost to patients.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:** Novel Multi-modal Imaging Marker for Regional Glymphatic Flow in Humans

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** Foundation of the ASNR

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2024

***Proposal/Active Project End Date: (MM/YYYY):** 06/2026

***Total Anticipated Proposal/Project Amount:** $120,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1.32

***Overall Objectives:** Previous efforts in motion artifact correction have used deep learning approaches that learn the pattern of motion artifacts from motion simulation data. However, these approaches have also identified a significant gap between real and simulated motion artifacts. Current simulation methods, such as manipulating K-space data (introducing Gaussian noises in K-space or deleting some specific frequency data), oversimplify the complex nature of head movements, lacking accuracy in capturing translational, rotational, multi-directional, and dynamic aspects. To overcome this limitation, we propose developing a Motion-Adaptive Diffusion Model (MADM), a generative deep learning model designed to learn more realistic and diverse patterns of the motion artifact seen in the image domain.

***Statement of Potential Overlap:** none

***Proposal/Active Project Title:** Complexity of fMRI in Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: R01AG066711

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California

***Proposal/Active Project Start Date: (MM/YYYY):** 04/2020

***Proposal/Active Project End Date: (MM/YYYY):** 03/2026

***Total Anticipated Proposal/Project Amount:** $3,269,805

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months

Year	Person Months
2025	2.4

***Overall Objectives:** Biomarkers for pre-clinical stages of Alzheimer's disease (AD) have become increasingly important for the development of preventative interventions. For neuroimaging, positron emission tomography (PET) imaging of amyloid beta (Aβ) and tau protein provide early imaging markers of AD and can track disease progression but are expensive and require the use of radioactive tracers. This project aims to develop and evaluate a noninvasive, quantitative and economical imaging marker of AD based on the complexity or regularity of resting state functional magnetic resonance imaging (fMRI).

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Global Brain Health Predictors of Post-Stroke Sensorimotor Recovery using AI-Enhanced Clinical MRIs
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	NIH
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	09/2025
***Proposal/Active Project End Date: (MM/YYYY):**	08/2026
***Total Anticipated Proposal/Project Amount:**	$493,456

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.96

***Overall Objectives:** Identifying global brain health predictors for stroke recovery through AI-enhanced MRI analysis.

***Statement of Potential Overlap:** none

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment

program.

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OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Kisler Elliott, Kassandra

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-4161-2498

Position Title: Assistant Professor of Research

Organization and Location:Physiology and Neuroscience/Keck School of Medicine of the University of Southern California, Los Angeles, California, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
Keck School of Medicine of the University of Southern California, Los Angeles, California, United States	Postdoctoral Fellow	01/2010	03/2016	Physiology and Biophysics/Neuroscience
Cornell University, Ithaca, New York, United States	Postdoctoral Fellow	07/2009	12/2009	Applied Physics/ Chem. and Biomolecular Engineering/ Microbiology and Immunology
Cornell University, Ithaca, New York, United States	Doctor of Philosophy (PHD)	08/2001	08/2009	Applied Physics
Cornell University, Ithaca, New York, United States	Master of Science (MS)	08/2001	05/2006	Applied Physics
University of California at Davis, Davis, California, United States	Bachelor of Science (BS)	09/1996	12/1999	Applied Physics
University of California at Davis, Davis, California, United States	Bachelor of Arts (BA)	09/1996	12/1999	Chemistry
San Jose City College, San Jose, California, United States	Associate in Arts (AA)	08/1994	05/1996	Physics

Appointments and Positions

2022 - present	Assistant Professor of Research, Physiology and Neuroscience/Keck School of Medicine of the University of Southern California, Los Angeles, California, United States
2022 - present	Assistant Director, Research Operations, Center for Neurodegeneration and Regeneration, Keck School of Medicine of the University of Southern California, Los Angeles, California, United States
2021 - present	Assistant Director, Optical Imaging Core, Center for Neurodegeneration and Regeneration, Keck School of Medicine of the University of Southern California, Los Angeles, California, United States
2016 - 2022	Research Associate, Laboratory of Berislav V. Zlokovic, MD, PhD, Keck School of Medicine of the University of Southern California, Los Angeles, California, United States
2012 - 2016	Postdoctoral Research Associate, Laboratory of Berislav V. Zlokovic, MD, PhD, University of Southern California Keck School of Medicine, Los Angeles, California, United States
2010 - 2012	Postdoctoral Research Associate, Laboratory of Robert H. Chow, MD, PhD, Keck School of Medicine of the University of Southern California, Los Angeles, California, United States
2009 - 2009	Postdoctoral Research Associate , Laboratories of Lois Pollack, PhD and Susan Daniel, PhD, Cornell University, Ithaca, New York, United States
2002 - 2009	Research Assistant, Laboratory of Manfred Lindau, PhD, Cornell University, Ithaca, New York, United States
2000 - 2001	Manufacturing Test Engineer, Coherent, Inc., Santa Clara, California, United States

Products

Products Closely Related to the Proposed Project

1. Kisler K, Sagare AP, Lazic D, Bazzi S, Lawson E, Hsu CJ, Wang Y, Ramanathan A, Nelson AR, Zhao Z, Zlokovic BV. Anti-malaria drug artesunate prevents development of amyloid-β pathology in mice by upregulating PICALM at the blood-brain barrier. Mol Neurodegener. 2023 Jan 27;18(1):7. PubMed Central PMCID: PMC9883925.
2. Barisano G, Kisler K, Wilkinson B, Nikolakopoulou AM, Sagare AP, Wang Y, Gilliam W, Huuskonen MT, Hung ST, Ichida JK, Gao F, Coba MP, Zlokovic BV. A "multi-omics" analysis of blood-brain barrier and synaptic dysfunction in APOE4 mice. J Exp Med. 2022 Nov 7;219(11) PubMed Central PMCID: PMC9435921.
3. Zhao Z, Sagare AP, Ma Q, Halliday MR, Kong P, Kisler K, Winkler EA, Ramanathan A, Kanekiyo T, Bu G, Owens NC, Rege SV, Si G, Ahuja A, Zhu D, Miller CA, Schneider JA, Maeda M, Maeda T, Sugawara T, Ichida JK, Zlokovic BV. Central role for PICALM in amyloid-β blood-brain barrier transcytosis and clearance. Nat Neurosci. 2015 Jul;18(7):978-87. PubMed Central PMCID: PMC4482781.
4. Rust R, Sagare AP, Zhang M, Zlokovic BV, Kisler K. The blood-brain barrier as a treatment target for neurodegenerative disorders. Expert Opin Drug Deliv. 2025 May;22(5):673-692. PubMed PMID: 40096820.
5. Kisler K, Lazic D, Sweeney MD, Plunkett S, El Khatib M, Vinogradov SA, Boas DA, Sakadži S, Zlokovic BV. In vivo imaging and analysis of cerebrovascular hemodynamic responses and tissue oxygenation in the mouse brain. Nat Protoc. 2018 Jun;13(6):1377-1402. PubMed Central PMCID: PMC6402338.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Kisler K, Nelson AR, Montagne A, Zlokovic BV. Cerebral blood flow regulation and neurovascular dysfunction in Alzheimer disease. Nat Rev Neurosci. 2017 Jul;18(7):419-434. PubMed Central PMCID: PMC5759779.
2. Kisler K, Nelson AR, Rege SV, Ramanathan A, Wang Y, Ahuja A, Lazic D, Tsai PS, Zhao Z, Zhou Y, Boas DA, Sakadžić S, Zlokovic BV. Pericyte degeneration leads to neurovascular uncoupling and limits oxygen supply to brain. Nat Neurosci. 2017 Mar;20(3):406-416. PubMed Central PMCID: PMC5323291.
3. Rust R, Sagare AP, Kisler K, Kim Y, Zhang M, Griffin C, Wang Y, Clementel V, Torres-Sepulveda C, Tcw J, Zlokovic BV, Coba MP. Molecular signature and functional properties of human pluripotent stem cell-derived brain pericytes. bioRxiv. 2025 Apr 28; PubMed Central PMCID: PMC12026417.
4. Nikolakopoulou AM, Montagne A, Kisler K, Dai Z, Wang Y, Huuskonen MT, Sagare AP, Lazic D, Sweeney MD, Kong P, Wang M, Owens NC, Lawson EJ, Xie X, Zhao Z, Zlokovic BV. Pericyte loss leads to circulatory failure and pleiotrophin depletion causing neuron loss. Nat Neurosci. 2019 Jul;22(7):1089-1098. PubMed Central PMCID: PMC6668719.
5. Babos KN, Galloway KE, Kisler K, Zitting M, Li Y, Shi Y, Quintino B, Chow RH, Zlokovic BV, Ichida JK. Mitigating Antagonism between Transcription and Proliferation Allows Near-Deterministic Cellular Reprogramming. Cell Stem Cell. 2019 Oct 3;25(4):486-500.e9. PubMed Central PMCID: PMC6778044.

Certification:

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OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Kisler Elliott, Kassandra

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-4161-2498

Position Title: Assistant Professor of Research

Organization and Location: Physiology and Neuroscience/Keck School of Medicine of the University of Southern California, Los Angeles, California, United States

Personal Statement

My work focuses on the study of neurovascular physiology, especially in the interaction with and possible contribution of neurovascular dysfunction to pathological conditions such as Alzheimer's disease (AD) and stroke. My work includes the development and application of advanced imaging techniques to investigate biological problems, tackle important questions related to blood-brain barrier (BBB) integrity, vascular regulation and function, and to explore molecular changes and risk genes (e.g. APOE4, PICALM) that underpin vascular dysfunction and neurodegenerative diseases such as AD, and therapeutic approaches to treat these diseases. Much of my graduate and early postdoctoral work focused on the investigation of cellular function using a combination of electrophysiology and fluorescence microscopy techniques to gain insights into the mechanism of neurotransmitter release via exocytosis and to characterize ALS patient-derived motor neurons. I also gained experience with ex vivo two-photon laser scanning microscopy (TPLSM) and super resolution microscopy. My current work builds upon my experience with these techniques to tackle important questions related to blood-brain barrier (BBB) integrity, vascular regulation and function, its relation to brain electrical activity, and ultimately the mechanisms underlying neurodegenerative diseases such as AD using transgenic model animals, ex vivo and in vitro approaches. I have led or contributed to multiple studies investigating brain microvascular and BBB changes with disease, microvascular dynamics, microvascular dysfunction and blood flow changes at the single capillary level in vivo using advanced optical imaging techniques. Additionally I have used a multi-omics approach (e.g transcriptomics and proteomics) to investigate neurovascular effects of the APOE4 AD genetic risk factor in humanized mouse models and AD relevant models of amyloid overexpression and tauopathy, explore the role of the AD risk gene phosphatidylinositol binding clathrin assembly protein (PICALM) in disease progression, and therapeutic approaches to restore the BBB using iPSC-derived pericyte cells to replace vessel-associated pericytes lost in disease. In conducting these studies, I have also developed advanced high resolution in vivo imaging techniques to study vascular dynamics and blood flow changes at the single capillary level. Relevant to this proposal, I have over 10 years of experience investigating the role of AD risk gene phosphatidylinositol binding clathrin assembly protein (PICALM), and over 5 years investigating APOE4 in disease progression. We found that PICALM plays an important role in clearance of toxic amyloid from the brain to blood across the BBB. Importantly, we observed that changes in PICALM expression levels affected brain amyloid load, cerebral blood flow and BBB integrity in AD model mice. In APOE4 models, we have observed BBB changes that precede neuronal changes and contribute to AD-like pathology. These and other studies form the basis to further explore the impact of PICALM and APOE4 in AD and cerebrovascular amyloid angiopathy (CAA) pathology often observed in AD patients and others with diseases that exhibit CAA pathology. Note that I typically publish under my maiden name, Kassandra Kisler.

Honors

2023	2nd Place Datablitz presentation, 6th Perivascular space in cerebral small vessel disease Leducq network meeting
2019	Winner, Postdoctoral poster competition, Alzheimer's Association International Conference
2018	John Douglas French Alzheimer's Foundation Award, John Douglas French Alzheimer's Foundation
2012	Postdoctoral Travel Scholarship, William Hansen Sandberg Memorial Foundation
2011	Higher Education Grant Finalist, Edmund Optics
2008	Successfully competed to attend the NIH Graduate Research Festival , NIH Graduate Research Festival
2008	Travel Grant recipiant, Cornell Graduate School
2007	Chosen to attend the Leadership Alliance National Symposium , Leadership Alliance
2007	Featured in the Leadership Alliance 2007-2008 Emerging PhDs Yearbook , Leadership Alliance
2006	Travel Scholarship, 13th International Symposium on Chromaffin Cell Biology
1999	Saxon-Patten Prize in Physics , UC Davis
1996	Outstanding Graduate in Math and Science , San Jose City College
1998 - 1999	UC Davis Departmental Citation for Outstanding Accomplishment in Physics, UC Davis
1996 - 1998	James & Lela Fulmor Scholarship, UC Davis

Contribution to Science

1. Understanding neurovascular effects of PICALM Alzheimer's disease genetic risk factor. One focus of my work has been to investigate the role of the highly-validated AD risk gene encoding phosphatidylinositol binding clathrin assembly protein (PICALM) in disease progression. Numerous AD associated PICALM single nucleotide polymorphisms (SNPs) have been identified, but their role in AD has not been determined. PICALM is highly expressed in brain vasculature and we and others have reported reduced PICALM protein levels in the brains of AD patients, but how that affects the disease is still a topic of debate. We identified a critical role for PICALM in brain endothelium, where the protein acts as an endocytotic adaptor and chaperone to direct amyloid clearance from the brain. Furthermore, we found that reduced PICALM expression in amyloid overproducing mice led to greater amyloid accumulation in their brains compared to littermates with normal PICALM expression. With a goal to develop a new AD therapy, I have developed an assay and screened over 2000 FDA-approved compounds for PICALM upregulation, ultimately identifying the anti-malaria drug artesunate as the top hit. Treatment of AD model mice with artesunate upregulated vascular PICALM and reduced pathology, indicating that PICALM expression may play an important role in disease progression, and that artesunate may be a promising therapeutic candidate. I continue to contribute this and other studies of PICALM function and it's role in AD pathophysiology. See relevant publications listed above, including: Kisler K, et al. Mol Neurodegener. 2023. PMID: 36707892. PMCID: PMC9883925. Zhao Z, ..., Kisler K, et al. Nature Neuroscience. 2015. PMID: 26005850. PMCID: PMC4482781. Rust R, ..., Kisler K. Expert Opinion on Drug Delivery. 2025. PMID: 40096820.
2. Understanding neurovascular effects of APOE4 Alzheimer's disease genetic risk factor. APOE4 is the most prevalent genetic risk factor for AD. Using humanized APOE4 (risk allele) and APOE3 (common allele) expressing mouse lines, I participate in investigations of the effect of APOE4 on vascular function and brain changes. APOE4 mice and human carriers have increased BBB permeability, which may contribute to neurodegenerative changes and increased susceptibility to developing Alzheimer's disease. We recently showed that APOE4 compared with APOE3 leads to an early disruption of the BBB transcriptome in APOE4 knock-in mice, followed by dysregulation in protein signaling networks controlling BBB related genes in brain endothelium, as well as transcriptional changes suggestive of DNA damage in vascular-associated pericytes using single-nucleus RNA-sequencing and phosphoproteome and proteome analysis, which preceded other APOE4-mediated changes. Currently I contribute to investigations of the neurovascular effects of APOE4 in mouse models that also express either amyloid-beta or tau to mimic Alzheimer's disease pathology. See relevant publications above, including: Barisano G, Kisler K, et al. Journal of Experimental Medicine. 2022. PMID: 36040482. PMCID: PMC9435921. Kisler K, et al. Nature Reviews Neuroscience 2017. PMID: 28515434. PMCID: PMC5759779. Rust R, ..., Kisler K. Expert Opinion on Drug Delivery. 2025. PMID: 40096820.
3. Vascular contributions to neurodegenerative diseases and in vivo Imaging. In my work studying vascular contributions to neurodegenerative diseases such as Alzheimer's, I have focused strongly on the application of in vivo imaging techniques to study vascular dynamics and cells that support vascular function in real time. Vascular and blood-brain barrier (BBB) functionality is thought to break down during AD progression. Pericyte cells have been shown to play a role in maintenance of the BBB and is reduced in the brains of AD patients, but its role in vascular dynamics is unclear. Using pericyte-deficient mice and multiple in vivo and other imaging techniques, we demonstrated that pericyte degeneration leads to diminished CBF responses, neurovascular uncoupling, reduced oxygen supply to brain and increased metabolic stress, which over time lead to impaired neuronal excitability and neurodegenerative changes. For these studies, I developed new techniques for high resolution multiphoton imaging and analysis of capillary vascular dynamics. My use of advanced in vivo imaging has also lead to new information regarding capillary vascular and blood flow changes after acute pericyte ablation. In recent work I have contributed to exploring iPSC-derived pericyte cells as a cellular-based treatment to restore BBB integrity. Our preliminary studies show that iPSC-pericytes can home to brain capillary vasculature, and when implanted into pericyte-deficient mice, improve BBB integrity and neuronal retention, indicating that restoring the BBB may benefit neuronal function. See relevant publications above, including: Kisler K, et al. Nature Neuroscience. 2017. PMID: 28135240.PMCID: PMC5323291. Kisler K, et al. Nature Protocols. 2018. PMID: 29844521. PMCID: PMC6402338. Nikolakopoulou AM, et al. Nature Neuroscience 2019. PMID: 31235908. PMCID: PMC6668719. Rust R, et al. bioRxiv. 2025. PMID: 40291694. PMCID: PMC12026417.
4. Electrophysiology and TIRF microscopy: Neuronal function, the exocytotic mechanism, cell surface trafficking. Exocytosis is a highly regulated cell process by which cells, most importantly neurons, release signaling molecules or neurotransmitters. Dysregulation of this process is implicated in multiple diseases. My work involved design and development of microfabricated planar electrode arrays sized to fit single cells on glass coverslips. These arrays enabled fluorescent observation of labeled proteins or compartments within the cells of interest simultaneously with amperometric electrophysiology measurement. Using this method, I and my colleagues were able to measure accurate transmitter release and diffusion from single cells, and changes in exocytosis with manipulation of synaptobrevin, a SNARE protein essential for exocytosis (PMID: 20937897). I also applied the TIRF imaging technique to the study of estrogen receptor-alpha (ERalpha) at the cell surface in live cells in real time, publishing one of the first papers detailing this technique (PMID: 24353903). Our experiments showed that ERalpha was present at the membrane in live cells, and that binding of ligands to ERalpha at the cell surface induced endocytosis of the receptor. More recently, I have mastered patch clamp electrophysiology and multielectrode array measurement techniques. In collaboration with the Justin Ichida Lab at USC, I applied these techniques to study motor neuron degeneration mechanisms in

ALS using patient-derived motor neurons, helping to elucidate the mechanism behind the C9orf72 ALS risk gene (PMID: 29400714), and characterizing mouse and human fibroblast-derived motor neurons using new techniques for near-deterministic cellular reprogramming. See relevant publications above, including: Babos KN, et al. Cell Stem Cell. 2019. PMID: 31523028. PMCID: PMC6778044.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Kisler Elliott, Kassandra in SciENcv on 2026-01-2122:30:38

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Kisler Elliott, Kassandra

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-4161-2498

*POSITION TITLE: Assistant Professor of Research

*ORGANIZATION AND LOCATION: Keck School of Medicine of the University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	PICALM: Role in the pathogenesis and treatment of Alzheimer vascular blood-brain barrier clearance dysfunction
***Status of Support:**	Current
Proposal/Award Number:	4R01AG076921-02
***Source of Support:**	NIH/NIA
***Primary Place of Performance:**	University of Southern California, Los Angeles
***Proposal/Active Project Start Date: (MM/YYYY):**	08/2022
***Proposal/Active Project End Date: (MM/YYYY):**	07/2027
***Total Anticipated Proposal/Project Amount:**	$1,635,915

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	3.6

***Overall Objectives:** We will study the effects of endothelium-specific and neuron-specific PICALM deficiency on vascular and neuronal function and AD pathology; and the effects of artesunate and AAV-PHP.B-Picalm gene therapy, and the H465R PICALM mutation on vascular and neuronal function and AD pathology. We will next identify molecular steps in Aβ and tau BBB transcytosis and neuronal toxicity regulated by PICALM using BBB models and neurons from human rs3851179 PICALM variants and Picalm465R mice.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Vascular Contributions to Dementia and Genetic Risk Factors for Alzheimer's Disease

***Status of Support:** Current

Proposal/Award Number: P01AG052350

***Source of Support:** NIH/NIA

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 05/2022

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** $16,093,863

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8
2027	1.8

***Overall Objectives:** The overall goal of this continuing program project is to test the 'neurovascular hypothesis' of AD and establish whether the neurovasculature has a major role in the pathogenesis of early cognitive decline, and therefore could be a key new therapeutic target to treat cognitive impairment, dementia and early AD.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:** Activated Protein C mechanisms of brain white matter protection and new therapies for brain white matter ischemic injury

***Status of Support:** Current

Proposal/Award Number: R01NS117827-02

***Source of Support:** NIH/NINDS

***Primary Place of Performance:** University of Southern California, Los Angeles

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2020

***Proposal/Active Project End Date: (MM/YYYY):** 04/2026

***Total Anticipated Proposal/Project Amount:** $3,496,410

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.8

***Overall Objectives:** The goal is to provide proof of concept for hypothesized mechanisms for protective activities of APC in the white matter (WM) and to characterize novel PAR1-related P1-47 and PAR3-related P3-42 APC-mimetic peptides, and improved 3rd generation APC R-46-selective biologics for treating and preventing ischemic WM injury and stroke.

***Statement of Potential Overlap:** None.

***Proposal/Active Project Title:**	Synergistic Impact of Alzheimer's Disease risk factors PICALM and APOE on cerebral amyloid angiopathy and vascular pathology
***Status of Support:**	Pending
Proposal/Award Number:	Pending
***Source of Support:**	USC ADRC
***Primary Place of Performance:**	University of Southern California, Los Angeles
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$70,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	2.4

***Overall Objectives:** This pilot study aims to investigate the potential synergistic impact of APOE4 carrier status and vascular PICALM expression levels, and PICALM SNP allele status on extent and severity of cerebral amyloid angiopathy in human postmortem tissues.

***Statement of Potential Overlap:** None.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Kisler Elliott, Kassandra in SciENcv on 2026-01-30 12:51:22

OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Roman Ortiz, Ciorana

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-3554-4943

Position Title: Postdoctoral Fellow

Organization and Location:University of Southern California, Los Angeles, CA, United States

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
University of California- Los Angeles , Los Angeles, CA, United States	Postdoctoral Fellow	09/2021	02/2023	Neurobiology
Icahn School of Medicine at Mount Sinai , New York, NY, United States	DOCTOR OF PHILOSOPHY	08/2015	09/2021	Neuroscience
University of Puerto Rico Medical Sciences Campus , San Juan, PR, United States	BACHELOR OF SCIENCE	08/2010	12/2012	Health Administration
University of Puerto Rico Medical Sciences Campus , San Juan, PR, United States	ASSOCIATE IN SCIENCE	08/2006	05/2010	Radiologic Technology

Appointments and Positions

2023 - present	Postdoctoral Fellow, University of Southern California, Los Angeles, CA, United States
2021 - 2023	Postdoctoral Fellow, University of California - Los Angeles, Los Angeles, CA, United States
2015 - 2021	Graduate Student, Icahn School of Medicine at Mount Sinai, New York, NY, United States
2013 - 2014	Post-baccalaureate Fellow, University of Puerto Rico- Medical Sciences Campus , San Juan, PR, United States
2011 - 2012	Undergraduate Research Assistant, University of Puerto Rico- Medical Sciences Campus , San Juan, PR, United States

Products

Products Closely Related to the Proposed Project

1. Roman-Ortiz C, Guevara JA, Clem RL. GABAergic basal forebrain projections to the periaqueductal gray promote food consumption, reward and predation. Sci Rep. 2021 Nov 22;11(1):22638. PubMed Central PMCID: PMC8608827.
2. Bravo-Rivera C, Roman-Ortiz C, Brignoni-Perez E, Sotres-Bayon F, Quirk GJ. Neural structures mediating expression and extinction of platform-mediated avoidance. J Neurosci. 2014 Jul 16;34(29):9736-42. PubMed Central PMCID: PMC4099548.
3. Bravo-Rivera C, Roman-Ortiz C, Montesinos-Cartagena M, Quirk GJ. Persistent active avoidance correlates with activity in prelimbic cortex and ventral striatum. Front Behav Neurosci. 2015;9:184. PubMed Central PMCID: PMC4502354.
4. Diehl MM, Bravo-Rivera C, Rodriguez-Romaguera J, Pagan-Rivera PA, Burgos-Robles A, Roman-Ortiz C, Quirk GJ. Active avoidance requires inhibitory signaling in the rodent prelimbic prefrontal cortex. Elife. 2018 May 31;7 PubMed Central PMCID: PMC5980229.
5. Lucas EK, Wu WC, Roman-Ortiz C, Clem RL. Prazosin during fear conditioning facilitates subsequent extinction in male C57Bl/6N mice. Psychopharmacology (Berl). 2019 Jan;236(1):273-279. PubMed Central PMCID: PMC6374171.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Juarez B, Morel C, Ku SM, Liu Y, Zhang H, Montgomery S, Gregoire H, Ribeiro E, Crumiller M, Roman-Ortiz C, Walsh JJ, Jackson K, Croote DE, Zhu Y, Zhang S, Vendruscolo LF, Edwards S, Roberts A, Hodes GE, Lu Y, Calipari ES, Chaudhury D, Friedman AK, Han MH. Midbrain circuit regulation of individual alcohol drinking behaviors in mice. Nat Commun. 2017 Dec 20;8(1):2220. PubMed Central PMCID: PMC5738419.
2. Calipari ES, Juarez B, Morel C, Walker DM, Cahill ME, Ribeiro E, Roman-Ortiz C, Ramakrishnan C, Deisseroth K, Han MH, Nestler EJ. Dopaminergic dynamics underlying sex-specific cocaine reward. Nat Commun. 2017 Jan 10;8:13877. PubMed Central PMCID: PMC5234081.

3. Bravo-Rivera C, Diehl MM, Roman-Ortiz C, Rodriguez-Romaguera J, Rosas-Vidal LE, Bravo-Rivera H, Quiñones-Laracuente K, Do-Monte FH. Long-range GABAergic neurons in the prefrontal cortex modulate behavior. J Neurophysiol. 2015 Sep;114(3):1357-9. PubMed Central PMCID: PMC4556857.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Roman Ortiz, Ciorana in SciENcv on 2026-01-20 17:16:10

OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Roman Ortiz, Ciorana

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0002-3554-4943

Position Title: Postdoctoral Fellow

Organization and Location: University of Southern California, Los Angeles, CA, United States

Personal Statement

Throughout my academic journey my research interest has been is to further understand how neural circuits encode and retrieve memories to guide behavior, and how faulty brain circuits can result in chronic illness. As an undergraduate I worked in Dr. Gregory Quirk's lab characterizing the structures involved in the learning and extinction of passive versus active fear responses. In addition, I conducted experiments to investigate possible mechanisms driving extinction failure in an active avoidance task as a postbac for 2 years. Understanding the mechanisms involved in the acquisition of active avoidance and how this behavioral response could become maladaptive would shed light to improve anxiety therapies. During my graduate work in Dr. Roger Clem's laboratory, I used optogenetic approaches to study how GABAergic populations modulate behavior during fear conditioning and extinction, and how changes in inhibitory transmission support these forms of learning. Furthermore, during this time I started studying ingestive behaviors, looking at the role of basal forebrain GABAergic neurons in food consumption. Uncovering how inhibitory neural mechanisms regulate aversive memory processing will provide a foundation for novel treatment of fear-related disorders. While in the University of California, I worked in Dr. Gina Poe's laboratory elucidating how specific sleep features directly facilitate insight gain using electroencephalogram (EEG) and single-unit recordings. Currently, for my postdoctoral research in Dr. Schier's lab, I focus on understanding how dietary and metabolic-induced changes affect sugar appeal. My project aims to examine how semaglutide (FDA-approved drug for blood glucose and weight management) affects peripheral taste and nutrient detection in a mouse model of Alzheimer's disease (AD). For this project I will employ immunohistochemistry, microscopy, qPCR, pharmacology, and meal pattern/preference analysis. Result from this project will characterize age-related APOE ε4 mutation taste pathophysiology and provide evidence on how chronic semaglutide treatment impact the APOE ε4-linked deficits in the taste system, and its implications for AD treatment. My long-term goal is to establish an independent research lab dedicated to advancing our understanding of taste-guided and promoting innovative research in the context of metabolic and age-related diseases.

Honors

2019	Scholar, Diversity Specialized Predoctoral to Postdoctoral Advancement in Neuroscience (DSPAN)
2018	Fellow, Yale Ciencia Academy (YCA)
2017	Research Supplements to Promote Diversity in Health-Related Research , National Institute of Mental Health
2017	Carl Storm Underrepresented Minority Travel Award, Amygdala Gordon Conference
2015	Associate, Neuroscience Scholars Program
2014	Fellow, Summer Program in Neuroscience, Ethics, and Survival (SPINES)
2013	Summer Research Internship, Silvio O. Conte Center for Research in OCD
2012	Magna Cum Laude Graduate , University of Puerto Rico-Medical Sciences Campus
2010	Cum Laude Graduate, University of Puerto Rico-Medical Sciences Campus
2006	Dean List, University of Puerto Rico-Rio Piedras Campus

Contribution to Science

1. Graduate Career: My graduate work helped to shine light on the involvement of basal forebrain GABAergic neurons during fear expression and non-homeostatic feeding. While most studies have focused on the function of basal forebrain (BF) cholinergic neurons, only a small portion of basal forebrain neurons are cholinergic, whereas those that release GABA are more abundant. Therefore, I sought to investigate the role of this population and their long-range projections fear and in learning feeding behaviors. Local and cortical afferents of BF GABAergic photosilencing reduced cued fear responses suggesting that contrary to the traditional held view that BF role in the fear expression is mediated merely by its cholinergic component. Moreover, in this work we demonstrate that activation of basal forebrain GABAergic neurons and its projections to the periaqueductal gray results in consumption of food as well as non-food items in well-fed mice, and performance of fictive eating in the absence of ingestible materials. Interestingly, BF GABAergic activity triggers skilled predatory attack of

live prey and prey-like objects, but not social targets. These results outline a potent circuit mechanism for increased feeding through recruitment of distinct but synergistic behaviors.

2. Postdoctoral Career: As a postdoctoral fellow, my research aims to elucidate how taste buds integrate metabolic status information to regulate taste-guided decisions. The risk of developing Alzheimer's disease (AD) is elevated in individuals with type 2 diabetes mellitus (T2DM), reflecting the shared contribution of insulin resistance to the pathogenesis of both disorders. Glucagon-like peptide 1 receptor (GLP1r) agonists have proven effective for glycemic control and show promising results in slowing AD progression. However, how AD pathophysiology alters the taste system to promote preference for sugar-rich diets remains unknown. Understanding how taste pathways reorganize following GLP1r agonist treatment may reveal new therapeutic targets for APOE ε4-linked AD treatment.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Roman Ortiz, Ciorana in SciENcv on 2026-01-2017:16:10

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Roman Ortiz, Ciorana

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-3554-4943

*POSITION TITLE: Postdoctoral Fellow

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	APOE ε4-linked taste pathophysiology and therapeutic modulation by semaglutide
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USC Alzheimer's Disease Research Center
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$40,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	4

***Overall Objectives:** The overall objective of this proposal is to elucidate how APOE ε4 pathophysiology presents in the taste system to disrupt metabolic health and how GLP1r signaling restores taste and nutrient signaling to reduce AD-related dietary habits. Additionally, this work will provide the first evidence of transcriptomic gustatory changes that are associated with the APOE4 allele, paving the way for future work aimed at understanding mechanistic links between metabolic disturbances and sensory deficits in the context of neurogenerative disease. Moreover, this work will reveal novel mechanisms through which incretin-based treatments exert neurometabolic benefits, including at the first site of nutrient sensing.

***Statement of Potential Overlap:** None

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Roman Ortiz, Ciorana in SciENcv on 2026-01-27 17:18:12

OMB No. 3145-0279 (Expiration Date 10/31/2026)

NIH Biographical Sketch Common Form

Name: Thorwald, Max A.

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-0095-5344

Position Title: N/A

Organization and Location:

PROFESSIONAL PREPARATION

INSTITUTION AND LOCATION	DEGREE	Start Date	Completion Date	FIELD OF STUDY
University of Southern California, Los Angeles, California, United States	Postdoctoral Fellow	09/2018	07/2025	Neurobiology and Aging
University of California Merced, Merced, California, United States	Doctor of Philosophy (PHD)	09/2013	08/2018	Physiology & Biochemistry
University of California Merced, Merced, California, United States	Bachelor of Science (BS)	09/2009	05/2013	Human Biology

Appointments and Positions

2025 - present Research Associate, University of Southern California, Los Angeles, California, United States
present ,

Products

Products Closely Related to the Proposed Project

1. Thorwald MA, Silva J, Head E, Finch CE. Amyloid futures in the expanding pathology of brain aging and dementia. Alzheimers Dement. 2023 Jun;19(6):2605-2617. PubMed Central PMCID: PMC10271937.
2. Thorwald MA, Godoy-Lugo JA, Garcia G, Silva J, Kim M, Christensen A, Mack WJ, Head E, O'Day PA, Benayoun BA, Morgan TE, Pike CJ, Higuchi-Sanabria R, Forman HJ, Finch CE. Iron-associated lipid peroxidation in Alzheimer's disease is increased in lipid rafts with decreased ferroptosis suppressors, tested by chelation in mice. Alzheimers Dement. 2025 Jan;21(1):e14541. PubMed Central PMCID: PMC11775463.
3. Thorwald MA, Sta Maria NS, Chakhoyan A, O'Day PA, Jacobs RE, Zlokovic B, Finch CE. Iron chelation by oral deferoxamine treatment decreased brain iron and iron signaling proteins. J Alzheimers Dis. 2025 Feb;103(4):1180-1190. PubMed PMID: 39894909.
4. Thorwald MA, Godoy-Lugo JA, Kerstiens E, Garcia G, Kim M, Shemtov SJ, Silva J, Durra S, O'Day PA, Mack WJ, Hiniker A, Vermulst M, Benayoun BA, Higuchi-Sanabria R, Forman HJ, Head E, Finch CE. Down syndrome with Alzheimer's disease brains have increased iron and associated lipid peroxidation consistent with ferroptosis. Alzheimers Dement. 2025 Jun;21(6):e70322. PubMed Central PMCID: PMC12177674.
5. Thorwald MA, Cacciottolo M, Hou X, Morgan TE, Finch CE. Chimpanzee and human ApoE isoforms differ in the stimulation of neurite differentiation consistent with structural predictions with relevance to brain development and aging. J Alzheimers Dis. 2025 Dec;108(3):1378-1387. PubMed PMID: 41105620.

Other Significant Products, Whether or Not Related to the Proposed Project

1. Shkirkova K, Maria NSS, Anson H, Aghaei Y, Badami MM, Chakhoyan A, Godoy-Lugo JA, Chung C, Durra S, Tang-Tan A, Zhao L, Demetriou A, Morales M, Daggupati S, Bent I, Park H, Franklin C, Chen S, Chahine G, Dodds-Lewis S, Morishita M, Jacobs RE, Gout JF, Mack WJ, Forman HJ, Benayoun BA, Vermulst M, Cohen MD, Campen M, Sioutas C, Mack WJ, Finch CE, Thorwald MA. Induction of ferroptotic and amyloidogenic signatures linked to Alzheimer's disease by chemically distinct air pollutants. bioRxiv. 2026 Jan 7; PubMed Central PMCID: PMC12803064.
2. Godoy-Lugo JA, Hicks D, Durra S, Massey ER, Shkirkova K, Sauri AI, Kerstiens E, Chen S, Zhao L, Sioutas C, Mack WJ, Finch CE, Thorwald MA. Air pollution decreases postsynaptic PSD-95 and NMDA receptor subunits in synaptosomes from mouse cerebral cortex. Environ Pollut. 2025 Oct 15;383:126845. PubMed Central PMCID: PMC12288758.
3. Thorwald M, Rodriguez R, Lee A, Martinez B, Peti-Peterdi J, Nakano D, Nishiyama A, Ortiz RM. Angiotensin receptor

blockade improves cardiac mitochondrial activity in response to an acute glucose load in obese insulin resistant rats. Redox Biol. 2018 Apr;14:371-378. PubMed Central PMCID: PMC5647524.

4. Thorwald MA, Godoy-Lugo JA, Rodriguez GJ, Rodriguez MA, Jamal M, Kinoshita H, Nakano D, Nishiyama A, Forman HJ, Ortiz RM. Nrf2-related gene expression is impaired during a glucose challenge in type II diabetic rat hearts. Free Radic Biol Med. 2019 Jan;130:306-317. PubMed PMID: 30316779.
5. Thorwald MA, Godoy-Lugo JA, Rodriguez R, Stanhope KL, Graham JL, Havel PJ, Forman HJ, Ortiz RM. Cardiac NF-κB Acetylation Increases While Nrf2-Related Gene Expression and Mitochondrial Activity Are Impaired during the Progression of Diabetes in UCD-T2DM Rats. Antioxidants (Basel). 2022 May 9;11(5) PubMed Central PMCID: PMC9137621.

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

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Certified by Thorwald, Max A. in SciENcv on 2026-01-20 22:54:31

OMB No. 0925-0001 (Expiration Date 12/31/2027) and No. 0925-0002 (Expiration Date 11/30/2027)

NIH BIOGRAPHICAL SKETCH SUPPLEMENT

Name: Thorwald, Max A.

Persistent Identifier (PID) of the Senior/Key Person: https://orcid.org/0000-0003-0095-5344

Position Title: N/A

Organization and Location:

Personal Statement

I'm a postdoctoral scholar deeply interested in iron mediated lipid peroxidation drives Alzheimer's pathology. I began my scientific career by working on models of metabolic syndrome and diabetes using pharmacological interventions. My PhD work was conducted under two mentors from different disciplines, allowing me to tackle research questions that integrated physiology and biochemistry. In my predoctoral studies, I conducted several animal studies abroad, resulting in first-authored publications focused on antioxidant responses to glucose challenges in the cardiovascular system. The majority of my postdoctoral work was under Dr. Caleb Finch, who expanded my interests to neurobiology and the impact of environmental pollution on brain health. Specifically, I examined how our brains respond to acute and chronic exposure to air pollution, and how metals such as iron are implicated in Alzheimer's neuropathology. While at USC, I've participated in numerous studies related to the topics presented in the proposed work. Due to Dr. Finch's recent Emeritus status, I have joined Dr. Marc Vermulst's lab, where I continue to integrate these themes. Dr. Vermulst has a deep interest in mutations on our genome, some of which occur in response to increases in brain metals. My work is complementary to his. In summary, I have the needed experience to carry out the proposed work on all fronts. My progress as a postdoc was hindered during the first year after being diagnosed with keratoconus, diagnosed after I had a sharp decline in my visual acuity. Keratoconus results in distorted vision due to structural changes in the cornea, leading to double vision and blurriness not correctable with standard eyewear. My keratoconus resulted in balance issues and difficulty reading, causing me to alter my daily routine. I no longer drove to work and switched to public transit. In January 2019, I underwent surgery with a 9-month recovery process. The most significant issue at this time was reading and writing, and I was able to do some experimental work. However, three months later, COVID-19 further slowed my progress as we were unable to work in the lab. I was able to get specialized contact lenses in late 2021 and now have near-perfect vision. I have worked hard to make up for lost time and have numerous publications on the subjects presented in my proposal since 2022. I will continue to work towards my long-term goal of becoming a tenure-track professor linking iron dysregulation and neurodegeneration.

Honors

2025	Young Investigator Award, Society for Free Radical Biology and Medicine
2024	Young Investigator Award, Society for Free Radical Biology and Medicine
2023	Alzheimer's Disease Research Center REC Scholar, University of Southern California
2018	NIH Minority Health and Health Disparities International Research Training, National Institutes of Health
2018	Graduate Dean's Dissertation Fellowship, University of California Merced
2015	NIH Minority Health and Health Disparities International Research Training, National Institutes of Health

Contribution to Science

1. I led and integrated a series of studies establishing iron-dependent lipid peroxidation and ferroptosis as central mechanisms contributing to Alzheimer's disease (AD) neuropathology. Using human postmortem cortex, I demonstrated that AD brains exhibit increased iron accumulation, elevated lipid peroxidation, and reduced antioxidant defenses consistent with ferroptotic vulnerability. By systematically profiling ferroptosis-associated proteins across lipid rafts, nuclei, and whole-cell lysates, I identified lipid rafts, the site of amyloid precursor protein processing, as a major locus of iron-mediated oxidative injury (Thorwald 2025 Alz & Dem). To directly test causality, I designed and conducted in vivo studies showing that iron chelation with deferoxamine reduces brain iron, attenuates lipid peroxidation, and alters amyloid processing and fibril burden in mouse models, supporting a mechanistic role for plaque-associated iron in driving oxidative damage. I extended this framework to Down syndrome with AD, demonstrating that increased APP gene dosage parallels elevated brain iron and ferroptotic markers, while rare mosaic or partial trisomy genotypes attenuate iron load and lipid peroxidation, linking genetic risk to iron-driven vulnerability (Thorwald 2025 Alz & Dem). In parallel, I led conceptual and experimental work integrating amyloid biology, lipid raft dysfunction, and iron-mediated oxidative stress to contextualize these findings within brain aging and dementia

(Thorwald 2023 Alz & Dem). I further expanded this program to evolutionary and functional studies of ApoE, demonstrating that chimpanzee and human ApoE isoforms differentially influence neurite differentiation in ways consistent with structural predictions and relevance to brain development and aging (Thorwald 2025 JAD). Collectively, these studies reposition iron dysregulation and ferroptosis as mechanistic contributors to AD progression and therapeutic response.

2. Epidemiologic studies link air pollution exposure to increased Alzheimer's disease (AD) risk, yet air pollution is chemically heterogeneous and the mechanisms connecting diverse exposures to neurodegeneration remain unclear. A key unresolved question is whether distinct air pollution sources converge on shared AD-relevant pathogenic pathways. I demonstrated that chemically distinct but metal-rich air pollution exposures induce convergent molecular signatures linked to ferroptosis, lipid peroxidation, and amyloidogenic processing. Using diesel exhaust particles and World Trade Center dust as representative metal-rich exposures, I showed that acute cortical exposure produces highly overlapping transcriptional responses enriched for oxidative stress, ferroptosis-associated pathways, inflammation, and amyloid precursor protein processing, despite major differences in particle composition and size. These responses were accompanied by impaired lipid peroxide detoxification, increased lipid peroxidation within lipid rafts, and increased generation of aggregation-prone amyloid-β species. In contrast, exposure to woodsmoke, a low-metal particulate source, elicited a distinct transcriptional and biochemical profile that did not induce ferroptotic priming or amyloidogenic vulnerability, identifying metal content as a determinant of neurotoxic convergence (Shkirkova, 2026 Preprint). In complementary work, I showed that chronic diesel exhaust exposure selectively disrupts postsynaptic scaffolding proteins and NMDA receptor subunits while engaging GPx4-, HMOX1, and glutathione-dependent antioxidant responses within synaptic compartments, linking air pollution-induced lipid peroxidation to ferroptosis-relevant synaptic vulnerability (Godoy-Lugo, 2025 Environ Pollut). As senior and corresponding author, I conceived these studies, directed experimental design, supervised data generation and analysis, and integrated transcriptomic, biochemical, and synaptic approaches.
3. I led a series of mechanistic studies defining how dysregulated redox signaling contributes to mitochondrial dysfunction during insulin resistance and type II diabetes. Using well-characterized rodent models of obesity and diabetes, I demonstrated that inappropriate activation of the renin angiotensin system impairs mitochondrial enzyme activity and antioxidant capacity in the heart, particularly during acute metabolic challenge. I showed that angiotensin receptor blockade restores mitochondrial function by enhancing Nrf2 signaling and glutathione-dependent antioxidant defenses, thereby limiting oxidative damage to mitochondrial enzymes (Thorwald 2018 Redox Biology). In complementary studies, I defined how diabetes progression disrupts dynamic antioxidant responses rather than static antioxidant abundance. I demonstrated that diabetic hearts exhibit impaired induction of Nrf2-regulated genes during glucose challenge, resulting in reduced glutathione synthesis, diminished mitochondrial resilience, and accumulation of oxidative damage (Thorwald 2019 FRBM). Extending this work across disease stages, I showed that advancing diabetes is marked by increased NF-κB mediated inflammation coupled with suppression of Nrf2 signaling and mitochondrial activity, establishing a mechanistic link between inflammation, redox imbalance, and metabolic organ dysfunction (Thorwald 2022 Antioxidants).

Certification:

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Certified by Thorwald, Max A. in SciENcv on 2026-01-2022:54:31

NIH Current and Pending (Other) Support Common Form OMB-3145-0279

CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Thorwald, Max A.

PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0003-0095-5344

*POSITION TITLE: Research Associate

*ORGANIZATION AND LOCATION: University of Southern California, Los Angeles, California, United States

Proposals and Active Projects

***Proposal/Active Project Title:**	Iron Chelation as a Strategy to Mitigate Amyloid-Related Imaging Abnormalities During Anti-Amyloid Immunotherapy
***Status of Support:**	Pending
Proposal/Award Number:	
***Source of Support:**	USC Alzheimer's Disease Research Center
***Primary Place of Performance:**	University of Southern California
***Proposal/Active Project Start Date: (MM/YYYY):**	04/2026
***Proposal/Active Project End Date: (MM/YYYY):**	03/2027
***Total Anticipated Proposal/Project Amount:**	$40,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.2

***Overall Objectives:** The overall objective of this project is to define whether short-term anti-amyloid immunotherapy induces iron-dependent vascular and membrane injury consistent with ferroptotic stress, and to determine whether iron chelation mitigates these effects in a sex-dependent manner. Building on strong evidence that iron accumulates at amyloid plaques and that vascular injury permits entry of highly concentrated blood-derived iron into the brain parenchyma, this study tests the hypothesis that amyloid engagement by Lecanemab destabilizes amyloid-laden microvasculature, promoting iron-driven lipid peroxidation and ARIA-like pathology even in the absence of extensive plaque clearance. Using ultra-high-field 7T MRI as the primary endpoint, complemented by targeted transcriptomic and biochemical analyses, this project will distinguish vascular and oxidative consequences of immunotherapy from nonspecific immune effects and determine whether adjunctive deferoxamine treatment attenuates ferroptotic injury. By explicitly evaluating sex as a biological variable, this work aims to establish mechanistic and imaging biomarkers relevant to ongoing clinical

trials and to inform safer, sex-informed combination strategies for anti-amyloid therapy.

***Statement of Potential Overlap:** None

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. § 6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Thorwald, Max in SciENcv on 2026-01-23 00:04:54

E. OVERALL IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

NOTHING TO REPORT

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

Not Applicable

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

NOTHING TO REPORT

F. OVERALL CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. OVERALL SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

NOTHING TO REPORT

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Sub-Project ID	Study ID	Study Title	Delayed Onset	Clinical Trial	NCT	NIH-Defined Phase 3	ACT
Clinical-Core-001	534996	USC Alzheimer Disease Research Cohort	NO	NO			

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

Are there personnel on this project who are newly involved in the design or conduct of human subjects research?

No

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Does this project involve vertebrate animals?

No

G.8 PROJECT/PERFORMANCE SITES

Organization Name	UEI	Congressional District	Address
Primary: University of Southern California	G88KLJR3KYT5	CA-034	Keck School of Medicine 1540 Alcazar Street Los Angeles, CA 900890186
Children's Hospital Los Angeles	DVL1CMRMWRN9	CA-030	4650 Sunset Blvd Los Angeles, CA 90027

G.9 FOREIGN COMPONENT

No foreign component

G.10 ESTIMATED UNOBLIGATED BALANCE

G.10.a Is it anticipated that an estimated unobligated balance (including prior year carryover) will be greater than 25% of the current year's total approved budget?

Yes

Estimated unobligated balance: $1,952,257.00

G.10.b Provide an explanation for unobligated balance:
The late receipt of NOA funds resulted in delayed hirings and account activation.

G.10.c If authorized to carryover the balance, provide a general description of how it is anticipated that the funds will be spent
Project implementation will adhere to the previously approved budget and scope, with no modifications to goals or spending.

G.11 PROGRAM INCOME

Is program income anticipated during the next budget period? No

G.12 F&A COSTS

Not Applicable

Section 1 - Basic Information (Study 534996)

1.1. Study Title *

USC Alzheimer Disease Research Cohort

1.2. Is this study exempt from Federal Regulations * ❍ Yes ● No

1.3. Exemption Number ❑ 1 ❑ 2 ❑ 3 ❑ 4 ❑ 5 ❑ 6 ❑ 7 ❑ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?	● Yes	❍ No
1.4.b. Are the participants prospectively assigned to an intervention?	❍ Yes	● No
1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?	❍ Yes	● No
1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?	❍ Yes	● No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 534996)

2.1. Conditions or Focus of Study

- All information on conditions are found in Protection of Human Subjects document.

2.2. Eligibility Criteria

The eligibility criteria is outlined in a detailed summary that can be found in the Overall Protection of Human Subjects statement in Section 3.1.1.

2.3. Age Limits	Min Age: 50 Years	Max Age: 90 Years
2.3.a. Inclusion of Individuals Across the Lifespan	Inclusion_Across_the_Lifespan.pdf	
2.4. Inclusion of Women and Minorities	Inclusion_Women_and_Minorities.pdf	
2.5. Recruitment and Retention Plan	Retention_and_Recruitment_Plan.pdf	
2.6. Recruitment Status	Recruiting	
2.7. Study Timeline	Study_Timeline.pdf	
2.8. Enrollment of First Participant (SEE SECTION 6.3)		

Inclusion of Individuals Across the Lifespan

Children are excluded because they don’t generally develop sporadic Alzheimer disease. We will include persons who are 18 or older who may be carriers of autosomal dominant AD or have Down’s syndrome.

Inclusion of Women and Minorities

The USC ADRC included under-represented groups and historical minorities in numbers consistent with the population and Los Angeles County local communities from which participants are drawn. The ORE core and the clinical core proactively recruit Latinx from our community using a range of programs (see project narratives for the clinical core, the ORE core, and Section 3. Protection of Human Subjects). We plan for 35% or more of our clinical core participants will be of Latinx heritage. Currently, in the ADRC women represent 60% of our participants in the clinical core. We plan to continue to recruit women at that ratio to reflect their somewhat greater prevalence in the population with late-life cognitive impairment. Our inclusion and exclusion criteria for enrollment are clearly defined in the narratives for the overall, clinical core, ORE core, and Section 3. Protection of Human Subjects.

Recruitment and Retention Plan

See ORE core and clinical core narratives.

The first from the ORE core and Hussain on the many recruitment approaches. The second from the CC on how we retain participants and have only a ?5% loss rate, and substantial Ns followed for 5+ and 10+ years (in the current research strategy).

Study Timeline

For the study timeline, see the Clinical, Neuropathology, and ORE Cores.

2.9. Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
IER 555816	Domestic	Los Angeles and surrounding counties
IER 555817	Domestic	Los Angeles and surrounding counties

Human Subject Report (USC Alzheimer Disease Research Cohort)

OMB Number: 0925-0770

Expiration Date: 09/30/2024

Inclusion Enrollment Report 555816

1. Inclusion Enrollment Report Title* : USC ALzheimer's Disease Research Center

2. Using an Existing Dataset or Resource* : ❍ Yes ● No

3. Enrollment Location Type* : ● Domestic ❍ Foreign

4. Enrollment Country(ies): USA: UNITED STATES

5. Enrollment Location(s): Los Angeles and surrounding counties

6. Comments: Continued participants from the previous cycle

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	1	0	1	1	3
Asian	26	16	2	0	44
Native Hawaiian or Other Pacific Islander	0	0	0	0	0
Black or African American	26	10	1	0	37
White	151	110	44	23	328
More than One Race	15	4	17	10	46
Total	219	140	65	34	458

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	0	0	0	0	0	0	0	0	0	0
White	0	0	0	0	0	0	0	0	0	0
More than One Race	0	0	0	0	0	0	0	0	0	0
Unknown or Not Reported	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

OMB Number: 0925-0770
Expiration Date: 09/30/2024

Inclusion Enrollment Report 555817

1. Inclusion Enrollment Report Title* : USC Alzheimer's Disease Research Center

2. Using an Existing Dataset or Resource* : ❍ Yes ● No

3. Enrollment Location Type* : ● Domestic ❍ Foreign

4. Enrollment Country(ies): USA: UNITED STATES

5. Enrollment Location(s): Los Angeles and surrounding counties

6. Comments: New enrollments for the requested cycle

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	1	0	1	0	2
Asian	12	12	1	0	25
Native Hawaiian or Other Pacific Islander	0	0	0	0	0
Black or African American	12	8	0	0	20
White	52	60	37	39	188
More than One Race	6	3	8	8	25
Total	83	83	47	47	260

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	0	0	0	0	0	0	0	0	0	0
White	0	0	0	0	0	0	0	0	0	0
More than One Race	0	0	0	0	0	0	0	0	0	0
Unknown or Not Reported	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

Section 3 - Protection and Monitoring Plans (Study 534996)

3.1. Protection of Human Subjects	Protection_of_Human_Subjects.pdf
3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?	❍ Yes ● No ❍ N/A
Single IRB plan attachment	
3.3. Data and Safety Monitoring Plan	Data_Safety_Monitoring_Plan.pdf
3.4. Will a Data and Safety Monitoring Board be appointed for this study?	❍ Yes ● No

3.5. Overall structure of the study team

3.1 Protection of Human Subjects

The human subjects research associated with the USC ADRC involves cohort, observational research with diagnostic intervention, and NIH-defined phase 2, phase 3, and other clinical studies. The clinical intervention trials, however, are affiliated with, but not a funded part of the ADRC. Protection of human subjects is a primary concern to the ADRC and its investigators. The policies and procedures described below pertain to all personnel conducting clinical research under the auspices of the ADRC. Although specific details may differ among the individual cores and projects of this application, this section reviews the general principles by which the ADRC operates for the protection of human subjects.

The USC ADRC is part of a network of approximately 30 Alzheimer Disease Centers (ADCs) funded by the National Institute on Aging. The ADCs contribute to a database, managed by the National Alzheimer Coordinating Center (NACC), and collaborate at national level: including the National Cell Repository for Alzheimer Disease (NCRAD), and the Alzheimer Disease Genetics Consortium (ADGC). Many ADRCs, including USC, collaborate with the Alzheimer's Disease Neuroimaging Initiative (ADNI) Alzheimer's Clinical Trials Consortium (ACTC), Alzheimer Disease Cooperative Study (ADCS), and the Dominantly Inherited Alzheimer's Network (DIAN). Participants in the USC ADRC are also approached about placing their de-identified clinical data, biospecimens, genetic and imaging data in national databases.

3.1.1. Risks to Human Subjects

a. Human Subjects Involvement, Characteristics, and Design

Participants for the ADRC may be from the ages of 50 and older, but for most studies will be from 60 to 90. As a product of the research strategies developed for the USC ADRC, a substantial proportion of participants are not cognitively impaired; another group has mild cognitive impairment; and a somewhat smaller group has mild dementia. Many participants in our cohorts will age into progressively more severe dementia. Because of their age as well as our recruiting emphasis on vascular, metabolic risk factors and *APOE* genotype, many will have chronic medical illnesses in addition to cognitive impairment. By design, some ADRC affiliated studies include individuals with disorders that place them at somewhat higher risk for stroke and cardiovascular illnesses such as hypertension and diabetes. Participants with serious unstable medical conditions, however, are excluded from these studies. Approximately 260 new participants to the ADRC and related studies will be enrolled over 5 years. Estimated numbers, recruitment methods for individual studies and inclusion and exclusion criteria are detailed in the Overall, Administrative, Clinical, ORE, Neuropathology, Biomarker, and Imaging Core narratives in addition to below.

Description and justification of the proposed involvement of human subjects in the work outlined in the research strategy sections

Participants are older adults with no cognitive impairment, age-related cognitive decline, mild cognitive impairment, or dementia who agree to participate in one or more of several studies undertaken, supervised, or affiliated with the USC ADRC. Participants are generally intended to be followed indefinitely and evaluated, tested for cognitive ability, approximately yearly, have MRIs, amyloid and tau PET scans, and blood specimens taken to assess for research biomarkers. Research lumbar punctures are offered but are optional.

The general purpose of the clinical research within the ADRC is to advance knowledge of at-risk, vascular, metabolic, and *APOE* genotype factors in the development of dementia and cognitive impairment. The purpose of the brain donation program is to enhance clinicopathological correlation, ante mortem diagnosis, and to obtain important biospecimens to support future research and to be shared across research centers. There are other observational studies, and experimental, clinical trials in which specific experimental therapeutic interventions will be made. For these studies participants with no cognitive impairment, mild cognitive impairment, and with dementia will be recruited and may be co-enrolled in the VCS and BRS if there is no contraindication.

Describe and justify the sampling plan, as well as the recruitment and retention strategies and the criteria for inclusion or exclusion of any subpopulation

As above, we recruit and enroll participants in the at-risk longitudinal cohort of the clinical core. We estimate we need to maintain a cohort of 500 to serve the needs of affiliated studies and to contribute effectively to the NACC UDS and NCRAD multicenter cohorts. We estimate that we need to recruit approximately 52 at-risk participants per year to increase and maintain the clinical core cohort at 500 (see clinical core research strategy for details). Our recruitment methods are described in the research strategy sections of the Overall, the ORE core, the clinical core, and below. Briefly, our recruitment strategies are multifaceted and involve community outreach, earned

media, the use of registries, outreach to physicians, public awareness, and education. For autopsy participants we also focus on our existing cohorts and the local community. We used actuarial methods to determine a minimum size for our brain donation (autopsy) subgroup. We estimate we need to enroll approximately 26 new participants per year, mainly from the clinical core cohort, to maintain or increase our autopsy rates (see Neuropathology and Clinical cores).

Our retention methods for ADRC clinical core participants and affiliated studies involve proactive contact by ADRC staff including study coordinators, the original intake coordinator, and nurses. We contact participants several times between their annual visits to assess their status, thank them, and encourage their continuing interest. The ORE core uses newsletters, updates, and participant appreciation days as well. Participants in affiliated clinical studies are often seen more frequently than general clinical core cohort participants which allows for more frequent interaction and encouragement. We can reimburse participants for travel and food in many of our studies under the IRB-approved protocols for those studies.

General eligibility criteria. Criteria for cognitively non-impaired: We generally use the “cognitively unimpaired” or “subjective memory complaint” criteria as established by the ADNI4 cohort study and now by the new UDSv4 criteria/guidelines (forms D.1, D.2). Criteria include nominally normal episodic memory or delayed recall; MMSE score ≥ 24 or MOCA ≥ 22, with no impairment of recall or orientation; no impairments on instrumental activities of daily living; no major depression, marked anxiety; adequate corrected vision, hearing, and communicative ability in order to allow for obtaining the outcomes assessments. Criteria for mild cognitive impairment: We generally follow Albert et al 2011 criteria, and further specify to the best of our judgment or with biomarker support whether MCI is due to Alzheimer disease. Criteria for dementia and Alzheimer disease generally follow 1984 and 2011 McKhann et al criteria. We use other published criteria and guidelines for other neurodegenerative and cognitive impairing conditions.

Eligibility criteria for brain donation (autopsy). (1) Informed consent for brain donation (autopsy) by participants with decisional capacity or by a legally authorized representative if not; (2) understanding and agreement with post-mortem procedures, (3) participation in the clinical core cohort, (4) evaluated and diagnosed with the UDS and initial data uploaded to NACC and the USC RedCap database.

General exclusion criteria for ADRC participants. Unwillingness to give informed consent; lack of assent by a participant without decisional capacity to give informed consent; unwillingness of the legally authorized representative or proxy decision-maker to provide surrogate consent; inability or unwillingness to undergo or contraindication to an MRI, functional impairment so severe that the participant cannot participate in required research activities; unstable medical conditions that impair participant’s ability to participate in the studies.

Criteria for ORE core resource of Enhanced Recruitment from the LA County Department of Health Services at the Rancho Los Amigos National Rehabilitation Center (RLANRC). Our overall aim is to provide patients cared for in public-safety net hospitals (e.g., RLANRC) opportunities to participate in ADRC and affiliated research studies (e.g., NACC UDS, SCAN, CLARiTI, clinical trials). The sample pool includes participants seen at DHS hospitals and clinics. After informed consent, the ADRC DHS participant will be followed as usual and will provide additional data as noted below.

Procedures for obtaining informed consent for research on cognitively impaired human subjects who may not have the capacity to consent and to ensure future participation in research studies if or when the participant becomes unable to consent

Consent is demonstrated when an individual (either verbally or non-verbally) indicates what they are willing to do or allow a third party to do this for them. To give valid consent, participants need to be able to access, understand and process information relating to the decision they are making. Informed consent is therefore a fundamental requirement for participating in the USC ADRC. The mental capacity of an individual is at the center of the law relating to consent. For the ADRC studies, the physicians, nurse practitioner, or research nurses who are experienced in the field of dementia will determine whether a potential study participant has decisional capacity and is competent to consent to research prior to beginning any consent procedures. Informed consent will be presented and obtained in the language preferred by the potential participant and by appropriately trained and certified staff. Competency is assessed at each of the follow up visits. At the beginning of each of the follow up visits, the participant or responsible party (legally authorized representative) and/or study partner are provided a copy of the informed consent to review. They are given as much time as they need to answer any questions they might have about their continued participation. The research team during this time assesses whether the participant and study partner understand and are willing to continue participation in the study. When there is any

indication to suggest diminished capacity, we begin by discussing the research directly with the participant to the extent possible to gain as fair an idea as possible of what their understanding of the study and their inclination to participate. If a participant lacks capacity, we still ask for assent to study participation. If they refuse, then we will not continue the participant in the study. Additionally, we review the consent form with the participant's legally authorized proxy decision-maker or guardian, next of kin, or familial caregiver according to California State Law and require that that person consent to the study as well. If either party refuses, then we do not continue the participant in the study. Participants and LARs are given the option to revoke consent. As above, in cases where the participant no longer has the capacity to consent, consent is obtained from the legally authorized representative and assent is obtained from the participant per California State law AB2325 (California Health & Safety Code section 24178).

Procedures specifying how, by whom, and with whom brain donation (autopsy) will be discussed

Potential brain donors may be approached several times during their participation in the ADRC: in telephone screening by the intake coordinators and recruiters, by physicians, research nurse, nurse practitioner, and by study coordinators at initial visit, family conference, or follow-up visits. They are told about autopsy and what they will be asked to do. This includes: (1) completing an initial questionnaire asking about contact and health information, (2) participating as necessary in annual follow-ups or by telephone where memory testing may be done, (3) receiving time of death instructions (mortuary arrangements and funeral plans) to facilitate transfer of the body and to avoid delays. Participants are asked to give consent for USC to keep and to distribute blood and body fluid samples for future research and sharing information with other researchers.

b. Sources of Materials

Research material is obtained from living and deceased individuals in the form of specimens, tissue, records, or data as follows: The protocols of the ADRC require gathering a large amount of data specifically for research purposes. The major classes of data include medical diagnoses, findings on medical examinations, behavioral data on psychiatric disturbances, neuropsychological data on cognitive function, data on functional ability, demographic data, measurements of brain structure obtained with MRI, personal and family medical history, CSF, and genetic data. All this is gathered on individually identifiable participants and thus great care is taken to protect their privacy and the confidentiality of this information. The information is de-identified by the clinical core and data core before being sent to NACC, NCRAD, and other investigators. The data core narrative describes procedures for protecting participants' privacy and de-identifying participant and PHI data.

c. Potential Risks

In general, the risks to participating in the ADRC and affiliated studies are directly associated with the clinical data collection as the studies directly supported by the ADRC are observational and non-interventional, and there are minimal or no known risks regarding the observational components of the studies *per se.* There are risks involved in preparing for and traveling to the clinic for the studies. The risks involved in being assessed with noninvasive clinical and neuropsychological tests are minimal. Some participants may feel stressed or anxious, but this usually resolves after testing. There are also very low risks involved in venipuncture, including bruising and minor pain, anxiety or fatigue associated with neuropsychological testing, etc. The risks for MRI without contrast are rare and include risks of injury when patients with ferrous objects in their bodies undergo MRI. We do not plan to use MRIs with contrast for clinical core participants in this cycle.

Most clinical core participants will undergo a non-contrast MRI, a non-invasive procedure. Thousands of patients have undergone studies at USC on the equipment proposed for this study, using similar data acquisition strategies without any known harmful consequences. Magnetic field strengths, radio frequency power levels, gradient strength, and switching times are within the FDA approved range. The data acquisition sequences that will be used are provided by the manufacturer and are FDA approved. Some people may become claustrophobic in the confines of the magnets; those participants will be removed from the study. An MRI may cause possible anxiety for people due to the loud banging made by the machine and the confined space of the testing area.

Patients with pacemakers, aneurysm clips, artificial heart valves, ear implants, metal/foreign objects in their eyes, or a history of ferromagnetic material near vital structures will be excluded from the study because of the risk of pacemaker malfunction or movement of ferromagnetic material within the magnetic field of the imaging unit. Except for these patients there have been no adverse effects demonstrated following exposure to the magnetic field gradients and radiofrequency pulses used to generate the images. Some participants will undergo a 7T MRI in affiliated studies, under a separate protocol from the ADRC.

Many clinical core participants will undergo PET-CT scans, which include exposure to greater radiation than most diagnostic x-ray procedures. Radiation exposure can increase one's risk of developing cancer. The risk for cancer from the dose of radiation received in this study is known to be low. Injection site reactions could occur consisting of rash, irritation or pain. There are also minor risks with. venipuncture. The injection of the radioisotopes might include pain and bruising. It is possible that a pre-existing or unrelated medical abnormality is apparent on PET or MRI scans. If so, we will inform and help the participant get follow up care.

An important risk associated with the ADRC observational studies relates to loss of privacy or the potential exposure of personal medical information, particularly data related to risks of Alzheimer disease or dementia. The improper release of such information could have negative effects for participants. For example, loss of privacy, adversely affecting their ability to obtain health or life insurance and could carry negative effects for blood relatives as well. Participating in these studies generally does not require adding or withholding medications used to treat a participant's cognitive-impairing condition. The exceptions are for selected treatment studies with experimental medicines. This is addressed with the participant in separate IRB-approved and NIH-funded protocols and informed consent interviews and documents, as the clinical trials involve interventions with experimental drugs and the risks here are specific to the drug involved. The USC IRB reviews each protocol.

There are risks involving the optional lumbar puncture for clinical core participants. They include infection, bleeding, and headache due to CSF leakage. For prolonged post-LP headache or leakage an epidural blood patch may be placed.

The ADRC clinical core cohort is not a treatment study. There are no known alternative treatments and procedures for the conditions being studied. Hence, the risks and potential benefits of the alternative treatments and procedures to participants in the proposed research are not known.

3.1.2 Adequacy of Protection against Risks

a. *Recruitment and Informed Consent*

Written informed consent will be obtained for all participants in the ADRC. The experimental procedure and length will be explained to the participants who will be given consent forms and the time to read it. Participants will be allowed to ask questions and when their concerns have been satisfied will be asked to sign the form. In the absence of written consent, no research procedures will be performed. The major principle for approaching the participant is to ensure that they fully understand 1) the potential risks involved, 2) that participation in the research is purely voluntary; and 3) that refusal to participate will not, in any way, adversely affect their medical care. Potential benefits are also explained, but realistically, with care being taken not to induce false expectations. Individuals who are not actively involved in the research will witness the signing of the consent form. A copy of the signed consent form will be given to the participant.

Plans for recruitment and consent. Participants will be recruited in several ways, from existing longitudinal cohorts, or by new outreach (see Aim 1 (Gene Screen) and 2 Aim 2 (LA DHS facilities) of the ORE core and Aim 4 of the clinical core). In either case, potential participants may be approached initially by telephone, email, or in person, and the general nature of the research is discussed following guidelines approved by the USC IRB. Our protocols for obtaining consent include specific procedures for explicitly determining both the understanding of the research study and the willingness to participate. When there is an indication to suggest diminished capacity, we begin by discussing the research directly with the participants to the extent possible to gain as fair an idea as possible of what their understanding of the study and their inclination to participate. If the participant lacks decisional capacity, then we ask that they assent to study participation; if they refuse then they will not be enrolled.

A special problem is obtaining consent from people of limited decisional capacity or from those who are clearly incapable of consent. While this capacity to consent will not be at issue during recruitment of our cognitively normal or early MCI participants, it becomes one with early dementia patients and for those who experience cognitive decline during their illness such as might occur during follow-up evaluations of patients with MCI. This is discussed with the IRB for each new study we have done over the years. We and the USC IRB follow California State law AB2325 (California Health & Safety Code section 24178).

In addition, we review the consent form with the patient's legally authorized representative or guardian (if there is one), next of kin, or familial caregiver according to California State Law and require that this person consent to the study as well. If either refuse, then we do not enroll the person. We rely on the continuing informed consents from the participant or LAR decision-maker for continuing autopsy permission and for current and future

use of biological samples by qualified investigators including permission for sharing of cells, DNA, and genetic and phenotypic information, and storage in repositories. If the participant loses capacity to continue to provide informed consent, then we rely on the proxy decision-maker as described.

We follow the Alzheimer Disease Center Biospecimen Task Force guidelines for guidance on consent forms; use the NIH sharing policy and guidance on genetics research for guidance on language regarding genetics that we use in consent forms; and follow NIH policy on genome wide association studies.

b. Protections Against Risk

The following are our procedures for protecting against or minimizing potential risks, including risks to privacy of individuals and confidentiality of data, and an assessment of their likely effectiveness. This research does not involve vulnerable populations as defined in the DHHS regulations, Subparts B-D, i.e., pregnant women, human fetuses and neonates, prisoners, and children. It does involve cognitively impaired and older participants some of whom many have diminished decisional capacity and may be considered vulnerable.

Plans for ensuring necessary medical or professional intervention in the event of adverse effects to participants.

Research participants are overseen by physicians and are seen for assessment of adverse effects. All participants concerned about or reporting adverse events are evaluated and treated as necessary by the physician; adverse events are reported to the IRB, the NIH and FDA when required to ensure the safety of participants.

Plans to protect against and to ensure necessary medical intervention in the event adverse events occur for potential risks associated with ADRC participation including risks to confidentiality, risks of neuropsychological testing, of genotyping, MRI scans, and lumbar punctures are as follows:

Confidentiality. The ADRC takes the risk to subject confidentiality very seriously and has implemented a variety of measures to protect the confidentiality of data and the identities of our participants. In addition to several layers of safeguards for the electronic data (electronic firewalls, password protection of files, limiting access privileges, structural isolation of particularly sensitive components of the data base, use of pseudo ID numbers for sensitive data), normal prudence is observed with regard to the participant's physical records as well. Patient charts and records are either in locked or directly supervised areas during business hours, and always under lock and key during off hours. New staff members are specifically instructed in confidentiality issues and undergo required research, human subjects, and HIPPA training. A Federal certificate of confidentiality has been and will be applied for specific studies to add another set of protections against legal intrusions on subject privacy.

Neuropsychological testing. While many participants in the ADRC and affiliated studies are expected to have generally good cognitive abilities and, therefore, unlikely to fatigue during cognitive testing, we know that many will be anxious about the extent of their disabilities. This will be particularly true during follow-up evaluations where conversion to dementia or progression of dementia will be determined. All evaluations are held in confidence and shared with the patient and family at the initial evaluation or if diagnostic category is changed during follow-up. The neuropsychologists and psychometricians are sensitive to patient fatigue and anxiety and may end testing if a participant expresses discomfort. If so then, testing is often completed at a second visit.

Blood Sampling. The process of blood sampling is discussed with each participant. Blood samples are obtained in laboratory space by certified phlebotomists, technicians, or nurses and physicians, if needed.

APOE genotyping. Disclosure of *APOE* genotype has been not recommended by professional consensus groups for asymptomatic persons. In practice, *APOE* genotyping is available commercially and becoming more frequent for both symptomatic and asymptomatic persons in both medical and non-medical situations. We inform potential research participants that *APOE* genotype is part of the biospecimens that are collected but that an individual's genotype will not be disclosed to them unless they ask (e.g., GeneScreen). When *APOE* genotyping is not obtained and performed to CLIA standards we will not share these test outcomes with participants. However, for participants who know their genotype from elsewhere, other studies, or from commercial companies (e.g, 23&me) or who request to know their genotype we offer clinical (CLIA) genotyping from the USC clinical laboratory. Participants are counselled by ADRC physicians and a genetic counsellor, if requested, and provided with written educational materials like those used in the REVEAL, Generation studies, and by 23&me.

Other genetic studies. ADGC currently performs GWAS, whole genome wide sequencing, whole exome sequencing on DNA obtained from circulating white blood cells and sends DNA to NCRAD where this is also performed. We do not reveal these findings.

MRI imaging. All participants will answer detailed questions about potential ferro-magnetic implants or shrapnel that could result in injury during MRI. In addition, participants are directly observed during the imaging procedure and are in constant voice communication with the MRI technician. The clinical and research MRI scanners at USC are FDA approved for human use with respect to field and gradient switching times. All data acquisition sequences used will be FDA approved. Should a participant experience discomfort during a procedure, the study will be stopped. Participant confidentiality is maintained and no data which can be identified with an individual will be released to anyone not connected with this project without written consent of the subject. The exposure time and experimental factor will be recorded and filed with the consent forms for future reference. All procedures on humans will be subject to prior IRB approval.

PET/CT imaging. Participants are assessed for possible pregnancy. Those who are pregnant will not undergo PET/CT. If the participant is a woman who could become pregnant, then a pregnancy test will be performed before each PET/CT procedure. The ^{18}F radioactive tracers for amyloid-β and tau to be used in the ADRC are FDA approved. The tracer injected via a venous catheter. Images are obtained in the scanner about 90 to 100 minutes afterwards. The scan takes about 90 minutes. The exposure time and experimental factor will be recorded and filed with the consent forms for future reference. Participant confidentiality is maintained and no data which can be identified with an individual will be released to anyone not connected with this project without written consent of the participant. All procedures on humans will be subject to prior IRB approval.

Lumbar puncture. Risks of the lumbar puncture include infection, bleeding, CSF leakage, and headache, and will be explained to the participants and legally authorized representatives verbally and in the consent form before the procedure. Appropriate precautions will be taken to minimize these risks. Standard sterile protocol will be strictly followed. A small-bore atraumatic needle (22- or 24-gauge Sprotte brand) will be employed to help decrease the incidences of CSF leak headaches. After the procedure, participants will lie down in the procedure room for about an hour to decrease the risk of headache; will be given light food and drink; and will be instructed not to do any strenuous physical activity for 24 hours. This includes lifting, bending, doing housework, gardening, or doing exercise such as jogging or bicycling. They will be given instructions before they leave as to how to contact the ADRC staff if they should have any problems or concerns after the procedure. They will be contacted by the ADRC team on the day after the lumbar puncture to assess their status. Those who develop persistent positional headaches because of CSF leak will return for an evaluation and a possible blood patch.

Potential, rare risks of lumbar puncture include allergic reaction to lidocaine, infection, damage to nerves in your back, and bleeding that may affect the spinal cord or brain. The risk of these is very small. To minimize these risks, the lumbar puncture procedure will be performed by a physician specifically trained in the procedure.

Some lumbar punctures may be performed under fluoroscopy based on physician judgment or participant's preference. For those done under fluoroscopy there are additional risks similar to risks from other X-ray procedures. The radiation dose received varies depending on the individual procedure. On average, radiation received during a fluoroscopic procedure is about one-third of the annual ambient x-radiation dose that people receive from nature and man-made sources. Risks associated with exposure to very high levels of radiation include radiation-induced injuries to the skin and underlying tissues which occur shortly after the exposure, and radiation-induced cancers, which may occur sometime later in life. The probability that a person will experience these effects from a fluoroscopic procedure is very small. To minimize the radiation risk, fluoroscopy will be performed with the lowest acceptable exposure for the shortest time necessary.

Clinical and safety outcomes, clinical safety results will be recorded on ADRC case record forms and USC medical records. All data will be collected specifically for the ADRC and affiliated studies. After completion of the studies and confirmation of the accuracy of the research records, personal identifying information, and protected health information, will be separated from the research records, such that the research records will not be identifiable with any study participant. Paper records will be stored in a locked cabinet. Only study personnel with completed human subject protection training will have access to the obtained data.

3.1.3 Potential Benefits of the Proposed Research to Human Subjects and Others

The research procedures of ADRC studies do not generally directly benefit participants except insofar as some evaluations may inform a new clinical diagnosis and possibly improve the quality of participants' medical care. As described above the potential risks of participation are low to moderate and can be effectively mitigated. Participants are told that they are unlikely to accrue personal, tangible benefit from their research participation. Nonetheless, for many, participation is intrinsically rewarding because it is a way that they can contribute to

better understanding of illnesses that may have potentially affected them. Thus, there is a low degree of personal benefit, a substantial degree of general societal benefit to be weighed in relation to lesser, controllable risks.

3.1.4 Importance of the Knowledge to be Gained

Alzheimer's disease is an incurable, progressive, fatal illnesses. Vascular dementia may have a more variable, but also often progressive course. The knowledge to be produced by the studies associated with this grant is of significant value, particularly regarding the investigation of at-risk factors for neurodegeneration, dementia etiology and the potential for cerebrovascular disease to interact with Alzheimer pathology to produce cognitive impairment. The new information collected will help to better define diagnosis, prognosis, and potential treatments for Alzheimer disease, vascular dementia, and the mixture of both. The affiliated translational studies may help to better define potential treatments. The importance is that we may better understand how to prevent and treat aspects of cognitive decline with aging.

The risks to participants are reasonable in relation to the importance of the knowledge that reasonably may be expected to result. There are no known risks to appropriately selected individuals from MRI studies at the energy levels proposed in this study. The potential benefit of these studies is the advancement of scientific technique and knowledge. The success of these studies may provide additional insight useful for clinical prediction, diagnosis, and management of patients.

Data and Safety Monitoring Plan

We follow NIH Data and Safety Monitoring Policy. We monitor by study physician and nurse participants' continued safety in the observational studies. For affiliated clinical intervention studies, we establish DSMBs in cooperation with NIH for the NIH-funded, single site studies, or independently. Two examples are the DHA study (Yassine) and allopregnanolone P1 (Schneider) that are covered under separate NIA grants from the ADRC. We use the established DSMBs for multicenter trials. For NIH-sponsored multicenter trials we rely on the NIH-approved central DSMB. In the instance of the AHEAD 3-45 preclinical AD lecanemab trial, we use the ACTC/NIA standing DSMB. For other multicenter studies we use the central DSMB established by the sponsor and approved by the USC IRB. The frequency of monitoring depends on potential risks, complexity, and the nature of the observational study or of the clinical intervention study. For all studies, the PD/PI will monitor the trial; the IRB will receive yearly reports and reportable adverse event reports. Detailed data and safety monitoring plans are submitted to the IRB and to NIH for NIH-sponsored studies for approval prior to accruing participants.

Clinicaltrials.gov requirements. Per Public Law 110-85 (the FDA Amendments Act (FDAAA) of 2007) and subsequent amendments, we register all intervention studies at the ClinicalTrials.gov Protocol Registration Information Web site no later than 21 days after the first participant is enrolled. Summary results including adverse events will be voluntarily reported no later than 1 year after the completion date. For the affiliated studies, we ensure that they are certified, the responsible party, and made all required submissions to ClinicalTrials.gov. Note that the ADRC, itself, does not sponsor or fund intervention studies.

Section 4 - Protocol Synopsis (Study 534996)

4.1. Study Design

4.1.a. Detailed Description

4.1.b. Primary Purpose

4.1.c. Interventions

Type	Name	Description

4.1.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ❍ Yes ❍ No

4.1.e. Intervention Model

4.1.f. Masking ❍ Yes ❍ No

❑ Participant ❑ Care Provider ❑ Investigator ❑ Outcomes Assessor

4.1.g. Allocation

4.2. Outcome Measures

Type	Name	Time Frame	Brief Description

4.3. Statistical Design and Power

4.4. Subject Participation Duration

4.5. Will the study use an FDA-regulated intervention? ❍ Yes ❍ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.6. Is this an applicable clinical trial under FDAAA? (SEE SECTION 6.6)

4.7. Dissemination Plan

Section 6 - Clinical Trial Milestone Plan (Study 534996)

6.1. Study Primary Completion Date		
6.2. Study Final Completion Date		
6.3. Enrollment and randomization		
Enrollment of the First Participant (Study Start Date)	04/01/2025	Anticipated
25% of planned enrollment recruited by		
50% of planned enrollment recruited by		
75% of planned enrollment recruited by		
100% of planned enrollment recruited by		
6.4. Completion of primary endpoint data analyses		
6.5. Reporting of results in ClinicalTrials.gov		
6.6. Is this an applicable clinical trial under FDAAA?	❍ Yes ❍ No	

Composite Application Budget Summary

Categories	Budget Period
Salary, Wages and Fringe Benefits	2,216,571
Equipment	35,000
Travel	50,857
Participant/Trainee Support Costs	0
Data Management and Sharing (DMS) Costs*	
Human Fetal Tissue (HFT) Costs*	
Other Direct Costs (excluding Consortium)	596,364
Consortium Costs	50,960
Direct Costs	2,949,752
Indirect Costs	1,781,031
Total Direct and Indirect Costs	4,730,783

*HFT/DMS Costs, if present, are included in the budget summary for information only; the values have not been subtracted from "Other Direct Costs (excluding Consortium)"

Component Budget Summary

Components	Categories	Budget Period
Cores-001-001 (Additional)	Salary, Wages and Fringe Benefits	186,289
	Equipment	0
	Travel	3,000
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	32,933
	Consortium Costs	0
	Direct Costs	222,222
	Indirect Costs	142,230
TOTALS	Total Direct and Indirect Costs	364,452
001-001 (Admin Core)	Salary, Wages and Fringe Benefits	212,465
	Equipment	0
	Travel	27,000
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	147,282
	Consortium Costs	0
	Direct Costs	386,747

	Indirect Costs	255,253
TOTALS	Total Direct and Indirect Costs	642,000
Core-001-001 (Biomarker)	Salary, Wages and Fringe Benefits	189,554
	Equipment	35,000
	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	135,311
	Consortium Costs	0
	Direct Costs	359,865
	Indirect Costs	165,535
TOTALS	Total Direct and Indirect Costs	525,400
Core-001-001 (Clinical)	Salary, Wages and Fringe Benefits	537,048
	Equipment	0
	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	23,706
	Consortium Costs	0
	Direct Costs	560,754
	Indirect Costs	370,097

TOTALS	Total Direct and Indirect Costs	930,851
Management-Core-001-001 (Data)	Salary, Wages and Fringe Benefits	261,053
	Equipment	0
	Travel	3,000
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	37,206
	Consortium Costs	0
	Direct Costs	301,259
	Indirect Costs	198,830
TOTALS	Total Direct and Indirect Costs	500,089
Core-001-001 (Neuropathology)	Salary, Wages and Fringe Benefits	295,166
	Equipment	0
	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	82,692
	Consortium Costs	50,960
	Direct Costs	428,818
	Indirect Costs	249,387
TOTALS	Total Direct and Indirect Costs	678,205

8231-001 (Project)	Salary, Wages and Fringe Benefits	17,560
	Equipment	0
	Travel	2,000
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	20,440
	Consortium Costs	0
	Direct Costs	40,000
	Indirect Costs	26,400
TOTALS	Total Direct and Indirect Costs	66,400
8232-002 (Project)	Salary, Wages and Fringe Benefits	11,474
	Equipment	0
	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	28,526
	Consortium Costs	0
	Direct Costs	40,000
	Indirect Costs	26,400
TOTALS	Total Direct and Indirect Costs	66,400
8228-003 (Project)	Salary, Wages and Fringe Benefits	52,551

	Equipment	0
	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	17,449
	Consortium Costs	0
	Direct Costs	70,000
	Indirect Costs	46,200
TOTALS	Total Direct and Indirect Costs	116,200
8229-004 (Project)	Salary, Wages and Fringe Benefits	24,600
	Equipment	0
	Travel	1,500
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	13,900
	Consortium Costs	0
	Direct Costs	40,000
	Indirect Costs	26,400
TOTALS	Total Direct and Indirect Costs	66,400
8230-005 (Project)	Salary, Wages and Fringe Benefits	35,000
	Equipment	0

	Travel	0
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	5,000
	Consortium Costs	0
	Direct Costs	40,000
	Indirect Costs	26,400
TOTALS	Total Direct and Indirect Costs	66,400
8212-006 (Project)	Salary, Wages and Fringe Benefits	32,429
	Equipment	0
	Travel	5,171
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	2,400
	Consortium Costs	0
	Direct Costs	40,000
	Indirect Costs	26,400
TOTALS	Total Direct and Indirect Costs	66,400
Core-001-001 (Recruitment)	Salary, Wages and Fringe Benefits	270,415
	Equipment	0
	Travel	4,020

	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	49,519
	Consortium Costs	0
	Direct Costs	323,954
	Indirect Costs	213,809
TOTALS	Total Direct and Indirect Costs	537,763
Education-001-001 (Research)	Salary, Wages and Fringe Benefits	90,967
	Equipment	0
	Travel	5,166
	Participant/Trainee Support Costs	0
	Data Management and Sharing Costs*	
	Human Fetal Tissue Costs*	
	Other Direct Costs (excluding Consortium)	0
	Consortium Costs	0
	Direct Costs	96,133
	Indirect Costs	7,690
TOTALS	Total Direct and Indirect Costs	103,823
TOTALS		**4,730,783**

Categories Budget Summary

Categories	Components	Budget Period
R&R Budget - Senior/Key Person Funds Requested	Cores-001-001 (Additional)	91,688
	001-001 (Admin Core)	74,401
	Core-001-001 (Biomarker)	42,800
	Core-001-001 (Clinical)	224,561
	Management-Core-001-001 (Data)	92,442
	Core-001-001 (Neuropathology)	97,479
	8231-001 (Project)	17,560
	8232-002 (Project)	11,474
	8228-003 (Project)	27,715
	8229-004 (Project)	24,600
	8230-005 (Project)	35,000
	8212-006 (Project)	3,387
	Core-001-001 (Recruitment)	64,379
	Education-001-001 (Research)	47,137
TOTALS		**854,623**
R&R Budget - Other Personnel Funds Requested	Cores-001-001 (Additional)	94,601
	001-001 (Admin Core)	138,064
	Core-001-001 (Biomarker)	146,754
	Core-001-001 (Clinical)	312,487
	Management-Core-001-001 (Data)	168,611

	Core-001-001 (Neuropathology)	197,687
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	24,836
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	29,042
	Core-001-001 (Recruitment)	206,036
	Education-001-001 (Research)	43,830
TOTALS		**1,361,948**
R&R Budget - Section A & B. Total Salary, Wages and Fringe Benefits (A+B)	Cores-001-001 (Additional)	186,289
	001-001 (Admin Core)	212,465
	Core-001-001 (Biomarker)	189,554
	Core-001-001 (Clinical)	537,048
	Management-Core-001-001 (Data)	261,053
	Core-001-001 (Neuropathology)	295,166
	8231-001 (Project)	17,560
	8232-002 (Project)	11,474
	8228-003 (Project)	52,551
	8229-004 (Project)	24,600
	8230-005 (Project)	35,000
	8212-006 (Project)	32,429
	Core-001-001 (Recruitment)	270,415

	Education-001-001 (Research)	90,967
TOTALS		**2,216,571**
R&R Budget - Section C. Total Equipment	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	35,000
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**35,000**
R&R Budget - Domestic Travel	Cores-001-001 (Additional)	3,000
	001-001 (Admin Core)	27,000
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	3,000
	Core-001-001 (Neuropathology)	0

	8231-001 (Project)	2,000
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	1,500
	8230-005 (Project)	0
	8212-006 (Project)	5,171
	Core-001-001 (Recruitment)	4,020
	Education-001-001 (Research)	5,166
TOTALS		**50,857**
R&R Budget - Foreign Travel	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0

TOTALS		**0**
R&R Budget - Section D. Total Travel	Cores-001-001 (Additional)	3,000
	001-001 (Admin Core)	27,000
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	3,000
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	2,000
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	1,500
	8230-005 (Project)	0
	8212-006 (Project)	5,171
	Core-001-001 (Recruitment)	4,020
	Education-001-001 (Research)	5,166
TOTALS		**50,857**
R&R Budget - Tuition/Fees/Health Insurance	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0

	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Stipends	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**

R&R Budget - Trainee Travel	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Subsistence	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0

	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Other Participants/Trainee Support Costs	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Section E. Total Participants/Trainee Support Costs	Cores-001-001 (Additional)	0

	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Materials and Supplies	Cores-001-001 (Additional)	5,988
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	80,592
	Core-001-001 (Clinical)	2,956
	Management-Core-001-001 (Data)	21,606
	Core-001-001 (Neuropathology)	8,892
	8231-001 (Project)	15,040
	8232-002 (Project)	6,060
	8228-003 (Project)	17,449

	8229-004 (Project)	8,400
	8230-005 (Project)	5,000
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	500
	Education-001-001 (Research)	0
TOTALS		**172,483**
R&R Budget - Publication Costs	Cores-001-001 (Additional)	1,600
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	1,600
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	3,000
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	1,000
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**7,200**
R&R Budget - Consultant Services	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	15,100

	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**15,100**
R&R Budget - ADP/Computer Services	Cores-001-001 (Additional)	25,000
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	14,000
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	2,400
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0

	8230-005 (Project)	0
	8212-006 (Project)	2,400
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**43,800**
R&R Budget - Subawards/Consortium/Contractual Costs	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	50,960
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**50,960**
R&R Budget - Equipment or Facility Rental User Fees	Cores-001-001 (Additional)	345
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	10,844

	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**11,189**
R&R Budget - Alterations and Renovations	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0

	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Other Direct Cost 1	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	30,000
	Core-001-001 (Biomarker)	5,000
	Core-001-001 (Clinical)	3,000
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	17,000
	8231-001 (Project)	0
	8232-002 (Project)	9,600
	8228-003 (Project)	0
	8229-004 (Project)	4,500
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	23,119
	Education-001-001 (Research)	0
TOTALS		**92,219**
R&R Budget - Other Direct Cost 2	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	102,182
	Core-001-001 (Biomarker)	3,000
	Core-001-001 (Clinical)	3,100

	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	7,000
	8231-001 (Project)	0
	8232-002 (Project)	12,866
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	300
	Education-001-001 (Research)	0
TOTALS		**128,448**
R&R Budget - Other Direct Cost 3	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	25,875
	Core-001-001 (Clinical)	4,400
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	10,000
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0

	Core-001-001 (Recruitment)	2,700
	Education-001-001 (Research)	0
TOTALS		**42,975**
R&R Budget - Other Direct Cost 4	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	10,000
	Core-001-001 (Clinical)	1,250
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	39,800
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	4,000
	Education-001-001 (Research)	0
TOTALS		**55,050**
R&R Budget - Other Direct Cost 5	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	9,000
	Management-Core-001-001 (Data)	0

	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	3,000
	Education-001-001 (Research)	0
TOTALS		**12,000**
R&R Budget - Other Direct Cost 6	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	1,500

	Education-001-001 (Research)	0
TOTALS		**1,500**
R&R Budget - Other Direct Cost 7	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	8,400
	Education-001-001 (Research)	0
TOTALS		**8,400**
R&R Budget - Other Direct Cost 8	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0

	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	3,000
	Education-001-001 (Research)	0
TOTALS		**3,000**
R&R Budget - Other Direct Cost 9	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	3,000
	Education-001-001 (Research)	0

TOTALS		**3,000**
R&R Budget - Other Direct Cost 10	Cores-001-001 (Additional)	0
	001-001 (Admin Core)	0
	Core-001-001 (Biomarker)	0
	Core-001-001 (Clinical)	0
	Management-Core-001-001 (Data)	0
	Core-001-001 (Neuropathology)	0
	8231-001 (Project)	0
	8232-002 (Project)	0
	8228-003 (Project)	0
	8229-004 (Project)	0
	8230-005 (Project)	0
	8212-006 (Project)	0
	Core-001-001 (Recruitment)	0
	Education-001-001 (Research)	0
TOTALS		**0**
R&R Budget - Section F. Total Other Direct Cost	Cores-001-001 (Additional)	32,933
	001-001 (Admin Core)	147,282
	Core-001-001 (Biomarker)	135,311
	Core-001-001 (Clinical)	23,706
	Management-Core-001-001 (Data)	37,206
	Core-001-001 (Neuropathology)	133,652
	8231-001 (Project)	20,440

	8232-002 (Project)	28,526
	8228-003 (Project)	17,449
	8229-004 (Project)	13,900
	8230-005 (Project)	5,000
	8212-006 (Project)	2,400
	Core-001-001 (Recruitment)	49,519
	Education-001-001 (Research)	0
TOTALS		**647,324**
R&R Budget - Section G. Total Direct Cost (A thru F)	Cores-001-001 (Additional)	222,222
	001-001 (Admin Core)	386,747
	Core-001-001 (Biomarker)	359,865
	Core-001-001 (Clinical)	560,754
	Management-Core-001-001 (Data)	301,259
	Core-001-001 (Neuropathology)	428,818
	8231-001 (Project)	40,000
	8232-002 (Project)	40,000
	8228-003 (Project)	70,000
	8229-004 (Project)	40,000
	8230-005 (Project)	40,000
	8212-006 (Project)	40,000
	Core-001-001 (Recruitment)	323,954
	Education-001-001 (Research)	96,133
TOTALS		**2,949,752**

R&R Budget - Section H. Indirect Costs	Cores-001-001 (Additional)	142,230
	001-001 (Admin Core)	255,253
	Core-001-001 (Biomarker)	165,535
	Core-001-001 (Clinical)	370,097
	Management-Core-001-001 (Data)	198,830
	Core-001-001 (Neuropathology)	249,387
	8231-001 (Project)	26,400
	8232-002 (Project)	26,400
	8228-003 (Project)	46,200
	8229-004 (Project)	26,400
	8230-005 (Project)	26,400
	8212-006 (Project)	26,400
	Core-001-001 (Recruitment)	213,809
	Education-001-001 (Research)	7,690
TOTALS		**1,781,031**
R&R Budget - Section I. Total Direct and Indirect Costs (G +H)	Cores-001-001 (Additional)	364,452
	001-001 (Admin Core)	642,000
	Core-001-001 (Biomarker)	525,400
	Core-001-001 (Clinical)	930,851
	Management-Core-001-001 (Data)	500,089
	Core-001-001 (Neuropathology)	678,205
	8231-001 (Project)	66,400
	8232-002 (Project)	66,400

	8228-003 (Project)	116,200
	8229-004 (Project)	66,400
	8230-005 (Project)	66,400
	8212-006 (Project)	66,400
	Core-001-001 (Recruitment)	537,763
	Education-001-001 (Research)	103,823
TOTALS		**4,730,783**

A. COMPONENT COVER PAGE

Project Title: Core A: Administrative Core
Component Project Lead Information: CHUI, HELENA Chang

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

On 7/30/2025, we received Notice of Award (NOA) from NIH to fund the competitive renewal of the USC ADRC (5-year budget period: 7/1/25 to 3/31/30). We submitted our interim RPPR for Years 1-5 of the previous funding cycle on 9/28/2025 - during Year 6. The present RPPR covers progress from 7/1/25 to 12/31/25 and our plans going forward into Year 7.
The major goals of the administrative core continue are to:
1. Guide the scientific direction of the ADRC: The Administrative Core will develop and foster three overarching scientific goals of the ADRC, working with our Internal Executive Committee, Internal Advisory Committee (IAC), External Advisory Board (EAB), and Community Advisory Board (CAB).

A. Recruit and follow a clinical cohort enriched for APOE e4, focusing on factors that modify the effect of APOE e4 on cognitive decline and AD biomarkers.
B. Promote local ADRD research at USC:
1. Integrate research programs by holding annual events, pilot fundings, and harmonization of protocols.
2. Share data, tissues, biofluids, and a participant registry through a common USC ADRC database.
3. Train the next generation of ADRD researchers and educate participants and the public.
C. Further national collaborations: Advance ADRC research through contributing USC-generated data and expertise in lipid metabolism, vascular disease, and comprehensive imaging to national initiatives.

2. Coordinate fiscal, operational, and regulatory functions across all cores to ensure appropriate, optimal, and effective utilization of center resources.

3. Promote participation across all cores in collaborative national research initiatives and resource sharing, including NACC, NCRAD, ADGC, ADNI, ACTC, DVCID, ACAD, SCAN, and CLARiTI playing a leadership role in research into APOE leveraging our metabolic, vascular disease, and neuroimaging expertise.

4. Attract early-stage and new investigators, and support training and education to ADRD research through development projects (DP), working with the REC/RCMAR and all core leaders. REC and ORE Core, respectively, sponsor training workshops, research symposia for investigators, and provide public education on the importance of understanding and addressing genetic, metabolic and vascular risk profiles for brain health.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

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B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

1. March 8, 2025: USC ADRC Latino Study Participant Appreciation Event (research participants)
-Organized by Aranda & Ringman (Delivered primarily in Spanish)
2. November 1, 2025: 9th Verdugo Hills Brain Health Forum (lay community)
-Organized by Danner and Chui

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

1. Spring 2026: Strategic planning with Cores and Internal Advisory Committee (IAC) related to vascular and metabolic contributions of apolipoprotein E4 to Alzheimer pathology (ATN (v) among diverse communities.
2. Fall 2026 External Advisory Board (EAB): meeting, including progress reports from Development Projects.
3. Enhance ADRC database with ATN(v) biomarkers.
4. Build APOE4 enriched cohorts for CLARITI (focused on Amyloid Targeting Treatment) and SCAN (focusing on communities that are representative of our catchment area.

B.2 – Administrative Core

1) Major activities

SA1 (Scientific direction). The overarching goal of the USC ADRC is to elucidate the **vascular and metabolic contributions of apolipoprotein E4 to Alzheimer pathology (ATN) among diverse communities.** Accordingly, the recruitment goals of the Clinical and ORE cores now focus on apoE genotype and biomarkers of AD pathology. At the present time, the Clinical Core has 573 participants. The inclusion criteria for new ADRC participants include: 1) signed ADRC informed consent, 2) ApoE genotype, and 3) NCRAD biomarkers.

ADRC investigators and resources support 41 NIH grants focused on ADRD (Table a1), of which 5 were newly awarded in 2024-2025 (Table a1: P30AG06530, U01AG094622, R01NS134712, R01AG092949, 1R01AG095017;1R01AG092662). The ADRC plays a complementary role to 4 California state-supported contracts/grants, including 2 California Alzheimer Disease Centers (CADC) (Table a2). In 2025, the ADRC participated in 20 industry-supported therapeutic trials (Table a3).

In lieu of an External Advisory Board (EAB) meeting, our competitive renewal was reviewed by NIH Study Section in October 2024 and underwent internal NIA review in May 2025. We are now in the process of assembling a new EAB, which will be chaired by Donna Wilcock PhD, Professor of Alzheimer's Disease Research, University of Indiana. The former chair of our Internal Advisory Committee (IAC), Caleb Finch PhD (Founding Director of the USC ADRC 1984-2003) retired in January 2025. We are in the process of assembling a new IAC, which will be chaired by Carolyn Meltzer (Dean Keck School of Medicine.

Summary of ADRC productivity 7/1/25 to 12/31/25 (Year 6 - six months)

A. **Publications:** 57 center-investigator peer-reviewed publications.
B. **Training Awards:** T32 award (Crimmins): Multidsciplinary Training in Gerontology (T32AG000037-42)
C. **6 New NIH-funded grant awards** (P30, 3R01, 2U01)

SA2 (Coordination across ADRC cores):

The Administrative Core meets in person the 4th Tuesday of each month. The executive committee comprised of Core leaders and co-leaders meets virtually the 1st Tuesday of each month. The Admin Core coordinates the internal distribution of funds, certifications, and submission of this RPPR.

As part of a 4th quarter CY2025 University-wide budget mitigation, all School of Medicine research administrators were laid off. While many were hired back into centralized shared services, we were saddened to lose Michelle Aranda, our ADRC administrator. Tiffany Kolchins has stepped into the lead administrator role and Lucy Montoya joined us as our new Program Manager.

SA3 (Collaborative national research initiatives)

We continue active participation in NACC, NCRAD, ADGC, ADNI, and ACTC. Dr. Yassine serves on the NACC Scientific Review Committee (2023-2025).

- **NACC UDS**: Total currently enrolled (n=742).
- **10 national NIA collaborations** (NACC, NCRAD, SCAN, ACTC, ADCS, ADNI, ADGC, GAP, ACAD, CLARiTI)
- **5 multi-center NIH collaborations** (VG-PPG, MarkVCID, DVCID, DVR, HABLE)
- **1 national Alzheimer Association collaboration** (ALZ-NET)

On 11/15/25, we received approval as a participating site from **CLARiTI** (USC Site PI: L Schneider) which expands our goals to 1) perform standardized amyloid-PET, tau-PET, and MRI neuroimaging studies and to upload these images to SCAN for 36 persons per year, 2) to correlate amyloid, tau and

vascular biomarkers with severity and progression of brain atrophy (neurodegeneration) and cognitive decline, and 3) to compare ATN(V) relationships stratified by APOE genotype, by pooling data from independently funded, longitudinal, affiliated studies at USC and other ADCs. We are also funded internally by the ADRC to follow a comparable protocol for an additional 24 subjects per year.

On 9/11/2025, we received IRB approval for **ALZ-NET** (USC Site PI: E Joe) to develop a database to gather regulatory grade, longitudinal clinical and imaging data from patients being evaluated for and treated with novel FDA-approved therapies for AD in real -world clinical practice. We are currently treating 155 patients with lecanemab and donanemab.

SA4 (Development Projects, training and education):

Development Projects: Each year, the ADRC seeks to fund 2 development projects at $40K - $70K/year, to support a REC scholar or faculty member. Because we did not receive notice of funding until 7/30/2025, we combined Yr 6 and 7 funding for development projects (Total = $300K) and selected 6 development projects for funding in Year 7.

The 2025–2026 Development Project cycle was highly competitive. We received 18 applications (8 faculty and 10 post-doctoral students). Applications were evaluated by the ADRC Executive Team (Directors, Assistant Director, Core Leads) and external scientific reviewers using an NIH-aligned framework emphasizing overall impact, innovation, rigor, feasibility, investigator potential, and alignment with ADRC priorities. As part of this RPPR, we submit 6 Development projects (2 faculty-led, 4 REC scholar-led) for NIA approval, representing a total commitment of $270,000.

	Investigator	**Project Title**
DP 7.1	Hosung Kim, PhD (Faculty)	Glymphatic Dysfunction as a Mechanistic Biomarker for Cardiometabolic Risk and Treatment Response in Alzheimer's Disease
DP 7.2	Kassandra Kisler, PhD (Faculty)	Synergistic Impact of Alzheimer's Disease Risk Factors PICALM and APOE on Cerebral Amyloid Angiopathy and Vascular Pathology
DP 7.3	Rijwan Ahammad PhD (REC Scholar)	APOE-Dependent Impact of α-Synuclein and Aβ Cross-Seeding on Cognitive and Molecular Outcomes
DP 7.4	Cioranna Roman-Ortiz PhD (REC Scholar)	APOE ε4–Linked Taste Pathophysiology and Therapeutic Modulation by Semaglutide
DP 7.5	Maryam Fotouhi PhD (REC Scholar)	Platelet-derived growth factor receptor-βin Plasma and Cerebrospinal Fluid as a Marker of Pericyte Injury: Association Analysis with White Matter Changes in Alzheimer's Disease
DP 7.6	Max Thorwald PhD (REC Scholar)	Iron Chelation as a Strategy to Mitigate Amyloid-Related Imaging Abnormalities During Anti-Amyloid Immunotherapy

Training and Education: The Research Education Component (REC core leaders: H Yassine, D Han) sponsors a biweekly Fall-Winter seminar series (10/25 to 3/26) for post-doctoral students and REC scholars. A panel of USC experts address major research issues in ADRD (e.g., career development, access to LONI, clinical research design, basic science, secondary public health data sources, clinical trial data, health equity, KL2 application, grant applications for observational studies. The seminars typically attract 40 attendees from a host of training grants (see Table 1b).

In addition, the USC ADRC sponsored or co-sponsored the following Educational Events:

- March 8, 2025: **USC ADRC Latino Study Participant Appreciation Event** (research participants)
    - Organized by Aranda & Ringman (Delivered primarily in Spanish)
- May 23, 2025 : **Southern California ADC Symposium** (investigators)
    - Organized by Grill, Yassine, Brewer, Tan
- November 1, 2025: 9th **Verdugo Hills Brain Health Forum** (lay community)
    - Organized by Danner and Chui
- April 14, 2026: CPBH Symposium: **Advances in Personalized Brain Health** (Target audience: scientists)
    - Organized by Yassine, Kolchins & Center for Personalized Brain Health (CPBH)

2) Specific objectives

Notable Core accomplishments from 7/1/2025 to 12/31/2025 are summarized below.

- The Admin Core received a $3 Million Gift from the Pattiz Foundation to support the ADRC. This includes $1 million to support an endowed Professorship in Neuropathology.
- The Clinical Core collected 331 PBMC samples, 348 buffy coat samples, and 126 plasma samples. We received plasma biomarker results from NCRAD for 283 Aβ40, 272 Aβ42, 250 GFAP, 250 NfL, and 270 ptau217.
- The Data Core completed the harmonization of 14 ADRC and ADRC-affiliated datasets and made 47 total clinical and image analysis datasets available. From 7/1/25 to 12/31/25, MR scans from 2,489 participants and 1,068 PET scans from 1,073 participants were uploaded into the LONI IDA repository for USC ADRC affiliated studies.
- The Neuropathology Core completed 9 autopsies, backfilled APOE genotyping on > 600 cases lacking clear genotype, implemented a new REDCap tissue request form, and sent brain tissues to 8 outside investigators (3 new).
- In the ORE Core, the online GeneScreen registered 39 new enrollees. To date, the registry has enrolled 668 members; of these, 188 (28%) are APOE ε4 carriers.
- The Biomarker Core: Processed 143 plasma and 143 WBC, 18 sera, and 3 CSF. The Core analyzed over 450 CSF sample for AD biomarkers and 500 DNA samples for APOE genotype.
- The Imaging Core completed quality control for 114 amyloid-PET scans, 75 Tau-PET scans, and 917 structural MRI, and added the computation of Centiloids for amyloid-PET and CenTauR for tau-PET.
- The REC Core organized 13 REC webinar seminars, sent 25 scholars to the May 2025 Southern California ADC symposium, and invited 2 USC ADRC speakers (L Joe and A Hiniker). In coordination with the Admin Core, the REC selected 4 post-doctoral scholars for Year 7 ADRC development projects.

3) Significant results, key outcomes, including major findings, developments, or conclusions.

In first 6 months of year 6, there were 50 peer-review publications by ADRC investigators of which 16 are highlighted here because of their significance and relevance to USC ADRC themes. See Overview for complete list and annotated summaries.

Apolipoprotein E4: Metabolism, Lipids, and Inflammation

Nedelkov D, Tsokolas ZE, Rodrigues MS, Sible I, Han SD, Kerman BE, Renteln M, Mack WJ, Pascoal TA, Yassine HN, Alzheimer's Disease Neuroimaging I. Increased cerebrospinal fluid and plasma apoE glycosylation is associated with reduced levels of Alzheimer's disease biomarkers. Alzheimers Res Ther. 2025;17(1):151. Epub 20250710. doi: 10.1186/s13195-025-01795-7. PubMed PMID: 40635030; PMCID: PMC12243426. (Cerebrospinal fluid (CSF, n = 181) and plasma (n = 178) samples from the *Alzheimer's Disease Neuroimaging Initiative (ADNI)*

Li B, Wang S, Kerman B, Hugo C, Shwab EK, Shu C, Chiba-Falek O, Arvanitakis Z, H NY. Microglial

States Are Susceptible to Senescence and Cholesterol Dysregulation in Alzheimer's Disease. Aging Cell. 2025:e70189. Epub 20250811. doi: 10.1111/acel.70189. PubMed PMID: 40790613. (*Brain tissue from Religious Order Study/Memory Aging Project (ROSMAP) and SEA-AD).*

Wang S, Li B, Li J, Cai Z, Hugo C, Sun Y, Qian L, Tcw J, Chui HC, Dikeman D, Asante I, Louie SG, Bennett DA, Arvanitakis Z, Remaley AT, Kerman BE, Yassine HN. Cellular senescence induced by cholesterol accumulation is mediated by lysosomal ABCA1 in APOE4 and AD. Mol Neurodegener. 2025;20(1):15. Epub 20250204. doi: 10.1186/s13024-025-00802-7. PubMed PMID: 39901180; PMCID: PMC11792374. (*Religious Order Study/Memory Aging Project (ROSMAP)* and *APOE4-TR mice).*

Weissberger GH, Yassine HN, Axelrod J, Fenton L, Noriega-Makarskyy DT, Molinare C, Williams JT, Oyen E, Kim P, Lai MHC, Han SD. Cognitive functioning predicts vulnerability to financial exploitation in APOE e4 carriers. Int Psychogeriatr. 2025:100162. Epub 20251107. doi: 10.1016/j.inpsyc.2025.100162. PubMed PMID: 41206326; PMCID: PMC12711538. (*Participants were 95 older adults without dementia aged 50 or older).*

Vascular and Genetic Contributions to Cognitive Impairment and Dementia

Marmarelis V, Billinger S, Joe E, Shin D, Hashem S, Rizko J, Hazen E, Cardim D, Burns J, Zhang R, Chui HC. Dysregulation of cerebral perfusion dynamics is associated with Alzheimer's disease. Alzheimers Dement (Amst). 2025;17(3):e70134. Epub 20250718. doi: 10.1002/dad2.70134. PubMed PMID: 40689334; PMCID: PMC12272300. (*90 MCI/AD patients from 77 controls* from *USC ADRC, U Kansas UDRC, and UT Southwestern)*

Lohman T, Shenasa F, Sible I, Kapoor A, Engstrom AC, Dutt S, Head E, Sordo L, Alitin JPM, Gaubert A, Nguyen A, Pahlevan N, Nation DA. Blood pressure dynamic instability and neurodegeneration in older adults. J Alzheimers Dis. 2025;108(3):1276-86. Epub 20251017. doi: 10.1177/13872877251386443. PubMed PMID: 41105529. (N*ormal older adults (N = 105)).*

Engstrom AC, Dutt S, Alitin JPM, Kapoor A, Lohman T, Sible IJ, Marshall AJ, Shenasa F, Joyce JL, Gaubert A, Ferrer F, Nguyen A, Shao X, Wang DJJ, Nation DA. Age differences in spontaneous cerebrovascular reactivity at rest. Geroscience. 2025. Epub 20251030. doi: 10.1007/s11357-025-01935-9. PubMed PMID: 41166037.
(24 healthy younger adults (ages 18-34) and 163 cognitively unimpaired older adults (ages 55-87)).

Autosomal Dominant Alzheimer Disease (ADAD)

Leone R, Kobeleva X, Rowe B, Choupan J, Ringman JM, Barisano G. Alterations in MRI-visible perivascular spaces precede dementia diagnosis by 18 years in autosomal dominant Alzheimer's disease. Alzheimers Dement. 2025;21(8):e70588. doi: 10.1002/alz.70588. PubMed PMID: 40851076; PMCID: PMC12375433. (*96 non-demented individuals (62 ADAD mutation carriers) from the USC ADRC).*

Fernandez SG, Oria CG, Petit D, Annaert W, Ringman JM, Fox NC, Ryan NS, Chavez-Gutierrez L. Spectrum of gamma-Secretase dysfunction as a unifying predictor of ADAD age at onset across PSEN1, PSEN2 and APP causal genes. Mol Neurodegener. 2025;20(1):48. Epub 20250426. doi: 10.1186/s13024-025-00832-1. PubMed PMID: 40281586; PMCID: PMC12032737. (*Biochemical assessment of gamma-secretase dysfunction on PSEN1, PSEN2 and APP mutations, including disease-associated, unclear and benign variants in mice).*

Biomarkers: Clinical, Blood, Brain & Retina

Jann K, Park G, Ringman JM, Kim H. Classifying mild cognitive impairment from normal cognition: fMRI complexity matches tau PET performance. Alzheimers Dement (Amst). 2025;17(3):e70159. Epub 20250812. doi: 10.1002/dad2.70159. PubMed PMID: 40799842; PMCID: PMC12340428. (*ADNI data and validated on USC ADRC cohort (Estudio de la Enfermedad de Alzheimer in Jalisciensas*).

Alishahi F, Beam CR, Gatz M, Schneider LS, Nation DA, Yassine HN, Kaplan H, Ganesan S, Pappas I, Davis DW, Zandi E. High precision and cost-effective multiplex quantification of amyloid-beta40, amyloid-beta42, p181Tau, p217Tau, neurofilament light chain, and glial fibrillary acidic protein from plasma and serum. J Alzheimers Dis. 2025;106(2):739-55. Epub 20250701. doi: 10.1177/13872877251340999. PubMed PMID: 40356374. (M*atched cerebrospinal fluid (CSF) and plasma or serum of a cohort of cognitively normal (CN: n = 35), with mild cognitive impairment (MCI: n = 17), and with AD (n = 11) individuals).*

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Lend Expertise to National and International Collaborations

Urich TJ, Tsiknia AA, Ali N, Park J, Mack WJ, Cortessis VK, Dinalo JE, Yassine HN. APOE epsilon4 and Dietary Patterns in Relation to Cognitive Function: An Umbrella Review of Systematic Reviews. Nutr Rev. 2025;83(7):e2113-e26. doi: 10.1093/nutrit/nuae156. PubMed PMID: 39499795; PMCID: PMC12166175. (*Umbrella review of systematic reviews).*

Yassine HN, Hugo C, O'Donovan B, Stephens IO, Johnson LA, Cole G, Tcw J, Johansson J, Kisler K, Chiba-Falek O. APOE-Targeted Therapeutics for Alzheimer's Disease. J Neurosci. 2025;45(46). Epub 20251112. doi: 10.1523/JNEUROSCI.1388-25.2025. PubMed PMID: 41224653; PMCID: PMC12614056. (*Review).*

Ferdows NB, Aranda MP. Decomposing Racial and Ethnic Disparities in Risk and Protective Factors of Dementia in the U.S. Clin Gerontol. 2025:1-14. Epub 20250717. doi: 10.1080/07317115.2025.2534651. PubMed PMID: 40673372; PMCID: PMC12349561. (*Data from the Health and Retirement Study (HRS) and the Harmonized Cognitive Assessment Protocol (HCAP; n=3,495).*

USC ADRC Year 6 Publications
Publication Dates: July 1, 2025 to December 31, 2025

1. Ferdows NB, Aranda MP. Decomposing Racial and Ethnic Disparities in Risk and Protective Factors of Dementia in the U.S. Clin Gerontol. 2025:1-14. Epub 20250717. doi: 10.1080/07317115.2025.2534651. PubMed PMID: 40673372; PMCID: PMC12349561.
2. Fernandez SG, Oria CG, Petit D, Annaert W, Ringman JM, Fox NC, Ryan NS, Chavez-Gutierrez L. Spectrum of gamma-Secretase dysfunction as a unifying predictor of ADAD age at onset across PSEN1, PSEN2 and APP causal genes. Mol Neurodegener. 2025;20(1):48. Epub 20250426. doi: 10.1186/s13024-025-00832-1. PubMed PMID: 40281586; PMCID: PMC12032737.

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Table A1: Federal Funded Grants Supported by Resources of the ADRC Year 6 (07/01/26 – 03/31/26)				
PI/PD	**Grant Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Koronyo-Hamaoui M Subcontract: Hiniker A	Alzheimer's Disease Hallmark Pathology and Associated Inflammation in the Retina **5R01AG0055865**	NIA $796,050/year	4/1/2018-6/30/2026	Pathology Core Retinal and Brain Tissue
Goldberg T Harvey P Schneider L	Novel Cognitive and Functional Measure for Alzheimer's Disease Prevention Trials **R01AG051**	NIA $2,109,311	9/1/2018-5/31/2022 6/1/1011-5/31/2026	Clinical Core
Marmarelis V Chui H Billinger S Zhang R	Model-based cerebrovascular markers extracted from hemodynamic data for diagnosing MCI or AD and predicting disease progression **R01AG058162**	NIA $2,144,503	8/1/2018-7/31/2025 (NCE)	Clinical and Imaging Core
Nation D	Cerebrovascular resistance in cognitive aging and Alzheimer disease risk **5R01AG060049**	NIA $494,996	8/1/2019-4/30/2026	Data Core
Brinton R Rodgers K Schneider L	Allopregnanolone as Regenerative Therapeutic for Alzheimer's: Phase 2 Clinical Trial **R01AG063826**	NIA $6,580,973	9/1/2018 – 5/31/2022 6/1/2022 – 5/3120/26	Clinical Core
Nation D	Endothelial progenitor cells and neurovascular injury in the aging **5R01AG0642283**	NIA $633,126	9/1/2019-5/31/2026	Data Core
Wang D	BBB Permeability Imaging in CADASIL **5R01NS114382**	NINDS $247,500–yr 2	2/1/2020-1/31/2025	Imaging Core
Jann K Wang D	Complexity of fMRI in Alzheimer disease **R01A066711-05**	NIA $653,961	4/15/2020-3/15/2026	Imaging Core
Yassine H	ApoE, ABCA1, and endosomal dysregulation in AD **5R01AG067063-05**	NIA $751,791	5/15/2020-2/28/2025	Admin core
Mack W	Activated protein C mechanisms of brain white matter protection and new therapies for brain white matter ischemic injury **1R01NS117827**	NIA $147,637 NINDS 843,669	7/15/2020-4/30/2025	Imaging Core
O'Bryant S Johnson L Rissman R Toga A	Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study **1AG058533-05**	NIA $8,926,202 / yr	8/1/2020-4/30/2025	Imaging Core
Ringman J	Phenotype and Genotype of Autosomal Dominant Alzheimer's Disease in Jalisco, Mexico **1R01AG069013**	NIA $672,193	9/15/2020-5/31/2025	Clinical Core
Thompson P Davatzikos C Huang H Saykin A	Ultrascale Machine Learning to Empower Discovery in Alzheimers Disease Biobanks **1U01 AG068057**	NIA $3,767,729	9/15/2020-8/31/2025	Data Core Imaging Core
DeCarli C Fornage M (Site PI: Chui)	Diverse Vascular Cognitive Impairment **1U19NS120384**	NINDS $3,236,413 / yr	9/30/2020-8/31/2026	ORE Core Clinical Core Imaging Core
Yan LR Chui HC	Quantitative cerebral blood vessel imaging biomarkers for AD and VCID **1RF1AG072490, 4R01AG072490**	NIA $2,408,375 / y1-3 $775,949 / y4	4/1/2021-3/31/2027	Clinical Core Imaging Core

PI/PD	Grant Number	Source & Amount	Project Period	Role of ADC
Han SD	Finance, Cognition, and Default Network in Aging (FCDNA) **5R01AG068166-03**	NIA $679,868/yr	6/1/2021- 4/30/2026	REC
Wang D	Laminar perfusion imaging **R01EB032169-04**	NIA $487,460	8/1/2021- 4/30/2026	Imaging Core
Hohman Cuccaro Toga	Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium **U24AG074855**	NIA $8,370,718	09/01/21- 08/31/26	Imaging Core
Wang D Kashani A Ringman J	Validation of Imaging and Blood-based Small Vessel VCID Biomarkers in Multiethnic Population **U01NS100614-09**	NINDS $1,282,792	9/30/2021- 7//31/2026	Clinical Core Imaging Core
Hohman T Cuccaro M Toga A	Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium **U24AG074855-04**	NIA $8,518,767 / yr	9/30/2021- 8/31/2026	Imaging Core
Agadjanyan M Schneider L Sultzer D	Safety/Tolerability/Immunogenicity of first-in-human Aβ DNA vaccine, AV-1959D Phase 1 trials in early-stage AD subjects: based on IND18953 cleared by FDA. **R01AG074983-04**	NIA $2,288,273	2/22/2022- 1/31/2027	Clinical Core
Coba M	PICALM: Role in the pathogenesis and treatment of Alzheimer vascular blood-brain barrier clearance dysfunction, neuronal dysfunction, and amyloid-beta, tau and neurodegenerative disorders **R01AG076921**	NIA $2,410,664	4/1/2022- 3/31/2025	Imaging Core
Toga A	Vascular contributions to dementia and genetic risk factors for Alzheimer's disease **P01AG052350-08**	NIA 3,135,849/ yr	1/1/2022- 12/3120/27	Clinical, Imaging, and Biomarker
Yassine H Arvanitakis Z	Brain cPLA2 as a mechanism for neuroinflammation in AD/DRD with and without APOE4 **1RF1AG076124-02**	NIA $2,417,667 / 3 yr $717/234 /yr	5/15/2022- 4/30/2025	Admin Core
Yassine H Borzage M Braskie M	Effect of diabetes and AD pathology on brain imaging and cognition in Latino adults **1R01 AG078362** **1RF1 AG078362**	NIA $3,755,733 / yr	9/30/2022- 8/31/2025	Clinical Core Imaging Core
O'Bryant S Johnson L, Rissman R, Toga A, Yaffe K	The Health & Aging Brain Study - Health Disparities (HABS-HD) **1U19AG078109**	NIA $27,685,482 / yr	9/30/2022- 8/31/2027	Imaging Core
Kaplan H Finch CE Gatz M	Testing Hypothesized Pathways Linking Infection, Physical Activity, APOE Genotype, and Biological Sex to Low Dementia Prevalence and Reduced Brain Atrophy in Two Native American Populations **R01AG054442**	NIA $2,883,093	4/15/2017- 8/31/2022 9/1/2022- 4/30/2027	Clinical Core
Yassine H	Alzheimer Abeta, Tau, Apolipoprotein and Blood brain barrier **5R01AG34467-25**	NIA $771,703	3/1/2023- 2/28/2025	Admin Core
Wang D Jiang XJ Shi YG	Imaging Small Vessels in Vascular Cognitive Impairment and Dementia **1RF1AG084072**	NINDS 2,487,668	9/1/2023- 8/31/2026	Imaging Core

PI/PD	Grant Number	Source & Amount	Project Period	Role of ADC
Nation D Bondi M Mather M	Locus ceruleus imaging markers in preclinical Alzheimer's disease, cerebrovascular disease, and cognitive decline **1R01AG082073**	NIA $1,607,325	4/1/2023- 3/31/2028	Data Core
Wang LS Chui HC Jun G, Park V	Asian Cohort for Alzheimer's Disease (ACAD) **1U19AG079774**	NIA $40,532,690 /5 yr $8,057687 /yr	7/1/23- 6/30/28	Clinical Core
Joe E	Barriers to early identification of dementia in a safety net healthcare system **1R03AG082997**	NIA $165,000	9/1/23- 5/31/25	Clinical Core ORE Core
Hiniker, A	Targeting Proteostatic Mechanisms to Inhibit LRRK2-Mediated Neurodegeneration and Neuropathology **1R01NS135607**	NIA $648,555	9/21/23- 8/31/28	Pathology Core
Aisen P Petersen R Sperling R	Alzheimer Clinical Trials Consortium **3U24AG057437**	NIA $28,125,414/ 5 yr	12/02/2017- 11/30/2028	Clinical Core Biomarker Core
Wang D	Non-invasive MRI mapping of BBB water exchange rate and permeability. **R01NS134712-02**	NINDS $689,514	8/1/2024- 5/31/2029	Imaging Core

Table A1: New Federal Grant Awards
Year 6 (07/01/26 – 03/31/26)

PI/PD	Grant Number	Source & Amount	Project Period	Role of ADC
Agadjanyan M Galvin JE Schneider LS Tosun-Turgut D	FDA approved IND 29644: Multicenter Phase 1 Trial to Evaluate Safety and Immunogenicity of Preventive Tau Vaccine, AV-1980R/A, in cognitively unimpaired preclinical AD participants. **R01AG092949-01**	NIA $4,472,279	8/1/2025- 7/31/2030	Clinical Core
Chui, H Toga, A	Alzheimer Disease Research Center **P30AG066530-06**	NIA $23,851,920/ 5 yr $4,770,384 / yr	9/1/2020- 3/31/2030	All Cores
Yassine H Chen K Katrich V Louie S	Drug Development in Alzheimer Neuroinflammation **U01AG094622-01**	NIA $1,480,265 / yr	8/1/2025- 7/31/2030	Admin Core
Sajjadi SA Ezzati A Glover C Grill J (Site PI: Ringman J)	Discovery and Trial Ready Cohort for Limbic Predominant Age Related TDP-43 Encephalopathy (TRC LATE) **1R01AG095017-01**	NIA $3,128,129/yr	9/1/2025- 8/31/2030	Clinical Core Imaging Core
Johnson S (Site PI: Schneider L – Approved 11/14/25)	ADRC Consortium for Clarity in ADRD Research Through Imaging CLARiTI) **U01AG082350**	NIA $29,556,545 (USC Site $416,576)	9/15/23- 8/31/29	Clinical Core Imaging Core Data Core
Bienkowski M	Multiscale Modeling of Hippocampus Vulnerability to Alzheimer's Disease Neurodegeneration **1R01AG092662-01**	NIA $695,789	1/1/2026- 12/31/2030	Pathology Core Imaging Core

Table A1: Federal Funded Grants Supported by Resources of the ADRC Year 6 (07/01/26 – 03/31/26)				
PI/PD	**Grant Title/Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Bienkowski, M USC – 25-116	A Translational 3D Map of Hippocampal Cell Types to Drive Investigators of Alzheimer's Disease **1K01AG066847**	NIA $628,830	2021 - 2026	Pathology Core AD low Tissue (Frozen)
Bienkowski, M USC – 25-147	A Translational 3D Map of Hippocampal Cell Types to Drive Investigators of Alzheimer's Disease **1K01AG066847**	NIA $628,830	2021 - 2026	Pathology Core Control & AD Tissue (Frozen)
Bienkowski, M USC – 25-130	A Translational 3D Map of Hippocampal Cell Types to Drive Investigators of Alzheimer's Disease **1K01AG066847**	NIA $628,830	2021 - 2026	Pathology Core Control & AD Tissue (Frozen)
Bienkowski, M USC – 25-129	A Translational 3D Map of Hippocampal Cell Types to Drive Investigators of Alzheimer's Disease **1K01AG066847**	NIA $628,830	2021 - 2026	Pathology Core PSEN1 Tissue (Frozen)
Cherqui,S UCSD – 25-104	Understanding the mechanism of rescue of Alzheimer's disease by hemopoietic stem cell **1RO1 AG086443**	NIH $1,975,000	04/2024 - 03/2029	Pathology Core Control & AD Tissue (Frozen)
Hiniker, A USC – 25-144	Brain Research Study: Alzheimer Disease Research Center (ADRC) **2P30AG066530**	NIH $1,824,226	07/01/2025 – 03/31/2026	Pathology Core AD Tissue (Frozen)

Table A2: Non-Federal Funding Supported by Resources of the USC ADC Year 6 (7/1/25-03/31/26)				
PI/PD	**Grant Title/Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Schneider LS	California Alzheimer's Disease Center **(CADC) 15-10291**	State of California $281,800 /yr	07/01/23 to 06/30/28	Clinical Services; Education and Training; Community Education
Segal-Gidan F	California Alzheimer's Disease Center **(CADC)** at Rancho Los Amigos National Rehabilitation Center **15-10295**	State of California $281,800 /yr	07/01/23 to 06/30/28	Clinical Services; Education and Training; Community Education
Rabinovici G Carillo M Rafii M (Site PI: Joe E)	Alzheimer's Network For Treatment and Diagnostics **(ALZ-NET)**	Alzheimer Association	2025-present	Clinical Core Imaging Core Data Core

Table 3 . Therapeutic Trials Supported by Resources of the USC ADRC
Year 6 (7/1/25-03/31/26)

This table only includes therapeutic trials. In the far-right columns, enter the number of participants referred and if, known, consented from your ADRC. Funding source codes are 1 = U.S. National Institutes of Health; 2 = Other U.S. Federal agencies (for example: Food and Drug Administration, Department of Defense, Centers for Disease Control and Prevention, or U.S. Department of Veterans Affairs); 3 = Industry (for example: pharmaceutical and device companies); 4 = All others (including individuals, universities, and community-based organizations).

PI Last Name, First	Funding Source of Trial	Grant Title	NCT #	Project Period 1/01/24-12/31/25	Number of participants referred	Number of participants consented
Schneider, LS	1	Alzheimer's Disease Neuroimaging Initiative 3 (ADNI 3) WEI2015-01	NCT02854033	As above		42
Schneider, LS	1	Alzheimer's Disease Neuroimaging Initiative (ADNI 4)	NCT00106899 NCT05617014	"		
Schneider, LS	1, 3	A placebo-controlled, double-blind, parallel-treatment arm, 216-week study to evaluate efficacy and safety of treatment with BAN2401 in subjects with preclinical Alzheimer disease and elevated amyloid (A45 trial) and in subjects with early preclinical Alzheimer disease and intermediate amyloid (AHEAD 3-45)	NCT04468659	"	143	22
Schneider, LS	1, 3	A placebo-controlled, double-blind, parallel-treatment arm, 216-week study to evaluate efficacy and safety of treatment with BAN2401 in subjects with preclinical Alzheimer disease and elevated amyloid (A3 trial) and in subjects with early preclinical Alzheimer disease and intermediate amyloid (AHEAD 3-45)	NCT04468659	"	143	6
Schneider, LS	1, 3, 4	Alzheimer's Plasma Extension (APEX)	NCT06416072	"	50	29
Schneider, LS	1, 3	A Randomized Double Blind, Placebo Controlled, Parallel group… START		"	18	8

PI Last Name, First	Funding Source of Trial	Grant Title	NCT #	Project Period 1/01/24-12/31/25	Number of participants referred	Number of participants consented
Schneider, LS	1, 3	Longitudinal Evaluation of Amyloid Risk and Neurodegeneration - LEARN Study. Companion Observational Study to Anti-Amyloid Treatment in Asymptomatic AD (A4) Trial	NCT02488720?	"	20	8
Schneider, LS	1, 3	Anti-Amyloid Treatment in Asymptomatic Alzheimer's Disease (A4 Study)	NCT02008357?	"	32	32
Schneider, LS	1, 3	Anti-Amyloid treatment in asymptomatic AD (A4 Tau)		"	32	32
Schneider, LS	1, 3	Anti-amyloid treatment in asymptomatic AD (A4 OLE)		"	22	22
Schneider, LS	1	TRC-PAD Program: in-clinic trial-ready cohort.	NCT04004767	"		20
Schneider, LS	3	Phase 2 randomized double-blind placebo-controlled trial to evaluate the efficacy and safety of BHV-4157 in patients with mild to moderate AD(T2)	NCT03605667?	"	12	3
Schneider, LS	1,4	Allopregnanolone as regenerative therapeutic for AD disease: phase 2 clinical trial	NCT03748303?	"	22	11
Schneider, LS	3	Phase 3b open-label, multicenter, safety study of BIIB037 (aducanumab) in subjects with AD who had previously participated in the Aducanumab studies	NCT04241068?	"	6	1
Schneider, LS	3	Safety, Tolerability, and Efficacy of Donanemab in Early Symptomatic AD	NCT04437511?	"	15	6
Schneider, LS	1	Phenserine inhibition of neuronal death in Alzheimer's disease and developing brain-labeled plasma exosomes assays as biomarkers for a phenserine phase 1b ascending dose trial (Exosome)	n/a	"	300	108

PI Last Name, First	Funding Source of Trial	Grant Title	NCT #	Project Period 1/01/24-12/31/25	Number of participants referred	Number of participants consented
Schneider, LS	1, 4	Development of novel measures of cognition and function for AD disease prevention trials (NOMAD)	NCT03900273	"	100	63
Schneider, LS	1 or 4	Escitalopram for Agitation in Alzheimer's Disease (S-CitAD)	NCT03108846	"	12	4
Pawluczyk, S	1, 3, 4	Phase II/III multicenter randomized, double-blind, placebo-controlled platform trial of potential disease modifying therapies using biomarker, cognitive, and clinical endpoints in dominantly inherited AD	NCT01760005 NCT05269394	"	10	2
Schneider, LS	1, 3 ADCS	Seamless phase 2A-phase 2B randomized double-blind placebo- controlled trial to evaluate the safety and efficacy of benfotiamine in patients with early AD (BenfoTeam)	NCT06223360	"	11	36
Schneider, LS	3	Randomized, placebo-controlled, double-blind study to evaluate safety, tolerability, pharmacokinetics and preliminary efficacy of multiple ascending intravenous doses of VY7523 in participants with early Alzheimer's disease	NCT06874621	-	18	6

Table A5: ADRC Collaborations – Multiple (>5) Institutions Year 6 (7/1/25-03/31/26)				
PI/PD	**Grant Number**	**Source & Amount**	**Project Period**	**Role of ADC**
Stephens K	National Alzheimer's Coordinating Center (NACC) **5U24AG072122-05**	NIA $9,270,394	6/15/21- 5/31/2026	All Cores
Foroud T	National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD) **3U24AG021886-24**	NIA $15,944,087	7/15/2002- 5/31/2026	Clinical Core Data Core Biomarker Core
Aisen P Petersen R Sperling R	Alzheimer Clinical Trials Consortium (ACTC) **3U24AG057437**	NIA $28,125,414/ 5 yr	12/02/2017- 11/30/2028	Clinical Core Biomarker Core
Johnson S (Site PI: Schneider L – Approved 11/14/25)	ADRC Consortium for Clarity in ADRD Research Through Imaging CLARiTI) **U01AG082350**	NIA $29,556,545 (USC Site $416,576)	9/15/23- 8/31/29	Clinical Core Imaging Core Data Core
O'Bryant S Johnson L Rissman R Toga A Yaffe K	The Health & Aging Brain Study - Health Disparities Project (HABS-HD) **5U19AG078109-04**	NIA $**31,789,574**	2022- 8/31/27	Biomarker Core Imaging Core Data Core
O'Bryant S Johnson L Rissman R Toga A	Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study **1AG058533-05**	NIA $8,926,202 / yr	8/1/2020- 8/314//2027	Biomarker Core Imaging Core Data Core
Thompson P Davatzikos C Huang H Saykin A	Ultrascale Machine Learning to Empower Discovery in Alzheimers Disease Biobanks **1U01 AG068057**	NIA $3,767,729	9/15/2020- 8/31/2025	Data Core Imaging Core
DeCarli C Fornage M (Site PI: Chui)	Diverse Vascular Cognitive Impairment **1U19NS120384**	NINDS $3,236,413 / yr	9/30/2020- 8/31/2026	ORE Core Clinical Core Imaging Core
Hohman Cuccaro Toga	Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium **U24AG074855**	NIA $8,370,718	09/01/21- 08/31/26	Imaging Core
Wang D Kashani A Ringman J	Validation of Imaging and Blood-based Small Vessel VCID Biomarkers in Multiethnic Population **U01NS100614-09**	NINDS $1,282,792	9/30/2021- 7//31/2026	Clinical Core Imaging Core
Hohman T Cuccaro M Toga A	Alzheimer's Disease Sequencing Project Phenotype Harmonization Consortium **U24AG074855-04**	NIA $8,518,767 / yr	9/30/2021- 8/31/2026	Imaging Core
Toga A	Vascular contributions to dementia and genetic risk factors for Alzheimer's disease **P01AG052350-08**	NIA 3,135,849/ yr	1/1/2022- 12/3120/27	Clinical, Imaging, and Biomarker
O'Bryant S Johnson L, Rissman R, Toga A, Yaffe K	The Health & Aging Brain Study - Health Disparities (HABS-HD) **1U19AG078109**	NIA $27,685,482 / yr	9/30/2022- 8/31/2027	Imaging Core

PI/PD	Grant Number	Source & Amount	Project Period	Role of ADC
Kaplan H Finch CE Gatz M	Testing Hypothesized Pathways Linking Infection, Physical Activity, APOE Genotype, and Biological Sex to Low Dementia Prevalence and Reduced Brain Atrophy in Two Native American Populations **R01AG054442**	NIA $2,883,093	4/15/2017-8/31/2022 9/1/2022-4/30/2027	Clinical Core
Wang LS Chui HC Jun G, Park V	Asian Cohort for Alzheimer's Disease (ACAD) **1U19AG079774**	NIA $40,532,690 /5 yr $8,057687 /yr	7/1/23-6/30/28	Clinical Core
Sajjadi SA Ezzati A Glover C Grill J (Site PI: Ringman J)	Discovery and Trial Ready Cohort for Limbic Predominant Age Related TDP-43 Encephalopathy (TRC LATE) **1R01AG095017-01**	NIA $3,128,129/yr	9/1/2025-8/31/2030	Clinical Core Imaging Core
Johnson S (Site PI: Schneider L – Approved 11/14/25)	ADRC Consortium for Clarity in ADRD Research Through Imaging CLARiTI) **U01AG082350**	NIA $29,556,545 (USC Site $416,576)	9/15/23-8/31/29	Clinical Core Imaging Core Data Core
Rabinovici G Carillo M Rafii M (Site PI: Joe E)	Alzheimer's Network For Treatment and Diagnostics (ALZ-NET)	Alzheimer Association	2025-present	Clinical Core Imaging Core Data Core

Table A6: At Risk Population Grants Year 6 (7/1/25-03/31/26)						
Grant Title and Number	**PI**	**Award Period**	**Total Award Amount**	**Health Disparities Topic**	**Funding Agency**	**Target group(s)**
The Health & Aging Brain Study - Health Disparities (HABS-HD) Project **5U19AG078109-04**	O'Bryant S Johnson L Rissman R Toga A Yaffe K	2022-8/31/27	$**31,789,574**	Mexican Americans at Risk	NIA	Mexican American
Health and Aging Brain among Latino Elders (HABLE-AT(N)) Study **5R01AG058533-05**	O'Bryant S Johnson L Rissman R Toga A	2020-8/31/27	$**31,789,574**	Mexican Americans at Risk	NIA	Mexican Americans
Pathways linking infection, physical activity and ApoE and biological sex to dementia in 2 Native American populations **R01AG054442**	Kaplan H Finch C Gatz M Gurven M Stieglitz S Thomas G Thompson R Trumble B	2022-4/30/27	$3,199,665	Tsimane Bolivia at Risk	NIA	Indigenous Bolivian population
Asian Cohort for Alzheimer's Disease (ACAD) Project **1U19AG079774-01**	Wang LS Chui HC Jun G Park V	7/1/23-6/30/28	$8,325,999	Asian GWAS	NIA	Asian Americans
Clinical significance of WMH (DVCID) **U19NS120384**	DeCarli C Chui H	9/30/20 - 8/31/26	$3,236,413	Latino MRI Biomarkers	NINDS	Latino
Imaging Cerebral Small Vessels in Vascular Cognitive Impairment and Dementia (VCID **1RF1AG084072**	Wang D Jiang XJ Shi YG	9/01/23 8/31/26	$2,487,668	Chinese Korean Clinical Core	NIA	Chinese Korean Vietnamese

Table A7: Biomakers USC ADRC P30.066530-07 Year 6 (7/1/25-03/31/26)	
Biomarker Category	**Explanation**
Neuroimaging Core	(Y) Center has a Neuroimaging Core ***in addition*** to the required Biomarker Core in new P30 application.
amyloid PET	(List ligands) Center collects *amyloid PET* from participants: *florbetaben, florbetapir*
tau PET	(List ligands) Center collects *tau PET* from participants: *flortaucipir*
other PET	(x) Center uses *other PET ligands* to examine targets other than amyloid and tau; [List other PET targets]
MRI	(Y) Center collects *MRI* from participants

Blood AB42	(Y) Center conducts tests to quantitate *AB42* from participant blood
Blood Ttau	(Y) Center conducts tests to quantitate *Ttau* from participant blood
Blood ptau	(Y) Center conducts tests to quantitate *ptau* from participant blood
APOE	(Y) Center completes *APOE genotyping* from participant blood
CSFAB40	(Y) Center conducts test to quantitate *AB40* from participant CSF
CSF AB42	(Y) Center conducts test to quantitate *AB42* from participant CSF
CSF T tau	(Y) Center conducts test to quantitate *Ttau* from participant CSF
CSF ptau	(Y) Center conducts test to quantitate *ptau* from participant CSF
CSF	(Y) To date, the Biomarkers core has received and analyzed a total of n=497 cerebrospinal fluid (CSF) samples from 2 sites (USC and HMRI), including 222 longitudinal samples.

Cell Pellet	(N) Center does not collect and store *cell pellet* from participant blood
DNA	(N) Center does not collect and store *DNA* from participant blood
RNA	(N) Center does not collect and store *RNA* from participant blood
Plasma	(Y) Center collects and stores *plasma* from the blood of 143 participants (July 2025 – December 2025). The number of aliquots available is 770. (Biomarker Core)
Serum	(Y) Center collects and stores *serum* from the blood of 18 participants (July 2025 – December 2025). The number of aliquots available is 389. (Biomarker Core)
White Blood Cells	(Y) Center collects and stores *white blood cells* from the blood of 143 participants (July 2025 – December 2025).
PBMC	(Y) Center collects and store *PBMC* from participant blood (Biomarker Core)
Whole Blood	(Y) Center collects and sends *whole blood* to NCRAD.
CSF	(Y) Center collects and stores *CSF* from 3 participants (July 2025 – December 2025). The number of aliquots available is 67. (Biomarker Core)
Lymphoblast Cell Lines	(N) Center creates and stores *lymphoblast cell lines* from participant blood
Fibroblast Cell Lines	(N) Center creates and stores *fibroblast cell lines* from participants
Stool	(N) Center collects and stores *stool* from participants
ELISA	(Y) Center uses *ELISA methods* for quantitation
Quanterix Simoa	(Y) Center uses *Quanterix Simoa methods* for quantitation
Roche Elecsys	(N) Center uses *Roche Elecsys methods* for quantitation
Mass Spec	(Y) Center uses *mass spectrometry methods* for quantitation

Y6_ADRC RPPR_5P30AG066530-07 (07012025-03312026)

Biomarker Category	Explanation
Other Biomarkers	(Y) List any other biomarkers Center is measuring in participants that is not accounted for in above categories (e.g. post mortem MRI).

	Other biomarkers in CSF and plasma: *albumin ratio, active MMP-9, sPDGFRβ, TNF-α, IFN-γ, IL-1β, IL-2, IL-6, IL-8, IL-10, IL-12p70, Neurofilament-L, GFAP, APOE* Other biomarkers in CSF: *Fibrinogen, Plasminogen, CypA, sVCAM-1, sICAM-1, PDGF-BB, S100B, MBP, MAG, NSE, VEGF-A, VEGF-C, VEGF-D, FLT-1, PlGF, bFGF, Tie-2, neurogranin*
Notes	List any special conclusions/considerations about the Center's Biomarker Core that is not accounted for in the above categories.

B.4 – Admin Core

- Core A – Administrative Core
  - May 23, 2025: Southern California ADC Symposium (USC ADRC presentations: E Joe, A Hiniker)
- Core B – Clinical Core: Presentation (E Joe) at Southern California ADC Symposium; REC Seminars
- Core C – Data Core: supported: Weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.
- Core D – Neuropathology Core: Presentation (A Hiniker) at Southern California ADC Symposium. Two neuropathology fellows were trained during the 2024-2025 academic year.
- Core E – ORE Core: provided training and professional development to 8 PhD or MD professionals.
- Core F – Biomarker Core: 1 technician trained for processing biofluids
- Core G – Imaging Core: Weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.
- Core H – Research Education Component: We held an educational, multi-disciplinary symposium with 26 speakers attended by 80+ people in person (150+ total); planning next symposium for April 2026; regular REC seminars (n=13) with attendance averaging (n=29) for the current reporting period.

Table A4: Training Awards Year 6 (7/1/25-03/31/26)			
PI	**Grant Title/Type of Award**	**Project Period**	**Funding Source/Amount**
Crimmins E Mathers, M Ailshire A	Multidisciplinary Research Training in Gerontology **T32AG000037-42**	9/1/77- 4/30/29	NIA $**691,774**
Toga A	Training in the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease (AD). **T32 AG058507-008**	5/1/18- 4/30/26	NIA $175,797
Zissimopoulos J Aguilar E Ailshire A	USC Alzheimer's Disease Resource Centers for Minority Aging Research (USC AD-RCMAR) **2P30AG043073**	9/1/18 6/30/28	NIA $**716,875**
Zissimopoulos J Battacharya J Goldman D Hayward M	Center for advancing sociodemographic & economic studies of Alzheimer and related dementias (CeASES-ADRD) **P30AG066589**	7/16/20 - 6/30/26	NIA $**872,914**
McEligot A Braskie M Toga A	Neurocognitive Aging & Analytics Research Education (NAARE)-05 **5R25AG069711**	6/1/21- 5/31/26	NIA $329,886
Bienkowski M	A translational 3D map of hippocampal cell types to drive investigations of Alzheimer's disease **K01AG066847**	8/1/21- 7/31/26	NIA $125,766
Han D	Mid-career development and mentoring on financial vulnerability and AD **K24AG081325-03**	3/21/23 1/31/26	NIA $199,219/yr

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

See the ORE and Neuropathology cores for details on clinical core resource sharing of clinical materials and research specimens. See Administrative Tables and overall component for description of resource sharing.

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Arthur		Toga		PI	225,700.00	0.8			15,122.00	4,794.00	19,916.00
2.	Helena		Chui		PI	225,700.00	1.6			30,086.00	9,537.00	39,623.00
3.	Hussein		Yassine		Co-I Associate Director	225,700.00	0.6			11,285.00	3,577.00	14,862.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **74,401.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Administrator	3.0	0.0	0.0	30,090.00	9,539.00	39,629.00
1	Program Manager	9.5	0.0	0.0	74,742.00	23,693.00	98,435.00
2	**Total Number Other Personnel**					**Total Other Personnel**	**138,064.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**212,465.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	27,000.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**27,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	0.00
2. Publication Costs	0.00
3. Consultant Services	15,100.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. Development Projects - budgets on individual awardee's budget	30,000.00
9. Imaging Costs	102,182.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**147,282.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**386,747.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	386,747.00	255,253.00
		Total Indirect Costs	**255,253.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**642,000.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	642,000.00

L. Budget Justification* File Name: Admin Core just REV.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – ADMINISTRATIVE CORE

Key Personnel:

Helena C. Chui, M.D., MPI/PD, Core Lead, 1.6 Calendar Months in Years 1 – 5. Dr. Chui is the Raymond and Betty McCarron Professor of Neurology at USC, Chairperson of the Department of Neurology at the USC Keck School of Medicine. Dr. Chui has worked in the field of dementia for over 40 years, specializing in clinical-pathological correlation of Alzheimer and cerebrovascular disease and the relationship between vascular-metabolic risk factors and cognitive impairment. She shares responsibility with Dr. Toga and Dr. Yassine for establishing the scientific aims of the Center in collaboration with all Core and Project leaders and affiliated investigators. Further, her expertise as a clinician provides oversight to assure that the interface between the research participants is developed with integrity of all procedures and data collected, ethical treatment of all participants, and assurance that all data is made available to the scientific community. In addition to guiding the overall aims, she administers the Center by leading all ADRC Executive Committee meetings as well as the annual External Advisory Board meetings for reviewing the Center's progress. As contact mPI, she is responsible for assuring that the Center's goals and objectives are met within the proposed timeframes and that additional objectives proposed by NIA and NACC are met as requested. Finally, she is responsible for the fiscal and ethical integrity of the project operations and as such, works with staff and University departments to assure that all requirements are being met in the operation of the Center. In this competitive renewal, Drs. Chui and Toga as mPIs, together with Dr. Yassine as Associate Director, have combined their expertise as a physician-scientist, neuroimaging-informatics scientist and translational physician-scientist to guide the center into the next phase of scientific growth and expansion of the local and national ADRC research community.

Arthur W. Toga Ph.D., Co-Core Lead, MPI/PD 0.8 Calendar Months in Years 1 – 5. Dr. Toga is the founder and director of the Laboratory of Neuro Imaging (LONI). He is director of the USC Stevens Neuroimaging and Informatics Institute (INI). His formal training in neuroscience, computer science and informatics makes him not only well suited to lead both the Imaging and Data Cores, but also to serve as Co-PI/PD of this Center. Dr. Toga has a career-long funding and publication record as the director of several national and international collaborative studies that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results. Currently, he is PI of several national efforts in neurodegenerative disease where data from multiple sites are aggregated and analyzed to identify usable biomarkers. He has also successfully recruited and mentored a vast number of young scientists, along with this leadership team, which is central to the aims of this Center. With this vast reach and well-earned respect as a renown scientist, Dr. Toga will serve as an essential collaborator in leading the Center into the next five-year period with Drs. Chui and Yassine.

Hussein Yassine, M.D., Associate Director/Co-Investigator, 0.6 Calendar Months in Years 1 & 2; 0.9 Calendar Months in Years 3 – 5. Dr. Yassine is Kenneth and Bette Volk Professor of Neurology, Medicine, Physiology & Neuroscience (With Tenure) and Director of the new USC Center for Personalized Brain Health (CPBH). Yassine is a NIA-funded translational scientist whose research combines imaging studies, clinical trials, observational cohorts and animal models with a focus on AD/APOE. During the past 10 years, Dr. Yassine has brought his deep knowledge related to apolipoproteins to bear on AD. Dr. Yassine is an inspirational leader and mentor, who currently leads the Research and Education (REC) core of USC ADRC. He chaired the national ADC REC steering committee. In July, 2023, Dr. Yassine was appointed Associate Director of the USC ADRC. He has immersed himself in all aspects of this competitive renewal. His contributions can be seen in our *APOE*-related scientific theme, innovative recruitment approaches (e.g., GeneScreen, incorporation of the EHR as a source of real-world data), close interactions between cores, weekly REC seminars and annual symposia. During the next funding cycle, he will be considered as a successor.

Other Personnel:

Tiffany Kolchins, MBA, ADRC Administrator, 3.0 Calendar Months in Years 1 – 5. Ms Kolchins will provide executive leadership for the ADRC, driving operational coordination to deliver against the ADRC's stated aims. They will work closely with the Center Directors (Drs. Chui, Toga, and Yassine) to plan and implement high-level strategic initiatives and drive internal and external key meetings. They will interface with University divisions and affiliated initiative leaders to advocate for the Center's research mission. A key component of this role involves spearheading the Center's institutional outreach and community engagement strategy to expand the ADRC's footprint. Additionally, they will strategize with project personnel to address overarching research operation and process improvements, ensuring that the Center's growth remains aligned with institutional compliance and the ethical standards mandated by federal, State, and University regulations.

Program Manager Lucy Montoya, 9.5 Calendar Months in Years 1 – 5. Ms. Montoya's will be responsible for the day-to-day administrative operations and reporting requirements of the Center. They will support the ADRC Directors and Administrators in tracking metrics across all Center components, which includes coordinating all communication across Cores, planning meetings, and implementing the External Advisory Meeting. They will interface with Research Administration teams on budgeting and budget reporting, preparing the annual Progress Reports (RPPR), and handling inquiries for administrative information from the National Alzheimer Coordinating Center (NACC). Furthermore, they will work directly with Dr. Chui to interface with University compliance offices to ensure that all research operations are conducted in an ethical fashion and meet all mandatory federal and State guidelines.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years. Fringe benefits do not apply to student employees.

Travel:

Travel costs for ADRC Directors, Dr. Helena Chui and Dr. Arthur Toga and Associate Director, Dr. Hussein Yassine, to attend the semi-annual ADC Directors' meetings ($1,500 x 2 meetings x 2 Core Leads & 1 Associate Director = $9,000/yr).

Travel costs for one Core Lead from each Core to attend one annual ADRC meeting: Drs. Schneider and Joe (Clinical Core); Drs. Hiniker and Hawes (Neuropathology Core); Dr. Rissman (Biomarker Core – will attend 2 meetings per year); Drs. Aranda and Ringman (ORE Core); (8 meetings x $1,500 = $12,000/yr). We request $12,000/yr.

Travel costs for Administrators to attend the semi-annual ADC meetings: Tiffany Kolchins and Lucy Montoya ($1,500 x 2 meetings/year = $3,000/yr x 2 personnel = $6,000/yr). We request $6,000/yr to cover these costs.

Other Costs: Development Projects – In order to attract and engage early career and new investigators, a budget of $300,000 is requested to fund 6 development projects in year 7. Awards are announced after initial review by the ADRC Executive Committee and secondary review by external consultants. Detailed budgets are located in each DP.

Annual External Advisory Meeting – A budget of $4,500 is requested for the three external advisors. Each advisor will receive $500 for an honorarium. We request $1,000 for travel expenses.

Imaging Costs – We request $102,182 for SCAN-Compliant imaging expenses, $10,357/participant (Clinical and Imaging Cores) Adjustments to this amount on budget to account for separate development budget.

Imaging	Cost
3T MRI Scan	$ 720
MRI Reading	$ 125
Amyloid PET Scan Reading	$ 125
Tau PET Scan Reading	$ 125
Amyloid PET Scan - Acquisition	$ 1,800
Tau Pet Scan - Acquisition	$ 1,800
PET Radioligand - Amyloid (florbetaben)	$ 1,200
PET Radioligand - Tau (flortaucipir)	$ 4,162
Transportation cost for florbetaben (Sophie)	$ 678
TOTAL cost per participant	**$ 10,735**

Facilities and Administrative Costs:

Facilities and Administration Costs are based on USC's federally negotiated rate (dated May 29, 2024) of 66% as of July 1, 2024. The F&A rate is 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

A. COMPONENT COVER PAGE

Project Title: Core B: Clinical Core
Component Project Lead Information: Schneider, Lon

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

In year 6, we oriented the clinical core toward apolipoprotein E4 and revised scientific aims for the new funding cycle. The overall aims of the ADRC are (1) recruit and follow a clinical cohort that reflects the population of Los Angeles and is enriched for APOE ε4, focusing on factors that modify the effect of APOE ε4 on cognitive decline and AD biomarkers; (2) promote local ADRD research at USC; and (3) further national collaborations.

The clinical core interacts with the other ADRC Cores to achieve its aims which are:
1) Participant recruitment and retention: The Clinical Core will recruit and maintain a clinical cohort of approximately 500 community-dwelling older adults, enriched for APOE ε4 carriers, for biomarker characterization, and post-mortem pathological study. The core works with the ORE core and GeneScreen to enrich the proportion of APOE ε4 carriers; and with USC's California Alzheimer Disease Centers at the Health Science Campus (CADC-USC) and Rancho Los Amigos National Rehabilitation Center (CADC-RLANRC) to ensure a cohort representative of the population of our catchment area.
2) Clinical assessment and phenotyping: The core medical teams will evaluate participants' cognitive status and diagnoses using NACC UDSv4 clinical evaluation and neuropsychological assessment; assess social determinants of health and vascular risk; characterize participants phenotypically and with ATN(v) biomarkers; and share research data with NIA-supported programs (e.g., NACC, NCRAD, ADGC, CLARiTI).
3) Tissue and biomarker collection and biobanking: The core will collect blood and CSF (optional) biospecimens, amyloid PET, tau PET, and MRIs; consent and follow participants to autopsy
4) Collaborate with the neuropathology, biomarker, imaging, and data cores to integrate clinical and biological data; facilitate the development of canonical and emerging molecular, genetic, and imaging biomarkers for prediction, diagnosis, and surrogate clinical progression, and share data with NIA-supported programs (as above).
5) Develop/maintain the research participant registry: Per this current RFA-AG-24-001, collaborating with the ORE and data cores, the Clinical Core will maintain a composite cohort of 'study-ready' who are managed to review their clinical, biomarker, and social determinants of health status for potential recruitment and participation in the ADRC, affiliated studies, and clinical trials.
6) Training and education: The core will collaborate with the administrative core, ORE core, REC and RCMAR to: 1) train next-generation researchers, 2) provide ongoing education and training for our current ADRC researchers and teams, and 3) educate the community by teaching study design, methods, GCP, clinical research quality, regulation, ethics, cultural sensitivity, and implicit bias in clinical research.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : B2 RPPR ADRC Yr 6_Clinical Core_ July-Dec 2025 .1.28.26.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ADRC Y6_Clincal Core_1.23.26.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

The clinical core has led and participated in the outreach activities of the ADRC including the Annual Brain Health Forum, USC Verdugo Hills MC, November 1, 2025 and the Weekly Research Education Component (REC). See outreach, recruitment, and education core for additional events and REC details.

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

We plan to prosecute the ADRC's new specific aims which are: Our recruitment goals are in accordance with the new theme of APOE4. We will enroll and follow participants in NACC rely on NCRAD biomarkers for AD pathology (β-amyloid, p-tau), and neurodegeneration (MRI, NfL). We will review core participants based on their activity in the core and affiliated studies. Continue to implement UDSv4, scan 24 clinical core participants as part of the new cycle; implement CLARiTi imaging for 36 participants in the first two years. Continue regular training in the UDSv4.

B.2 – Clinical Core

- Major activities
- Specific objectives
- Significant results, key outcomes, including major findings, developments, or conclusions:
- Key outcomes or other achievements

The clinical core contributed to the ADRC's focus on factors that modify the effect of APOE ε4 on cognitive decline and AD biomarkers. The core addresses important goals of the continuing 2011 National Alzheimer Project Act: developing translational tools, infrastructure and capabilities (4.G), trial design and conduct (10), targets (6.A and 5.B), care and caregiver support research (13.G), recruitment and citizen engagement (12.H), biomarkers, health disparities, data sharing, enabling technologies, and disease monitoring.

Aim 1: Participant Recruitment and Retention

The clinical core cohort is enriched for APOE ε4 carriers**.** Clinical core participants undergo annual UDSv4 assessment and NCRAD-assayed Aβ, tau, neurodegeneration, and glial biomarkers. Clinical core participants are approached annually to consider autopsy participation (Brain Research Study – Autopsy Program) and are recruited to participate in affiliated imaging studies (SCAN, CLARiTi) as well as several ADRC-affiliated, independently funded studies and pilot studies including research studies, investigator-initiated clinical, biomarker and phase 1 -3 studies, and ACTC and ADCS multicenter clinical studies.

Table 1: Demographic and clinical characteristics of the clinical core

	Overall	Hispanic	White*	Black*	Asian*
N	573	192 (33.5%)	242 (42.2%)	36 (6.3%)	65 (11.3%)
Age, y m (SD)	72.7 (11.3)	68.7 (13.2)	76.1 (10.4)	73.2 (8.7)	70.9 (6.8)
Educ, y med [Q1, Q3]	16 [14, 18]	14 [11, 16]	17 [16, 18]	16 [14, 18]	16 [16, 18]
Male (%)	207 (36.1%)	65 (33.9%)	99 (40.9%)	10 (27.8%)	24 (36.9%)
MoCA (n=527)	25.4 (4.5)	24.9 (4.5)	25.1 (4.7)	26.6 (3.8)	26.5 (4.1)
CDR = 0	317 (55.3%)	95 (49.5%)	142 (58.7%)	22 (61.1%)	38 (58.5%)
CDR = 0.5	217 (37.9%)	83 (43.2%)	79 (32.6%)	13 (36.1%)	24 (36.9%)
CDR ≥ 1	39 (6.8%)	14 (7.3%)	21 (8.7%)	1 (2.8%)	3 (4.6%)
APOE ε4 ((n=442)	36.7%	20.0%	42.8%	36.7%	36.0%
Follow up, y m (SD)	5 (4.5)	4.1 (3.6)	6.7 (4.9)	4.2 (4.5)	3.1 (3.8)
*Non-Hispanic; 38 (6.6%) not classified who identified as others or unknown					

During the shortened Year 6, the core maintained its recruitment and follow-up activities, characterized our participants by APOE genotype and core plasma biomarkers, clinical and neuropsychological assessments.

Aim 2: Clinical assessment and phenotyping.

We completed the transition to UDSv4 for English speaking participants in late June/July 2025. We conducted a series of training sessions on UDSv4 implementation for clinicians, with emphasis on major changes from UDSv3. We transitioned to a locally hosted REDCap instance for UDSv4, including direct data entry. From July 2025 to December 2025, we conducted 128 in-person, yearly follow-up visits and 7 remote follow-ups; and 6 initial evaluations. (UDS packets pending are 30, 2, and 1, respectively). We initiated use of the Spanish UDSv4 in January 2026 and made preparations for the Simplified Chinese UDSv4.

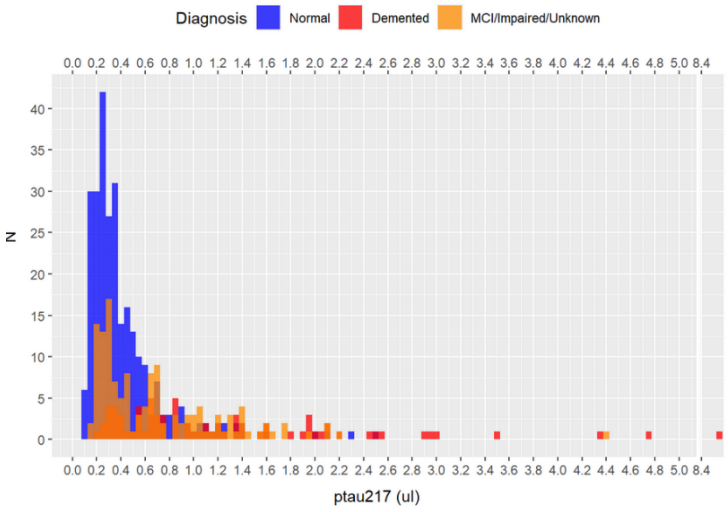
We maintained our operational meeting schedules: Monday (ADRC administrative, R01s, affiliated studies), Tuesday (CADC, education, recruitment BRS administration), weekly VCS administrative meetings; Wednesday (UDS case reviews, administration), Thursday (ADNI, ACTC, trials), Thursday (administration, Data Core database development), Friday (clinical core research conference, REC teaching).

We completed the startup process to become an active site for CLARiTi and are consenting and scheduling existing clinical core participants for imaging. We have completed submissions for all required institutional approvals for the ADRC-funded imaging studies (MRI, amyloid and tau PET) and expect to have final approval shortly.

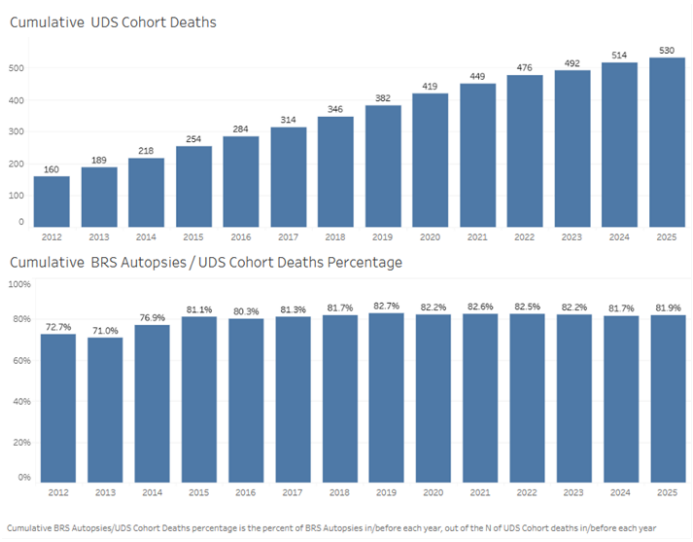
We were activated as a site for ALZ-NET and have started to consent and submit data and imaging for clinical core participants and clinical patients on amyloid targeting therapies. In addition to SCAN and CLARiTi as above, we continue to participate in other multi-site studies and collaborations including DiverseVCID.

Aim 3: Tissue and biomarker collection and biobanking

From July 2025 to December 2025, we collected 331 PBMC samples, 348 buffy coat samples, and 126 plasma samples. We received plasma biomarker results from NCRAD for 283 Aβ40, 272 Aβ42, 250 GFAP, 250 NfL, and 270 ptau217. Figure 1shows the returned ptau217 distribution across our at-risk cognitively unimpaired, MCI, and dementia clinical core participants. In general there was threshold separation at approximately 0.4 µL and 0.6 pg/ µL, consistent with external and other ADRC NCRAD samples, defining high-likelihood of absent plaques and high plaques, with a significant uncertainty interval.



Between July 2025 and December 2025, 10 people were newly consented and enrolled in the autopsy program. We had 2 withdrawals, 9 cohort deaths, and obtained 9 autopsies for an autopsy rate of 100%. Our cumulative autopsy rate as of December 31, 2025, was 434 autopsies in 530 total ADRC/UDS participant deaths, 81.9%.



The Neuropathology and Clinical Cores share specimens from our deceased ADNI participants with Dr. Richard Perrins at Washington University, St. Louis. This will be a long-term commitment as we currently follow approximately 40 ADNI participants, most for over 15 years. We also plan to share tissue from deceased participants in the AHEAD 3-45 trial with MGH/BWH.

Aim 4. Develop/maintain the research participant registry.

Collaborating with the ORE and DMSC, the Clinical Core maintains a composite 'study-ready' ready cohort that are managed to review their clinical, biomarker, and social determinants of health status for potential recruitment and participation in the ADRC, affiliated studies, and clinical trials.

To develop participation in the core and the registry, our community outreach during the July to December reporting period, the clinical core delivered a coordinated set of community outreach activities across Los Angeles County, providing brain-health education at senior centers and community fairs in Pico Rivera, Irwindale, Lincoln Heights, Montebello, Huntington Park, Santa Monica, and Verdugo Hills. Programs addressed evidence-based brain-health practices, cardiovascular health, and culturally tailored cognitive-health education for Spanish-speaking older adults, strengthening engagement with diverse communities.

Dr. Ringman contributed to both public education and workforce development. He delivered a Spanish-language presentation for Alzheimer's Los Angeles on the benefits and limitations of current Alzheimer's treatments, presented on the genetics of Alzheimer's disease at the Verdugo Hills Brain Health Forum, and led a grant-writing session on observational research at the USC ADRC weekly

research conference. These activities advanced community awareness, supported underserved populations, and enhanced research training consistent with ADRC and NIH priorities.

Aim 5. Training and education. The core will collaborate with the administrative core, ORE core, REC and RCMAR to: 1) train next-generation researchers, 2) provide ongoing education and training for our current ADRC researchers and teams, and 3) educate the community by teaching study design, methods, GCP, clinical research quality, regulation, ethics, cultural sensitivity, and implicit bias in clinical research. Dr. Ringman also gave a talk at the Spanish-language symposium targeting the Latino community sponsored by Alzheimer's Los Angeles. Titled, "Beneficios, Riesgos y Limitations de los Tratamientos actualmente disposable para la Enfermedad de Alzheimer's (Benefits, Risks, and Limitations of currently available treatment for Alzheimer's Disease). He also spoke on the Genetics of Alzheimer's disease to a lay audience at the Verdugo Hills Brain Health Forum on November 1, 2025.

On October 25, 2025, in association with the ORE Core, Dr. Ringman spoke at the USC ADRC's weekly research conference on "How to Write Grants for Observational Research," attended by students, fellows, and junior faculty members as well as more senior researchers.

B.4 What opportunities for training and professional development has the project provided?

The clinical core hosted a comprehensive series of trainings during the transition to UDSv4, on topics such as the neurologic exam, the UPDRS, and distinguishing clinical stage from etiological diagnosis. Clinical core members also widely attend the weekly REC seminar series and support the series as speakers and hosts.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022
FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.		Elizabeth		Joe		Clinical Co-Core Lead	199,500.00	2.4			39,900.00	12,648.00	52,548.00
2.		Freddi		Segal-Gidan		Geriatrician	200,000.00	0.48			8,000.00	2,536.00	10,536.00
3.		John		Ringman		Co-I	225,700.00	1.2			22,570.00	7,155.00	29,725.00
4.		Lina		D'Orazio		Neuropsychologist	127,050.00	2.8			29,935.00	9,489.00	39,424.00
5.		Lon		Schneider		Clinical Core Lead	225,700.00	1.2			22,570.00	7,155.00	29,725.00
6.		Michael		Harrington		Co-I	142,000.00	1.2			14,200.00	4,501.00	18,701.00
7.		Nasim		Sheikh-Bahaei		Co-I	225,700.00	0.6			11,285.00	3,577.00	14,862.00
8.		Sonia		Pawlucyzk		Research Associate	183,751.00	1.44			22,050.00	6,990.00	29,040.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **224,561.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Research Nurse	2.4	0.0	0.0	25,773.00	8,170.00	33,943.00
5	Study/Research Coordinators	10.8	0.0	0.0	104,290.00	33,060.00	137,350.00
1	Research Associate	3.6	0.0	0.0	32,121.00	10,182.00	42,303.00
3	Data Manager/Project Assist	9.0	0.0	0.0	50,087.00	15,878.00	65,965.00
1	Genetics Counselor	3.0	0.0	0.0	25,000.00	7,926.00	32,926.00
11	**Total Number Other Personnel**					**Total Other Personnel**	**312,487.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**537,048.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	2,956.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. Lab Costs	3,000.00
9. Participant Incentives	3,100.00
10. Transportation for participants	4,400.00
11. CTSI Data Extraction Costs	1,250.00
12. A1C and Lipid lab costs	9,000.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**23,706.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**560,754.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	560,754.00	370,097.00
		Total Indirect Costs	**370,097.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**930,851.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	930,851.00

L. Budget Justification* File Name: Clinical Core Just.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – CLINICAL CORE

The clinical core operates at the USC Health Science Campus (HSC), Keck Medical Center and Keck School of Medicine. An affiliated site is Rancho Los Amigos National Rehabilitation Center (RLANRC) in Downey (14 miles south of HSC). USC has 2 California Alzheimer Disease Centers (CADC), one located at the Health Science Campus (CADC-HSC: Director: Lon Schneider, MD, MS) and the other at RLNRC (CADC-Rancho; Director: Freddi-Segal-Gidan PA, PhD)

Key Personnel:

Lon S. Schneider M.D., Clinical Core Lead, Co-Investigator, 1.2 Calendar Months in Years 1 – 5. - Dr. Schneider is the Della Martin Endowed Professor of Psychiatry, Neurology and Gerontology. He has directed the USC ADRC Clinical Core and the CADC-HSC (CA Dept. of Health Services contract 21-10291) for many years Dr. Schneider is responsible for accomplishing the specific aims of the core in collaboration with clinical core co-lead Elizabeth Joe, MD. This includes overseeing the professional medical staff that engage and enroll participants, clinicians and psychometricians that complete the UDS. provide neuropsychological testing, and the implementation and efficiency of the enrollment, retention, and diagnostic accuracy of all procedures. He works with the neuropsychologists, Drs. Lina D'Orazio and McCleary to oversee the clinical psychology research assistants and clinical technicians. In this role, he reviews ADRC operations and new proposals (research projects and pilots) regarding the use of core resources including staffing availability and use of the data. Dr. Schneider works closes with the leaders of all the other Cores: starting with ADMIN Core and DMSC overall, the ORE Core on subject recruitment, Imaging Core on standardized MR and PET acquisitions, the Biomarker Core on plasma and CSF biomarkers, the NP core on autopsy, and the REC on training. and this includes holding regular core meetings in collaboration with the DMSC to assess data collection progress and quality across sites. Together with Drs. Chui, Toga, and Yassine, he is responsible for responding to NACC, NCRAD, and other NIA initiatives (ACTC, ADCS, ADNI, CLARiTI).

Elizabeth Joe, M.D., Clinical Core Co-Lead, Co-Investigator, 2.4 Calendar Months in Years 1 – 5. – Dr. Joe is an Assistant Professor of Clinical Neurology and Co-Lead of the Clinical Core. She completed a CTSI K2-Award and was recently awarded a R03AG082997. Her research focuses on diagnosis and treatment of dementia in the DHS safety net healthcare system. Dr. Joe has supervised recruitment, study coordinators and clinical databases for several longitudinal observational studies (e.g., PPG [Toga], DVR [Marmarelis],). During the next funding cycle, she will work closely with Dr. Schneider on the selection of participants for SCAN and CLARitI (enriched for APOE ε4) working closely with both GeneScreen and clinics in the DHS safety net. She is site PI for ALZ-NET and neurology liaison with primary-specialty care initiative using the Linus Digital Cognitive Assessment tool. Dr. Joe oversees several clinical study coordinators (including Diana Lopez, and Margarita Jasso), completes UDSv4, and study procedures (e.g., LP). Dr. Joe will also review data collection operations, the resultant integrity of the data, and participate in the interpretation of data and in the drafting of manuscripts.

John Ringman, M.D., M.S., Co-Investigator, 1.2 Calendar Months in Years 1 – 5. – Dr. Ringman is the Helene and Lou Galen Endowed Professor of Neurology. Dr. Ringman is the Chief of the Division of Memory Disorders in the Department of Neurology. He has been funded for many years by NIA for research studies of Autosomal Dominant Alzheimer Disease (ADAD). In the Clinical Core, Dr. Ringman will continue to conduct UDSv4 assessments and perform LPs. He also writes final letters interpreting autopsy reports to families. As the Co-Lead of the ORE Core, he will also serve as a bridge between the Clinical and ORE cores. , particularly in the outreach to the Latinx community Dr. Ringman will continue with collecting and interpreting data and writing manuscripts.

Nasim Sheikh-Bahaei, MD, PhD, Co-Investigator, 0.60 Calendar Months in Years 1-5. Dr. Sheikh-Bahaei is Assistant Professor of Radiology and Neurology, Keck School of Medicine. She is the Chief of Neuroradiology in the Department of Radiology. Her PhD was earned at Cambridge University in Alzheimer disease advanced imaging using MRI and PET. Dr. Sheikh-Bahaei will read research MRI and PET scans for participants enrolled in the ADRC (e.g., SCAN, CLARitI, ALZ-NET). Our research participants greatly y appreciate receiving results of their imaging studies, which contributes importantly to participant satisfaction and retention.

Freddi Segal-Gidan, P.A., Ph.D., Co-Investigator, 0.48 Calendar Months in Years 1 – 5. – Dr. Segal-Gidan is Associate Professor of Clinical Neurology and Director of the California Alzheimer Disease Center (CADC) located at Rancho Los Amigos National Rehabilitation Center (Rancho) which is operated by the

Los Angeles County Department of Health Services (CADC-Rancho). She is a licensed Physician Associate, PA, holds a doctoral degree in Health Policy, and was a former Leader of the ADRC ORE Core. The CADC-Rancho is a referral source to the ADRC for persons with memory concerns, who receive their healthcare from the DHS. , Dr. Segal-Gidan will work closely with colleagues at other DHS facilities (e.g Heather Schickedanz MD, Chair Family Medicine, Harbor UCLA and with Drs Joe and Towfighit at Los Angeles General Hospital). [Drs. Joe and Towfighi] .

Michael Harrington, M.D., Co-Investigator, 1.2 Calendar Months in Years 1 – 5. – Dr**.** Harrington is a Professor of Research in Neurology at the University of Southern California (USC). He is a senior NIH-funded neuroscientist with expertise in disease pathophysiology and biomarker studies. He assists with the standardized completion of the UDSv4.

Lina D'Orazio, Ph.D., Co-Investigator, Neuropsychologist, 2.8 Calendar Months in Years 1 – 5. – Dr. D'Orazio is an Assistant Professor of Clinical Neurology who is a bilingual (English and Spanish) and bicultural licensed. clinical neuropsychologist who currently performs, oversees, and guides Spanish- and English-language cognitive assessments at the USC ADRC. Dr. D'Orazio ensures that the Spanish-language cognitive testing is performed as per protocol and is responsible for ongoing scoring, quality control, and interpretation of these tests. Dr. D'Orazio is also involved with the selection and training of all graduate student testers as well as psychometricians hired as staff. She is particularly qualified to recruit, select and train Spanish speaking testers.

Sonia Pawluczyk, M.D., Co-Investigator, 1.44 Calendar Months in Years 1 – 5. – Dr. Pawluczyk is a geriatric psychiatrist, assistant professor of psychiatry, bilingual Spanish, who has worked in the field of dementia for the over 30 years. She has worked with Dr. Schneider in the pharmacology program during this time and has a major role in the operation of the clinical core and in affiliated studies supported by the ADRC. As a senior clinician, she is responsible for assessing participant eligibility for the clinical core, general research pool, clinical trials, and external requests for participant resources. As part of the evaluation she completes all research documentation, including the UDSv4 in collaboration with the Senior Research Nurse and the UDS Coordinator. She is responsible for ensuring quality patient care.

Other Personnel:

Diana Lopez, Research Coordinator, 6.0 Calendar Months in Years 1 – 5. The lead Research Coordinator for SCAN and CLARiTI isfully bilingual (Spanish-English speaking). Her responsibilities include recruitment and tracking for research participants in several longitudinal observational protocols. In addition, she is the primary liaison between recruiters and clinical coordinators for other affiliated studies, as well as between the ADRC Clinical Core and the potential participants seen at Rancho and Los Angeles General Hospital. She schedules initial evaluations and follow-ups, maintains communications between visits, fields participant's inquiries and guides the participants through the facilities on the day of their appointment. In addition, she maintains all research documents for participants and facilitates data queries using this information. The Research Coordinator assists the Investigators with IRB documentation.

Genetics Counselor TBD, 3.0 Calendar Months in Years 1 – 5. Given our goal of recruiting a sample enriched for APOE ε4, we anticipate that many research participants will have questions about their or their family member's risk of AD. For clinical, ethical, and educational reasons we propose to hire a part-time genetics counselor. The counselor will meet with patients who have been informed they have *APOE* Ɛ4.

TBD, USC GeneScreen Liaison and Linus Digital Cognitive Assessment, 4.20 Calendar Months in Years 1 – 5: Ms. Jasso is an experienced recruiter, having worked with the ADRC for 5 years. She is familiar with the design and procedures associated with longitudinal observational studies. She will serve as the liaison between GeneScreen and clinical coordinators by referring participants from GeneScreen for study participation in the Clinical Core and ADRC affiliated studies. In addition, she will administer the Linus Digital Clock and Recall (DCR) to potential student participants.

Diana Ferman, P.A.C. , Research Associate, 3.6 Calendar Months in Years 1 – 5. – Ms. Ferman is a physician assistant who participates on the clinical core team. As a team member she assesses participant eligibility for the clinical core, does initial and follow-up NACC-UDS history and physical evaluations, liaises with participants' primary care physicians/staff, help to refer patients to ADRC-affiliated studies and external requests for research participant resources. She completes research documentation, including the UDSv4 in collaboration with the Senior Research Nurse, the UDS Coordinator, and attends team meetings.

Liberty Teodoro, R.N., Research nurse, 2.4 Calendar Months in Years 1 – 5. – Ms. Teodoro is an experienced senior research nurse (trilingual with Tagalog and Spanish) and has been with the ADRC for over 15 years. Her responsibilities are to assist the physicians with clinical histories and examinations for normal and cognitively impaired study participants. She provides clinical support to the participants in her capacity as RN, including pre- and post-visit interviews, and interacts with the multidisciplinary team members. She maintains medical chart documentation and UDSv4 research documentation for study participants, and for the research pool as well as additional Center supported studies. She serves as the coordinator for participants who are interested in or have consented to brain autopsy. In this role, she is responsible for patient enrollment with the clinician staff and follow-up, tracking enrollment and clinical data, and maintaining current and accurate study documentation (i.e., consent forms, conservatorship documentation, and instructions at time of death).

Mauricio Becerra, M.D., Study Coordinator, 1.2 Calendar Months in Years 1 – 5. – Mr. Becerra, M.D. is bilingual in Spanish and English, has many years of experience coordinating and directing clinical studies and research programs. He works with the clinical team to assure recruitment targets are met, manages the review for potential clinical trial participants as well as some aspects of the overall visits for ADRC Clinical core participants in pilots and affiliated projects such as ADNI, ATRI, ADCS, CLARiTI, and observational studies. He interfaces with the IRB and the funding agencies to assure study compliance.

Maya Slowinska, Clinic and UDS Coordinator, 3.0 Calendar Months in Years 1 – 5. – Ms. Slowinska is the clinic coordinator and, in this role, coordinates all aspects of the patient visits, including general UDS enrollment and follow-up visits, clinical trial enrollment and follow-ups; is responsible for coordinating the entire clinic schedule, and all aspects of patient visits. She also manages clinical core clinical records with Julia Boudreau. She also coordinates the California Alzheimer's Disease Center. Ms. Slowinska is the point person for referral of current ADRC clinical core research participants to additional studies. She provides patient contact and, scheduling. She organizes patient documents including consent forms, charts and cumulative patient data for tracking. She is also responsible for managing the clinic staffing including hourly student workers and neuropsychology graduate students and technicians, scheduling to accommodate the daily clinic needs.

Julia Boudreau, Data manager, 2.4 Calendar Months in Years 1 – 5. – Ms. Boudreau is involved in data entry, checking, and retrieval of clinical UDS data before and after initial and annual visits. Her primary responsibility is to coordinate the flow of the UDS data collection forms and implement related quality control. This includes being the point person for receipt of all UDS documents on all study participants, final review for correct form completion, range checks and data consistency across forms. In this capacity, she works as a liaison between the Clinical and Data Cores; and is knowledgeable of the ADRC and related databases.

Emily Fernandez, Neuropsychology (NP) tester/project assistant, 2.4 Calendar Months in Years 1 – 5. – Ms. Fernandez is fully bilingual Spanish and is responsible for completing the UDS neuropsychology testing per protocol for both cognitively impaired and normal participants. He has been trained by senior psychologists to assure standardization in the administration of tests and in both languages.

Cynthia Ayala, Psychometrician, Neuropsychology (NP) tester/research coordinator, 2.4 Calendar Months in Years 1 – 5. – Ms. Ayala is fully bilingual Spanish and is responsible for completing the UDS neuropsychology testing per protocol for both cognitively impaired and normal participants and has been trained by senior psychologists to assure standardization in the administration of tests and in both languages.

Adriana Pedrosa, Neuropsychology (NP) tester, 4.2 Calendar Months in Years 1 – 5. – Ms. Pedrosa is fully bilingual Spanish and is responsible for completing the UDS neuropsychology testing per protocol for both cognitively impaired and normal participants. He/she will be fully trained by senior psychologists to assure standardization in the administration of tests. Ms. Pedrosa will also provide UDSv4 neuropsychological testing to clinical core patients at Rancho Los Amigos, under the direction of Freddi Segal-Gidan, P.A., Ph.D., two half days per month.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years. Fringe benefits do not apply to student employees.

Other Costs

1. **Materials and Supplies**: Funds are requested for supplies needed for laboratory studies.
2. **Lab Costs**: $3,000/year is requested per year to cover lab costs for SCAN participants ($125 x 24 participants per year). Total requested: $15,000
3. **Project related expenses**: **Funds are requested** for project related supplies to cover services on equipment, computer programs and software updates, tablets, web maintenance, and patient telephones.
4. **Study participant incentives**: $3,100 is requested per year to cover incentives for study participants:$50 per study visit x 3 visits x 24 SCAN participants. Total requested: $18,000
5. **Transportation costs**: $4,000 is requested per year to cover transportation costs (Uber/Lyft) for study participants: $75 (RT) x 3 trips x 24 SCAN participants. Total requested: $27,000
6. **A1C and Lipid Costs**: $9,000 per year is requested to cover costs for non-SCAN Clinical Core participants; hemoglobin A1C and lipids panel.
7. **CTSI Data Extraction Costs**: USC Clinical and Translational Science Institute (CTSI)'s Clinical Research Informatics Service – To support linkage of medical records with clinical data for consented Keck Medicine and DHS patients enrolled in the clinical core, we request funds for annual data extraction costs from the USC CTSI. A total of $6,250 (10 hours per year (5 hours per year at each of 2 health systems) at $125 per hour, or $1,250 total per year) is requested.

Facilities and Administration Costs:

Facilities and Administration Costs are based on USC' federally negotiated rate (dated May 29, 2024) of 65.5% as of July 1, 2023. The rate increases to 66% as of July 1, 2024. The F&A rate is 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

A. COMPONENT COVER PAGE

Project Title: Core C: Data Management and Statistical Core
Component Project Lead Information: TOGA, ARTHUR W

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

B.1 What are the major goals of the project?

Aim 1. Data management. Provide a F.A.I.R. data management framework for ADRC-related research and supporting the Clinical, Neuropathology, Biomarker and Imaging Cores, with a centralized data system for:
a. A unified ADRC archive that houses and integrates ALL ADRD research projects and programs at USC.
b. Participant tracking dashboards to facilitate monitoring and scheduling assessments.
c. Standardized collection of local assessments, scales, and enrollment status.
d. Integration of local assessments, scales and measures data with Uniform Data Set (UDS) in a common database infrastructure, facilitating analyses across local and broader community levels.
e. De-identifying, storing, and tracking all data types - genetics, clinical, imaging, digital pathology and biosample data.
f. Controlled access to all UDS and Neuropathology data, genetic (APOE, GWAS, ES, WGS) data, images and sample availability including data system linkage to the biomarker repository and pathology core.
g. Comprehensive search across AD projects for cohort discovery and data examination.
h. Download and integrated processing with access control by project.

Aim 2. Program Coordination. Ensure timely and accurate data entry and management for the NACC-related UDS and Neuropathology databases as well as other data efforts such as NCRAD, ADGS, SCAN, CLARiTI, and GAAIN (Global Alzheimer's Association Interactive Network).

Aim 3. Biostatistical Support. Provide biostatistical consulting in the design, coordination, and analysis of ADRC-related projects, including development projects supporting the REC Core, design of clinical studies and trials, integration and analysis of multimodal data, and support for data interpretation and manuscript preparation.
a. Assist with study design, data collection, and statistical analyses of multimodal data.
b. Assist with image analysis and associations with other phenotype and genotype data.
c. Provide access to computational resources (see Resources & Environment) available in INI.

Aim 4. Education & Training. Provide education and training (in concert with the ORE Core and REC) via practical and didactic experiences on relevant study designs, statistical models, and informatics concepts and practices to trainees and investigators involved in dementia-related research. Provide communications and documentation via the ADRC website.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : B2_ADRC Y6_DATA Core_FINAL.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ADRC Y6_DATA Core_FINAL.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

- Continue management of the central data repository infrastructure using the LONI IDA
- Continue development of a central umbrella project for all ADRC study/sub-study data.
- Continue upload of de-identified imaging, biospecimen and clinical data through the LONI IDA
- Complete migration of blood samples and autopsy tissue tracking from legacy platforms into a linked REDCap ecosystem
- Completion of the data exploration and cohort creation interface, the IDA Analysis Ready Cohort (ARC) Builder
- Compile and consolidate A, T, N, and vascular biomarker data across multiple sources for clinical classification of participants in the ADRC
- Continue to provide study coordination across NACC, NCRAD, and other study efforts
- Continue to provide consultation in study design, coordination and analysis
- Continue to provide education and training through the dissemination of materials on the website and through the USC NIIN Masters Program.

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

C DATA Core
In addition to managing the multiple study-based electronic data collection systems utilized for capturing clinical, biomarker and other data, the Data Core works cooperatively with study investigators to define necessary modifications and improvement to align with study protocol updates and return of analysis data

Major activities by aim:

Specific Aim 1: Data Management – Provide a F.A.I.R. data management framework and centralized data system

Centralized data repository and infrastructure. To ensure data is findable, accessible, interoperable and reusable (F.A.I.R.), our data management framework includes several data collection systems and a central data repository where all data—such as Assessment, Imaging, Pathology, and Biospecimen—from retrospective and ongoing studies are collected and made available to investigators (Figure 1). This infrastructure has been expanded to incorporate data from external EDCs, including demographic, neuropsychological, health history, CSF/Blood biomarker, and genetic data, using secure data transfer methods (i.e. APIs developed by the Data Core).

Our primary data management goals remain to ensure timely, accurate, and complete data collection, along with the development and support of a secure and reliable infrastructure that enables centralized data aggregation and flexible sharing options. We also provide support to investigators collecting and utilizing data from the ADRC studies. All data collected through various systems are stored centrally in the IDA repository and are accessible to ADRC investigators. The LONI USC group continued to provide a secure data repository and sharing infrastructure through the LONI Image & Data Archive (IDA) for data related to the USC ADRC study.

The Laboratory of Neuro Imaging (LONI) Image and Data Archive (IDA) was founded in 2002 and contains data from over 150 studies. The IDA is a secure, long-term data repository supporting a wide range of neuroscience research. It receives data from 190 sites worldwide and shares data with more than 25,000 scientists in 215 countries. The IDA's data sharing system allows studies to combine data across institutions and share it according to their own policies. It features integrated data use request and review systems, preserves study branding, promotes research via the IDA website, and offers management tools. Study logos are displayed in the user interface, and data sharing is facilitated through an online request system where access decisions are managed by the study PI. A de-identification process for imaging data replaces patient ID and name, removes non-standard strings, and encrypts UIDs. The IDA's sharing options are flexible to meet each study's needs. The data access process involves investigators submitting a request, which is reviewed by the access committee, during which additional information may be requested, an account may be created, and access is granted.

Development of a Centralized Umbrella Project. The Data Core is in the planning stages to create a central umbrella project for all ADRC study/sub-study data. This will improve our ability to search and combine data across studies helping to ensure data may be the most widely used with minimal effort.

During this reporting period, the USC LONI group conducted extensive evaluations of data collection assessments across all USC ADRC affiliated projects, identifying opportunities for data reconciliation and if feasible, the development of harmonized datasets. The USC LONI group successfully completed the

harmonization of 14 datasets and made 47 total clinical and image analysis datasets available. Additionally, the USC ADRC Data Use Agreement underwent a comprehensive review, and the revised version has been finalized and is now available on the USC ADRC website.

In Q1 2026, we intend to launch the automated data use application feature to allow for external user application of the data in the USCADRC study.

Development of the USC ADRC Data QuickStart Guide. To further enhance data accessibility and standardize user training, the Core has developed the **USC ADRC QuickStart Guide**. This comprehensive resource was created to orient new investigators, trainees, and collaborators to the full breadth of ADRC data resources, including the Image Data Archive (IDA) and National Alzheimer's Coordinating Center (NACC) datasets. The guide provides a centralized index of available study files (e.g., Clinical, Imaging, Biospecimen), explicit instructions on navigating the Data Dictionary, and step-by-step protocols for querying and downloading data. By consolidating documentation on data provenance, variable definitions, and access procedures into a single reference document, we aim to lower the technical barrier to entry for junior investigators, ensure reproducible data usage across projects, and facilitate the rigorous integration of multi-modal datasets by the broader research community.

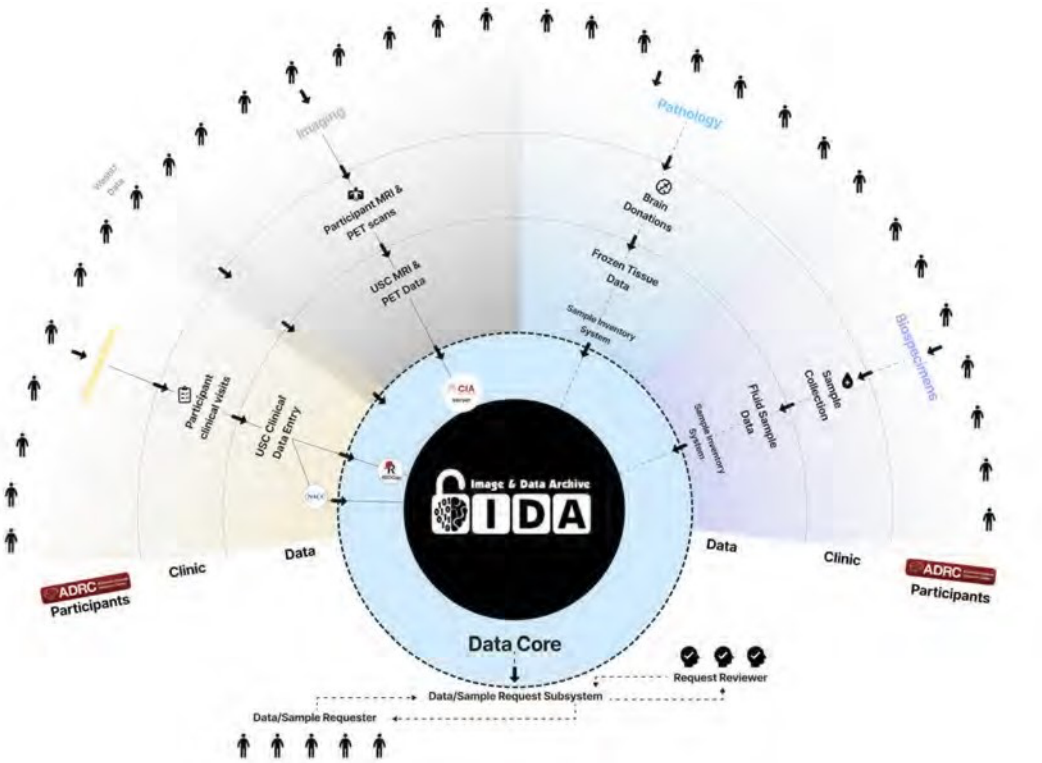


Figure 1: Data Management Systems/Data Flow

Imaging data management (MRI and PET). The current system for de-identifying and archiving MRI scans from the Center for Image Acquisition 3T scanner continues to be used for scans performed locally at USC and for the repository set up to receive scans from external sites. This system accepts MRI uploads from both the Center's 3T and 7T scanners. PET scans are still being acquired and uploaded into the repository using the same processes developed for MRI, including de-identification.

From July 1, 2025 to December 31, 2025, 2,618 MR scans from 2,489 participants and 1,068 PET scans from 1,073 participants were uploaded into the LONI IDA repository for USC ADRC affiliated studies.

Biospecimen and clinical data integration. This infrastructure has been expanded to incorporate data from external EDCs, including demographic, neuropsychological, health history, CSF/Blood biomarker, and genetic data, using secure data transfer methods (i.e. APIs) developed by the Data Core. Data from the clinical systems are routinely exported from the respective system and imported into the central archive using customized software that manages the data synchronization processes.

Biospecimen results data are either entered into the Informatics Core's REDCap database or programmatically integrated. These systems employ data validation routines to help ensure data quality. Accommodating plasma and CSF biospecimen results required the creation of new fields in existing data collection instruments and modifications to branching logic. Additionally, new database tables were established to accommodate one-time analysis results data.

Participant tracking and dashboard development (work in progress). The Data Core continues to enhance its data management infrastructure, focusing on the centralization of participant registration, BRS enrollment, and autopsy logs within REDCap. Work is currently proceeding on migrating the tracking of blood samples and autopsy tissues from legacy platforms into this linked REDCap ecosystem, with future plans to integrate sample request workflows. Using linked REDCap databases, we are also experimenting with piping data from the centralized participant registration to project-specific participant enrollment tracking database to reduce data entry burden. Concurrently, we maintain and refine interactive Tableau dashboards to visualize UDS data. Recent updates include revised BRS autopsy tracking dashboards (**Figure 2**) and expanded dynamic displays of ATNv data availability. These tools allow researchers to efficiently filter cohorts, assess study feasibility, and monitor data quality.

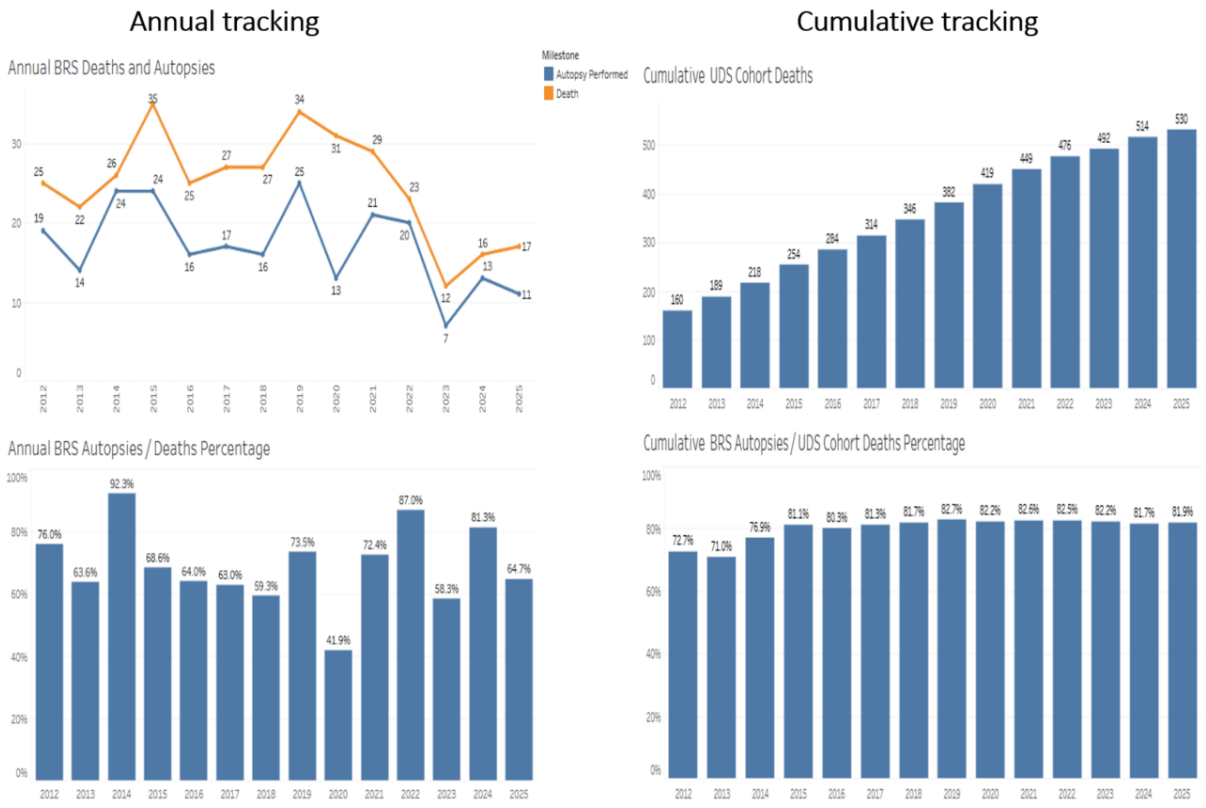


Figure 2. Data dashboard for BRS autopsy tracking.

Cohort discovery tools (ARC Builder) (work in progress). In the second quarter of 2025, the IDA Analysis Ready Cohort (ARC) Builder was deployed. The ARC Builder is a data exploration and cohort creation interface allowing investigators to visualize data and create cohorts using variables stemming from examination, assessment and imaging variables. Investigators may build customized datasets containing variables of interest and the associated MRI and PET images. See Figure 3. The ARC Builder system continues to be optimized, reviewed, and improved as part of the ongoing rollout of the new platform. Key components are performing as expected, and refinements are being implemented to enhance stability, performance, and user experience. Progress remains on track, with nine clinical assessment datasets pending integration and an anticipated completion in the second quarter of 2026.

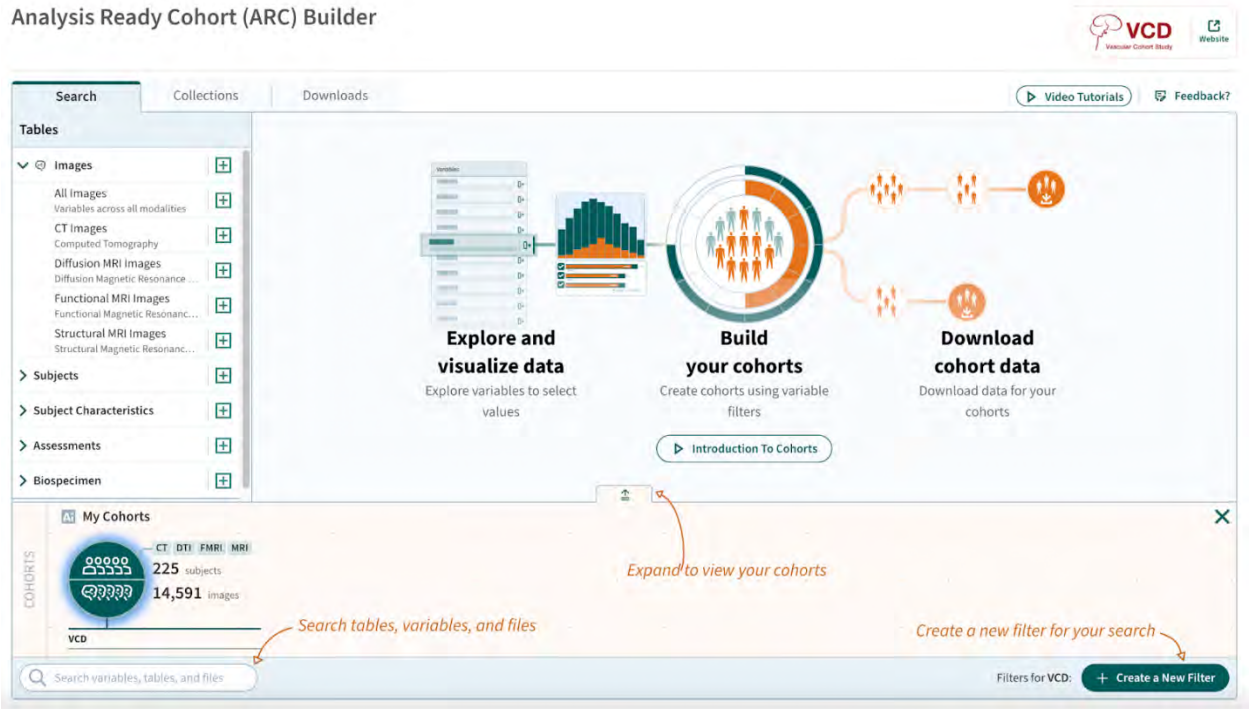


Figure 3. Analysis Ready Cohort (ARC) builder on the IDA.

ATNv data consolidation (Work in Progress). To better understand the ATNv biomarker data for our ADRC clinical participants, the Data Core began compiling and consolidating A, T, N, and vascular biomarker data from multiple sources. Specifically, we extract data from NACC UDS (demographics), NCRAD (plasma and APOE genotype), IDA (MRI/PET scans), and additional data from project-specific databases (e.g., VCS/PPG, DVR, etc.) stored in the IDA system. This aims to track the availability of ATNv data and characterize ADRC clinical participants who are actively being followed.

To characterize the baseline Alzheimer's disease (AD) pathology within the PPG cohort, we applied the NIA-AA Research Framework (ATN) criteria to participants with available cerebrospinal fluid (CSF) and structural MRI data. Amyloid (A) and Tau (T) status were determined using established cut-offs for the MSD V-PLEX platform: A+ was defined as a CSF Aβ42/Aβ40 ratio < 0.060, and T+ was defined as pTau181 > 52 pg/mL. Neurodegeneration (N) was assessed via FreeSurfer (v7) processing of T1-weighted MRI, with N+ defined as an AD-signature cortical thickness < 2.67 mm (Figure 4). In the sub-cohort with baseline CSF data (n=217), 87 subjects (40%) were classified as Amyloid-positive (A+) and 118 (54%) as Tau-positive (T+). Neurodegeneration (n=339 MRI scans) was identified in 135 subjects (40%) at baseline. Among the subset of participants with concurrent multimodal data (n=143), 43 subjects (30%) met the biological criteria for Alzheimer's disease (A+T+), with 28 of these (20%) displaying full multidomain pathology (A+T+N+). These frequencies indicate a robust representation of AD pathologic change within the study sample at study entry.

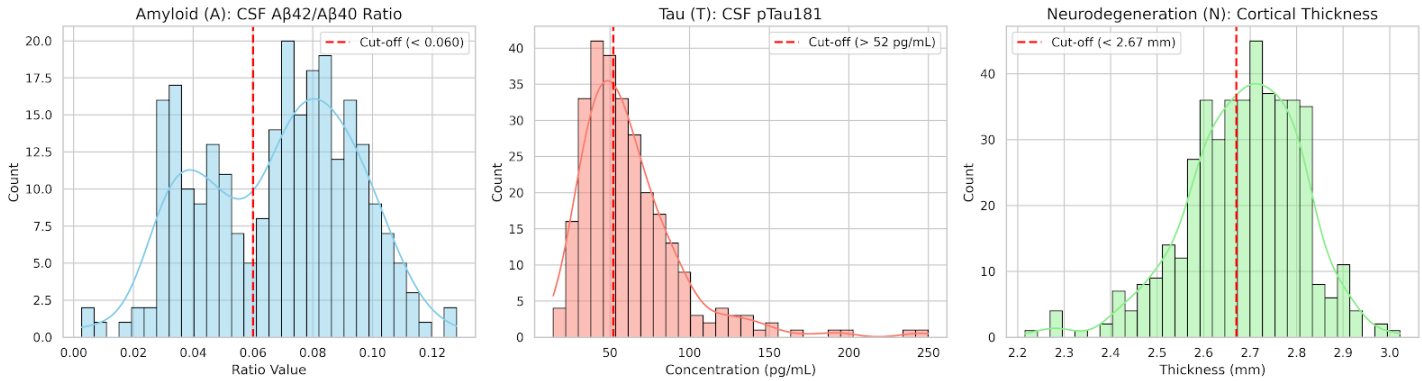


Figure 4. Amyloid, tau and neurodegeneration distributions with clinical cut-offs

<u>Specific Aim 2:</u> Program Coordination – Ensure timely data entry and management for NACC, NCRAD, and other efforts

NACC data coordination and UDSv4 transition. USCADRC participants' UDS data from NACC continues to be synced in the database on an ongoing basis. A new dataset containing NACC UDS freeze data was successfully integrated into the existing data environment. Additionally, LONI provided support for the transfer of new non-imaging data via API including database schema development for MRI and PET image analysis results.

Additionally, we constructed a local UDSv4 REDCap database incorporating NACC-provided standard UDSv4 data dictionary and our center-specific data instruments. The UDSv4 data collection and submission to NACC has also been established. To facilitate data entry and database utilization, we will continue monitoring feedback from our investigators and data coordinators and updating the database following monthly NACC UDSv4 updates. We provide periodic training and assistance as needed. Following the transition of UDSv3 to UDSv4, we also built a local NACCID enrollment & Participant transfer REDCap database and established the new NACCID application process. Two additional local REDCap databases, for tracking NACC Milestones form and NACC Neuropathology form respectively are also built as transition from the legacy system. Data collection and submission using the new system have been established as well. We will keep collaborating with investigators to develop new reconciliation reports needed for the evolving requirements of the ADRC.

Biospecimen and external coordination (NCRAD). Per the Alzheimer's Disease Center Fluid Biomarker initiative, plasma samples from the USCADRC are analyzed by the NCRAD Biomarker Assay Laboratory utilizing Quanterix Simoa HD-X Neurology 2-Plex B kits, the ALZpath pTau217 Plus kits, and the Fujirebio Lumipulse G beta amyloid 1-40 and Lumipulse G beta amyloid 1-42 kits. Plasma concentrations of neurofilament light chain (NfL), glial fibrillary acidic protein (GFAP), amyloid beta 42 (Abeta 42), amyloid beta 40 (Abeta 40), and phosphorylated tau 217 (Ptau 217) were measured and returned to us. In the second quarter of 2024, these results were integrated in the USCADRC repository and made available for internal researchers. Updates to this data are made on a quarterly basis, following releases from NACC.

Moreover, we developed an online USC ADRC Biobank Request form through surveys linked to multiple REDCap databases. These help facilitate the request and acquisition of CNS tissue and biofluid samples for research purposes. All these enhance administrative oversight in tracking internal data flow and external requests for data, tissues and biosamples.

<u>Specific Aim 3:</u> Biostatistical Support – Provide consulting in design, coordination and analysis

Statistical consulting and analysis. In 2025, the DMS core continued providing statistical modeling assistance and data analysis support for various studies led by ADRC or ADRC-affiliated investigators. We also provided statistical consultations for grant applications at the pre-award stage, assisting with hypotheses formation, study design, and statistical modeling approaches, among other areas.

Methodological development. Regarding statistical method application, we continue to develop statistical tools to implement commonly used modeling approaches in AD research and collaborate with faculty in the Biostatistics Division and LONI to develop new techniques for high dimensional mediation analyses involving neuroimaging data

Outcomes. These efforts resulted in three abstracts accepted at conferences, two papers published, and several others in submission or preparation.

Specific Aim 4: Education and Training – Provide education, training and communications

Website and communication resources. The USC ADRC website has been enhanced with the launch of a new Learning Hub that centralizes valuable resources to support continued education about Alzheimer's disease. The hub includes education and training materials, along with online seminars and symposia, making it easier for users to access and engage with ongoing learning opportunities in one place.

Within this learning hub, the website has been enhanced through the development of a dedicated Online Seminars section that enables users to explore past, current, and upcoming events, along with speaker biographies and topic descriptions to help them identify sessions aligned with their interests. The section also includes clear instructions on how to join the online seminars.

Additionally, the USCADRC website has been enhanced with a dedicated Symposium page, providing a centralized location where interested participants can access comprehensive information and stay up to date on event details.

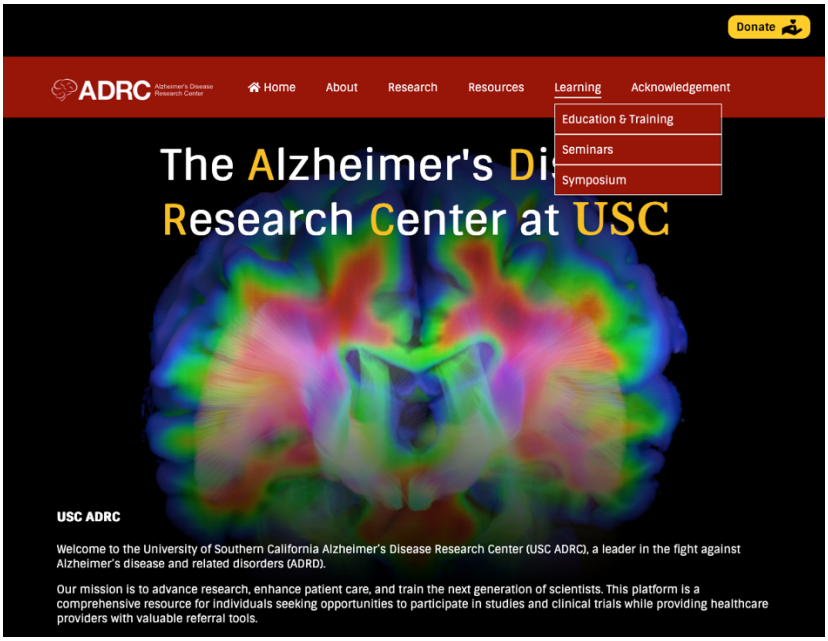


Figure 5.
USC ADRC learning hub providing valuable resources for education and training.

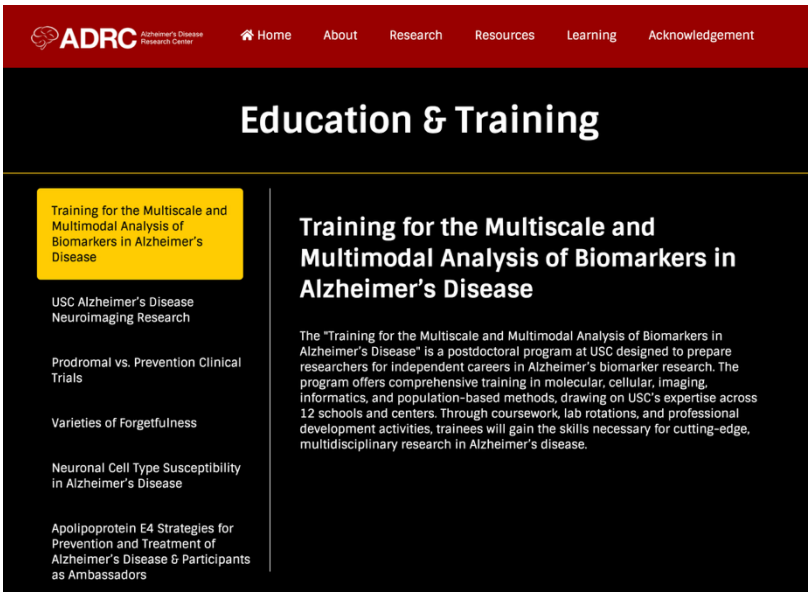


Figure 6. Education and Training page within the USCADRC learning hub.

Figure 7. Seminars page within the USCADRC learning hub, featuring

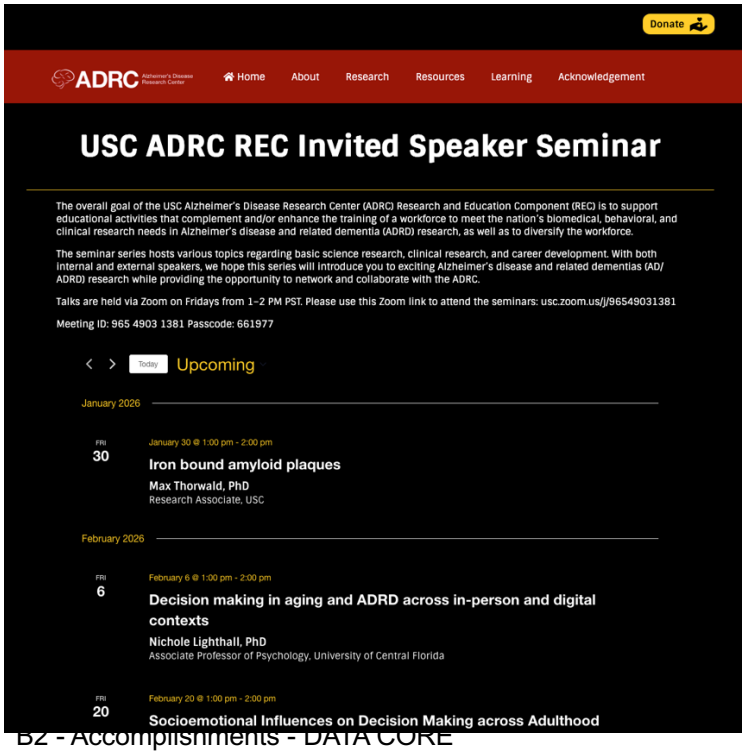


Figure 8. Symposium page within the USCADRC learning hub, featuring upcoming events.

The Neuropathology Teaching Set website was updated to provide a more accurate and user-friendly experience. Redundant and erroneous slides were removed, and the remaining content was reorganized and enhanced with more detailed information. A new grouping feature was also implemented to help users easily locate and navigate related slides.

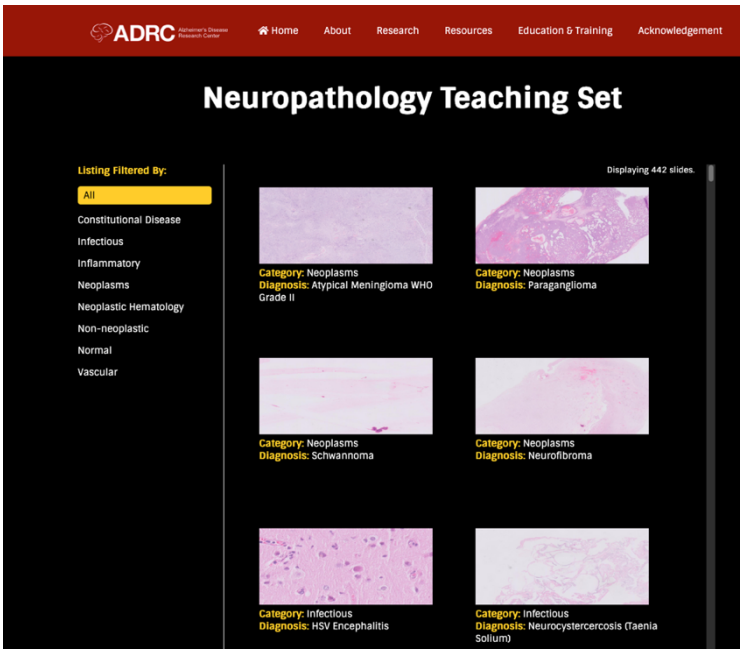


Figure 9. Updated Neuropathology Teaching Set.

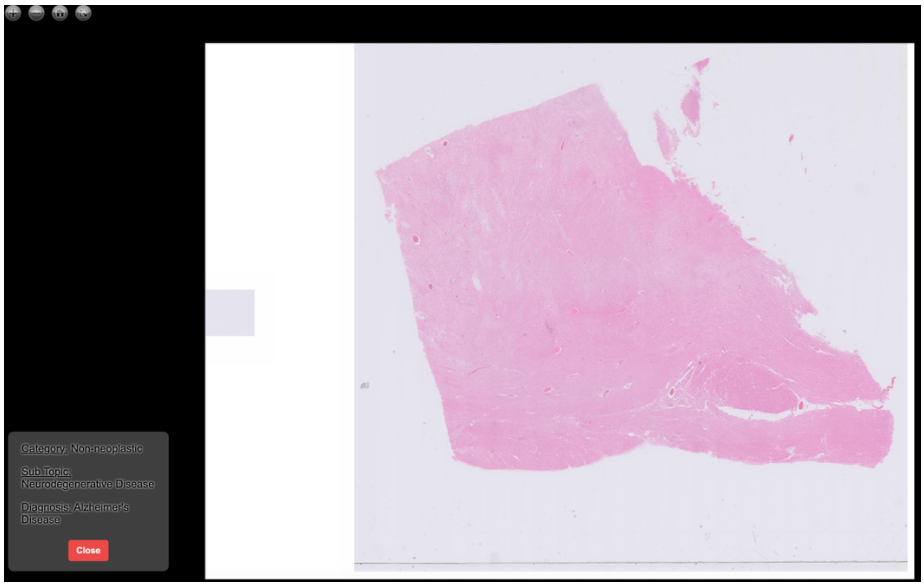


Figure 10. Example of the slide viewer. The updated version now provides users with more comprehensive information about each slide.

Finally, A dedicated content management system (CMS) was implemented to enable the neuropathology group to independently maintain and update slide information and thumbnails, supporting timely updates and sustained accuracy of the teaching set content.

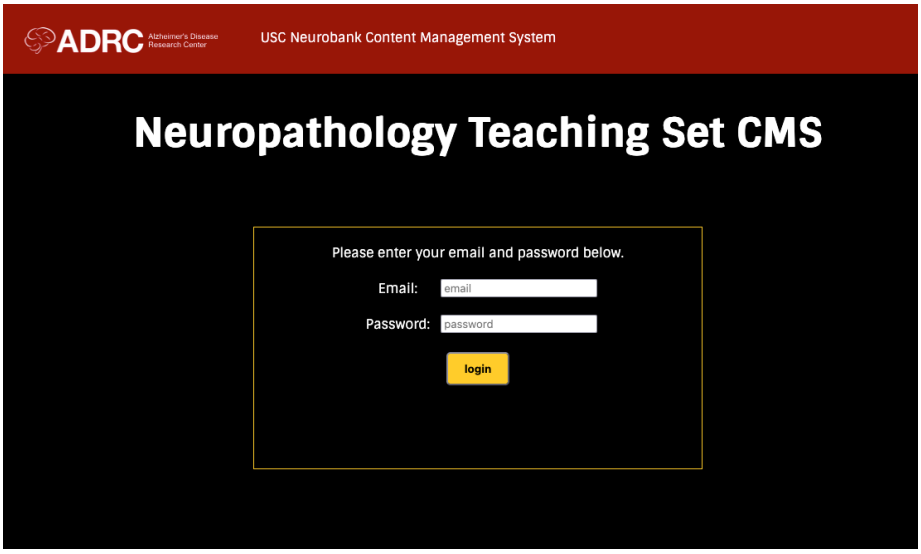


Figure 11. Neuropathology Content Management System (CMS).

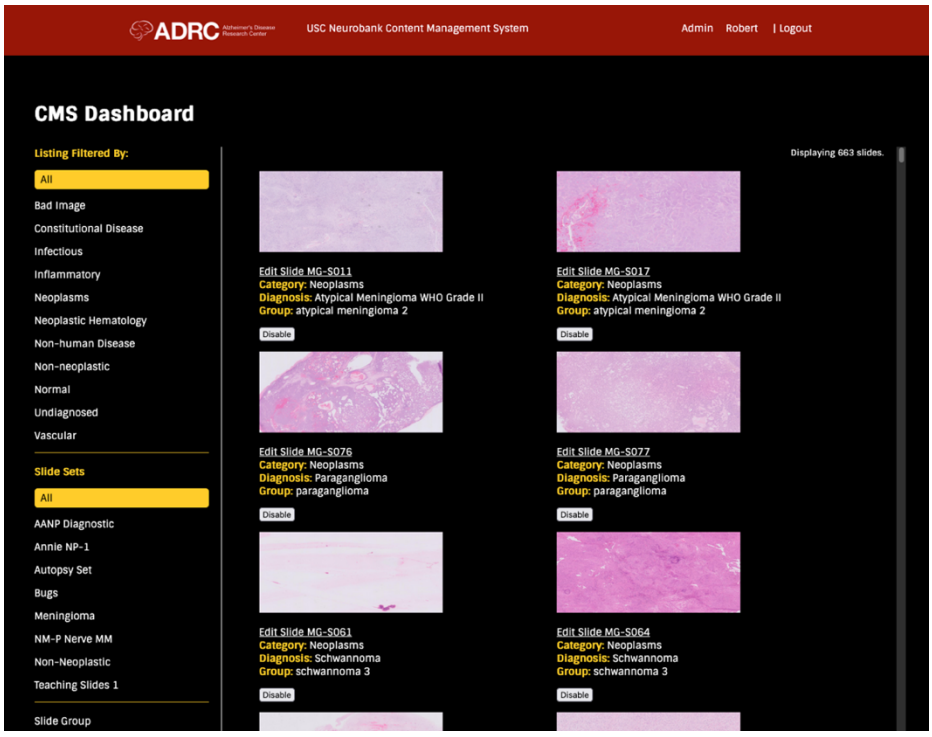


Figure 12. Slide page for the Neuropathology CMS.

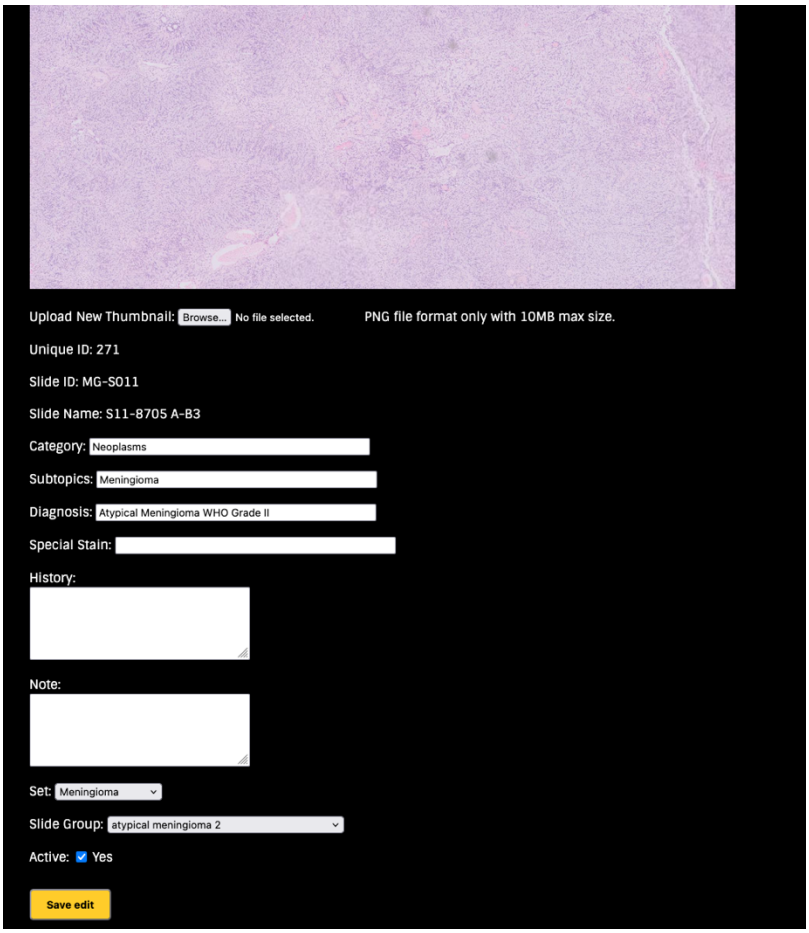


Figure 13. Editing page for a slide in the CMS.

Training and professional development. We continue to provide training and professional opportunities for trainees in various labs across USC. These include weekly seminars, videos, as well as code and hands-on practice sessions through INI's NIIN master program. We provide periodic training and assistance as needed.

USC ADRC Data Counts

The table below shows the counts of cognitive assessments, biomarker data, and imaging modalities conducted per patient ID.

Counts per Patient	CDR	CSF BIOMARKERS	CT	FAQ	GDS	MRI	NEUROPSYCHIATRIC	NEUROPSYCHOLOGICAL	PET	PLASMA BIOMARKERS
Grand Total	141	0	0	140	136	2618	141	135	1068	170

Table 1. Counts of modalities within the 07/01/2025 – 12/31/2025 period.

CDR	Clinical Dementia Rating
CSF BIOMARKERS	Cerebrospinal Fluid Biomarkers
CT	Computed Tomography
FAQ	Functional Activities Questionnaire
GDS	Geriatric Depression Scale
MRI	Magnetic Resonance Imaging
NEUROPSYCHIATRIC	Neuropsychiatric Inventory Questionnaire
NEUROPSYCHOLOGICAL NEUROPSYCHIATRIC	Neuropsychological Battery Scores Neuropsychiatric Inventory Questionnaire
PET NEUROPSYCHOLOGICAL	Positron Emission Tomography Neuropsychological Battery Scores
PLASMA BIOMARKERS PET	Plasma biomarkers Positron Emission Tomography
PLASMA BIOMARKERS	Plasma biomarkers

Table 3. Legend for tables.

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

DATA MANAGEMENT & STATISTICAL CORE

We continue to provide training and professional opportunities in trainees in various labs across USC. These include weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

See the ORE and neuropathology cores for details on clinical core resource sharing of clinical materials and research specimens. See Administrative Tables and overall component for description of resource sharing.

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022
FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Arthur		Toga		Data Core Lead	225,700.00	1.2			22,570.00	7,155.00	29,725.00
2.	Daniel		Nation		Data Core Co-Lead	225,700.00	0.8			15,122.00	4,794.00	19,916.00
3.	Xinhui		Wang		Co-I	122,639.00	3.18			32,499.00	10,302.00	42,801.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **92,442.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Research Associate	2.16	0.0	0.0	14,459.00	3,268.00	17,727.00
1	Database Manager	0.3	0.0	0.0	5,735.00	1,818.00	7,553.00
1	Project Assistant	2.58	0.0	0.0	19,844.00	6,291.00	26,135.00
1	Research Associate	3.99	0.0	0.0	42,844.00	13,581.00	56,425.00
1	Systems Analyst	6.72	0.0	0.0	46,144.00	14,627.00	60,771.00
5	**Total Number Other Personnel**					**Total Other Personnel**	**168,611.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**261,053.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**3,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	21,606.00
2. Publication Costs	1,600.00
3. Consultant Services	0.00
4. ADP/Computer Services	14,000.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**37,206.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**301,259.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	301,259.00	198,830.00
		Total Indirect Costs	**198,830.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**500,089.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	500,089.00

L. Budget Justification* File Name: Data Core justification.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – DATA CORE

Key Personnel:

Arthur W. Toga, Ph.D., Core Lead, 1.20 Calendar Months in Years 1 - 5. Dr. Toga is the founder and director of the Laboratory of Neuro Imaging (LONI). He is director of the USC Stevens Neuroimaging and Informatics Institute. His formal training in neuroscience, computer science and informatics makes him well suited to lead this project. Dr. Toga has a career-long funding and publication record in informatics. He directs several national and international collaborative studies that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results. Dr. Toga has experience and expertise in image analysis, brain atlasing, multimodal integration, large-scale multisite trials, databasing and scientific visualization. He has developed a comprehensive supercomputing facility for neuroscience incorporating the latest technology in the analysis of structure and function of the brain in health and disease. Dr. Toga has the skills, knowledge, and resources necessary to participate in the proposed project. He will coordinate activities in data management, computing infrastructure, and dissemination.

Daniel Nation, Ph.D., Co-Core Lead, 0.80 Calendar Months in Years 1 - 5. Dr. Nation will lead implementation of blood pressure and capnography monitoring during brain MRI studies and will make available relevant processed and analyzed physiological data to ADRC-affiliated investigators to facilitate studies of cerebrovascular factors contributing to cognitive decline. Dr. Nation will also lead the coordination of cross-study data and statistical analysis projects that leverage common clinical, imaging and biomarker data elements in order to test hypotheses regarding vascular factors in cognitive decline and AD. Dr. Nation will also support the Data Management and Statistical Core by contributing to biostatistical consultations and assisting other investigators in navigating and analyzing ADRC study cohort data and cross-study data analysis for their projects.

Xinhui Wang, Co-Investigator, 3.18 Calendar Months in Years 1 - 5. Dr. Wang is an Assistant Professor of Research Neurology. She is an expert in epidemiology and statistical analysis. Dr. Wang will assist in designing the statistical analysis framework for this project and will perform quality control.

Other Personnel:

Francesca Sibilia, PH.D. Research Associate 2.16 Calendar Months in Years 1 - 5. Ms. Sibilia has experiences in computational neuroimaging analysis and dementia research. She will extend her work on perivascular space measurements by combining imaging and physiological data with the aim of investigating the effect of cardiovascular risk factors such as stress and hypertension-related biomarkers on brain perivascular space. Ms. Sibilia has a background in both medical biotechnology and neuroscience; her knowledge on the physiological processes occurring during aging and degeneration, both at cellular and molecular levels, plays a key role for the development of the study hypothesis. Her background helps her to research on how different components (both imaging and physiological factors) are crucially entangled in explaining the neurophysiological changes primarily, and the cognitive change secondarily, seen with the onset of Alzheimer's disease and what role perivascular spaces play. Combining multi-modal biomarkers will add knowledge and help explaining the complex mechanisms in the brain perivascular spaces during neurodegenerative processes.

Karen Crawford, M.S., Database Manager, 0.312 Calendar Months in Years 1 - 5. Ms. Crawford has expert knowledge of database management systems and development of informatics systems. Ms. Crawford will oversee systems used to archive and catalog data acquired from investigators and collaborators. She will conduct data analyses, construct data models, develop and document data access requirements and strategies. She will also design databases, administer and supervise the metadata representation, determine feasibility of database applications, monitor and evaluate database performance, modify database design and/or implementation when appropriate, assist users in appropriate/ approved use of database applications and functions, develop client-server and web-based user interfaces.

Monica Estrada, Project Assistant, 2.58 Calendar Months in Years 1 - 5. Ms. Estrada will assist in coordinating and managing the data upload and download activities and user access control systems. While in that capacity, she will provide support to study site personnel performing data upload activities, assist with user access and security and provide statistical reports on data upload and download activities.

Leon Aksman, Ph.D., Research Associate, 3.996 Calendar Months in Years 1 - 5. Dr. Aksman is an Assistant Professor of Research Neurology at the USC Stevens Neuroimaging and Informatics Institute. His background is in engineering and neuroimaging, with core expertise in statistical modeling, machine learning and longitudinal analysis. He has doctoral and postdoctoral training in the development and application of advanced computational models for understanding the progression of neurodegenerative diseases such as Alzheimer's and Parkinson's. He has developed novel biomarker trajectory methods and has codeveloped pySuStaIn, a widely used software package for modeling the progression of neurodegenerative diseases.

James Luo, Systems Analyst, 6.72 Calendar Months in Years 1 - 5. Mr. Luo will help refine data management systems to meet the needs of ADRC investigators, clinicians and staff. In this capacity he will analyze and revise existing system logic and documentation as necessary to reduce operating problems, expedite processing and maximize systems efficiency and works with users to analyze functional areas where computing technology will offer benefits. This is a chronic effort as data needs evolve and platforms become more sophisticated.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all employees, 22.60% for post-docs for all 5 years.

Other Direct Costs:

Materials and Supplies: We request $21.605 in Years 1 - 5 for the purchase of computer supplies, printing supplies, and research supplies. There is a significant need on this project for hard drive and computer disk backup to ensure the safe archival and failsafe backup of our datasets.

Travel: In Years 1 – 5 we request $3,000 per year ($1,500 per meeting x 2) for attendance at the ADRC semi-annual meeting.

Publication costs: We request $1,600 in Years 1 – 5 for publication costs that include page charges and open access charges imposed by publishers. Publication costs may also be required to publish abstracts or posters at conferences.

Computer Services Costs. We request $14,000 in Years 1 – 5 for high-performance supercomputing time and maintenance services, which guarantees high throughput of the very large computational image datasets. The costs listed here are standard data analysis costs associated with imaging projects. These

costs help defray the depreciation costs for the equipment. These computational resources make its use essential for complicated algorithms typically used during the calculation of population-based statistics. This, coupled with access to the large database, makes it an efficient system for many brain analytic problems. The hardware support provides 24/7 response coverage for Isilon, Cisco, Quantum and F5 equipment discussed under the Resources and Environment section. Because of concerted efforts eliminating single points of failure in our computing

environment, this model is by far the most cost effective. The software maintenance provides our facility with immediate vendor software support, software licensing and version upgrades for sophisticated clustering, visualization, development and hierarchical file management software. Such upgrades become critical when new features that can facilitate daily operations are released.

Facilities and Administration Costs:
Facilities and Administration Costs are based on USC' federally negotiated rate (dated May 29, 2024) of 66% as of July 1, 2024. The F&A rate is 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

A. COMPONENT COVER PAGE

Project Title: Core D: Neuropathology Core
Component Project Lead Information: Hiniker, Annie E

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

Aim 1: Rapid Autopsies from Our Clinical Populations with a focus on APOE e4. With the Clinical Core, we will increase autopsies from our Clinical Core and prioritize collecting brain tissue from cases with the APOE e4 allele, in keeping with Center mission.

Aim 2: Comprehensive Neuropathological Diagnosis. In addition to the standard stains for diagnosis of AD and full staging for ADNC, we utilize comprehensive evaluation for co-pathologies including alpha-synuclein and pTDP-43 for LATE, and tau for ARTAG at NACC11 standards. There will be independent review of all cases by both Dr. Hiniker and Hawes with re-review to reach consensus diagnosis. New markers will be developed and implemented as needed for diagnostic accuracy and research purposes.

Aim 3: Distribute Tissue to Catalyze ADRD Research. In addition to fixed hemibrains, eyes, and spinal cord the biorepository will continue to collect blocks, slides, and frozen aliquots and distribute our unique collection of neuroanatomic and systemic regions to researchers to further ADRD diagnosis and therapeutics.

Aim 4: Develop a freely accessible de-identified digital library. Dr. Hiniker and Dr. Bienkowski, an expert in digital neuropathology and spatial transcriptomics, will lead the creation of an online, open-access digital pathology library of all >1,100 cases in our Center at NACC 11 standards, in collaboration with the DMS Core. This digital library will expand the NP Core's ability to provide high-resolution AD pathology data to the wider AD community as both a research and education resource.

Aim 5: Interrogate the effect of the ApoE4 allele on CAA, AD neuropathology and perivascular inflammation. Using our digital pathology machine learning approach, we will quantify and map the distribution of fibrinogen leakage, CAA, and microgliosis in the hippocampus and cortex of 80 ADRC patients (40 APOE e4+ and 40 APOE e4-). In a smaller cohort of 20 patients, we will perform Multiplexed Error Robust Fluorescent In Situ Hybridization (MERFISH) spatial transcriptomics on hippocampal tissue sections (n=10 APOE e4+ and n=10 APOE e4-) with a 1000 gene panel to determine APOE differential cell-type specific changes to brain cell types in low and high stage AD (n=5 per group). This study will provide new insights into the effect of APOE genotype on the microglial response to fibrinogen and amyloid deposition in AD and ADRD.

Aim 6: Mentor the Next Generation of ADRD Researchers and Neuropathologists. The NP Core will increase engagement with the REC Core and expand the formal lectures on neurodegenerative neuropathology available to neuropathology and neurology fellows and resident, and medical and graduate students

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : B2_ADRC RPPR Y6_Neuropath.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ADRC RPPR Y6_Neuropath.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Due to the truncated reporting period, our aims for Year 7 of the grant are highly similar to Year 6:

Aim 1: Rapid autopsies from our clinical populations with a focus on APOEε4. Continue with subject recruitment via the Clinical Core BRS for performance of autopsies – including specifically, recruitment of new clinical patients with APOE ε4 for our Center's major focus. We will continue our increased recruitment of normal controls to provide sufficient numbers of tissues for current and future studies. We also want to maintain the number of disease autopsies to provide samples for the current list of labs using tissues. We harvest fresh and frozen tissue and aim to improve on our Core's average postmortem intervals.

Aim 2: Comprehensive neuropathological diagnosis. Perform tissue procurement, preservation, and diagnosis. Tissues (including eyes and dura) are rapidly obtained and frozen at autopsy for research projects. Based on the increasing number of requests for frozen tissue, we continue to harvest highly requested regions (hippocampus, cortical regions) at autopsy; however, we have also implemented coronal slabbing the right hemisphere for additional frozen requests. The left hemibrain is fixed in 10% formalin until dissected by the neuropathologist and stored for future allocations. These protocols will continue. We will continue to implement comprehensive neuropathologic diagnosis

Aim 3: Distribute tissue to catalyze ADRD research. We will continue to prioritize rapid allocation of tissue to investigators, including ongoing fresh tissue harvesting for multiple researchers, as well as autopsy frozen tissue, fixed tissue, and deidentified data. Through our new online tissue request form we continue to increase both the number of requests (by 1.7-fold per month over the prior cycle) and number of new investigators/institutions who had not previously used our bank.

Aim 4: Develop a freely accessible de-identified digital pathology library. We are cataloging all stained slides in our 1,120-case archive and will continue to backfill stains up to NACC 11 as described below; this is an ongoing and large project. We have begun working with members of the Data Core to design the website hosting our digital slide archive and will continue to build this website. This reporting period we will focus on digitizing a large number of the slides in our collection and optimizing our protocols for getting them into the digital library.

Aim 5: Interrogate the effect of the APOE ε4 allele on AD neuropathology and perivascular inflammation. This cycle, we will continue to optimize the histology workflow for MERFISH spatial transcriptomics. Once this is established we will order the gene panel. In parallel we anticipate working on the vascular pathology staining and beginning the analysis. We hope to begin to analyze FFPE human post-mortem tissue sections from APOE4 patients to investigate transcriptomic changes.

Aim 6: Mentor the next generation of ADRD researchers and neuropathologists. Drs. Hiniker and Hawes will continue to be active participants in the USC Neuropathology Fellowship. The fellowship program was reviewed by the ACGME and granted continuous approval without citations. Dr. Bienkowski and Dr. Hiniker will also continue to train graduate students and postdoctoral fellows in neuropathology. Dr. Hiniker will continue to lead weekly gross brain cutting sessions, including ADRC cases, attended by residents, fellows, med students, and faculty. Dr. Hiniker will continue to co-direct the national AANP Neurodegenerative Scholars program.

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

OVERVIEW: During the period 07/01/25-12/31/25, the first 6 months of our renewed grant, the ADRC NP Core continued to perform rapid autopsies, neurodegenerative disease diagnosis, tissue banking, and tissue distribution for researchers. We continued to modernize and improve the NP Core, with additional re-evaluation, staining, and backfill of copathology and other diagnoses in our bank of >1,100 cases up to NACC 11 standards. We continued to increase the number and types of cases distributed to researchers and we further solidified cross-core work on data infrastructure and digital pathology.

Table 1. Brain Research Study (BRS) 07/01/25 - 12/31/2025

	# of participants	BRS enrolled (%)	F	M
New participants enrolled	10	4.33%	5	5
Participant withdrawal	2	0.87%	2	0
Current enrollment*	231	–	146	85
Participant died *with* autopsy	9	3.90%	3	6
Participant died *w/o* autopsy	0	0.00%	0	0

Aim 1: Rapid autopsies from our clinical populations with a focus on APOEε4

Key updates this reporting period:

- We performed autopsies on all participants who died this cycle.
- We now have two on-call dieners, allowing us to have robust backup coverage for autopsies.
- We continue to prioritize decreasing the PMI of all cases and updated our call center protocols with this goal in mind.
- We put into place a detailed prion disease screening protocol to decrease likelihood of inadvertently performing an autopsy on a CJD case.
- In conjunction with the biomarker core, we backfilled ApoE genotyping on >600 cases lacking clear ApoE genotyping, expanding the cohort of tissue available for ApoE E4 research studies.

The Brain Research Study (BRS) enrollment for 07/01/2025 to 12/31/2025 was 231 subjects with 10 new participants enrolled and consented in the BRS for autopsy (Table1). During this 6 months, a total of 9 patients died, 9 participants came to autopsy (100%), and 0 died without autopsy (0%) (Table 1). From 07/01/2025 to 12/31/2025, the NP Core performed 9 diagnostic brain autopsies (Table 1); 6 cases were male (67%); and 3 was female (33%).

Since Dr. Hiniker took over as core director, we have prioritized tissue distribution with increasing numbers of requests for frozen tissue and fresh tissue with PMI time restrictions. We now have multiple ongoing studies involving researchers internally and at outside institutions who require immediate fresh tissue or short PMI. We are therefore working to increase rapidity of autopsies to minimize PMI (postmortem interval). More than half (55%) of our autopsies this cycle had PMIs <7.5 hours vs our average PMI of 8.5 hours over the 40-year history of the bank. The new diener we recruited and trained last reporting period has now performed sufficient proctored cases and is performing independent brain autopsy and frozen tissue harvest for us and is now covering approximately half the call on the schedule. We updated the protocols for our call center (the first contact for families to report participant deaths) to ensure rapidity of autopsy. We also updated protocols for prion screening. Finally, in conjunction with the biomarker core, we received data back to confirm ApoE genotyping of >600 historical cases with unclear ApoE genotype (either not performed or not performed by NACC), greatly increasing the reliability and number of ApoE genotyped cases for human tissue studies.

Aim 2: Comprehensive neuropathological diagnosis.

Key updates this reporting period:

- We continue to perform our updated and comprehensive immunostaining protocol introduced in our last RPPR, with routine staging of LATE and α-synuclein staging on all cases.
- We continue to backfill co-pathology diagnoses and staging to current NACC 11 standards including: pTDP-43 LATE hippocampal neuropathology and FTLD diagnosis now up to 584 banked cases; backfilled a-synuclein staging (162 banked cases), tau (115 banked cases) and Abeta (279 banked cases).

This cycle, we continued to perform up-to-date tissue procurement, preservation, and diagnosis. Tissues (including eyes and dura) are rapidly obtained and flash frozen at autopsy for research projects. Based on the increasing number of requests for frozen tissue, we continue to harvest highly requested regions (hippocampus, cortical regions) at autopsy as well as performing coronal sectioning of the entire remaining right hemisphere for additional frozen requests. We fix the left hemibrain in 10% formalin until dissected by Dr. Hiniker and store both cassettes and the remaining fixed hemibrain for future allocations. We continue to implement the prior improvements made last reporting period including routine pTDP-43 LATE staging on all cases; routine sampling of olfactory bulb; full α-synuclein staging on all cases regardless of clinical history; and dual, independent review of all cases by both Drs. Hiniker and Hawes.

For AD diagnosis per NIA-AA guidelines, we use H&E, phospho-Tau antibody (AT8), Aβ antibody (4G8), phospho-α-synuclein (81A), and phospho-TDP-43. We continue to backfill the immunostaining and diagnostics of our banked older cases as above, with pTDP-43 hippocampal neuropathology and FTLD diagnosis now up to 584 banked cases, a-synuclein staging on 162 banked cases, additional tau staining on 115 banked cases, and additional Abeta staining on 279 banked cases since Dr. Hiniker's recruitment. Regarding diagnoses during this time period, given the short reporting period, some cases from this period are still undergoing neuropathologic evaluation but neuropathologic diagnoses included ADNC; aging-related tau astrogliopathy (ARTAG), and small vessel pathologies.

Aim 3: Distribute tissue to catalyze ADRD research.

Key updates this reporting period:

- We continue to utilize our new REDCap tissue request form and Tissue Request Committee with sustained increases in request rate (see below).
- The Biomarker core is now using the same REDCap request form for biofluid requests and we have improved integration of neuropathology and biomarker requests through this system.
- 3 new institutions not previously using our bank received tissue from us (U of Pittsburgh, Amprion Diagnostics, VIB KU Leuven Center for Brain and Disease Research)
- NP Core continues to collaborate with Data Core in biweekly meetings to assist in construction of the REDCap database to track specimens and data (see Data Core section), contributing data and collaborating on tissue requests requiring data.

The NP Core continues to coordinates all aspects of tissue distribution to researchers. This 6-month cycle, we distributed frozen brain tissue for transcriptomics, lipidomics, metabolomics, biochemistry, RT-QuIC, and DNA sequencing; fresh brain tissue for cell culture and RT-QuIC studies; and fixed tissue for immunostaining and spatial transcriptomics. We also supplied eye tissue and systemic tissue and filled requests for data, protocols, and letters of support. This cycle, we distributed samples to investigators both at USC and outside institutions including UCLA, Cedars Sinai, Case Western, the Mayo Clinic, UC San Diego, VIB KU Leuven Center for Brain and Disease Research, Amprion Diagnostics, and University of Pittsburgh. Within USC, we provided tissues to multiple individuals pursuing our ADRC's APOE themes as well as other studies.

We continue to provide stewardship of these precious tissues by assessing scientific rigor of

tissue requests. We assess each request in our new Tissue Request Committee including leaders from NP, Biomarker, and ORE/Clinical Core. We utilize a new online REDCap tissue request form, allowing easier allocation to researchers. This cycle, Biomarker Core also implemented this REDCap request form, allowing easier integration of requests requiring matched tissue and biofluids. Grant-funded projects supported by the Neuropathology Core are listed in overall Administrative Tables; however, we also support rigorous pre-award work. This cycle, we continued to increase our rate of requests filled by 1.7x, filling 25 requests in the 6 months reported in this RPPR.

The NP Core actively fosters innovative areas of research including the use of state-of-the art techniques by collaborating with USC and outside investigators—this cycle we mediated new spatial transcriptomic studies from fixed tissue in collaboration with the USC Histology lab. The NP Core also continues to collaborate with the Data Core and Clinical Core to update and integrate our data. We continue to assist the Data and Clinical Core in their implementation of a Research Electronic Data Capture (REDCap) database tracking both clinical and neuropathologic data. NP Core collaborates with Data core in biweekly meetings to assist in construction of this REDCap database. NP Core also continues to provide data to the Data Management and Statistical Core regularly and collaborates on tissue requests requiring data.

<u>Aim 4: Develop a freely accessible de-identified digital pathology library</u>.

Key updates this reporting period:

- **With Data Core, successfully piloted techniques necessary to create our ADRC digital pathology database (see next bullet)**
- **Created a new digital teaching set (Figure 1) of >500 neuropathology slides**

In collaboration with Data Core (see their complementary description), we created a pilot proof-of-concept neuropathology digital teaching library of >500 deidentified slides ranging from neurodegenerative to tumor to infectious pathology from Dr. Hiniker's personal teaching set. The goals of this project were to troubleshoot the scanning and conversion to real-time zoomable images of the slide set as well as to create the website interface using the expertise of Data Core. An additional longstanding goal of the project is to create a teaching library for residents and fellows (first, at USC, later available nationally) to learn neuropathology.

Briefly, we collected and organized by diagnostic category >500 deidentified neuropathology slides ranging from neurodegenerative to tumor to infectious pathology. These slides were scanned at 40x using a Philips SG300 Ultra-Fast Scanner with Image Management System (IMS) files and converted to TIFFs. TIFFs and associated non-PHI metadata were then transferred to Data Core who processed and uploaded them to their server. Data Core created the website, which is now accessible

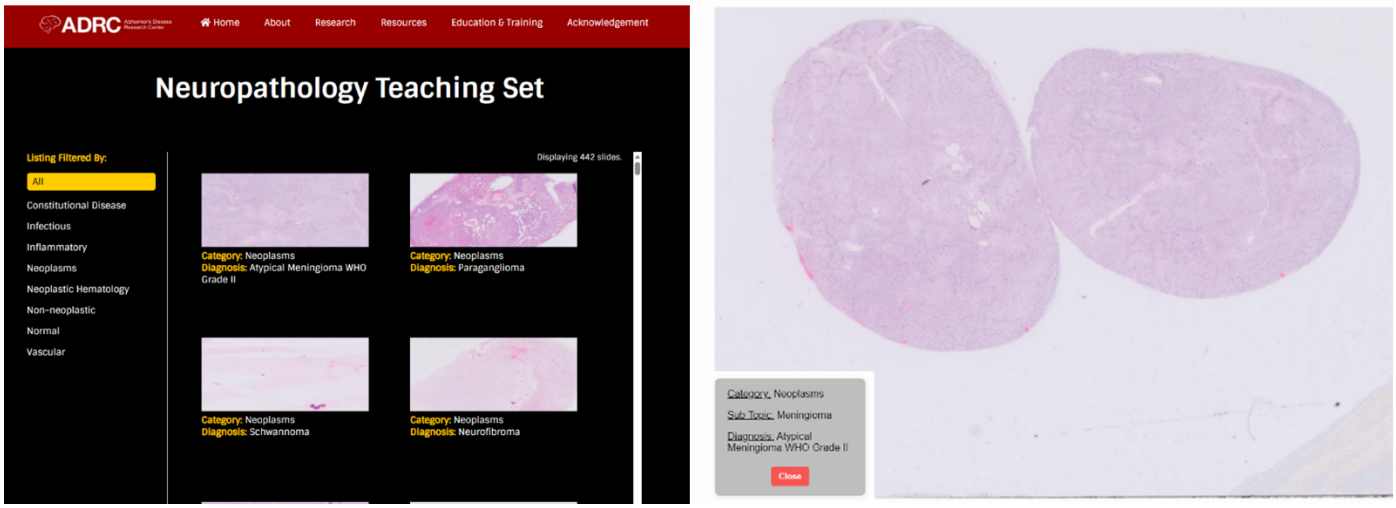


Figure 1. Neuropathology teaching set homepage (left) and example of slide viewer (right).

internally via password and soon will be live without password. The website and viewer show impressive functionality and rapid near real-time zoom. Slides are organized by topic (infectious, neoplastic, neurodegenerative, vascular, etc.) and the diagnosis is displayed. NP Core is able to independently maintain the content and update slide information, diagnoses, and thumbnails of the library. We received permission to include additional deidentified case content from Keck Medicine of USC and Children's Hospital of Los Angeles and will continue to add content to grow this teaching library. Accomplishing this library gives us confidence in our workflow to begin creation of the larger neurodegenerative digital library for our ADRC. As above (Aim 2) we continue to backfill our immunostaining of old cases to NACC11 standards in preparation for scanning and inclusion in our digital pathology library.

Aim 5: Interrogate the effect of the APOE ε4 allele on AD neuropathology and perivascular inflammation.

Key updates this reporting period:

- We selected the genes in the 1,000 gene panel for MERFISH.
- We chose the case cohort, including evaluating cases for DV200.
- Dr. Bienkowski's team has been performing pilot experiments to evaluate MERFISH on fixed human tissues.

This cycle, we focused on selecting our cohort, assessing cases for RNA quality, designing our MERFISH panel, and building the pipeline for our spatial transcriptomic studies looking at APOE *ε4+* and 40 APOE *ε4-* cases. We intend to use Multiplexed Error Robust Fluorescent *In Situ* Hybridization (MERFISH) spatial transcriptomics on hippocampal tissue sections (n=10 APOE *ε4+* and n=10 APOE *ε4-)* with a 1000 gene panel to determine APOE differential cell-type specific changes to brain cell types in low and high stage AD (n=5 per group). We anticipate that this study will provide new insights into the effects of APOE genotype.

This cycle, we created the gene panel to examine transcriptional changes in vascular, inflammatory, and proteostasis markers in the setting of APOE *ε4+* vs APOE *ε4-* genotypes. We scoured the literature and worked with collaborators across the ADRC (Yassine, Bienkowski) as well as the scientific advisers at Vizgen (maker of merscope) to identify the most useful genes for this study. We identified a cohort of patients with appropriate genotypes and demographics for this work and performed pilot DV200 on these cases. For these cases, we found DV200 of 50 (range 35-67) with Vizgen stating that the technique works on tissue with DV200 >30. The USC Histology Core has successfully mounted our fixed tissue onto the special Vizgen slides required for this technique. Using this tissue, the Bienkowski lab ran pilots of our ADRC fixed hippocampal tissue and was able to visualize and quantify RNA signals spatially. They are now troubleshooting the technique as runs have shown variable levels of RNA signals, suggesting RNA integrity of each individual sample is not fully captured by DV200 score and additional preliminary steps to determine highest quality tissue for analysis and appropriateness of fixed vs frozen tissue may be necessary.

Aim 6: Mentor the next generation of ADRD researchers and neuropathologists.

Key updates this reporting period:

- Drs. Hiniker, Hawes, and Bienkowski continue to train USC and visiting scholars including neuropathology fellows, residents, medical and grad students in neurodegenerative neuropathology
- Dr. Hiniker continues to co-direct the American Association of Neuropathologists (AANP) Neurodegenerative Scholars (R13) program

Our NP team is highly involved in mentoring, specifically we continue to train USC and visiting scholars including neuropathology fellows, residents, medical and grad students in neurodegenerative neuropathology.

Y6_ADRC RPPR_5P30AG066530-07 (07012025-03312026)

Dr. Hiniker co-directs the American Association of Neuropathologists (AANP) Neurodegenerative Scholars (R13) program (renewed in 2023 for 5 years, 4th percentile) with a goal to increase the pool of highly trained neurodegenerative neuropathologists. The USC Keck ACGME-accredited Neuropathology fellowship (in which Drs. Hiniker and Hawes are teaching faculty) as well as the Pathology and Neurology residencies include training in gross/microscopic findings of neurodegenerative diseases and criteria for diagnosis. Dr. Hiniker continues to lead a weekly gross brain cutting sessions, including ADRC cases, attended by residents, fellows, med students, and faculty. She also holds histology slide reviews attended by medical students, residents and clinical fellows.

B.4 What opportunities for training and professional development has Neuropathology Core provided?

The NP Core provides opportunities for trainees, including students, postdocs, and junior faculty, to participate in research projects and utilize neuropathology core specimens and data resulting in publications, poster and oral presentations. The NP Core has supported trainees and junior faculty to present at national and international meetings, which this cycle included accepted abstracts for local conferences including GLAM, national conferences including Society for Neuroscience, and international conferences including AD/PD in Copenhagen. The following trainee-driven abstracts and manuscripts were also presented or accepted during the 6 months reported here:

Presentations:

- Yang, W.*, Simon, S.*, Eichhorner, S., Mertens, J., & Hiniker, A. Investigating the role of cellular senescence in LRRK2 mutation-driven Parkinson's Disease. 2025 Geroscience Los Angeles Meeting (GLAM). September 5th 2025
- Qiu-Lan Ma, Bilal E. Kerman, Boyang Li, Shaowei Wang, Michael Renteln, Ashley Sanchez, Stan G. Louie, Zoe Arvanitakis, Hussein N. Yassine. Cytosolic phospholipase A2 mediates synaptic loss and cognitive dysfunction in Alzheimer's disease. Society for Neuroscience (SFN), November 15, 2025, in San Diego Convention Center
- Rodriguez R, Bauman AJ, Roberts A, Kim J, Knott A, Baez-Fortunato B, Chaney S, Church C, Hiniker A, Nelson A. Investigating the role of caldesmon in pericyte-mediated Alzheimer's disease pathology. Neuroscience 2025, LBP071.33, November 18, 2025, San Diego, CA.
- Yang, W.*, Simon, S.*, Eichhorner, S., Mertens, J., Hiniker, A. Interaction Between Cellular Senescence and LRRK2 Kinase Activity. AD/PD™ 2026 International Conference on Alzheimer’s and Parkinson’s Diseases and Related Neurological Disorders. March 17th-21st 2026 (accepted) Copenhagen, Denmark
- Galvan C*, Avina M*, Shavarebi F*, Unapanta A*, Shen Y*, Liu M*, Ramani B, Kampmann M, Hiniker A. Mechanistic Intersection of LRRK2 mutations and proteostasis in Parkinson’s Disease. AD/PD™ 2026 International Conference on Alzheimer’s and Parkinson’s Diseases and Related Neurological Disorders. March 17th-21st 2026 (accepted) Copenhagen, Denmark
- Peng I, Shin C, Avina M*, Hiniker A, Kraus A. Widespread distribution of tau seeds in brains lacking neurodegeneration pathology. AD/PD™ 2026 International Conference on Alzheimer’s and Parkinson’s Diseases and Related Neurological Disorders. March 17th-21st 2026 (accepted) Copenhagen, Denmark
- Coopman E, Faura J, Vicente CT, Bijnens B, Bidhan V, Alidadiani S, Wynants S, De Witte L, Meese T, Van den Broeck M, Cacace R, Sieben A, De Vil B, Baker M, van Blitterswijk M, Ghayal NB, King A, Bodi I, Donker Kaat LL, Mancuso R, Nicolas G, Deramecourt V, Hiniker A, Rissman RA, Murray ME, Petersen RC, Josephs KA, Whitwell JL, Nguyen AT, Reihard RR, Grinberg LT, Seeley WW, Kofler J, Lopez OL, Ang L, Al-Sarraj S, Troakes C, Cras P, Engelborghs S, Graff-Radford NR, Boeve BF, Ghetti B, Seelaar H, van Swieten JC, White CL, Dickson DW, Mackenzie IRA, De Coster W, Rademakers R. Unique and shared transcriptional signatures across FTLD-TDP subtypes: insights from single-nucleus RNA sequencing. . AD/PD™ 2026 International Conference on Alzheimer’s and Parkinson’s Diseases and Related Neurological Disorders. March 17th-21st 2026 (accepted) Copenhagen, Denmark.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022
FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Anne		Hiniker		Pathology Core Leader	225,700.00	3.6			67,710.00	21,464.00	89,174.00
2.	Michael		Bienkowski		Co-I	126,126.00	0.6			6,306.00	1,999.00	8,305.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **97,479.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Research Lab Tech	12.0	0.0	0.0	56,160.00	17,803.00	73,963.00
1	Project Assistant	12.0	0.0	0.0	57,544.00	18,241.00	75,785.00
1	Research Lab Tech	8.4	0.0	0.0	36,400.00	11,539.00	47,939.00
3	**Total Number Other Personnel**					**Total Other Personnel**	**197,687.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**295,166.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	8,892.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	50,960.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. Histology Services	17,000.00
9. Transportation	7,000.00
10. Diener	10,000.00
11. Additional staging histology and dig path	39,800.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**133,652.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**428,818.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	377,858.00	249,387.00
		Total Indirect Costs	**249,387.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**678,205.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	678,205.00

L. Budget Justification*	File Name: NP Core budget justification.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Budget Justification Neuropathology Core D Year 7

The fringe benefit rates are in accordance with the current University fringe rates calculated at 31.7%.

A. Senior/Key Personnel:

Annie Hiniker, M.D., Ph.D., Core Leader (30% effort, 3.6 cal. months): Dr. Hiniker, an Associate Professor at the University of Southern California Department of Pathology, is a physician-scientist who specializes in the pathology and biology of neurodegenerative disorders. Dr. Hiniker is a board-certified neuropathologist who will oversee the daily operations of the USC Neuropathology Core including supervising Core personnel, coordinating tissue storage and allocation, and diagnostic neuropathology. She was the neuropathologist (since 2018) and Co-Director (since 2022) of the UCSD Neuropathology Core before moving to USC. Dr. Hiniker will oversee autopsy protocol dissections, final diagnostic reports, tissue storage and distribution, digital pathology, and Core facilities maintenance. She will coordinate the approval process and distribution of brain tissue to other ADRC investigators, provide leadership for maintaining the ADRC biorepository, lead the biospecimen resource review committee and be the direct/initial contact for tissue requests. She will work with co-Core leader Dr. Debra Hawes (effort on Children's Hospital of Los Angeles subaward) and Dr. Michael Bienkowski to accomplish the aims of the core, specifically diagnosing, storing, and distributing tissue for ADRD research, building a sharable digital pathology database of the 1,100 USC neuropathology cases, testing the effect of ApoE genotype on inflammation and vascular pathology using digital pathology and transcriptomic methods, and mentoring the next generation of neuropathologists.

Michael Bienkowski, Ph.D. (5% effort, 0.6 cal. Months) Dr. Bienkowski is an Assistant Professor of Physiology and Neuroscience within the USC Stevens Neuroimaging and Informatics Institute and Zilkha Neurogenetic Institute. His formal training in neuroanatomy, connectomics, and microscopy image informatics and research experience in digital pathology and collaborative team science projects like the Mouse Connectome Project and NIH Brain Initiative Cell Census Network (BICCN) makes him well-suited to lead the proposed creation of the digital pathology image library. Dr. Bienkowski has contributed to several high impact studies published in Cell, Nature Neuroscience, and Nature Communications. His work creating the mouse Hippocampus Gene Expression Atlas (HGEA) is highly-regarded (99.9th percentile all-time and ranked #31 of all Nature Neuroscience articles ever tracked by Altmetric) and featured by the NIH Director's blog. Dr. Bienkowski will help supervise team members to digitize ADRC neuropathology slides, manage digital pathology image database, develop online pathology image viewer website, and perform MERFISH spatial transcriptomics experiments.

B. Other Personnel:

TBD, Research Lab Tech II (100% effort, 12 cal. months**):** This RLT II is responsible for fulfillment of the Neuropathology core tissue requests and assists in cutting and preparation of brain tissue for pathologic analysis. They are also responsible for coordinating the staining and processing of all 1,100 cases in the NP Core to NACC 11 diagnostic standards as well as backfilling ApoE E4 genotyping and facilitating the laboratory databasing and digitization of these cases performed by the designated RLT I (below). The technician maintains laboratory stocks and relevant supplies as well as all Neuropathology Core facilities and equipment, including biorepository freezer farm and fixed tissue bank, and assist in updating records of these repositories. They also maintains and updates standard operating procedures for all neuropathology core processes.

Carol Church, Project Assistant (100% effort, 12 cal. months**):** Ms. Church is responsible for the direct administrative oversight of the core including managing NACC reporting, tissue distribution to researchers, and interactions with Keck Hospital. She receives and manages the intake calls for brain donors and communicates with the diener, transport, and neuropath core teams to successfully obtain donor brains and ensure that they are transported successfully to the lab for preparation and storage. She assists the Core Director in fulfilling material transfer agreements (MTAs) for data, tissue, and biomarker samples. She assists with lab organization, ordering lab supplies, ensuring proper regulatory oversight occurs, and tissue inventory and procurement as needed.

TBD, Research Lab Tech I (70% effort, 8.4 cal,months): This RLT I will digitize, organize, track, and begin to work with the ADRC Data Core digitize ADRC pathology tissue, organize and manage digital pathology image files, and begin initial studies to facilitate analysis of digital pathology and spatial transcriptomics data. This tech will be trained by Dr. Hiniker to evaluate select newly performed immunohistochemical stains for tau (AT8), Aβ (GT-8), α-synuclein, and phospho-TDP-43 to stage all 1,100 cases for staging at NACC 11 standards. This tech will also assist in creating the detailed neuropathologic database of all cases at NACC 11 standards.

Detailed Budget: Equipment Cost:

Equipment: $0.

Detailed Budget: Other Direct Costs:

Diagnostic Histology Services: $17,000/year is budgeted for cost of histology services and supplies for an estimated 20 new cases a year, with full NACC 11 staging (17-20 regions processed and embedded for H&E, immunostains for tau (AT8), Aβ (GT-8), α-synuclein, and phospho-TDP-43) at ~$750/case as well as for the development of 1-2 new immunohistochemical and histochemical stains/year such as RD4 and RD3 tau and TAF-15.

General Lab Supplies: Supplies are budgeted at $8,892 per year. This includes costs of the following:

- General lab supplies (formalin, tubes, antibodies, liquid nitrogen, etc.) at an estimated cost of $3,892/year.
- General supplies for autopsy (body bags, gowns, masks) estimated at $2,000/year.
- Freezer repair, maintenance, and monitoring $3,000/year

Transportation: Transportation of ADRC autopsy patients at an estimated cost of $350/patient and 20 patients for an estimated cost of $7,000 for the year.

Dieners: Cost of on-call diener based on an estimated 20 cases at $10,000.

Scanning and digital pathology: $39,800. Cost for backfilling staining, scanning and digitizing some of bank of >1,100 cases. This cost is estimated at $40/slide to cut and immunostain, $20/slide to scan through the USC histology core.

Data Management and Sharing Costs: $0 Data management and storage will occur in the Data Core and is part of their budget.

Subaward: CHLA subaward will be utilized for Dr. Debra Hawes' salary.

Indirect costs:

Facilities and Administration/Indirect Costs were calculated using the modified total direct costs, which is the total direct costs less the equipment, tuition remission and only on the first $25,000 of subaward. The calculations are based on the USC DHHS Federal Rate Agreement from February 8, 2023. Per the agreement effective July 1, 2024 through June 30, 2026, the predetermined rate is 66%.

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: DVL1CMRMWRN9

Budget Type*: ❍ Project ● Subaward/Consortium

Enter name of Organization: Children's Hospital Los Angeles

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Debra		Hawes		Co-Core Lead	225,700.00	1.2			22,190.00	7,611.00	29,801.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **29,801.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
0	**Total Number Other Personnel**					**Total Other Personnel**	**0.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**29,801.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: DVL1CMRMWRN9

Budget Type*: ❍ Project ● Subaward/Consortium

Enter name of Organization: Children's Hospital Los Angeles

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: DVL1CMRMWRN9

Budget Type*: ❍ Project ● Subaward/Consortium

Enter name of Organization: Children's Hospital Los Angeles

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	0.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**0.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**29,801.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	71.0	29,801.00	21,159.00
		Total Indirect Costs	**21,159.00**

Cognizant Federal Agency

(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**50,960.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	50,960.00

L. Budget Justification* File Name:

RESEARCH & RELATED Budget {F-K} (Funds Requested)

A. COMPONENT COVER PAGE

Project Title: Core E: Outreach, Recruitment, and Engagement (ORE) Core
Component Project Lead Information: Aranda, Maria

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

The Outreach, Recruitment, and Engagement Core (ORE Core) serves as the primary bilingual and bi-cultural resource and liaison between the USC ADRC, our study participants, their caregivers/partners, and the larger community, i.e., the general public, people living with dementia and caregivers, professional care providers, and the lay community in Los Angeles County. Our aims are as follows:

Aim 1: Engage in participant recruitment & develop a research registry. It is well documented that among some populations the APOE e4 genotype is the strongest genetic risk factor for late-onset AD and that homozygotes for this allele may be at a twelve times higher risk for AD (2), though the risk conferred among the population in general and in persons tracing their roots to regions other than Western Europe is less clear, with conflicting results having been reported (3, 4). The ORE Core will maintain, characterize, and further develop the USC Alzheimer's Prevention Registry, "USC GeneScreen," a registry of persons in the LA County area interested in participating in studies and enriched for APOE e4 carriers. We will augment Latino participation (English, Spanish, low SES) in the registry given the higher prevalence and disease burden in this population and our catchment area composition. Through online enrollment/consent available in English and Spanish and APOE genotyping, USC GeneScreen will genotype and register a diverse pool of 1500 participants who are "study-ready" to enroll in AD/ADRD research.

Aim 2: Enhance recruitment from the LA County Department of Health Services (DHS) at the Rancho Los Amigos National Rehabilitation Center (RLANRC). The USC ADRC hosts two California Alzheimer's Disease Centers (CADCs), one of which is at RLANRC, part of the DHS safety-net community health system that provides health services to socioeconomically disadvantaged populations. The CADC at RLANRC will expand recruitment of this population into the Clinical Core and associated studies at the USC ADRC. In addition, the USC ADRC will partner with the newly established Dementia Care Aware (DCA) program within the DHS, the goal of which is earlier identification of older patients with cognitive impairment in the primary care setting and refer to specialists as appropriate.

Aim 3: Continue to engage in bi-directional education, information and resource sharing: In collaboration with our extensive network of community partners, we will continue to serve as a community resource in targeted engagement and educational outreach to the public-at-large, healthcare and human service providers, potential study participants, and current participants. We will communicate the latest ADRC- and AD/ADRD research findings and timely information on brain health, aging and wellness, ADRC study participation opportunities, and community resources for the general population in English and Spanish.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : B2_ORE Core_FINAL_1.27.26.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ORE Core_Prefinal_20JAN25.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

Dr. Maria Aranda (ORE core co-lead) provided a community presentation to over 300 Hispanic/Latino participants on the following:
Clinical Trials: Past and Present = Investigación Clínica: Pasado y Presente
Co-Sponsor: Alzheimer's Los Angeles
Conference: Advances in Alzheimer's Research
Downey, CA. September 26, 2025.

Dr. Maria Aranda also co-led a Community Advisory Board meeting with our CAB members on August 5, 2025. Our racially- and ethnically diverse CAB met at the California Endowment to discuss study projects, roles of members in recruitment, etc.

Dr. John Ringman gave a talk at the Spanish-language symposium targeting the Latino community sponsored by Alzheimer's Los Angeles. Titled, "Beneficios, Riesgos, y Limitations de los Tratamientos actualmente disposable para la Enfermedad de Alzheimer's (Benefits, Risks, and Limitations of currently available treatment for Alzheimer's Disease). He also spoke on the Genetics of Alzheimer's disease to a lay audience at the Verdugo Hills Brain Health Forum on November 1, 2025, which was attended by 250 community members.

On October 25, 2025, in association with the ORE Core, Dr. Ringman spoke at the USC ADRC's weekly research conference on "How to Write Grants for Observational Research," attended by students, fellows, and junior faculty members as well as more senior researchers.

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Recently, donor funds have been identified (the Pattiz Fundation) that can be utilized to enhance recruitment into the ADRC from the Rancho CADC (e.g. by covering the salary of a recruitment-focused research associate) going forward. We will continue with our Community Advisory Board meetings, community engagement, and referrals to the ADRC.

B2. What was accomplished under these goals?

Specific Aim #1:

The USC GeneScreen registry has continued to enroll community members in the greater Los Angeles area who are interested in participating in Alzheimer's disease and related dementias research. For the current reporting period, we have added 19 new members, 39 new online enrollments. The registry continues to mail APOE genotype testing kits to prospective members, including both English- and Spanish-speaking participants, as part of its recruitment efforts. Study personnel have initiated contact with 1507 leads since registry inception. To date, the registry has enrolled 668 members, including 201 males and 467 females; of these, 188 are APOE ε4 carriers.

As part of our expansion efforts to connect community members with Alzheimer's disease and related dementia studies, we have received Institutional Review Board approval for studies to partner with GeneScreen for APOE genotype screening and recruitment. Such efforts will expand the registry and enhance recruitment of diverse participants into Alzheimer's disease and related dementia research, thereby advancing the field.

Specific Aim #2:

In the reporting period of this RPPR, USC ADRC faculty continue to see patients at Rancho Los Amigos in the Geriatric and Neurobehavior Clinic there and hold weekly consensus conferences during which cases are discussed and appropriate work-up, diagnoses, treatment recommendations, and when appropriate, referral to research initiated. Among the referrals from Rancho enrolled in research at the ADRC at the USC Health Sciences Campus (HSC) campus is an additional case of a patient suffering from the otherwise rare A431E mutation in *PSEN1*. This enrollment will allow future involvement of the patient and family members in additional studies at the ADRC, including ultimate neuropathological examination in the Brain Research Study.

Also, during this reporting period, despite the more limited relationship between USC and the Los Angeles County General Hospital, Dr. Elizabeth Joe from the ADRC has established and staffs a weekly memory disorders specialty clinic at the LAC General which is a short walk from the USC HSC. With Dr. Joe's intimate involvement in the Clinical Core, this will undoubtedly lead to enhanced recruitment of underserved and underrepresented persons into the ADRC.

Specific Aim #3:

We continue to mount extensive community engagement and educational activities and events In collaboration with our wide network of community partners comprised of persons living with dementia, their care partners and families, community residents, human services organizations, public sector departments and units, health care and social services providers, small community groups, faith-based organizations, voluntary health organizations, universities, among others. Given our extensive networking and longstanding community relationships, we will continue to serve as the primary community resource in targeted engagement and educational outreach to the public-at-large, healthcare and human service providers, potential study participants, current participants, etc. We will communicate the latest ADRC- and AD/ADRD research findings and timely information on brain health, aging and wellness, ADRC study participation opportunities, and community resources for diverse audiences in English and Spanish, and other languages where we are able to do so.

Table E1: ORE Core Group Events and Activities – Year 6 (7/1/25-03/31/26)				
	Event or Activity	# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
1	Trial Match: Alzheimer's Association	1	Hispanic / Latina/o / Latinx 100%	100% female
2	Mexican American Opportunity Foundation	1	Hispanic / Latina/o / Latinx 100%	100% female
3	Into the Dark Blue	1	Hispanic / Latina/o / Latinx 100%	100% male
4	Dementia Springs Foundation	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male
5	Previous Study Participant	1	Hispanic / Latina/o / Latinx 100%	100% female
6	Previous Study Participant	1	Hispanic / Latina/o / Latinx 100%	100% female
7	Previous Study Participant	1	Hispanic / Latina/o / Latinx 100%	100% female
8	City of La Puente	1	Hispanic / Latina/o / Latinx 100%	100% female
9	City of Santa Monica Human Services Division	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
10	Solano County: Health & Social Services Public Health Division Older & Disabled Adult Services	1	Asian / Asian American 100%	100% female
11	Stanislaus County: Community Services Agency	1	Hispanic / Latina/o / Latinx 100%	100% female
12	Marin Aging and Disability Institute	1	Hispanic / Latina/o / Latinx 100%	100% female
13	SEIU Local 2015	1	African American / Black	100% female
14	City of Santa Monica, Homelessness Prevention and Intervention	1	Asian / Asian American 100%	100% female

	Event or Activity	# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
15	Easton Center for Alzheimer's Research and Care @UCLA	1	Hispanic / Latina/o / Latinx 100%	100% female
16	Palmview Park Community Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
17	Teamsters Union Local 572	1	Hispanic / Latina/o / Latinx 100%	100% female
18	Aging Life Care Association	1	Hispanic / Latina/o / Latinx 100%	100% male
19	Aging Life Care Association	1	Hispanic / Latina/o / Latinx 100%	100% male
20	West Covina Community Center: Cool Kids of Summer Event	200	Hispanic / Latina/o / Latinx 100%	20% male 80% female
21	Estrada Courts Community Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
22	Estrada Courts Community Center	50	Hispanic / Latina/o / Latinx 100%	50% male 50% female
23	Human Services Association (HSA)	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
24	YWCA in Covina	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
25	West Covina Adult Health Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
26	La Puente Adult Day Health Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
27	Forever Young Adult Day Health Care	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
28	City of Azusa	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
29	City of Vista	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
30	Madera County	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
31	Merced County Human Services Agency	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
32	OC Social Services Agency	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
33	San Luis Obispo	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
34	Sacramento Agency on Aging \ Area 4	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
35	Alliance on Aging, Inc.	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
36	Maywood Healthcare and Wellness Centre	1	Hispanic / Latina/o / Latinx 100%	100% female
37	League of United Latin American Citizens	1000	Hispanic / Latina/o / Latinx 100%	30% male 70 female
38	Vista Adult Day Health Care Center	1	Hispanic / Latina/o / Latinx 100%	100% female
39	Raymundo Zacarias, Los Angeles County Department of Public Health	1	Hispanic / Latina/o / Latinx 100%	100% male
40	Marie Torres, PhD	1	Hispanic / Latina/o / Latinx 100%	100% female
41	Participant Inquiry	1	Hispanic / Latina/o / Latinx 100%	100% female
42	Participant Inquiry	1	Hispanic / Latina/o / Latinx 100%	100% female
43	City of Santa Monica Senior Housing Task Force Monthly Meeting	20	Hispanic / Latina/o / Latinx 10% Non-Latino White/ White, not of Latino ethnicity 90%	20% male 80% female
44	USC Suzanne Dworak-Peck School of Social Work	1	Hispanic / Latina/o / Latinx 100%	100% female
45	George Caballero	1	Hispanic / Latina/o / Latinx 100%	50% male 50% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
46	USC Suzanne Dworak-Peck School of Social Work	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male
47	Global Alzheimer's Platform & Mobile Research Unit	1	Hispanic / Latina/o / Latinx 100%	100% female
48	California Department of Public Health Chronic Disease Control Branch- Alzheimer's Disease	2	Hispanic / Latina/o / Latinx 50% Asian / Asian American 50%	100% female
49	Rosa Rangel, Southern California Clinical and Translational Science Institute	1	Hispanic / Latina/o / Latinx 100%	100% female
50	LAPD Southeast Division	1	Hispanic / Latina/o / Latinx 100%	100% female
51	Dr. Rodrigo Rodriguez, Keck Medicine of USC	2	Hispanic / Latina/o / Latinx 100%	100% female
52	USC Suzanne Dwork Peck School of Social Work: Student Life and Engagement	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male
53	Participant Inquiry	1	Hispanic / Latina/o / Latinx 100%	100% female
54	Cell-Ed	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
55	Department of Public Social Services	1	African American / Black 100%	100% female
56	Los Angeles Police Department	1	Biracial / Mixed Race 100%	100% male
57	Adventist Health White Memorial Charitable Foundation	1	Hispanic / Latina/o / Latinx 100%	100% male
58	Los Angeles County Fire Department	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male

	Event or Activity	# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
59	Quiosquito Comunitario De Bell Gardens: Bell Gardens Veterans Park	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
60	Alzheimer's Los Angeles: Cal State University Dominguez Hills	100	Hispanic / Latina/o / Latinx 5% African American / Black 80% Asian / Asian American 5% Non-Latino White/ White, not of Latino ethnicity 5% Biracial / Mixed Race 5%	50% male 50% female
61	Alzheimer's Los Angeles: Skirball Cultural Center	275	Hispanic / Latina/o / Latinx 5% African American / Black 5% Asian / Asian American 20% Non-Latino White/ White, not of Latino ethnicity 70%	30% male 70 female
62	Alzheimer's Los Angeles: Rio Hondo By Wedgewood	250	Hispanic / Latina/o / Latinx 100%	20% male 80% female
63	City of Los Angeles Office of the Mayor	1	African American / Black 100%	100% female
64	Los Angeles County Aging & Disabilities Department	1	Hispanic / Latina/o / Latinx 100%	100% female
65	USC Suzanne Dwork Peck School of Social Work	1	Hispanic / Latina/o / Latinx 100%	100% female
66	USC Suzanne Dwork Peck School of Social Work	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
67	USC Suzanne Dwork Peck School of Social Work	1	Hispanic / Latina/o / Latinx 100%	100% female
68	Linda Lopez: Memory Guide Solutions	1	Hispanic / Latina/o / Latinx 100%	100% female
69	Merrill Gardens at West Covina	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
70	California Mission Inn	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
71	Regency Grand at West Covina - Cascade Living	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
72	Inspired Elderly Care Living, Assisted Living Facility, Memory Care, Board and Care, and Retirement Community	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
73	Atria Senior Living	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
74	Placer Independent Resource Services	2	Hispanic / Latina/o / Latinx 50% Non-Latino White/ White, not of Latino ethnicity 50%	100% female
75	AltaMed	1	Hispanic / Latina/o / Latinx 100%	100% female
76	Whittier Area Community Church	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
77	Morningstar Thrift Shop	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
78	Park Regency Care Center	1	Hispanic / Latina/o / Latinx 100%	100% female
79	Whittier Heights Assisted Living &Memory Care	1	Hispanic / Latina/o / Latinx 100%	100% female
80	Oakmont Memory Care	2	Hispanic / Latina/o / Latinx 100%	100% female
81	Our Lady of Guadalupe Catholic Church	1	Hispanic / Latina/o / Latinx 100%	100% female
82	La Mirada Heights Sky Active Living	1	Hispanic / Latina/o / Latinx 100%	100% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
83	ONEGeneration Wilkinson Center: Wilkinson Center	300	Hispanic / Latina/o / Latinx 10% African American / Black 5% Asian / Asian American 60% Non-Latino White/ White, not of Latino ethnicity 25%	25% male 75% female
84	Community Resource Center for Aging USC Verdugo Hills Hospital	1	Hispanic / Latina/o / Latinx 100%	100% female
85	Christine Kaneshige, USC Keck School of Medicine	1	Hispanic / Latina/o / Latinx 100%	100% female
86	USC Suzanne Dwork Peck School of Social Work	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male
87	USC Suzanne Dwork Peck School of Social Work	19	Hispanic / Latina/o / Latinx 100%	10% male 90% female
88	Arboleta Senior Apartments	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
89	Walnut Valley Senior Living	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
90	Claremont Manor Retirement Community	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
91	Meridian at Chino	100	Hispanic / Latina/o / Latinx 50% Non-Latino White/ White, not of Latino ethnicity 50%	50% male 50% female
92	The Terrace at Villa Verde	100	Hispanic / Latina/o / Latinx 50% Non-Latino White/ White, not of Latino ethnicity 50%	50% male 50% female
93	El Descanso Retirement Home	100	Hispanic / Latina/o / Latinx 50% Non-Latino White/ White, not of Latino ethnicity 50%	50% male 50% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
94	White Memorial Hospital	100	Hispanic / Latina/o / Latinx 80% Biracial / Mixed Race 20%	50% male 50% female
95	AltaMed	1	Hispanic / Latina/o / Latinx 100%	100% male
96	Sun Valley Research Center	3	Hispanic / Latina/o / Latinx 100%	100% female
97	Wash & Wear	2	Hispanic / Latina/o / Latinx 70% Non-Latino White/ White, not of Latino ethnicity 30%	50% male 50% female
98	Braille Institute	1	Hispanic / Latina/o / Latinx 100%	100% female
99	OneGeneration	1	Hispanic / Latina/o / Latinx 100%	100% female
100	City of Gardena	1	Hispanic / Latina/o / Latinx 100%	100% female
101	City of La Habra Senior Center	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% female
102	SoCal Premiere Lifestyle Health and Active Aging	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
103	Alzheimer's Los Angeles: City Of Bell Gardens Recreation and Community Services Neighborhood Youth Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
104	Lincoln Heights Senior Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
105	Our Lady of Guadalupe Church: City of La Habra	100	Hispanic / Latina/o / Latinx 40% African American / Black 5% Asian / Asian American 5% Non-Latino White/ White, not of Latino ethnicity 50%	50% male 50% female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
106	Department of Recreation & Community Services - Senior Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
107	Alzheimer's Los Angeles: Huntington Park Community Center	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
108	Mexican American Opportunity Foundation (MAOF): AARP Los Angeles	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
109	Montebello Senior Center	100	Hispanic / Latina/o / Latinx 90% Asian / Asian American 5% Non-Latino White/ White, not of Latino ethnicity 5%	50% male 50% female
110	County of Los Angeles Department of Parks & Recreation: Las Palmas Park	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
111	YWCA Community Center: Glendale CA	100	Hispanic / Latina/o / Latinx 10% Non-Latino White/ White, not of Latino ethnicity 90%	50% male 50% female
112	Sterling Organization: Plaza Mexico	500	Hispanic / Latina/o / Latinx 100%	50% male 50% female
113	Alzheimer's Los Angeles: Las Palmas Park	150	Hispanic / Latina/o / Latinx 95% African American / Black 1% Asian / Asian American 1% Non-Latino White/ White, not of Latino ethnicity 3%	50% male 50% female
114	Participant Inquiry	1	Hispanic / Latina/o / Latinx 100%	100% female
115	AltaMed: Southeast Rio Vista YMCA	50	Hispanic / Latina/o / Latinx 99% Non-Latino White/ White, not of Latino ethnicity 1%	30% male 70 female

Event or Activity		# Attendees/ Participants	Audience Breakdown or Target Group(s)	Gender (%)
116	City of Maywood	1	Non-Latino White/ White, not of Latino ethnicity 100%	100% male
117	Participant Inquiry	1	Hispanic / Latina/o / Latinx 100%	100% female
118	97.9 FM La Raza	1	Hispanic / Latina/o / Latinx 100%	100% male
119	97.9 FM La Raza	1	Hispanic / Latina/o / Latinx 100%	100% female
120	97.9 FM La Raza	1	Hispanic / Latina/o / Latinx 100%	100% female
121	97.9 FM La Raza	1	Hispanic / Latina/o / Latinx 100%	100% female
122	SELA Collaborative	1	Hispanic / Latina/o / Latinx 100%	100% female
123	Southeast Rio Vista YMCA at Maywood Center	300	Hispanic / Latina/o / Latinx 90% Non-Latino White/ White, not of Latino ethnicity 10%	30% male 70 female
124	Sterling Organization: Plaza Mexico	300	Hispanic / Latina/o / Latinx 90% African American / Black 5% Non-Latino White/ White, not of Latino 5%	50% male 50% female
125	YMCA Maywood	100	Hispanic / Latina/o / Latinx 100%	50% male 50% female
126	YMCA Maywood	150	Hispanic / Latina/o / Latinx 100%	50% male 50% female
127	Sterling Organization: Plaza Mexico	500	Hispanic / Latina/o / Latinx 90% Biracial / Mixed Race 10%	50% male 50% female
128	Estrada Courts Community Center	100	Hispanic / Latina/o / Latinx 90% Biracial / Mixed Race 10%	50% male 50% female

Table 2: Summary Table of ORE Core Activities – Year 6 (7/1/25-03/31/26)	
Type of Activity – REPORTED BY ADRC ORE CORE	**Number**
Community/Population -targeted events – (*total number of items in EIT Table 1*) [1]	128
Community/Population - related publications[1] See narrative	
Community/Population -related grants – (*total number of items in Admin Table 6*) [2]	
Community/Population -targeted materials (brochures, newsletter articles, videos) [2] narrative (B2)	0
Community/Population groups contacted (churches, community organizations) [1] See Table 1	155

Community/Population n groups collaborating [1]	128
Advisory committee meetings [1]	1
Scientists in Cores and Projects (Faculty and post-doctoral fellows) [1] (*report as ratio with all scientists*)	
Staff (non-faculty) in Cores and Projects [1] *(report as ratio with all staff)*	
Satellites [2]	n/a

[1] = current reporting period
[2] = cumulative

Community/Population - Related Activities Narrative
Refer to Section E. for this information.

B.4 What opportunities for training and professional development has the project provided?

ORE Core provided training and professional development to the following trainees:

Dr. Aranda provided training/advising/mentoring to the following scholars:

1. David Camacho, PhD, Assistant Professor, University of Illinois Chicago; disabilities and sexual-gender minorities; pain; loneliness; cognition; culturally adapted mental health interventions; Latinx aging and mental health; qualitative methods.
2. Mirella Diaz-Santos, PhD, Assistant Professor-in-Residence, Department of Neurology, David Geffen School of Medicine, University of California, Los Angeles: prevention, and early detection/diagnosis of Alzheimer's disease and related dementias, science of recruitment and retention for clinical trials, mixed-method Community Partnered Participatory Research
3. Marianne Chanti-Ketterl, PhD, Assistant Professor, Duke University, North Carolina: the role of exposome, environmental exposures such as pesticides on cognition; social determinants of health; traumatic brain injuries; metabolic biomarkers; resilience; and cognitive training; ADRD outreach and engagement with historically marginalized populations; engagement and recruitment of underrepresented groups in ADRD research.
4. Stacey Schepens Niemiec, PhD, University of Southern California: occupational therapy; cognition; culturally adapted mental health interventions for low-income Latinx populations; promotora-mediated community interventions
5. Angela Gutierrez, PhD, Western University of Health Sciences, Pomona, CA: stress and cognitive decline; minority aging; health disparities; psychosocial resources; promotoras and health interventions; mental health; Latinx populations.
6. Janet Choi, PhD, Assistant Professor, Caruso Department of Otolaryngology-Head and Neck Surgery, Otology, Neurotology, Lateral Skull Base Surgery, Keck Medicine of USC: hearing loss, hearing aid use, Alzheimer's Disease and Related Dementias; mortality; medical and surgical treatments for hearing loss
7. Jiaming Liang, PhD, UT Houston Health, Houston, Tx. Assistant Professor, School of Public Health, aging and mental health; dementia caregiving; brain health adaptations; intervention development and healthy aging
8. Elizabeth Ambriz, PhD, University of California, San Francisco: successful aging; ADRD knowledge and illness perceptions; health disparities; rural aging; culturally adapted interventions for successful aging in rural settings

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Amytis		Towfighi		Co-I	221,900.00	0.3			5,548.00	1,759.00	7,307.00
2.	John		Ringman		ORE Co-Lead	225,700.00	0.8			15,122.00	4,794.00	19,916.00
3.	Maria		Aranda		ORE Core Lead	225,700.00		0.9	0.6	28,213.00	8,943.00	37,156.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **64,379.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Project Assistant	12.0	0.0	0.0	60,000.00	19,020.00	79,020.00
1	Project Assistant	12.0	0.0	0.0	60,000.00	19,020.00	79,020.00
1	Programmer	1.2	0.0	0.0	12,444.00	3,944.00	16,388.00
1	Project Assistant (GeneScreen)	7.2	0.0	0.0	24,000.00	7,608.00	31,608.00
4	**Total Number Other Personnel**					**Total Other Personnel**	**206,036.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**270,415.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	4,020.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**4,020.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	500.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. GeneScreen Recruitment and Genotyping	23,119.00
9. Community Advisory Board	300.00
10. Community Advisory Board Meetings	2,700.00
11. Participant Incentives- Brain Health	4,000.00
12. Printing costs	3,000.00
13. Community events/vendor fees for recruitment, education and information	1,500.00
14. Retention, recruitment and dissemination event	8,400.00
15. Education/informational/promotional/recruitment/retention items	3,000.00
16. Advertisement costs	3,000.00
17.	0.00
Total Other Direct Costs	**49,519.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**323,954.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	323,954.00	213,809.00
		Total Indirect Costs	**213,809.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**537,763.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	537,763.00

L. Budget Justification* File Name: ORE Core Just.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – ORE CORE

Key Personnel:

María P. Aranda, PhD, MPA, MSW, LCSW, Core Lead/Co-Investigator, 0.90 Academic Month, 0.60 Summer Month in Years 1 -5. Dr. Aranda is Professor with joint appointments in Social Work, Gerontology and Neurology. Dr. Aranda is Executive Director of the USC Edward R. Roybal Institute on Aging at the USC Suzanne Dworak-Peck School of Social Work. She is an internationally recognized sociobehavioral scholar and licensed clinician in the areas of Alzheimer's disease and related dementias, depression, psychosocial interventions for individuals and family caregivers, community engagement, health disparities, and recruitment of underrepresented populations in clinical trials, observational research, and qualitative studies. She has extensive experience in the integration of a health equity lens in randomized trials, qualitative methods, large-scale epidemiological research, stakeholder engagement, and innovative recruitment and retention protocols. Her publications on health equity, intervention development and testing, inclusion of underrepresented groups in ADRD research, and mentoring of early career scholars have informed the national research and mentoring agenda.

Along with Dr. Ringman, she will provide oversight for all research activities including research personnel training and supervision, scientific integrity, programmatic/operational activities, as well as specified timeline, administrative and financial aspects of the USC ADRC Outreach, Recruitment and Engagement Core. She is a family caregiver and is bilingual (English/Spanish) and bicultural. Dr. Aranda's effort level will be 10% in YRS 1 and 5. We also request three months of summer salary in YRS 1 through 5 to devote to project implementation and reporting.

John Ringman, MD, MS, Co-Core Lead/Co-Investigator, 0.80 Calendar Months in Years 1 – 5. Dr. Ringman is a Professor of Clinical Neurology in the USC Department of Neurology who has been involved in AD research since his fellowship in behavioral neurology in 1999. His research focus is on dementia among persons of Latino origin, both of autosomal dominant origin and of late onset, and he is site co-lead investigator on the MarkVCID project at USC. He sees patients both at the USC Health Science Center in East Los Angeles and at the Rancho Los Amigos Rehabilitation Center (RLARC) within the LAC DHS system. He speaks Spanish moderately well and is actively involved in recruiting and enrolling Spanish-speaking participants into the ADRC. He and Dr. Aranda have collaborated on community-based outreach activities and provider events since 2019. In addition to identifying and enrolling participants through his clinical activities, he will continue to work closely with Dr. Aranda, and faculty and staff at the RLARC on Gene Screen, the development of educational materials, and on other outreach activities. Dr. Ringman's effort level will be 2% in YRS 1 through 5.

Amy Towfighi, M.D, Co-Investigator, 0.30 Calendar Months in Years 1 – 5. Dr. Towfighi is the Associate Chair for USC Department of Neurology and the Director of Neurological Services for the Los Angeles County (LAC) Department of Health Services (DHS) and Chief of Neurology at the Los Angeles General Medical Center. In her role as Chief of Neurological Services for LAG, she will support the ORE Core's plan to enhance diverse perspectives (Specific Aim #2) and facilitate forward an increased partnership with the USC ADRC and support efforts to streamline the appropriate recruitment of DHS patients for potential research participation.

Other Personnel:

Project Assistant (2) – TBD, 12 Calendar Months in Years 1 – 5. We request support for two project assistants to assist with the project aims. Each person will work closely with Drs. Aranda and Ringman to manage day-to-day operations, training and supervision, data gathering and study flow activities, coordinating research activities with service providers, community collaborators, and others to ensure recruitment and intervention goals are attained. The persons will be responsible for monitoring and conducting recruitment strategies, subject remuneration, study flow follow-up, scheduling and conducting participant interviews and related activities, translating, and interpreting for bilingual audiences and participants, disseminating information and findings in diverse formats to both scientific and lay audiences, handling and troubleshooting of challenges that may emerge, and administering alternative options when necessary. Effort will be 12 months at 100% for each 1.0 FTE in YRS 1 through 5.

Adrienne Mackay, MS, Programmer, 1.2 Calendar Months in Years 1 – 5. Ms. Mackay will be responsible for setting up recruitment advertisements, mail saliva kits, triage phone calls, and maintain the REDCap database. She has worked with Dr. Yassine on the USC GeneScreen since its and has deep project and cohort knowledge. Among the 300 persons who will be genotyped, we estimate that we will recruit about 100 individuals per year to the ADRC cohort and affiliated studies.

Project Assistant (GeneScreen), TBD, 7.2 Calendar Months in Years 1 – 5. This person will verify registry eligibility and receive participants into the ADRC cohort and affiliated studies.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years.

Other Costs:

GeneScreen Recruitment and Genotyping: We will recruit and analyze genotype on a cohort of 300 participants through GeneScreen per year and add an estimated 100 participants to the ADRC Active Cohort.

Community Advisory Board (CAB): We request funds to remunerate our x CAB members for their work advising the ORE Core and ADRC. We have letters of support from all board members.

Nora Manzanilla, Chair. Ms. Manzanilla will serve as the CAB Chair during YRS 1 through 5 for planning meetings. Planning meetings will consist of developing, reading, writing meeting materials including agenda, handouts, translations, and technology planning for online meetings with the full board. Remuneration will be $100 per three planning meetings per year (1.5 hours each meeting), ($300 for YRS 1 through 5).

CAB Members. Up to nine CAB members will receive remuneration for their work on the project by attending 3 meetings per year. During their service they will 1) gain new knowledge about AD/ADRD, brain health, aging, caregiving, center findings, recruitment/retention challenges, 2) advise on matters related to ongoing needs assessment and strategic planning on outreach, recruitment and engagement, and 3) co-design community events and materials. Our esteemed CAB members submitted letters in support of our application. See Letters of Support. Remuneration: 9 members, 3 meetings per year at $100 per meeting (1.5 hours each meeting) for board members during YRs 1 through 5. ($2,700 for YRS 1 through 5).

Participant Incentives: We request funds to remunerate participants who participate in the Brain Health for Me Study/Registry. These participants will be approached to participate in ADRC studies. Such remuneration incentivizes participants to participate and maintain participation over the course of the study. Prior research suggests that 1) establishing relationship is a key to study participation, and 2) monetary incentives are effective at eliciting participant participation and retention. We will target a total of 50 subjects who will receive a $20 gift card for three observation assessments: T1-baseline, + T2 through T4 follow-ups (50 x 4 x $20=$4,000 for YRS 1 through 5).

Printing Costs: Funds are requested to pay for cost related to printing large quantities of recruitment and retention materials, informed consents, certificates of study completion, and miscellaneous distributions and mailings of flyers, brochures, surveys, etc. Although much of our work is conducted online, we acknowledge that a significant part of our outreach and engagement necessitates hands-on and in-person recruitment activities with printed materials especially for people with low SES, low health and digital literacy, and no/low access to the internet. Lastly, we host many onsite community events and activities that require printed materials, posters, signage, and business cards. ($3,000 for YRS 1 through 5).

Community events/vendor fees for recruitment, education and information: Our extensive experience in promoting community events and activities to support the ADRC underscores the importance of partnering with local groups, agencies, organizations, and other non-profit companies. We request funds to pay for vendoring fees that these partners charge to promote our studies and distribute educational and informational materials to their patrons and consumers that we approach to engage and recruit study/registry participants. ($1,500 for YRS 1 through 5).

Retention-recruitment-dissemination event: Funding is needed to cover the costs of a large community event intended to showcase the latest findings in cognitive aging, brain health, and clinical trials advances. Intended for a bilingual (English-Spanish) audience with simultaneous interpretation services, this half-day event will be sponsored by the ORE Core to highlight these advances—many from our own USC ADRC—and to provide enhanced opportunities for study participation for community residents. We also will leverage the event to invite our current study participants and their close other to attend for retention and recruitment purposes. Event budget items include: venue fee: $1,500; event catering: $2,800; IT/AV tech services: $2,200; simultaneous interpretation services: $1,900. ($8,400 for YRS 1 through 5).

Education/informational/promotional/recruitment/retention items: We request promotional items to assist with recruitment and retention. Based on our extensive experience with diverse stakeholders and the principle of "giving before we receive," we will provide promotional items for the purpose of recruitment and retention of study participants. Therefore, we request funding to purchase items that will assist with recruitment of potential participants, and retention of participants associated with the ADRC, e.g., inexpensive items such as exercise bands, pedometers, healthy eating information, stress busters, etc. We request funding for $3,000 for YRS 1 through 5.

Advertisement costs for research recruitment, including print and online advertisement to cast a wide net in order to reach projected recruitment numbers. We found in our previous work that relying on single-focus, recruitment strategies were not effective to recruit the numbers and diversity of subjects. Drawing from lessons learned, we will be devoting significant time and resources to generate interest in study participation. This entails generating multiple strategies (print, broadcast media, ads in large-subscription newspapers, videography to post on social media, YouTube, etc.). We need to ramp up quickly and sustain our efforts with these recruitment strategies to generate interest and research participation/retention of diverse and underrepresented groups, including Spanish-language participants. Our prior networks include strong relationships with local TV and radio stations which enhance participation from local audiences in under-resourced communities, particularly Spanish-speakers who rely heavily on traditional ethnic TV and radio outlets for broadcasting of community events. We request a total of $3,000 for YRS 1 through 5.

Miscellaneous office supplies: We request funds to purchase miscellaneous project supplies for the day-to-day operational tasks of the project. These include items such project binders, thank you cards, postage stamps, secure USB keys, and paper shredder. ($500 for YRS 1 through 5.)

Travel:

Mileage reimbursement for project related travel is requested at the current IRS reimbursement rate. Funds will be used for research staff to implement recruitment activities, dissemination, and travel to community locations for usual and typical project activities to meet study aims, recruit participants and enhance relationships with community agencies and groups. Our local region covers a wide, highly urbanized area that necessitates travelling long distances and incurring parking fees. For 2024, the IRS standard mileage rate for business use of a vehicle will be between 66 and 68 cents per mile. We estimate the following: ~6,000 miles/year x $.67/mile=$4,020.

Facilities and Administrative Costs: Facilities and Administration Costs are based on USC' federally negotiated rate (dated May 29, 2024) of 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

A. COMPONENT COVER PAGE

Project Title: Core F: Biomarker Core
Component Project Lead Information: Rissman, Robert

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

The Biomarker Core's (Core F) primary goal is to serve as a central resource for the ADRC to collect, store, track and share plasma, serum, DNA and cerebrospinal fluid (CSF) biosamples. In addition, the Biomarker Core contributes to the goals of the ADRC by performing genetic and biochemical phenotyping of participants followed by the Clinical Core through analyses of APOE genotype and Amyloid, Tau and Neurodegeneration (A/T/N) biomarkers. Data from these gold standard analyses will facilitate characterization of longitudinal staging and progression of Clinical Core participants and provide a solid foundation to enable and support national ADRC research efforts and the rich environment of investigator-initiated research at USC. Biomarker phenotyping at this level of detail can allow the ADRC Clinical Core to personalize the analysis of risk, mechanisms and contributing pathologies, which has major implications for development of treatments and advancement of National Alzheimer's Project Act (NAPA) goals. Our large biobank and foundational biomarker data has historically been and continues to serve as a critical vehicle to attract and establish the careers of young investigators as well as support ongoing, multi-year collaborations with established USC ADRC-affiliated investigators. Aim 1: Biofluid collection, processing, storage and sharing. 1) Coordinate collection, processing and storage of CSF, PBMCs, blood, plasma and DNA from ADRC Clinical Core participants using national ADRC Best Practices SOPs. 2) Perform analyses of CSF, plasma and DNA relevant to ADRD using validated methods to facilitate characterization of the ADRC cohort. 3) Provide ADRC samples and core resources (bioassay, DNA analyses, extractions, processing pipelines) to a broad array of investigators in support of new research projects and foster career development of scientists in biomarker research. Provide expertise for training in biomarker methods. Share plasma and PBMCs with the National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD). Aim 2: Facilitate biomarker research through an integrated informatics toolkit. Maintain a state-of-the-art biosample inventory linked with the ADRC DMSC to provide internal and external investigators with CSF and plasma samples, DNA and derived data in support of ADRD research and novel biomarker discovery. Aim 3: Accelerate local, national and international biomarker discovery efforts. Leverage ADRC biospecimens to develop new methods that inform on neuropathology, diagnosis, progression, prognosis and other research applicable to ADRD. 1) Leverage ADRC specimens for discovery projects at USC and nationally. 2) Work with Industry partners to develop and validate novel assays and platforms.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ADRC Y6_Biomarker Core.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

Conference and other presentations

1. Core F PI - Robert Rissman: Speaker, AC immune Clinical Advisory Board, San Diego, California Biomarker-based recruitment and endpoints in preclinical AD.
2. Core F PI - Robert Rissman: Speaker, Clinical Trials in Alzheimer's Disease (CTAD), San Diego, California Biotech Showcase Session – Blood-Based Diagnostics and Biomarker
3. Core F PI - Robert Rissman: Speaker, Clinical Trials in Alzheimer's Disease (CTAD), San Diego, California, Biomarker and Bench-to-Bedside Research
4. Core F PI - Robert Rissman: Speaker, Alamar Precision Biomarker Summit, La Jolla, California Alpha-Synuclein mechanisms and therapeutics in Alzheimer's disease
5. Core F PI - Robert Rissman: Speaker, Biogen AD Emerging Insights Day, Boston, Massachusetts
6. Core F PI - Robert Rissman: Speaker, University of Wisconsin ADRC External Advisory Board
7. Core F PI - Robert Rissman: Speaker, IMPACT-AD, San Diego, California ADRD Trial Outcomes: Fluid Biomarkers, Institute Methods and Protocols for Advancement of Clinical Trials in ADRD
8. Core F PI - Robert Rissman: Session Chair, Alzheimer’s Research Day, Alzheimer’s Therapeutic Research Institute, University of Southern California, San Diego, California
9. Core F PI - Robert Rissman: Speaker, Alzheimer’s Association International Conference (AAIC), Toronto, Canada, Global Biomarker Standardization Consortium (GBSC), Toronto, Canada. Using blood-based biomarkers for clinical trial eligibility: Platform Selection & Target Populations
10. Sagare A.P., E.B. Joe, H.C. Chui, J.M. Ringman, M.G. Harrington, L.S. Schneider, H.N. Yassine, C. Torres-Sepulveda, G. Spillard, B.V. Zlokovic, D.A. Nation, A.W. Toga, R.A. Rissman. (2025) A specific and sensitive assay for the detection of APOE4 carrier status and protein levels in biofluids. Neuroscience Meeting Planner. Society for Neuroscience, 2025 San Diego. Online.
11. Ortega N., J. Pa, Y. Shi, J. Yue, A. Sagare, H. Chui, J. Ringman. (2025) Sex differences in the relationship between tau pathology, neurodegeneration and depression in carriers of the PSEN1 A431E mutation causing early onset autosomal dominant Alzheimer’s disease. Organization for the Study of Sex Differences (OSSD) 19th Annual Meeting, Albuquerque, NM.
12. Ortega NE., J. Pa, Y. Shi, J. Yue, A.P. Sagare, H.C. Chui, J.M. Ringman. (2025) Neuropsychiatric symptoms are associated with tau pathology in autosomal dominant early onset Alzheimer’s disease caused by PSEN1 and APP mutations. Alzheimer’s Association International Conference (AAIC) July 27-31, 2025, Toronto, Canada.
13. Zhang M, R. Rust, A.P. Sagare, K. Kisler, Y. Kim, C. Griffin, Y. Wang, V. Clementel, C. Torres-Sepulveda, J. TCW, B.V. Zlokovic, M.P. Coba. (2025) Molecular signature and functional properties of human pluripotent stem cell-derived brain pericytes. Neuroscience Meeting Planner. Society for Neuroscience, 2025 San Diego. Online.
14. Kisler K., Z. Dai, A. Sagare, A. Grond, B. Zlokovic, M.P. Coba. (2025) Rare PICALM coding mutation (murine H465R) protects against amyloid-beta pathology in a murine 5xFAD model. Neuroscience Meeting Planner. Society for Neuroscience, 2025 San Diego. Online.
15. Torres-Sepulveda C, M. Zhang, A. Sagare, R. Rust, A. Chakhoyan, K. Kisler, V. Clementel, G. Spillard, D.M. Holtzman, B. Zlokovic, H.N. Yassine, M.P. Coba. (2025) APOE4 and Tau P301S act synergistically to increase the severity of vascular pathology and neurodegeneration. Neuroscience Meeting Planner. Society for Neuroscience, 2025 San Diego. Online.
16. Petkus AJ., B. Rowe, L. Montoya, A. Sagare, J. Ringman. (2025) Associations between motor and sensory abilities with memory and executive function in Latinos: Comparison between autosomal dominant and late-onset Alzheimer’s disease. Alzheimer’s Association International Conference (AAIC) July 27-31, 2025, Toronto, Canada.

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Under the leadership of Dr. Rissman and the resources available at USC’s Alzheimer’s Therapeutic Research Institute (ATRI) and USC Zilkha Neurogenetic Institute (ZNI), Biomarker Core F will continue to achieve all AIMS and provide standardized methods for biofluid sample acquisition and storage, biomarker analysis, resources, and expertise on biomarkers to all USC ADRC investigators.

CORE F – Biomarker Core, B.2

The Biomarker Core has continued to work closely with ADRC investigators. Specific progress towards each aim during the reporting period is detailed below.

AIM 1. Biofluid collection, processing, storage and sharing.
The ADRC Biomarker Core F has standardized the collection and storage of biofluid samples for each participant for analysis, cohort characterization, and sample sharing. During the reporting period, Biomarker Core processed and stored plasma and CSF samples (**Table 1**). The core shared an additional 50 plasma samples with Dr. Ebrahim Zandi for further validation of a Luminex-based plasma AD biomarker assay (PMID: 40356374).

Table 1. Summary of biofluid samples processed

Biofluid	Number of participants	Number of aliquots stored
Plasma	143	770
White Blood Cells	143	283
Serum	18	389
CSF	3	67

AIM 2. Facilitate biomarker research through an integrated informatics toolkit.
The Biomarker Core analyzed over 450 CSF samples for standard AD biomarkers (e.g., Aβ, total tau, p-tau) and performed *APOE* genotyping on over 500 DNA samples. The core provided samples for national and international collaborations. Plasma samples processed in the clinic lab are shipped to NCRAD (Indiana University) for Aβ40, Aβ42, pTau217, NfL, and GFAP (Quanterix Simoa HD-X) and Aβ40, Aβ42, pTau281 (Fujirebio Lumipulse) assays. APOE genotypes are obtained from NCRAD. The Biomarker core maintains a sample inventory to provide internal and external investigators with CSF and plasma samples, DNA, and derived data, and supports ADRD research and novel biomarker discovery. The Core utilizes the REDCap centralized data system to manage data and requests in collaboration with other cores, including Administrative, Clinical, Imaging, Data Management, and Statistics.

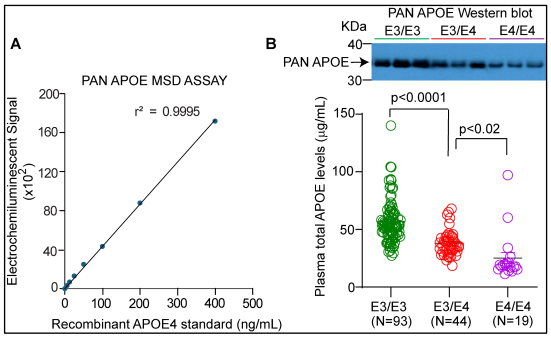

FIGURE 1. Performance of the MSD PAN APOE assay. (**A**) Representative standard curve showing recombinant APOE4 standard concentration and the corresponding ECL signal exhibits a linear curve fit over a tested concentration range from 6.25 to 400 ng/mL; r^2 = 0.9995. (**B**) Western blot and scatterplot of plasma APOE levels in APOE4 non-carrier and APOE4 carriers. Error bars (mean ± SEM). E3/E3 carriers (56.75 ± 1.92 µg/mL), E3/E4 carriers (38.11 ± 1.65 µg/mL), and E4/E4 carriers (25.2 ± 4.68 µg/mL).

The Biomarker Core has continued to refine its novel biomarker battery, including new plasma biomarkers such as PAN APOE (**Fig. 1**) and APOE4-specific ultrasensitive MSD assays for measuring levels in plasma, serum, and CSF (**Fig. 2**). The ratio of plasma total APOE levels and APOE4 levels allows for the identification of APOE4 homozygous (E4/E4) and heterozygous (E3/E4) carriers with high accuracy (**Fig. 3**). The core presented the data on the developed assay at the annual SFN meeting in 2025.The optimized APOE4-specific MSD assay may be helpful for: rapid screening of patients in the clinic for APOE4 carrier status using one microliter plasma or CSF in less than two hours; understanding AD risk in different ethnicities; and analyzing steady-state levels of APOE4 in the CSF or plasma after therapeutic intervention aimed at. reducing *APOE4* expression. The core has recently analyzed total APOE levels in 188 CSF and 188 plasma samples and provided the data to Dr. H. Yassine for analyses and manuscript preparation. We will use the developed methods to analyze APOE and APOE4 levels in biofluids and study their association with lipids, cerebrovascular

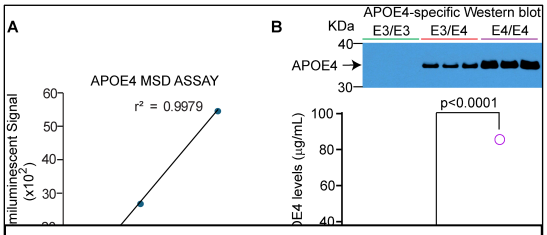

FIGURE 2. Performance of the MSD APOE4 assay. (A) Representative standard curve showing recombinant APOE4 standard concentration and the corresponding electrochemiluminescence signal exhibits a linear curve fit over a tested concentration range from 6.25 to 400 ng/mL; r^2 = 0.9979. (B) Western blot and scatterplot of plasma APOE4 levels in E3/E4 and E4/E4 carriers. Error bars (mean ± SEM). E3/E4 carriers (8.26 ± 0.34 µg/mL) and E4/E4 carriers (23.59 ± 3.66 µg/mL).

and blood-brain barrier dysfunction, inflammation, AD biomarkers, correlated positively with the plasma Aβ42/40 ratio (**Fig. 4A**) and correlated negatively with the amyloid positivity score (APS) provided by C2N diagnostics and cognitive decline. For example,
plasma total APOE levels (**Fig. 4B**). sPDGFRβ has recently been proposed as a vascular injury (v) marker that should become part of the Aβ/tau/neurodegeneration ATN matrix to read as ATN(vi) matrix that allows Aβ, tau, and vascular injury to contribute to neurodegeneration and early cognitive impairment independently of each other or by acting together (PMID:34239130; PMID:36040482). Plasma sPDGFRβ levels were higher in *APOE4* vs *APOE3* individuals and further increased in *APOE4* individuals in the CDR 0.5 group (**Fig. 5A,B**). The present study suggests that plasma sPDGFRβ is a useful biomarker of brain pericyte/BBB damage in *APOE4* carriers, and its elevated levels may predict cognitive dysfunction in *APOE4* carriers, as we previously showed for CSF sPDGFRβ (PMID: 32376954). ADRC investigators have communicated these findings for publication.

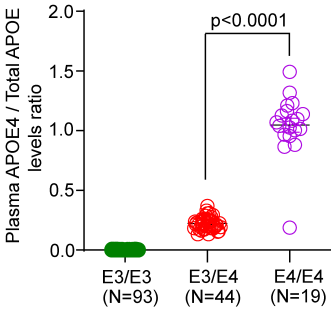

FIGURE 3. ADRC participants with an apoe4 allele had higher total plasma APOE4 using our newly developed assay.

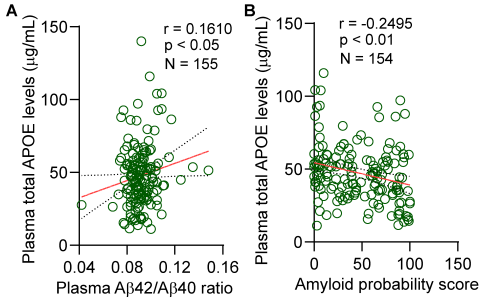

FIGURE 4. Correlation between plasma total APOE levels and plasma Aβ42/40 ratio (**A**), and plasma total APOE levels and C2N amyloid probability score (**B**)

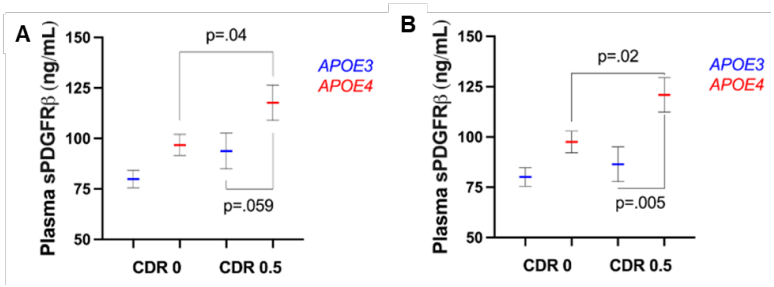

FIGURE 5. Plasma sPDGFRβ levels after controlling for plasma $A\beta_{42/40}$ ratio and plasma pTau217 levels. (**A**) Estimated marginal mean ± standard error of measurement (SEM) of plasma sPDGFRβ after controlling for age, sex, education, VRF burden and plasma $A\beta_{42/40}$ levels and (**B**) in a separate model after controlling for age, sex, education, VRF burden and plasma pTau217.

AIM 3. Accelerate local, national and international biomarker discovery efforts.
For this reporting period, Dr. Robert Rissman has been recruited to lead the Biomarker Core. Leveraging ADRC and ACTC biospecimens, the Biomarker Core has expanded its biomarker assay and biorepository infrastructure to accelerate biomarker discovery by developing and validating novel assays and platforms at USC and USC ATRI, and through collaborations with Industry partners. The Biomarker Core has been working with Meso Scale Discovery scientists to develop new and sensitive assays for novel biomarkers of interest, which besides sPDGFRβ and p-Tau217 in CSF and plasma, include cyclophilin A (CypA), matrix metalloprotease-9 (MMP9), pleiotrophin (growth factor expressed by brain capillary pericytes), ferritin, soluble LRP1 (sLRP1), active TGF-β1, Aβ oligomers and tau oligomers in CSF, aminopeptidase N, as well as recently optimized APOE, APOE4, total pTau217 and brain-derived pTau217 and pTau181 (BD-pTau217 and BD-pTau181) plasma assays.

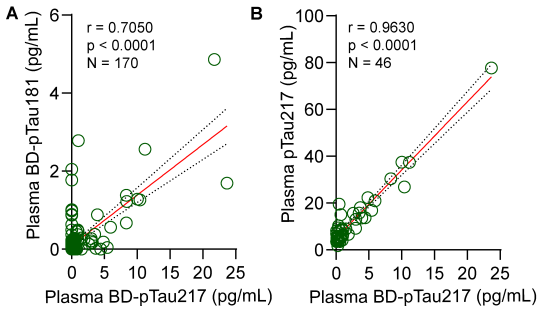

FIGURE 6. BD forms of p-Tau181 and pTau217 are positively correlated to their respective plasma levels in ADRC participants.

Our recent analysis revealed that plasma BD-Tau181 and BD-Tau217 (**Fig. 6A**) and total pTau217 and BD-pTau217 (**Fig. 6B**) show significant positive correlation. The ADRC Biobank continues to maintain PBMCs and iPSCs. The Biomarker Core successfully differentiated and characterized pericyte-like cells derived from human iPSCs to test their therapeutic potential in preclinical studies as a cell therapy for alleviating neurovascular dysfunction in neurodegenerative diseases by implanting them in brain regions of mice. A revised manuscript is currently under consideration for publication (PMID: 40291694).

The Biomarker core will continue working with Dr. Toga and the Data Core to manage data and requests in collaboration with other cores, including Administrative, Clinical, Imaging, and Statistics. The Data Core provides secure data access to the local research community and external users for data analysis.

B.4 What opportunities for training and professional development has the project provided?

A variety of training opportunities are available at USC's Alzheimer's Therapeutic Research Institute (ATRI) and USC Zilkha Neurogenetic Institute (ZNI) Biomarker Core for biofluid processing, APOE, APP, and PSEN1 genotyping, performing MSD and ELISA assays, and assay development. Below, we highlight the specific training opportunities as well as attendance and presentations made at conferences over the reporting period.

Dr. Sagare trained Nikaya Coleman, a Biomarker Core technician for processing and storage of biofluid samples.

Dr. Sagare presented a poster entitled "A specific and sensitive assay for detection of APOE4 carrier status and the protein levels in biofluids" at the Society for Neuroscience Conference in San Diego from November 15-19, 2025, and at the Zach Hall Lecture and Symposium held at Zilkha Neurogenetic Institute, University of Southern California on December 8, 2025.

Dr. Sagare attended the ARGOnaut Club Voyager Summit 2025, an annual event dedicated to advancing biomarker science and translational innovation in healthcare, held in person/virtually in Napa, California on October 13-15, 2025.

Dr. Rissman (PI Biomarker Core) presented data at many conferences and events in the reporting period: see C.1: Conferences and Other Presentations.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

Category	Explanation
Other	We developed a sensitive assay to measure total APOE and APOE4 protein levels in CSF and plasma.
Other	We developed a sensitive assay to detect a brain microvascular pericyte injury biomarker sPDGFRβ in the cerebrospinal fluid (CSF) that indicates BBB disruption (See section C.5).

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

Category	Explanation
Other	Protocols: Additionally, Biomarkers Core has developed new assays to measure CSF and plasma levels of soluble platelet-derived growth factor-β (sPDGFRβ), CSF cyclophilin-A (CypA), and MMP9 levels. We have identified a combination of capture and detecting antibodies for the development of the Meso Scale Discovery (MSD) assay to detect soluble sPDGFRβ and CypA (Sweeney et al., Alzheimer's & Dementia, 2020; 16(6):821–830).
Other	We recently developed a sensitive assay to measure total APOE and APOE4 protein levels in CSF and plasma.
Other	Protocols: The Biomarkers Core has developed a novel panel of molecular biomarkers in CSF/plasma that integrate neurovascular unit (NVU) cell- and system-specific biomarkers with established AD biomarkers such as Aβ, total tau, and pTau, to simultaneously analyze changes in vascular/BBB injury markers (Qalb, fibrinogen, Cyclophilin-A, active MMP9 and ferritin) in relation to inflammatory, neuronal injury, Aβ and tau biomarkers.
Other	We have previously validated sPDGFRβ as a sensitive and novel biomarker of pericyte injury (Sagare et al., Neuroscience Letters, 2015; 607:97-101) in the brain that reflects BBB disruption in MCI individuals (Montagne et al., Neuron, 2015; 85, 296–302), independent of Aβ and tau (Nation et al., Nature Medicine, 2019; 25(2):270-276). We have found that high baseline levels of sPDGFRβ predicted future cognitive decline in APOE4 carriers (Montagne et al., Nature, 2020; 581:71-76).

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

See the ORE and neuropathology cores for details on clinical core resource sharing of clinical materials and research specimens. See Administrative Tables and overall component for description of resource sharing.

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Abhay		Sagare		Co-I	99,277.00	1.2			9,928.00	3,147.00	13,075.00
2.	Robert		Rissman		Biomarker Core Lead	225,700.00	1.2			22,570.00	7,155.00	29,725.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **42,800.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Lab Tech	12.0	0.0	0.0	62,400.00	19,780.00	82,180.00
1	Lab Manager	2.4	0.0	0.0	36,626.00	11,610.00	48,236.00
1	Lab Assistant	1.2	0.0	0.0	12,405.00	3,933.00	16,338.00
3	**Total Number Other Personnel**					**Total Other Personnel**	**146,754.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**189,554.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
1. Freezers	35,000.00
Total funds requested for all equipment listed in the attached file	
Total Equipment	**35,000.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	80,592.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	10,844.00
7. Alterations and Renovations	0.00
8. Equipment Maintenance/Service	5,000.00
9. Data Management server fees	3,000.00
10. HDL particle biomarkers	25,875.00
11. Communications	10,000.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**135,311.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**359,865.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	209,722.00	138,417.00
2. MTDC - Off Campus	26.0	104,299.00	27,118.00
		Total Indirect Costs	**165,535.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**525,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	525,400.00

L. Budget Justification*	File Name: Biomarker Core just.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDJET JUSTIFICATION – BIOMARKER CORE

KEY PERSONNEL

Robert Rissman, Ph.D., Core Leader/Co=Investigator 1.20 Calendar Months, in Years 1-5: Dr. Rissman leads the USC ADRC Biomarker Core, is Professor of Physiology and Neuroscience at USC, and Director of the Neuroscience Translational Research Division at Alzheimer's Therapeutic Research Institute (ATRI). As the ADRC Biomarker Core Lead, Dr. Rissman will be responsible for the management of biosamples, bioassays and genetic analyses that will be performed by the core and will oversee the daily repository operations, and will supervise personnel who will receive, biobank, analyze, and share biofluids (plasma, CSF, and DNA) from the ADRC cohort. Under Dr. Rissman's Leadership, the ADRC Biomarker Core will perform analysis of ATN biomarkers in CSF and ApoE genotype from DNA and specific vascular related biomarkers in plasma will be measured in collaboration with MARK VCID. Dr. Rissman will work closely to other members the study team to ensure timely and accurate assay of samples, QA/QC the obtained results, provide appropriate analysis and interpretation of the data, and collaborate with the USC ADRC executive team to interpret the results.

OTHER PERSONNEL

USC ADRC Biomarker Core, HSC Los Angeles

Abhay Sagare, PhD, Research Associate, 1.20 Calendar Months in Years 1-5. Dr. Sagare will oversee day to day operations for Dr. Rissman's ADRC Biomarker Core space in Los Angeles and liase directly with the clinical core to receive all biospecimens. He will work with Mr. Ramirez to process blood samples to plasma and aliquot both plasma and CSF samples for analyses. In terms of analyses, Dr. Sagare will execute vascular biomarkers as part of the MARK VCID program to characterize the ADRC cohort. Dr. Sagare will work closely with Dr. Rissman's ATRI personnel, Barlow and Mullen, to coordinate the shipment of biosamples for long-term storage at the San Diego facility. Dr. Sagare reports to Dr. Rissman.

TBN, Lab Technician, 12.0 Calendar Months in Years 1-5. The individual will work with Dr. Sagare for biosample processing, aliquoting and will manage freezer inventories, as well as, prepare sample inventory lists for analysis for long-term storage. He will provide overall support to complete all work associated with this proposal. The individual reports to Drs. Rissman and Sagare.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years. Fringe benefits do not apply to student employees.

USC-ATRI Research and Biomarker Laboratories and Proj Management, ATRI San Diego

James Barlow, MT, Research Lab Manager, 2.40 Calendar Months in Years 1-5. Barlow is a CLIA-certified medical technician and manager of Dr. Rissman's research lab at USC ATRI. Barlow will coordinate with Dr. Sagare and Mr. Ramirez to manage the receipt and transfer of biofluids for long-term biobanking and sharing. Mr. Barlow will also complete ATN bioassays in CSF and plasma, perform DNA extractions and ApoE genotyping. Mr. Barlow reports to Dr. Rissman.

Kimberley Mullen, Research Program Manager, 1.20 Calendar Months in Years1-5. Mullen will manage overall operational functions of the project including procurement of study related materials and supplies, coordinate preventative maintenance of research lab instruments and equipment, submit and manage Material Transfer Agreements associated with sharing ADRC cohort biofluids, compose regular progress reports, and assists Dr. Rissman with presenting data for publications. Ms. Mullen reports to Dr. Rissman.

EQUIPMENT

$140,000 (YR2-5: $35,000 per year)
Four Panasonic Freezer TwinGuard Series -86°C/-80°C Ultra-Low Temperature Freezer with monitoring system.

OTHER DIRECT COSTS

Materials and Supplies:

General lab supplies ($12,592 YR2): including serological multichannel pipette and filter tips (10 ul, 20ul, 200ul, 1000ul), barcoded cryovials, Nalgene plastic boxes, Eppendorf tubes, centrifuge tubes, SealPlate films for assays, biohazard safety supplies, TTP-343C label printer and thermal poly labels for storage.

Molecular Biology materials ($68,000/yr, YRS 1-5): Materials necessary to run Lumipulse CSF assays, plates, and antibodies, reagentson 40% of participants. Unique ApoE vascular measures to complement those of NCRAD.

HDL particle biomarkers: ($25,875/yr, YRS 1-5): Cost per sample - $225 x 115 samples (75 plasma and 40 CSF).

Data Management ($3,000/yr, YR 1-5): Dell G5 15-5590 Laptop and Optiplex 7070 Minitower Core i7-9700 Server

Equipment or facility rental/user fees:
The ATRI is housed in off-campus leased space and is subject to the 26% off-campus MTDC rate. For the associated pro-rated effort on this project occurring in off campus space, we anticipate the cost to be $10,844 ($2.03x1272sqftx12x0.325 FTE) per year.

Communication:
Remote access, phone/data line access, software licenses necessary for this study are expected to be $10,000 per year: The Coordinating Center pays for communications charges which are directly related to the individuals working on the project. Based on adjusted historical comparative usage averages, costs include: all phone line usage, all fax line usage, voice mail, weekly conference calls between the Coordinating Center and the project leader, and other long-distance charges required to conduct Coordinating Center business; internet connection, mobile internet access necessary for personnel serving as primary business continuity access plan and necessary for travel and remote work. There are recurring charges for the various necessary data manipulation programs, data storage and transmission, and resource software and licenses such as Atlassian, BOX, Adobe, BPM, SAS, Sigma Plot, National Drug Data File, Pegasus, MedDRA, WHODrug, etc. for the staff involved with the project.

Other Project Costs:
$5,000 per year for Preventative Maintenance: A portion of costs for maintenance and repair of all related equipment for the duration of this project, specifically for biosample processing (centrifuges, plate readers, pipette, etc.) and biorepository (Twin Guard freezer and freezer monitoring system).

FACILITIES & ADMINISTRATIVE COSTS
Facilities and Administration Costs are based on USC' federally negotiated rate (dated May 29, 2024) of 66% as of July 1, 2024. The F&A rate is 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

USC ATRI San Diego: Facilities and Administration costs are calculated at the USC institutional rate 26% for the activities taking place at USC in off-campus space.

A. COMPONENT COVER PAGE

Project Title: Core G: Research Education Component (REC)
Component Project Lead Information: Yassine, Hussein N

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

The mission of the University of Southern California (USC) Alzheimer's Disease Research Center (ADRC) Research Education Component (REC) is to direct the scientific careers of scientists from multiple disciplines into a career in Alzheimer's disease and related dementias (ADRD) research in general, with an emphasis on age associated vascular/metabolic processes that affects brain health and ADRD risk in populations in Los Angeles (LA) County. We achieve this mission with the following aims:
1. Support ADRC scientists through multidisciplinary training program, pilot funding and a symposium. Our goal is to make ADRC scientists competitive in securing an award for an independent, investigator-initiated research project.
2. Support the career development of post-doctoral investigators who have shown remarkable professional promise for a 2-year post-doctoral fellowship.
3. Monitor the progress of ADRC scientists during the period of the award as well as through their early career milestones (e.g., receipt of an independent grant). ADRC scientist alumni will be supported in continued involvement in activities that support and improve both their career options and the experience of the next cohort of pilot grantees.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : REC_B2_PreFinal_5P30AG066530-07_07012025-03312026.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : REC_B4_Final.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

To accomplish project goals, the ADRC will: (1) support the training of ADRC scientists in collaboration with the University of Southern California Resource Center for Minority Aging Research (RCMAR), including pilot funding and the hosting of an annual symposium; (2) support postdoctoral career development to help investigators secure career development awards; and (3) track metrics for ADRC scientists throughout their early careers.

B.2 What was accomplished under these goals?

Progress for Aim 1: Over the past three years, we created the ADRC REC Webinar Series (Table G2. ADRC REC EVENTS) to foster thought-provoking discussions of current research and provide new perspectives on emerging developments in the field. We also supported several REC scholars (Table G1. Summary table of REC SCHOLARS and TRAINEES).

During the reporting period we held 13 REC Webinar seminars each attended by an average of 29 attendees. Example topics include: “Alzheimer's Disease risk factors APOE4 and PICALM at the blood-brain barrier,” and “Targeting brain insulin signaling in Alzheimer’s: A focus on the BBB.” To further scientific progress and promote collaborative research, a research symposium titled *Advances in Personalized Brain Health* was held (February 2025) and attended by 150+ individuals, including 26 featured speakers. The second annual symposium is planned for April 14, 2026. ADRC scholars are encouraged to apply to serve as featured speakers or poster presenters. We also sent a contingent of 25 scholars from USC to the Southern California ADC Symposium. Our representative cohort included five poster presenters and two featured speakers (Neuropathology Core Leader, Annie Hiniker and Clinical Core Co-Leader, Elizabeth Joe). This was successful opportunity for advancing career development in Alzheimer’s Disease research.

Progress for Aim 2:
Several scholars and trainees (Table G1) received funding to support their research development across diverse topic areas, including omics, AD biomarkers, genetic risk, and modifiable risk factors, among others. Additionally, the USC Alzheimer’s Disease Research Center (ADRC) Development Project (DP) Program is a competitive pilot funding mechanism designed to stimulate innovative, high-impact research in Alzheimer’s disease and related dementias (ADRD). The program supports projects that are novel, mechanistically grounded, and well aligned with ADRC scientific priorities, while also fostering career development for early-stage faculty and postdoctoral investigators. We selected and supported the career development of three scholars:

Qiulan Ma: This project investigated whether cPLA2 and its eicosanoid products are linked to mitochondrial stress, neurotransmitter changes, and Alzheimer’s pathology in synaptosomes and gliosomes from postmortem brain tissues across NCI, MCI, and AD groups. Using metabolomics and advanced imaging, Dr. Ma assessed mitochondrial energy metabolism, lipid profiles, and cPLA2 colocalization with MAVS. They expect higher eicosanoids and cPLA2 activity in AD, correlating with synaptic loss, mitochondrial dysfunction, gliosis, and tau pathology.

Tobias Ulmer: This project focused on understanding the structure and signaling of CD33m, a minor isoform of the immune receptor CD33 that lacks the ligand-binding IgV domain but shows unexpected functional activity. Dr. Ulmer will produce mutant IgC2 domains, characterize their oligomerization states, and determine structural details to uncover how CD33m mediates signaling. The goal is to reveal novel mechanisms by which CD33m contributes to immune regulation, with implications for Alzheimer’s and related diseases.

Isaac Asante: This project explored how APOE3 versus APOE4 genotypes affect cPLA2-driven lipid and metabolite changes in brain and plasma, hypothesizing that APOE4 causes hyperactivation of cPLA2 leading to neuroinflammatory profiles. Specific Aim 1 characterizes genotype-linked lipidomic signatures and potential biomarker pathways, while Specific Aim 2 tests whether ASB14780, a cPLA2 inhibitor, reverses these changes in APOE4 mice. The study combined metabolomics, statistical modeling, and drug-response analysis to establish cPLA2 as a therapeutic target.

For the next 2025–2026 Development Project cycle launched September 2025, separate tracks were offered for faculty investigators and postdoctoral scholars, and expectations tailored to career stage. There were 8 faculty track and 10 postdoctoral track applicants. Applications were evaluated by the ADRC Executive Team (Directors, Assistant Director, Core Leads) and external scientific reviewers using an NIH-aligned framework

emphasizing overall impact, innovation, rigor, feasibility, investigator potential, and alignment with ADRC priorities.

At the time of writing, six Development Projects have been selected for funding: 2 Faculty-led projects and 4 Postdoctoral-led projects.

Progress for Aim 3:
We continue to support the career development of our scholars. For example, Olega Eremina secured a K99 from NIH to pursue Nanoparticle- based Raman imaging of AD neuropathology and Janet Choi KL2 from USC-SCIT to pursue the connection between hearing loss and cognition. Janet Choi is now competing for an NIH K23.

Specific Barriers and Steps to Overcome:
Our next goal is to increase participation from ADRD scholars, track those trainees as well as their training needs, participation, and contributions to AD research within a system that is accurate and low burden and this will be achieved by working with ADRCMAR. We are collaborating with other programs to identify optimal methods for tracking, supporting, and retaining trainees. Furthermore, we will continue to bring new investigators from across campus into the AD research field.

B.4 What opportunities for training and professional development has the project provided?

The ADRC has provided professional development opportunities through symposia and has sponsored a weekly seminar series to foster cross-disciplinary collaboration and discussion in the field of Alzheimer's disease and related dementias.

Table G1: Summary Table of REC Scholars and Trainees
Year 6 (07/01/25 – 03/31/26)

Last name, First name	Current or Past Trainee	Scholar Trainee	Level of Training	Area of Research	Progress	Currently Affiliated with an ADRC?	Working/ collaborating within the ADRD field?
Elizabeth Joe	Current	Scholar	MD	Dementia diagnoses in a safety net health system	CTSI KL2 Scholar	Yes	Yes
Jieran Choupan	Current	Scholar	PhD	Volumetric changes of perivascular spaces in early cognitive decline	Oral presentation at the Spring National ADC meeting 2021	Yes	Yes
MariCarmen Pachicano	Current	Scholar	PhD candidate	Neuropathology	Diversity Supplement	Yes	Yes
Angela Gutierrez	Past	Scholar	PhD	Social Work/ Recruitment Science	Post Doc Provost Scholar	Yes	Yes
Isaac Asante	Current	Scholar	PhD	Omics, AD biomarkers	Assistant Professor, Diversity Supplement	Yes	Yes
Amy Christiansen	Current	Scholar	PhD	AD mouse models, ketogenic diet	Assistant Professor	Yes	Yes
Olga Eremina	Current	Scholar	PhD	Nanoparticle-based Raman imaging of AD neuropathology	K99 applicant	Yes	Yes

Table G1: Summary Table of REC Scholars and Trainees
Year 6 (07/01/25 – 03/31/26)

Last name, First name	Current or Past Trainee	Scholar Trainee	Level of Training	Area of Research	Progress	Currently Affiliated with an ADRC?	Working/ collaborating within the ADRD field?
Max Andrew Thorwald	Current	Scholar	PhD	APOE nuclear translocation in AD	K99 applicant	Yes	Yes
Ashim Ahuja	Current	Scholar	MD	Clinical and Biochemical Markers of Lewy Body disease	Clinical Fellow	Yes	Yes
Aaron Lim	Past	Scholar	PhD	Alcohol and financial vulnerability in AD	K23 applicant (scored and resubmitted)	No	No
Laura Fenton	Current	Trainee	PhD student	Neuroimaging and financial vulnerability in AD	Presentation at NACC REC poster session, now intern at	Yes	Yes
Daisy Noriega-Makarskyy	Current	Trainee	PhD student	Cognition and cognitive decline in Latinos	Graduate student	Yes	Yes
Cassidy Molinare	Current	Trainee	PhD student	Memory and executive functioning in decision making	Graduate student	Yes	Yes
Belinda Zhang	Current	Scholar	PsyD	Neuroimaging and cognitive correlates of financial vulnerability in aging	Postdoctoral fellow (planned K99/R00 submission)	Yes	Yes
Emma Oyen	Current	Trainee	PhD student	Structural neuroimaging and early cognitive impairment	Graduate student	Yes	Yes

Table G1: Summary Table of REC Scholars and Trainees
Year 6 (07/01/25 – 03/31/26)

Last name, First name	Current or Past Trainee	Scholar Trainee	Level of Training	Area of Research	Progress	Currently Affiliated with an ADRC?	Working/ collaborating within the ADRD field?
Melanie Leguizamon	Current	Trainee	PhD student	Physical activity, cognition, and diffusion tensor imaging	Graduate student	Yes	Yes
Juyoun Lee	Current	Visiting Scholar	Md,PhD. Assistant Professor	Glymphatic	Visiting Faculty Scholar working with Helena Chui MD and Hosung Kim PhD	Yes	Yes
Janet Choi	Current	Scholar	Assistant Professor	Hearing Loss & Cognition	Received CTSI KL2 award, working on K23 application	No	Yes
Cristelle Hugo	Current	Scholar	PhD student	Effects of mitochondrial antioxidants on inflammatory infectious diseases		Yes	

Table G2: ADRC REC EVENTS Y6 (07/01/25 – 03/31/26)			
Event, Date(s), Sponsors	**Brief Description, Intended Audience**	**Attendance**	**Platform, Advertising, Access**
"Career development in Alzheimer's Research," presented by Hussein Yassine, M.D., September 19, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	55	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Targeting brain insulin signaling in Alzheimer's: A focus on the BBB," presented by Elizabeth Rhea, Ph.D., September 26, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	34	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"The intersection of FDG-PET and dementia associated biology in clinical decision making," presented by David Jones, M.D., October 3, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	27	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Structural biology of receptors controlling innate immunity in Alzheimer's disease," presented by Tobias Ulmer, Ph.D., October 10, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	31	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"What Can We Learn from High-Frequency Digital Assessments?" presented by Jason Hassenstab, Ph.D., October 17, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	30	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1

Table G2: ADRC REC EVENTS Y6 (07/01/25 – 03/31/26)			
Event, Date(s), Sponsors	**Brief Description, Intended Audience**	**Attendance**	**Platform, Advertising, Access**
"Writing Grant Applications for Observational Studies," presented by John Ringman, M.D., October 24, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	32	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Pericyte-based therapy to treat Alzheimer's disease," presented by Ruslan Rust, Ph.D., October 31, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	30	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Diabetes, related strategies and AD/ADRD," presented by Jose A Luchsinger, M.D., November 7, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	29	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Mechanisms of cytosolic phospholipase A2-Meditated Synapse Loss in Alzheimer's Disease," presented by Quilan Ma, M.D., Ph.D., November 14, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	21	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Research After Lockup: Why ADRD Scientists Should Care About Incarceration," presented by Rodlescia Sneed, Ph.D., November 21, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	22	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1

Table G2: ADRC REC EVENTS Y6 (07/01/25 – 03/31/26)			
Event, Date(s), Sponsors	**Brief Description, Intended Audience**	**Attendance**	**Platform, Advertising, Access**
"Alzheimer's Disease risk factors APOE4 and PICALM at the blood-brain barrier," presented by Kassandra Kisler Elliott, Ph.D., December 5, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	32	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Updates from CTAD 2025," presented by Hussein Yassine, M.D., December 12, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	46	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1
"Microglial lactate dynamics link amyloid pathology to slow-wave sleep loss in Alzheimer's disease," presented by Macauley-Rambach, Shannon L., Ph.D., Ph.D., December 19, 2025, 1:00-2:00 p.m. Sponsored by the Research and Education Component (REC).	REC scholars and attendees	19	This event was a Zoom Meeting and open to registrants. https://usc.zoom.us/j/91356077711?pwd=dtcxhgN4xvlH0nynttbXQtilBZyYKD.1

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022
FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Duke		Han		Co-REC Lead	225,700.00	0.8			15,122.00	4,793.00	19,915.00
2.	Hussein		Yassine		REC Lead	225,700.00	0.8			15,122.00	4,793.00	19,915.00
3.	Julie		Zissimopoulos		Co-I	225,700.00	0.3			5,548.00	1,759.00	7,307.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **47,137.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Administrator	6.0	0.0	0.0	33,280.00	10,550.00	43,830.00
1	**Total Number Other Personnel**					**Total Other Personnel**	**43,830.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**90,967.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	5,166.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**5,166.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	0.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**0.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**96,133.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	8.0	96,133.00	7,690.00
		Total Indirect Costs	**7,690.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**103,823.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	103,823.00

L. Budget Justification* File Name: REC Core just.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – RESEARCH EDUCATION COMPONENT (REC)

Key Personnel:

Hussein Yassine, REC Leader, 0.8 Calendar Months in Years 1 – 5. Dr. Yassine will be responsible for leading the REC training program, organizing the training course, the pilot funding program, and annual Finch AD Symposium. He has mentored many undergraduate, graduate, and post-doctoral fellows in the areas of obesity, diabetes, lipids and AD in the past 10 years.

Duke Han, Co-REC Leader, 0.8 Calendar Months in Years 1 – 5. Dr. Han is Professor of Psychology, Dornsife School of Letters, Arts and Sciences with secondary appointments in Neurology, Psychiatry and Gerontology. Dr. Han joins the ADRC and Co-Leader of the Research Education Component (REC), focusing on mentoring post-doctoral students and early-career faculty members in ADRD research. Dr. Han's own research focuses on financial and healthcare decision making in older age using multi-modal neuroimaging, behavioral economic, neuropsychological, and epidemiologic approaches. He is the PI of multiple NIH grants (R01, RF1, and K24) and co-investigator/consultant on a wide range of research topics (including cholesterol mechanisms and cerebrovascular markers in preclinical Alzheimer's Disease).

Julie Zissimopoulos, Co-Investigator, 0.30 Calendar Months in Years 1 - 5. Dr. Zissimopoulos is a Professor in the Sol Price School of Public Policy at the University of Southern California (USC) and co-director of the Aging and Cognition Program at the Schaeffer Center for Health Policy & Economics. She is Principal Investigator of the NIH-funded Resource Center for Minority Aging Research (RCMAR) focused on Alzheimer's disease and other dementias, Principal Investigator of the NIA-funded Center for Advancing Sociodemographic and Economic Study of Alzheimer's Disease and Related Dementias (CeASES-ADRD). Dr. Zissimopoulos has trained 45 junior scientists, including 11 assistant professors, 17 postdoctoral fellows, and 17 predoctoral fellows. Dr. Zissimopoulos will assist in the training activities for this grant.

Other Personnel:

TBD, Program Administrator, 6.0 Calendar Months in Years 1 – 5. Collects data supplied by all ADRC scientists and faculty mentors to track the success of participating scientists. The PIs will use this information to aid the scientists, inform the program and make appropriate changes to the program. All evaluations and feedback will be entered into an on-going evaluation table kept current by the Project Coordinator. The program coordinator will also assist with recruitment of ADRC scientists, planning and administering the training course and the annual symposium.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years.

Other Direct Costs:

Travel Funds: A budget of $1,166 per year is requested for traveling to the semi-annual ADRC meetings for the 2 REC leaders. In addition, $4,000 is requested for up to 3 ADRC REC scholars to attend the ADRC meeting.

Facilities and Administration Costs: 8% F&A is requested per RFA.

A. COMPONENT COVER PAGE

Project Title: Core H: Imaging Core
Component Project Lead Information: TOGA, ARTHUR W

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

Aim 1: Provide an imaging framework for all ADRC. The Imaging Core will provide a standardized imaging data acquisition protocol in compliance with SCAN requirements. In addition, the Imaging Core will assist and tracking of all raw images in conjunction with the DMSC. Finally, the Imaging Core will assist with data dissemination of images to authorized users under the USC ADRC project.

Aim 2: Perform high quality control on all imaging data and derive MRI analytics. State-of-the-art workflow for the review and assessment of all imaging data. Quantitative MRI analytics including white matter hyperintensities load, perivascular spaces, volumetrics, vessel density, BBB integrity, and cerebral blood flow.

Aim 3: Develop technical and scientific approaches for ultra-high field 7T MRI. The Imaging Core will provide sequence development of novel 7T MRI sequences. The Imaging Core will assist with protocol development and study design for 7T MRI research that will allow structural, vascular, and functional mapping with unprecedented resolution.

Aim 4: Assist in PET protocol development and quality control and provide PET analytics. The Imaging Core will provide regulatory guidance and standardized protocol deployment for AD PET studies. In addition, the Imaging Core will provide qualitative review of PET scan data quality and PET data analytics including Standard Uptake Value Ratios (SUVRs).

Aim 5: Conduct education and training in modern imaging techniques. The Imaging Core will provide didactic and neuroimaging training for AD research across departments and campuses.
Impact: The Imaging Core serves an essential role on the theme of USC ADRC on APOE e4 and dementia. It provides a comprehensive set of imaging markers for studying the differential effects of the APOE e4 genotype on dementia and disease progression. The Imaging Core will provide guidance on compliance with national imaging initiatives as well as conduct education and training across USC.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : B2_ADRC Y6 RPPR_Imaging Core_Prefinal.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

File Uploaded : B4_ADRC Y6 RPPR_Imaging Core_Prefinal.pdf

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

- We will continue to provide a well-integrated and updated framework for all ADRC related imaging data collection and storage in the LONI IDA (Image Data Archive)
- We will continue to provide a standardized quality control (QC) pipeline and imaging data analytics for PET and structural MRI data as part of the IDA data workflow
- We will develop a unified minimum standard for the neuroimaging acquisition that can serve as a guide for future ADRC-affiliated projects.
- We are develop a single, harmonized processing stream for neuroimaging data such that resulting metrics are methodologically consistent
- We will compute standardized scales for ADRC-affiliated PET data, including Centiloids for amyloid-PET and CenTauR (Centiloid-like Tau Ratio) for tau-PET
- We will continue to provide educational and training support for the T32 postdoctoral training program, collaborations with California State University, Fullerton, and USC Masters Program in Neuroimaging and Informatics (NIIN).

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

H. OTHER --- IMAGING CORE

Major activities by Aims:

Specific Aim 1: Provide an imaging framework for all ADRC

The Imaging Core continues to provide a well-integrated and updated framework for all ADRC related imaging data collection. Data is collected on the MRI scanners and transferred to the LONI IDA (Image Data Archive) server. In turn, data is immediately available to authorized users and PIs. The IDA supports several other large, national initiatives (e.g., ADNI) and undergoes hardware updates as needed. The IDA contains all the imaging data for the ADRC and associated studies with a user interface for downloading the data by authorized investigators. This infrastructure is also used by the Imaging Core to upload data related to the SCAN initiative. The IDA serves as a centralized data system that support uniform image preprocessing and image storage.

Effective this reporting period, the ADRC Imaging Core no longer provides direct support for the Vascular Cohort Study (VCS). The Program Project Grant (PPG) P01AG052350 has assumed full responsibility for the longitudinal follow-up of pre-existing participants. A key focus of this continued longitudinal surveillance includes the monitoring of participants currently receiving anti-amyloid therapeutics, with specific attention to vascular outcomes and safety profiles in this population, and imaging data will continue to be released through the IDA by the ADRC.

Specific Aim 2: Perform high quality control on all imaging data and derive MRI analytics

The Imaging Core provides a standardized quality control (QC) pipeline as part of the IDA data workflow. All imaging data flowing into the IDA undergo QC using in-house software (LONI QC). This system employs automated pre-processing to generate vector statistics and derived images. QC derivatives and metrics are distributed to investigators and users through the IDA.

PET QC Pipelines
A dedicated PET processing pipeline has been developed to ensure high-quality data control.
Documentation is available to automate both the amyloid-PET and tau-PET pipelines:

- Amyloid-PET:
  - Uses the Desikan-Killiany atlas to extract regions of interest (ROIs): frontal, lateral parietal, lateral temporal, and anterior/posterior cingulate.
  - ROIs and reference regions are registered to PET space using the Advanced Normalization Toolbox (ANTs) for uptake analysis.
  - Standard Uptake Value Ratios (SUVRs) are calculated, with amyloid status determined by established thresholds.
  - Scripts have been created to streamline each step of this workflow.
- Tau-PET:
  - Follows the FTP protocol from the Jagust Lab (UC Berkeley, ADNI).
  - All tau-PET scans are acquired on a Siemens Biograph64 TruePoint system at USC.
  - ROIs are defined for Braak stages I, III–IV, and V–VI, as well as a meta-temporal composite.
  - ROIs and reference regions are registered to PET space with ANTs to facilitate SUVR analysis.
  - Both unweighted and volume-weighted SUVRs are computed for all composite ROIs.

MRI Structural QC Pipeline
The structural MRI QC pipeline has also been automated:

- ROIs include the middle temporal gyrus, inferior temporal gyrus, fusiform gyrus, and entorhinal cortex.
- QC is conducted in FreeSurfer, using both 2D and 3D imaging for internal and external review.
- Final QC ratings are categorized as pass, moderate, or fail, with detailed notes indicating problematic images or hemispheres.
- QC ratings follow the ENIGMA Cortical QC 2.0 protocol.

QC Summary to Date:

- 114 amyloid-PET scans quality controlled
- 75 tau-PET scans quality controlled
- 917 structural MRIs quality controlled

Recent activity includes the upload of 14 new raw PET datasets, primarily supporting the PPG cohort.

Imaging Data Analytics
The Imaging Core also provides data analytics for ADRC imaging data, currently applied to a subset of the dataset:

- MRI: Volumetric values and total intracranial volume (TIV) using FreeSurfer.
- PET: SUVRs for amyloid and tau using in-house “ADNI-like” pipelines.
    - Amyloid composite ROI includes frontal, parietal, lateral temporal, and cingulate cortices.
    - Tau composite ROIs correspond to Braak stages 1–2 and 3–4.
- Biospecimen Data Integration: To support multi-modal analysis, the Imaging Core now facilitates the management of non-imaging datasets. This includes the regular upload and organization of biospecimen analysis results received from NCRAD into the centralized "Study Data" files on the ADRC server.

Analytics are integrated into the IDA server and shared with the Data Core to support ATN table collation. All pipelines are implemented in the LONI Pipeline system and can be distributed as shareable scripts, ensuring reproducibility by external investigators. Integration of multi-modal data enables statistical comparison between imaging data and other phenotypic, clinical, and demographic data.

MRI Data Harmonization and Standardization Initiatives (Ongoing Work)
Recognizing that heterogeneity in acquisition parameters and analytics across different ADRC-affiliated studies hinders aggregation across projects, the Imaging Core has initiated a focused harmonization strategy. This ongoing work addresses two critical areas:

- Protocol Unification: We are currently conducting a review of existing imaging protocols to identify a "core" set of necessary acquisitions. The goal is to define a unified minimum standard that can serve as a guide for future ADRC-affiliated projects. By encouraging the adoption of these standardized parameters at the study design phase, we aim to ensure that data collected across independent projects remains interoperable and comparable.
- Analytic Pipeline Standardization: Concurrently, the Core is working to ensure that all aggregated data is analyzed through a single, uniform processing stream. While many datasets are already processed via standard IDA workflows, we are actively addressing inconsistencies arising from studies that submit tabulated data derived from divergent external processing schemes. Our objective is to transition toward a model where all raw data, regardless of origin, is re-processed through the Core’s harmonized pipelines. This ensures that downstream metrics are methodologically consistent, maximizing the statistical power of the aggregate ADRC dataset. Currently, PET reconstruction processing pipelines are harmonized across all ADRC projects.

Specific Aim 3: Develop technical and scientific approaches for ultra-high field 7T MRI

Ultra-high field 7T MRI methods continue to be developed at USC INI and as a part of the ADRC Imaging Core.
Our focus has been on developing 7T methods for imaging the blood brain barrier (BBB) using DCE MRI. We have obtained preliminary data on whole-brain, high-resolution DCE-MRI images (results shown in the EAB presentation). Our new development allows us to obtain blood-brain-barrier maps with unprecedented spatial resolution. Future improvements to this method include better techniques for accounting magnetic field inhomogeneities (improving B1 mapping) as well as improving the T1 mapping estimation that is necessary for calculating BBB maps. To this end we are experimenting with variable flip angle techniques as well using MP2RAGE image acquisitions.

We developed a brain imaging technique that reveals how microvessels in the brain pulse with each heartbeat—changes that may hold clues to aging and diseases such as Alzheimer's.

We recently published in Nature Cardiovascular Research, a noninvasive method for measuring "microvascular volumetric pulsatility"—the rhythmic expansion and contraction of the brain's smallest vessels— in living humans. Using ultra-high-field 7T MRI, we showed that these microvessel pulses increase with age, especially in the brain's deep white matter, a region critical for communication between brain networks and is particularly susceptible to reduced blood supply of distal arteries—the blood vessels that carry blood away from the heart and into the farthest parts of the body—with aging. innovation combines two advanced MRI approaches—vascular space occupancy (VASO) and arterial spin labeling (ASL)—to track subtle volume changes in microvessels over the cardiac cycle. Older adults show heightened microvascular pulsations in deep white matter compared to younger adults, and that hypertension further amplifies these changes.

Specific Aim 4: Assist in PET protocol development and quality control and provide PET analytics

PET imaging continues to progress well at USC and as a part of the ADRC Imaging Core. We have developed a database that keeps track of the PET protocols across ADRC. The database allows to control for differences and similarities between protocols and we work in conjunction with the PET technicians to ensure proper data acquisition. Obtained PET data undergo image quality using in-house QC pipelines. Data analytics are provided in terms of SUVRs as explained in Aim 2.

To further enhance the interoperability of our PET data, the Imaging Core is initiating the computation of **Centiloids** for amyloid-PET and **CenTauR** (Centiloid-like Tau Ratio) for tau-PET using the computational models provided by the Global Alzheimer's Association Interactive Network (GAAIN). These standardized scales differ from traditional SUVRs by converting tracer-specific values into a universal unit, effectively normalizing differences between the various radiotracers and acquisition protocols used across multi-site studies. By adopting the GAAIN-based harmonization pipelines, we aim to ensure that USC ADRC imaging data can be directly compared and aggregated with wider national datasets, overcoming the significant barrier of tracer heterogeneity in Alzheimer's research.

Specific Aim 5: Conduct education and training in modern imaging techniques.

The Imaging Core continues to provide ample training opportunities for students interested in neuroimaging techniques and analysis. We continue to provide support to the T32 postdoctoral training program entitled "Training for the Multiscale and Multimodal Analysis of Biomarkers in Alzheimer's Disease" (AD).
Additionally, we continue to support a training collaboration with a traditionally underfunded R01 university,

California State University, Fullerton. The Imaging Core has also provided educational support for the USC Masters Program in Neuroimaging and Informatics (NIIN). The USC ADRC Speaker Series is available to NIIN students and other trainees, and ADRC members actively teach courses in the program and provide guest lectures. We have been working closely with the Global Alzheimer's Association Network (GAAIN) to provide imaging biomarker training as part of a funded R25 Supplement study, including computation of centiloid data used by GAAIN for training purposes. We also created freely available training videos earlier this year. Current and past trainees have successfully matriculated to PhD and MD programs, in addition to full-time careers in data science and medical imaging.

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

SUMMARY

- Core H- Imaging Core: Weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.

H. OTHER – IMAGING CORE

We continue to provide training and professional opportunities in trainees in various labs across USC. These include weekly seminars, videos, as well as code and hands on practice sessions through INI's NIIN master program.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

See the ORE and neuropathology cores for details on clinical core resource sharing of clinical materials and research specimens. See Administrative Tables and overall component for description of resource sharing.

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Arthur		Toga		Imaging Core Lead	225,700.00	1.2			22,570.00	7,155.00	29,725.00
2.	Kristen		Lynch		Imaging Core Co-Lead	111,395.00	3.0			27,849.00	8,828.00	36,677.00
3.	Meredith		Braskie		Co-I	159,765.00	0.6			7,988.00	2,532.00	10,520.00
4.	Yonggang		Shi		Co-I	224,248.00	0.6			11,212.00	3,554.00	14,766.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **91,688.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Imaging Tech	9.0	0.0	0.0	55,905.00	17,722.00	73,627.00
1	Researcher	1.6	0.0	0.0	15,925.00	5,049.00	20,974.00
2	**Total Number Other Personnel**					**Total Other Personnel**	**94,601.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**186,289.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	3,000.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**3,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	5,988.00
2. Publication Costs	1,600.00
3. Consultant Services	0.00
4. ADP/Computer Services	25,000.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	345.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**32,933.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**222,222.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	211,356.00	139,495.00
2. MTDC - Off Campus	26.0	10,521.00	2,735.00
		Total Indirect Costs	**142,230.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**364,452.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	364,452.00

L. Budget Justification*	File Name: Imaging Core just.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION – IMAGING CORE

Key Personnel:

Arthur W. Toga, Ph.D., Co Lead, 1.2 Calendar Months in Years 1 - 5. Dr. Toga is the founder and director of the Laboratory of Neuro Imaging (LONI). He is director of the USC Stevens Neuroimaging and Informatics Institute. His formal training in neuroscience, imaging and informatics makes him well suited to lead this project. Dr. Toga has a career-long funding and publication record in brain imaging focused on neurodegenerative diseases. He directs several national and international collaborative studies that involve multi-site acquisition of data, coordination of analysis, and dissemination of data and results. Dr. Toga has experience and expertise in image analysis, brain atlasing, multimodal integration, large-scale multisite trials, databasing and scientific visualization. He has developed a comprehensive supercomputing facility for neuroscience incorporating the latest technology in the analysis of structure and function of the brain in health and disease. Dr. Toga has the skills, knowledge, and resources necessary to participate in the proposed project. He will coordinate activities in image acquisition, analysis and data aggregation.

Kirsten Lynch, Ph.D., Co-Core Lead/Co-Investigator, 3.0 Calendar Months in Years 1 - 5. Dr. Lynch is Assistant Professor of Research in the Laboratory of Neuro Imaging at the University of Southern California Mary and Mary Stevens Neuroimaging and Informatics Institute. Dr. Lynch's research focuses on the development, evaluation, and application of computational tools for visualizing, modeling, and analyzing neuroimaging datasets, intersecting computer science, neuroscience, and medicine. She has experience in software development related to neuroimage analysis and has used a variety of neuroinformatics platforms in structural brain imaging studies to advance understanding of longitudinal brain changes. As co-investigator, Dr. Lynch will apply computational models to quantitatively analyze the MRI data collected in this study. She will also participate in the development of publications and presentations. Her background and publications record will provide the knowledge and tools necessary for this project.

Yonggang Shi, Ph.D., Co-Investigator, 0.60 Calendar Months in Years 1 - 5. Dr. Shi is an Associate Professor of Neurology and Electrical Engineering at the Stevens Neuroimaging and Informatics Institute at the Keck School of Medicine of USC. His research focuses on the development of novel mathematical frameworks and algorithms in mapping neuroanatomical structures and function. He has extensive experience in analysis methods for diffusion imaging data as well as mathematical models for vessel segmentation. He will apply his novel algorithms to analyze brain connectivity using the multi-shell diffusion imaging data collected in this project and vessel integrity using the obtained MRI modalities.

Meredith Braskie, Ph.D., Co-Investigator, 0.60 Calendar Months in Years 1 - 5. Dr. Meredith Braskie is an Assistant Professor at the University of Southern California. She will perform analyses the MRI data and will help to interpret the MRI findings in the context of this project. She also will be responsible for overseeing data flow and analyses related to grant aims. She will oversee and edit drafts of the MRI analysis-related methods sections for manuscripts. Dr. Braskie has an outstanding track record of publications in neuroimaging, with work on structural MRI (including high resolution hippocampal focuses analyses), functional MRI, diffusion tensor imaging (DTI), and PET analyses. Her research focus is Alzheimer's disease (AD) and cognitive aging.

Other Personnel:

TBN, Researcher, 1.59 Calendar Months in Years 1 - 5. TBN will be a researcher at the Institute of Neuroimaging and Informatics in the Keck School of Medicine at USC. He/She will assist with analysis of MRI and PET imaging data focusing on estimation of microbleed burden and white matter hyperintensities among other signals at 3T and 7T. This person will also help monitor data progression as it is scanned, databased. QC'd, analyzed and made available to subsequent researchers for further study.

TBD, Imaging Coordinator, 9.00 Calendar Months in Years 1 – 2 and 11.54 Calendar Months in Years 3 – 5. Nick will work closely with the Core Lead and Co-investigators. He will be responsible for subject recruitment, scheduling MRI scans, and consenting subjects.

Fringe Benefits: The personnel costs also include fringe benefits that are currently calculated with the rate of 31.70% for all 5 years.

Travel:

Funds will be requested for the Co-Core Lead to attend one annual ADRC meeting and for another scientific meeting to present research results from the proposed project: $1,500 x 2 = $3,000 per year.

Other Direct Costs:

Materials and Supplies: We request $5,988 for the purchase of computer supplies, printing supplies, and research supplies. There is a significant need on this project for hard drive and computer disk backup to ensure the safe archival and failsafe backup of our datasets

Publication costs: We request $1,600 per year for publication costs that include page charges and open access charges imposed by publishers. Publication costs may also be required to publish abstracts or posters at conferences.

Computer Services Costs. We request $25,000 per year for high-performance supercomputing time and maintenance services, which guarantees high throughput of the very large computational image datasets. The costs listed here are standard data analysis costs associated with imaging projects. These costs help defray the depreciation costs for the equipment. These computational resources make its use essential for complicated algorithms typically used during the calculation of population-based statistics. This, coupled with access to the large database, makes it an efficient system for many brain analytic problems. The hardware support provides 24/7 response coverage for Isilon, Cisco, Quantum and F5 equipment discussed under the Resources and Environment section. Because of concerted efforts eliminating single points of failure in our computing environment, this model is by far the most cost effective. The software maintenance provides our facility with immediate vendor software support, software licensing and version upgrades for sophisticated clustering, visualization, development and hierarchical file management software. Such upgrades become critical when new features that can facilitate daily operations are released.

Rental Costs. We request $345 per year for the off-campus rental and facilities costs, which are calculated based on the project FTEs. The 26% off campus indirect cost rate is used for part of the project.

Facilities & Administrative Costs:
Facilities and Administration Costs are based on USC' federally negotiated rate (dated May 29, 2024) of 66% as of July 1, 2024. The F&A rate is 66% of the modified total direct costs (MTDC) for On-Campus Research. MTDC consists of all salaries and wages, fringe benefits, materials and supplies, services, travel and the first $25,000 of each subaward (subgrant and subcontract) regardless of the period covered by the subgrant or subcontract. Equipment, alterations and renovations, charges for patient care and tuition remission, space rental costs, scholarships and fellowships, as well as the portion of each subaward in excess of $25,000 are excluded from the MTDC calculation.

A. COMPONENT COVER PAGE

Project Title: 7.1_Glymphatic Dysfunction as a Mechanistic Biomarker for Treatment Response and Cardiometabolic Risk in Alzheimer's Disease_Kim_new
Component Project Lead Information: Kim, Hosung

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

This project's overall goal is to establish MRI-derived glymphatic function as a mechanistic biomarker linking CMS risk to AD pathology and treatment-related adverse events.

Aim 1: Elucidate the pathway linking CMS, blood-brain barrier (BBB) permeability, and glymphatic dysfunction. We will use the deeply phenotyped longitudinal USC-VC cohort (n=212), which includes quantitative BBB imaging (DCE-MRI),5glymphatic imaging (DTI-ALPS, T1-ePVS, fMRI-g BOLD; details in IV. Methodology), and biomarker data. We will test the hypothesis that the effects of cardiometabolic risk factors on Aβ accumulation and cognitive decline are mediated by BBB breakdown and subsequent glymphatic impairment. Hypothesis: Greater CMS burden(defined in IV. Methodology) leads to increased BBB leaks (K-trans), which in turn reduces glymphatic clearance (lower ALPS), ultimately inducing Aβ deposition and cognitive decline.

Aim 2: Determine if baseline glymphatic dysfunction predicts the incidence of ARIA-H in preclinical AD trial.We will leverage the A4 dataset (n=1,169), the largest preclinical AD prevention trial with longitudinal multimodal imaging.2Hypothesis:Preclinical AD individuals with lower baseline glymphatic clearance will have a significantly higher incidence of ARIA-Hafter anti-amyloid body treatment as Aβmay not be fully cleared from the brain tissue.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : H.Kim_Proposal_ADRC DP Y7_2026.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Methodology

Datasets:
1. USC-VC Cohort (Aim 1): A longitudinally followed cohort (n=212) with 1-7 years of follow-up, containing MRI (DCE-MRI for K-trans, DTI for ALPS, fMRI), PET, plasma Aβ40/42, and cognitive data.

2. A4 Trial Cohort (Aim 2): A large-scale preclinical AD trial (n=1,169) with longitudinal structural MRI, DWI, fMRI, amyloid PET, and cognitive outcomes up to 240 weeks.2

Cardiometabolic (CMS) burden. We will operationalize cardiometabolic syndrome (CMS) burden using a primary continuous severity score and multiple secondary definitions to ensure robustness as in previous studies.7 Medication use will be included as a covariate and explored in sensitivity analyses.

MRI Biomarker Processing:
All imaging data will be processed using our established, automated pipelines at the USC Mark and Mary Stevens Neuroimaging and Informatics Institute (INI).
• Glymphatic Function: Three features from different modal MRI will be extracted to address for different aspect of glymphatic function. 1) Enlarged perivascular space (ePVS) volume will be automatically
segmented from T1w scans8 as a structural marker of impaired periarterial clearance;9 2) Global BOLD signal amplitude (gBOLD) serves as an indirect marker of glymphatic clearance efficiency, reflecting
global fluctuations associated with cerebrospinal fluid (CSF) flow and vascular pulsatility;10 3) The DTI-

ALPS index will be computed from DWI scans to quantify perivenous clearance efficiency.4,11
• BBB Permeability (Aim 1 only): K-trans will be derived from DCE-MRI to quantify BBB leakage.
• ARIA-H Evaluation (Aim 2 only): ARIA-H will be identified from T2* and FLAIR scans as per the A4 trial protocol, distinguishing microhemorrhage vs. edema patterns as in a prior study.
• Data Harmonization: All MRI data will undergo standard preprocessing (FSL, ANTs) with ComBat harmonization across scanners.12 Visual quality control and motion exclusion will be implemented.

USC-ADRC Development Project Proposal
Title of Study: Glymphatic Dysfunction as a Mechanistic Biomarker for Treatment Response and Cardiometabolic Risk in Alzheimer's Disease
Investigator(s):

- **Principal Investigator:** Hosung Kim, PhD, **Co-Investigators:** Yaqiong Chai, PhD

ADRC Collaborator/Sponsor: Helena Chui, Admin Core, USC ADRC
Study Duration: 12 months (March 2026 - February 2027)

I. Scientific Rationale

Alzheimer's disease (AD) pathology, including amyloid-β (Aβ) accumulation, begins decades before the onset of dementia.[1] Recent anti-Aβ antibody therapies have shown promise in clearing amyloid plaques but have yielded limited clinical benefits and are associated with a high incidence of Amyloid-Related Imaging Abnormalities (ARIA), particularly microhemorrhages (ARIA-H).[2] A critical unanswered question is why some amyloid-positive individuals are vulnerable to ARIA while others are not. Current biomarkers like amyloid PET and plasma tau can identify risk but fail to explain the mechanistic heterogeneity of treatment response.[3]
Emerging evidence points to the glymphatic system—the perivascular network that clears interstitial solutes—as a key determinant of both Aβ accumulation and microvascular injury. Glymphatic dysfunction, quantifiable by non-invasive MRI (e.g., Diffusion Tensor Imaging Along the Perivascular Space [DTI-ALPS] index),[4] is associated with both Aβ positivity and small-vessel pathology. Our preliminary data show that impaired glymphatic clearance correlates with amyloid burden, reduced vascular pulsatility, and cognitive decline.
Cardiometabolic syndrome (CMS), such as hypertension, obesity, and insulin resistance, is a major contributor to AD (see Preliminary data). These factors increase blood-brain barrier (BBB) permeability, which may in turn impair perivascular clearance and promote amyloid deposition. Yet, no study has comprehensively modeled the cardiometabolic → BBB → glymphatic → amyloid pathway in humans.
This project will leverage two unique, powerful datasets: the **USC ADRC Vascular Cohort (USC-VC)** and the **A4 clinical trial**[10]. By first elucidating the pathway from cardiometabolic risk to glymphatic failure in the deeply phenotyped USC-VC cohort, and then testing whether these glymphatic metrics predict ARIA in the landmark A4 trial, this project will establish a novel, clearance-based biomarker with immediate translational potential.

II. Specific Aims

This project's **overall goal** is to establish MRI-derived glymphatic function as a mechanistic biomarker linking CMS risk to AD pathology and treatment-related adverse events.
Aim 1: Elucidate the pathway linking CMS, blood-brain barrier (BBB) permeability, and glymphatic dysfunction.
We will use the deeply phenotyped longitudinal USC-VC cohort (n=212), which includes quantitative BBB imaging (DCE-MRI),[5] glymphatic imaging (DTI-ALPS, T1-ePVS, fMRI-gBOLD; details in IV. Methodology), and biomarker data. We will test the hypothesis that the effects of cardiometabolic risk factors on Aβ accumulation and cognitive decline are mediated by BBB breakdown and subsequent glymphatic impairment.
Hypothesis: Greater CMS burden (defined in IV. Methodology**)** leads to increased BBB leaks (K-trans), which in turn reduces glymphatic clearance (lower ALPS), ultimately inducing Aβ deposition and cognitive decline[12].

Aim 2: Determine if baseline glymphatic dysfunction predicts the incidence of ARIA-H in preclinical AD trial.
We will leverage the A4 dataset (n=1,169), the largest preclinical AD prevention trial with longitudinal multimodal imaging.[2]
Hypothesis: Preclinical AD individuals with lower baseline glymphatic clearance will have a significantly higher incidence of ARIA-H after anti-amyloid body treatment as Aβ may not be fully cleared from the brain tissue.

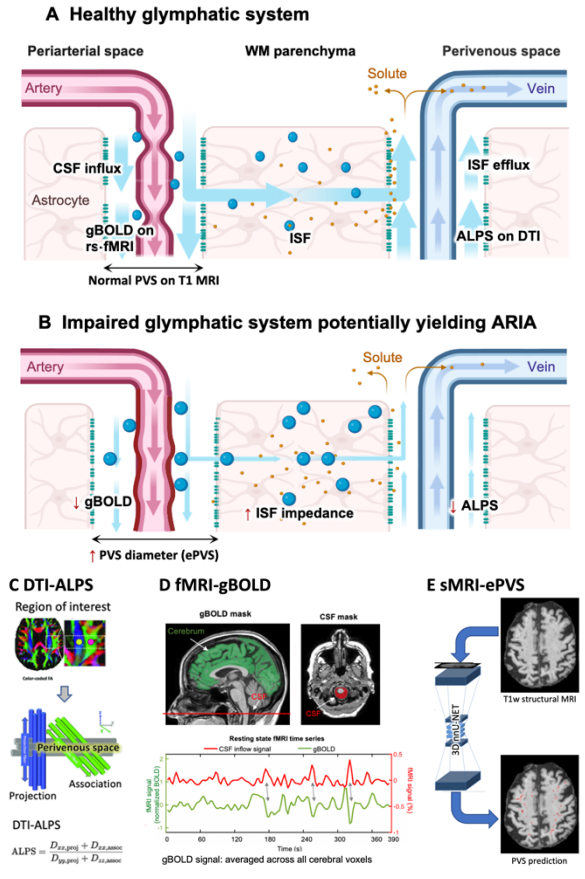

Fig. 1. MRI-based glymphatic features to predict ARIA-H in response to amyloid anti-body medication

III. Preliminary Data

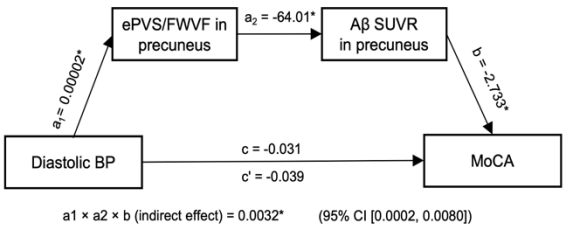

Fig 2. In subsample of ADNI cohort (n=120), a mediation analysis shows that higher blood pressure and consequent reduced arterial pulsatility leads to increased free-water volume and fluid impedance in perivascular spaces within the precuneous and AD-signature regions,[6] which are considered areas where Aβ is deposited early in AD progression. This relationship further leads to more Aβ deposition in these brain regions and further results in declined cognitive performance. This supports a mechanistic link between weak arterial dynamics, potentially led by CMB, and impaired perivascular clearance of Aβ and subsequent cognitive impairment (submitted to Nature Aging). However, given that ADNI does not hold full CMS profiles (only BP or history of cardiovascular diseases), analysis of USC-VC cohort will allow us to better understand the CMS contribution to AD via glymphatic dysfunction.

IV. Methodology

Datasets:

1. **USC-VC Cohort (Aim 1):** A longitudinally followed cohort (n=212) with 1-7 years of follow-up, containing MRI (DCE-MRI for K-trans, DTI for ALPS, fMRI), PET, plasma Aβ40/42, and cognitive data.
2. **A4 Trial Cohort (Aim 2):** A large-scale preclinical AD trial (n=1,169) with longitudinal structural MRI, DWI, fMRI, amyloid PET, and cognitive outcomes up to 240 weeks.[2]

Cardiometabolic (CMS) burden. We will operationalize cardiometabolic syndrome (CMS) burden using a primary continuous severity score and multiple secondary definitions to ensure robustness as in previous studies.[7] Medication use will be included as a covariate and explored in sensitivity analyses.

MRI Biomarker Processing:

All imaging data will be processed using our established, automated pipelines at the USC Mark and Mary Stevens Neuroimaging and Informatics Institute (INI).

- **Glymphatic Function:** Three features from different modal MRI will be extracted to address for different aspect of glymphatic function. 1) **Enlarged perivascular space (ePVS)** volume will be automatically segmented from T1w scans[8] as a structural marker of impaired periarterial clearance;[9] 2) **Global BOLD** signal amplitude (**gBOLD**) serves as an indirect marker of glymphatic clearance efficiency, reflecting global fluctuations associated with cerebrospinal fluid (CSF) flow and vascular pulsatility;[10] 3) The **DTI-ALPS index** will be computed from DWI scans to quantify perivenous clearance efficiency.[4,11]
- **BBB Permeability (Aim 1 only): K-trans** will be derived from DCE-MRI to quantify BBB leakage.
- **ARIA-H Evaluation (Aim 2 only):** ARIA-H will be identified from T2* and FLAIR scans as per the A4 trial protocol, distinguishing microhemorrhage vs. edema patterns as in a prior study.[2]
- **Data Harmonization:** All MRI data will undergo standard preprocessing (FSL, ANTs) with ComBat harmonization across scanners.[12] Visual quality control and motion exclusion will be implemented.

Statistical Analysis:

Because longitudinal follow-up in the USC-VC cohort is limited (~20% with mostly 1-year imaging), Aim 1 will primarily use baseline cardiometabolic, BBB, glymphatic, amyloid, and cognitive measures to test mechanistic mediation hypotheses, with the 1-year subset used for pre-specified validation.

Aim 1 – Primary analysis (baseline mediation): We will test both serial and parallel mediation models using baseline measures:

- S*erial mediation* model will follow the biologically plausible order, testing whether BBB breakdown leads to impaired perivascular clearance: CM burden → BBB permeability (K-trans) → glymphatic dysfunction (DTI-ALPS) → cognitive decline/Aβ accumulation
- *Parallel mediation* will be tested if K-trans and DTI-ALPS are not significantly correlated, assessing independent indirect effects of each mediator on CM-related outcomes.

Mediation will be estimated using structural equation modeling (lavaan or Mplus) and **bootstrapped indirect effects** (5,000 resamples) to produce bias-corrected CIs. All baseline models will adjust for age, sex, education, APOE ε4, intracranial volume (for MRI volumetrics), and site/scanner harmonization factors (ComBat). For clinical interpretability we will report standardized path coefficients, direct and indirect effects, and proportion mediated.

Limitation: We acknowledge that cross-sectional mediation cannot establish temporal causality. Therefore, baseline mediation results will be interpreted as evidence consistent with the proposed mechanism rather than proof of temporal ordering.

Validation using available longitudinal data: The subset (~20%) with 1-7 year follow-up will be used to validate key causal inferences: we will test whether **baseline** K-trans and DTI-ALPS predict **annual** Δ of

Aβ/cognition, and whether baseline mediation paths are consistent with observed association between baseline CMS burden and **annual** Δ of Aβ/ cognition. Validation analyses will be framed as supportive/exploratory given smaller n. If results in the longitudinal subset are consistent with baseline mediation (e.g., baseline BBB/glymphatic dysfunction predicts ΔAβ/cognition and mediates the relationship between CMS and ΔAβ/cognition), this will strengthen causal interpretation.

Aim 2 – Prediction of ARIA-H: Primary analysis will test whether baseline glymphatic indices (DTI-ALPS, ePVS burden, and gBOLD amplitude) predict the occurrence of ARIA-H following anti-amyloid treatment in the A4 trial.

- **Primary endpoint:** incident ARIA-H within **12 weeks** of first treatment—given that ARIA most often occurs shortly after antibody dosing, representing an acute treatment response. Cumulative dose up to 12 weeks will be modeled as a continuous covariate in logistic regression.
- **Secondary analyses: time-to-first ARIA-H** across all available imaging follow-ups (up to 240 weeks) using Cox proportional hazards models. Treatment and dose will be modeled as **time-varying covariates** in Cox proportional hazards models.

All models will adjust for **treatment arm (active vs placebo),** age, sex, APOE ε4, baseline centiloid amyloid load, and vascular risk. Nonlinearities will be also evaluated via restricted cubic splines; multiple comparisons will be FDR-corrected.
Interaction terms (**Glymphatic × Arm, Glymphatic × Dose**) will test whether clearance metrics modify treatment-related ARIA risk.

Power Calculation: Formal power for complex mediation models will be assessed through **Monte Carlo simulations**. Simulations will model small (β = 0.10), moderate (β = 0.25), and large (β = 0.40) standardized path coefficients under the planned serial and slope mediation structures.

- For **Aim 1 (USC-VC, n = 212)**, power exceeds 80% to detect **moderate** indirect effects in baseline serial/parallel mediation under plausible parameter settings. The 1-year longitudinal subset (≈20%; n ≈ 40–45) is underpowered for formal mediation testing and will be used as a **validation** sample only.
- For **Aim 2 (A4, n = 1,169)**, assuming 27% ARIA-H incidence and an odds ratio (OR) = 1.6 per SD decrease in DTI-ALPS, power exceeds 90% for logistic regression analyses. Final power simulations and analytic specifications will be finalized with the ADRC Data Core prior to study initiation.

V. Synergy with USC ADRC, Innovation, and Impact
Synergy: This project directly aligns with the ADRC's mission to understand modifiers of APOE ε4-related risk and cerebrovascular contributions to dementia. It leverages ADRC-supported datasets (USC-VC) and mentorship from **Dr. Helena Chui**.
Innovation:

1. **Mechanistic Integration:** First study to integrate quantitative BBB imaging and glymphatic MRI to explain cardiometabolic contributions to AD risk.
2. **Predictive Translation:** Applies clearance-based MRI biomarkers to predict ARIA in an ongoing anti-Aβ trial.
3. **Multimodal Fusion:** Combines vascular, clearance, and amyloid data using advanced mediation and survival models, providing a new mechanistic framework for treatment safety.

Impact:
Findings will yield a non-invasive biomarker for ARIA risk stratification and trial optimization, improving safety and therapeutic targeting in emerging AD treatments. By linking vascular health to brain clearance capacity, this work may identify modifiable pathways for AD prevention. The one-year results will establish the foundation for an NIH R01 proposal expanding this framework to multi-site validation and intervention.

VI. Study Timeline

Months	Activities
0	Data Use Agreements for A4 – Approved; Transfer all data to USC-INI secure server – Complete; USC-VC downloaded from USC-IDA under the approval by Dr. Chui
1–2	Harmonize MRI data, perform QC, extract DTI-ALPS, ePVS, gBOLD, and K-trans metrics
3–6	**Aim 1:** Mixed-effects and mediation analyses on USC-VC; draft abstract for 2026 ADRC symposium
7–10	**Aim 2:** Logistic and Cox analyses on A4; replicate findings and cross-validate
11–12	Integrate findings, prepare manuscript for submission, and draft follow-up NIH R01 proposal

References

1 Mormino, E. C. & Papp, K. V. Amyloid Accumulation and Cognitive Decline in Clinically Normal Older Individuals: Implications for Aging and Early Alzheimer's Disease. *J Alzheimers Dis* **64**, S633-s646, doi:10.3233/jad-179928 (2018).

2 Sperling, R. A. *et al.* Trial of Solanezumab in Preclinical Alzheimer's Disease. *N Engl J Med* **389**, 1096-1107, doi:10.1056/NEJMoa2305032 (2023).

3 Sperling, R. A. *et al.* Amyloid and Tau Prediction of Cognitive and Functional Decline in Unimpaired Older Individuals: Longitudinal Data from the A4 and LEARN Studies. *J Prev Alzheimers Dis* **11**, 802-813, doi:10.14283/jpad.2024.122 (2024).

4 Taoka, T. *et al.* Reproducibility of diffusion tensor image analysis along the perivascular space (DTI-ALPS) for evaluating interstitial fluid diffusivity and glymphatic function: CHanges in Alps index on Multiple conditiON acquIsition eXperiment (CHAMONIX) study. *Jpn J Radiol* **40**, 147-158, doi:10.1007/s11604-021-01187-5 (2022).

5 Sourbron, S., Ingrisch, M., Siefert, A., Reiser, M. & Herrmann, K. Quantification of cerebral blood flow, cerebral blood volume, and blood–brain-barrier leakage with DCE-MRI. *Magnetic Resonance in Medicine* **62**, 205-217, doi:https://doi.org/10.1002/mrm.22005 (2009).

6 Xie, L. *et al.* Higher intracranial arterial pulsatility is associated with presumed imaging markers of the glymphatic system: An explorative study. *NeuroImage* **288**, 120524, doi:https://doi.org/10.1016/j.neuroimage.2024.120524 (2024).

7 Soldatovic, I., Vukovic, R., Culafic, D., Gajic, M. & Dimitrijevic-Sreckovic, V. siMS Score: Simple Method for Quantifying Metabolic Syndrome. *PLoS One* **11**, e0146143, doi:10.1371/journal.pone.0146143 (2016).

8 Huang, P. *et al.* Development and validation of a perivascular space segmentation method in multi-center datasets. *NeuroImage* **298**, 120803, doi:https://doi.org/10.1016/j.neuroimage.2024.120803 (2024).

9 Smeijer, D., Ikram, M. K. & Hilal, S. Enlarged Perivascular Spaces and Dementia: A Systematic Review. *J Alzheimers Dis* **72**, 247-256, doi:10.3233/JAD-190527 (2019).

10 Han, F., Liu, X., Mailman, R. B., Huang, X. & Liu, X. Resting-state global brain activity affects early beta-amyloid accumulation in default mode network. *Nat Commun* **14**, 7788, doi:10.1038/s41467-023-43627-y (2023).

11 Liu, X. *et al.* MRI free water mediates the association between diffusion tensor image analysis along the perivascular space and executive function in four independent middle to aged cohorts. *Alzheimers Dement* **21**, e14453, doi:10.1002/alz.14453 (2025).

12 Middleton, D. M. *et al.* Harmonization of multi-site diffusion tensor imaging data for cervical and thoracic spinal cord at 1.5 T and 3 T using longitudinal ComBat. *Sci Rep* **13**, 19809, doi:10.1038/s41598-023-46465-6 (2023).

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Hosung		Kim		PI	128,587.00	0.24			2,572.00	815.00	3,387.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **3,387.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Research Associate	3.6	0.0	0.0	22,050.00	6,992.00	29,042.00
1	**Total Number Other Personnel**					**Total Other Personnel**	**29,042.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**32,429.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	5,171.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**5,171.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	0.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	2,400.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**2,400.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**40,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	40,000.00	26,400.00
		Total Indirect Costs	**26,400.00**

Cognizant Federal Agency (Agency Name, POC Name, and POC Phone Number) DHHS Jeanette Lu 415-437-7820

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**66,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	66,400.00

L. Budget Justification*	File Name: Budget Justification Kim FINAL.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION

SENIOR/KEY PERSONNEL

Hosung Kim, PhD, Principal Investigator, 0.24 Calendar Months: Dr. Kim is an Associate Research Professor of Neurology, Neuroscience, and Biomedical Engineering at USC Mark and Mary Stevens Neuroimaging and Informatics Institute (USC-INI), Keck School of Medicine. His formal training in neuroscience, computer science and informatics makes him well suited to lead the proposed project. He has developed unique mathematical and artificial intelligence techniques for multimodal MRI analysis (structural, functional, perfusion and connectivity), and multidimensional modeling of imaging biomarkers for glymphatic clearance and Alzheimer's disease as well as machine-learning techniques for the prediction of disease progression and outcome in sleep and neurodegenerative disorders. Dr. Kim has a close working relationship not only with the clinical neuroscience departments (neurology, gerontology, psychiatry, and neuroradiology) but also with the basic science programs throughout campus (e.g., psychology, physiology, computer science, electrical engineering, and biostatistics). He will be the corresponding PI responsible for overseeing the proposed project, communicating with other investigators, and supervising the postdoc and research coordinator to ensure the fulfillment of the study.

Yaqiong Chai, Ph.D., Research Associate, 3.60 Calendar Months: Dr. Chai has been a senior research associate who finished a postdoc training under Dr. Kim's supervision, with extensive experiences in MRI acquisition and computational neuroimaging analysis. She will focus on optimization of glymphatic measurements in Aim 1 and 2, working closely with Dr. Kim. She will also perform all the proposed statistical analyses.

OTHER PERSONNEL

Juyoun Lee, MD, Visiting Research Scholar, N/A Calendar Months.: Dr. Lee is a visiting scholar of Dr. Helena Chui and has been a researcher collaborator with Dr. Kim's team. Dr. Joo is a Professor specializing in the field of Alzheimer's Disease at the Department of Neurology, Chungnam University School of Medicine, Korea. With a strong focus on vascular contribution to AD and related neurodegenerative diseases, she has worked in QC'ing and analyzing ADRC VC Cohort while having worked with Drs Kim and Chai. She will provide her insights and interpretation on the findings while meeting Dr. Kim's team weekly.

Fringe Benefits: The fringe benefits were calculated on salaries using the USC-DHHS Federal Rate Agreement from May 29, 2024, that set the faculty and staff employees' fringe benefit rate at 31.7% and the postdoc rate at 22.60%.

OTHER DIRECT COSTS:

Travel: We request $5,171 for 2 domestic/international travels. The travel costs include abstract fees, conference registration, airfare, hotel, and meals. Travel is budgeted to allow for the presentation of project results and/or to interface with other investigators working on related research. Funds will be used by the PI or by the postdoctoral researcher to attend scientific conferences held either in the United States and/or internationally. Conferences attended by investigators on this project might include meetings such as the Alzheimer's Association International Conference (AAIC), Society for Neuroscience (SfN), International Society for Magnetic Resonance in Medicine (ISMRM), the Organization for Human Brain Mapping (OHBM) and Alzheimer's & Parkinson's Diseases Conference (AD/PD^{TM}).

ADP/Computer Services: We request $2,400 per year for supercomputer time/service for high-performance supercomputing time and maintenance services, which guarantees high throughput of the very large computational image datasets. The costs listed here are standard data analysis costs associated with imaging projects. These costs help defray the amortization costs for the equipment. These computational resources make

its use essential for complicated algorithms typically used during the calculation of population-based statistics. This, coupled with access to the large database, makes it an efficient system for many brain analytic problems. The hardware support provides 24/7 response coverage for Isilon, Cisco, Quantum and F5 equipment discussed under the Resources and Environment section. Because of concerted efforts eliminating single points of failure in our computing environment, this model is by far the most cost effective. The software maintenance provides our facility with immediate vendor software support, software licensing and version upgrades for sophisticated clustering, visualization, development and hierarchical file management software. Such upgrades become critical when new features that can facilitate daily operations are released.

A. COMPONENT COVER PAGE

Project Title: 7.2_Synergistic Impact of Alzheimer's Disease risk factors PICALM and APOE on cerebral amyloid angiopathy and vascular pathology_Kisler_new
Component Project Lead Information: Kisler, Kassandra

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

We will test this hypothesis in the following Aims:

Aim 1: Investigate whether (a) vascular PICALM expression spatially coincides and correlates with CAA accumulation and BBB breakdown in human cortex, (b) whether vascular PICALM expression in cortex differs by APOE genotype (APOE4 carriers vs. non-carriers), and (c) whether the association between PICALM expression and CAA pathology differ by APOE genotype.
• We expect that participants with reduced vascular PICALM expression will exhibit increased CAA pathology, BBB integrity loss and permeability, particularly in APOE4 carriers compared to non-carriers, and that APOE4 carriers will have reduced vascular PICALM expression vs non-carriers.

Aim 2: Explore (a) whether vascular PICALM expression and CAA pathology correlate with specific PICALM SNP alleles, and (b) if the associations differ by APOE genotype. We will utilize human post-mortem AD brain tissues from APOE4 carriers and non-carriers from USC's Alzheimer's Disease Research Center (ADRC) repository. We will determine vascular PICALM expression, CAA accumulation and extent, and BBB integrity via tight junction changes and parenchymal fibrinogen, using immunohistochemistry similar to previous work by us and others. Patient samples will be genotyped for prominent PICALM AD SNPs using commercially available assays. Data will also be correlated to APOE genotype and other demographics and measures of AD pathology and cognitive function available from the ADRC for these patients, and controlled for measures such as age, sex, PMI, etc.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Methodology. We will study vascular-specific changes in PICALM expression relative to CAA accumulation in APOE4 carriers vs non-carriers in postmortem occipital or frontal cortical tissue, where CAA pathology most frequently occurs, using immunostaining techniques. We will also investigate vascular tight junction expression (e.g. claudin 5, occludin) and perivascular fibrinogen as

markers of BBB integrity and leakage.

We will explore the impact of seven prominent PICALM SNP alleles (rs3851179, rs1237999, rs510566, rs7480193, rs41458, rs561655, rs642949) that occur in about 30-45% of the population, on PICALM expression and other measures. SNPs were chosen based on literature reporting correlative effects in cohorts ranging from cognitively normal to AD. Specifically:

Tissue. Human postmortem occipital or frontal cortical tissues will be sourced from ~1100 banked brains available in the USC ADRC tissue bank (courtesy of Dr. Hiniker). Samples will be sourced from APOE4 carriers and non-carriers at high and low Braak stages, in age-, sex-, PMI- and Braak stage-matched groups. We will evaluate four groups: high or low Braak stage in APOE4 carriers, and high or low Braak stage in APOE4 non-
carriers.

Tissue availability and demographics: Dr. Hiniker (ADRC Neuropathology Core) and her team reviewed the ADRC brain bank (~1,100 cases) to estimate availability and demographics for the four proposed groups: Braak I/II APOE4−, Braak I/II APOE4+, Braak V/VI APOE4−, and Braak V/VI APOE4+ (target n=9–12/group). The following counts are conservative because the database is currently being updated, and additional eligible cases are expected. Nonetheless, we identified >20 cases in the rarest group (Braak I/II APOE4+; ~1:1 male:female; mean age 78 years, range 58–96 years), >40 cases in Braak I/II APOE4−, and >60 cases in each high-Braak stratum. Across groups, there is sufficient age and sex distribution to support close matching to the Braak I/II APOE4+ cases. Therefore, we anticipate no difficulty assembling a robust, well-matched cohort for these studies. Tissue availability in the proposed cortical regions (occipital cortex or frontal cortex) is sufficient to support the planned analyses.

Immunostaining. Tissues will be sectioned and immunostained similar to previous work by us and others (Fig.1). Tissue will be stained for PICALM expression, amyloid, markers of BBB integrity (e.g. tight junction proteins and fibrinogen), and vascular endothelium marker (e.g. CD31 or lectin), and imaged on our Nikon A1R confocal microscope equipped with objectives ranging from 4-60x magnification and 6 laser lines. Vascular PICALM expression, amyloid/CAA, and BBB integrity markers will be quantified. PICALM expression will be evaluated relative to CAA extent and localization, APOE4 carrier status, and Braak stage. Similar analyses will be performed for BBB integrity markers. Images will be quantified using ImageJ/Fiji using our established protocols (e.g.32,39,53,61,62). Other image analysis software tools including Imaris and Nikon Elements are also
available.

PICALM SNP genotyping. Genomic DNA will be isolated from the tissue samples using our well-establishedprotocol with Zymo miniprep kits (Cat# D3025) and genotyped for 7 PICALM SNPs using validated commercially available TaqMan RT-PCR SNP genotyping assays (ThermoFisher Cat# 4351379) on our Bio-Rad CFX Connect RT-PCR system. PICALM SNP information will be correlated with PICALM vascular expression, APOE genotype,
CAA accumulation, and BBB integrity markers. Statistical analysis. Currently no data exists to our knowledge that compares immunostaining for vascular
PICALM, CAA, and APOE genotype. This proposal will establish preliminary data on these comparisons that will inform future sample size estimates.

Therefore, sample size estimations were calculated using means and SD from previous studies that explored key proposed comparisons via immunostaining, using GraphPad Prism assuming a two-sided $\alpha=0.05$, 80% power, and homogeneous variances for 2 or 4 conditions to be compared. Group size estimates varied from approx. 3-12 per group, depending on marker analyzed, with many estimates falling between 7-12 per group. Therefore, to err on the conservative side, we estimate a sample size for human tissue immunostaining studies of 9-12 per group (e.g. high/low Braak scores, APOE4 carrier/non-carrier). Demographic information for human tissues will be de-identified by the tissue repository. We will use the Kolmogorov-Smirnov test to evaluate the normality of the data, use t-test or ANOVA for normally-distributed data or Mann-Whitney U test or Kruskal-Wallis test as nonparametric alternatives for descriptive statistics, and linear regression models (or alternative non-parametric regression models) followed by false discovery rate correction to control for multiple tests. $p<0.05$ will be considered significant. Experiments will be blinded.

Title: ***Synergistic Impact of Alzheimer's Disease risk factors PICALM and APOE on cerebral amyloid angiopathy and vascular pathology***

Investigator: Kassandra Kisler, Assistant Professor of Research, Dept. of Physiology and Neuroscience

ADRC collaborator/sponsor: Annie Hiniker

Study Duration: 1 year. Work will be distributed throughout the duration of the study, with expectation to prepare extramural funding applications based on results towards the end of the study period.

Background, rationale and Aims.

Alzheimer's disease (AD)[1–6] is a progressive neurodegenerative disorder characterized by accumulation of toxic amyloid β-peptide (Aβ)[7–10], tau pathology[11–14], neuronal loss[7,8,11,12,15,16] and neurovascular dysfunction[15,16]. It is the most common cause of dementia[17], and a growing socioeconomic burden as the population ages[18]. Carriers of the most prominent **AD genetic risk factor apolipoprotein ε4 (*APOE4*)**, are at greater risk of developing AD and are more likely to exhibit neurovascular dysfunction, vascular blood-brain barrier (BBB) damage and increased permiability[9,15,19–22], but the reasons for this are not well understood.

A common feature observed in 80-100% of AD patients[23] is **cerebral amyloid angiopathy (CAA)**, the accumulation of Aβ, including Aβ oligomers, amyloid fibrils, protein aggregates, in and around brain vasculature. CAA is associated with increased vascular stiffness, reduced BBB integrity and microbleeds[23,24]. CAA is more common in AD patients carrying *APOE4* allele compared to the common *APOE3* allele[23,25,26], displaying amyloid deposits around capillaries as well as larger vessels[27,28]. Recent reports have also identified both *APOE4* carrier status[23] and CAA[23,29] as major risk factors for potentially life-threatening vasculotropic side-effects associated with the new FDA-approved AD anti-amyloid immunotherapies, such as edema and microhemorrhage, collectively called amyloid-related imaging abnormalities (ARIA). In sporadic cases, CAA is thought to develop in part due to lack of effective amyloid clearance from the brain[25] but the mechanisms underlying pathogenesis of CAA and related vascular changes are poorly understood. Evidence from amyloid precursor protein (APP) mutations linked to familial CAA and resulting in mutations located near Aβ residues 21-23, including Flemish A692G, Arctic E693G, Dutch E693Q, Italian E693K, and Iowa D694N, can present without significant parenchymal Aβ deposits[24,25] further implicating disrupted clearance mechanisms may be at fault. Currently no treatments for CAA exist.

PICALM, phosphatidylinositol-binding clathrin assembly protein[30,31], plays a key role in clathrin-mediated endocytosis and regulates the internalization and intracellular trafficking of cell surface receptors[19,32–36]. PICALM is highly expressed in brain vascular endothelium[32,37,38] that makes up the BBB, and plays an essential role in regulating *soluble* Aβ clearance from the brain to blood across the BBB[32,39,40]. However, PICALM expression is significantly reduced in AD brain vasculature[32,37,38] (**Fig. 1**).

Genome wide association and alternative splicing studies indicate *PICALM* is highly associated with late onset AD. In humans, some 20 noncoding *PICALM* single nucleotide polymorphisms (SNPs) have been identified and correlated with differential risk of development or progression of AD (e.g. SNPs *rs3851179, rs541458, rs7480193, rs510566, rs1237999, etc)*[1,41–47] and in some instances differences in PICALM expression[37,48,49]. Furthermore, human studies reported presence of protective *PICALM* SNP *rs3851179*^{A} allele in *APOE4* carriers correlated with slower clinical and pathological disease progression[50] (**Fig. 2**), while *APOE4* carriers with risk *PICALM rs3851179*^{G} allele associated with increased brain atrophy and cognitive impairment[51]. This implies that *PICALM* and *APOE4* may play a synergistic role in AD risk and pathology, but the details of this coactivity remain unknown.

Importantly, our recent work in a mouse β-amyloidosis model revealed that vascular PICALM upregulation using the anti-malaria drug artesunate could restore BBB integrity, reduce Aβ pathology, and *reduce CAA pathology,* likely via improved vascular PICALM-mediated Aβ clearance from brain[39]. Consistently, our preliminary data in mice indicates loss of vascular PICALM *increased CAA pathology* (**Fig. 3**), indicating that PICALM-mediated vascular clearance may play an important role in regulating CAA accumulation and severity.

While PICALM is ideally functionally and spatially positioned to influence the development and severity of CAA and vascular pathology, the spatial and genetic relationship between vascular PICALM, *APOE4* carrier status, and CAA vascular pathology has not been explored. Here we propose to investigate the synergistic role of PICALM in *APOE4* carriers and CAA, and related BBB pathologies. ***We hypothesize that endothelial PICALM expression, SNP status, and APOE4 carrier status synergistically contribute to the extent of CAA pathology and BBB breakdown.*** We will test this hypothesis in the following Aims:

Aim 1: Investigate whether **(a)** vascular PICALM expression spatially coincides and correlates with CAA accumulation and BBB breakdown in human cortex, **(b)** whether vascular PICALM expression in cortex differs by *APOE* genotype (*APOE4* carriers vs. non-carriers), and **(c)** whether the association between PICALM expression and CAA pathology differ by *APOE* genotype.

- We expect that participants with reduced vascular PICALM expression will exhibit increased CAA pathology, BBB integrity loss and permeability, particularly in *APOE4* carriers compared to non-carriers, and that *APOE4* carriers will have reduced vascular PICALM expression vs non-carriers.

Aim 2: Explore **(a)** whether vascular PICALM expression and CAA pathology correlate with specific *PICALM* SNP alleles, and **(b)** if the associations differ by *APOE* genotype.

We will utilize human post-mortem AD brain tissues from *APOE4* carriers and non-carriers from USC's Alzheimer's Disease Research Center (ADRC) repository. We will determine vascular PICALM expression, CAA accumulation and extent, and BBB integrity via tight junction changes and parenchymal fibrinogen, using immunohistochemistry similar to previous work by us and others[32,38,52,53]. Patient samples will be genotyped for prominent *PICALM* AD SNPs using commercially available assays. Data will also be correlated to *APOE* genotype and other demographics and measures of AD pathology and cognitive function available from the ADRC for these patients, and controlled for measures such as age, sex, PMI, etc.

This study will provide new information on the synergistic impact of *PICALM* with *APOE*, two of the most prevalent risk factors for AD, on CAA burden and related vascular pathologies. This will also establish PICALM as a therapeutic target for alleviating *APOE4*-mediated CAA burden, BBB damage, and potentially risk reduction or prevention of ARIA immunotherapy side effects. This data will form the basis for future (e.g. within a year or less of project completion) extramural NIH and foundation (e.g. Alzheimer's Association) grant applications that will further explore the mechanism(s) and interaction of PICALM and *APOE* in AD and CAA pathologies in human tissues, mouse models, and *in vitro* human (e.g. iPSC-derived cells, organoids and/or assembloids) models.

Preliminary data. We and others[32,37,38] found that PICALM is highly expressed in human brain microvessels in control (Braak 0-I) brains, illustrated by immunohistochemistry, immunoblotting of isolated cortical microvessels (**Fig. 1a-c**), and immunostaining colocalized to lectin-positive vascular endothelium (**Fig. 1d-e**). In AD patient brains (Braak V-VI), there was a ≥50% loss of vascular PICALM compared to Braak 0-I controls (**Fig. 1a-e**)[32,37,38]. Endothelial PICALM levels in control and AD cases (Braak 0-VI) were highly inversely correlated with Aβ load (**Fig. 1f**), Braak stage, CDR score (not shown), and positively correlated to Mini Mental Sate Examination scores (**Fig. 1g**). Together this demonstrates a pattern of endothelial PICALM loss with progression and severity of AD.

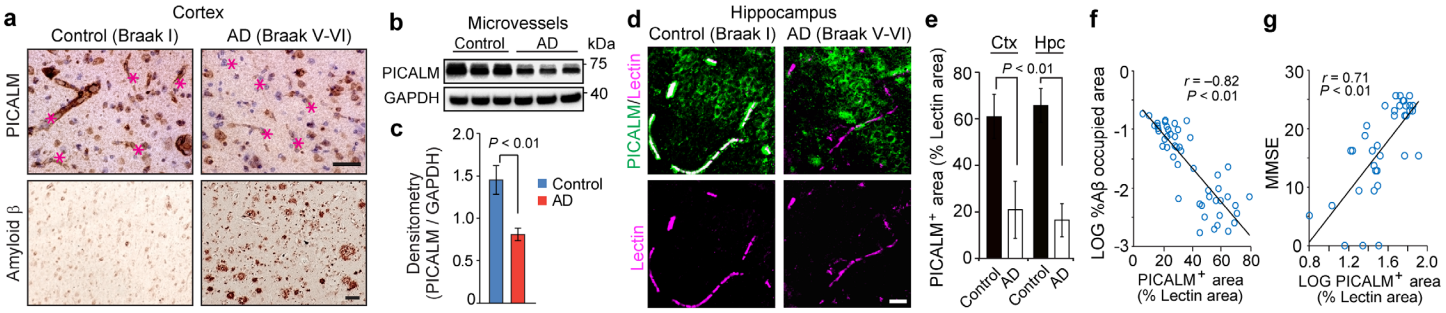

Fig. 1. PICALM is reduced in brain capillary endothelium in Alzheimer's disease. a. PICALM and Aβ immunostaining in the prefrontal cortex of an age-matched control (Braak I) and AD case (Braak V–VI). Stars mark several vessels; bar = 20 µm. **b-c.** Immunoblotting for PICALM and GAPDH loading control in cortical microvessels from controls (Braak 0-1) and AD cases (Braak V-VI) (**b**), and quantification relative to GAPDH (**c**). Mean ± SD, n=6 per group; Student's t-test. **d-e.** PICALM (green) and lectin–positive endothelial capillary vessels (magenta) in the hippocampus (CA1) of age-matched control (Braak I) and AD cases (Braak V–VI). Bar=20 µm (**d**) and quantification of PICALM-positive area (percentage) occupying lectin-positive capillary endothelium in the prefrontal cortex (Ctx) and the CA1 hippocampal (Hpc) subfield (**e**). Mean ± SD n=9 controls (Braak 0–I) and 7 AD cases (Braak V–VI); p<0.01 by Student's t–test. **f-g.** Correlations between PICALM⁺ area occupying lectin⁺-endothelial capillary (<6 mm diameter) profiles (percentage) and brain parenchymal Aβ load (**f**), and Mini Mental Sate Examination (MMSE, **g**). In **f-g**, individual values from 50 (**f**) and 37 (**g**) controls and AD cases. Pearson and Spearman rank correlation analysis. (Zhao…Kisler, et al [32]).

Liu et al[50] recently reported association of protective *PICALM* SNP *rs3851179*^{A} allele moderates the effect of *APOE4* on cognitive measures in a cross-sectional cohort of non-demented individuals **(Fig. 2)**. Followed longitudinally, the presence of *rs3851179*^{A} allele was associated with slower memory decline and CSF AD biomarker changes among *APOE4* carriers in this cohort (not shown)[50].

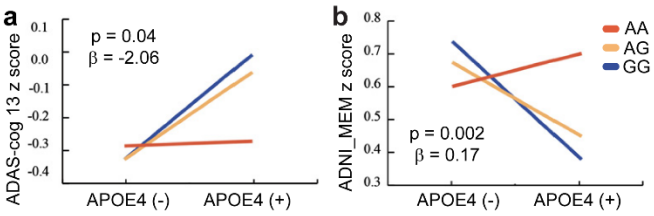

Fig. 2. Protective *PICALM rs3851179 A* allele moderates the relationship of *APOE4* with cognition. The SNP *rs3851179*×*APOE4* interaction accounted for a significant variance in Alzheimer's Disease Assessment Scale (ADAS)-cog13 **(a)**, and composite memory (MEM) **(b)** scores in a cohort of cognitively normal and MCI participants. Participants were stratified by number of copies of the protective *A* or non-protective *G* SNP allele. In **(a)**, lower scores indicate better cognition; in **(b)**, higher scores indicate better cognition. Estimate (β), the amount of change in the dependent variable corresponding to each one-unit change in the independent variable. Panels adapted from Liu et al [50].

In our preliminary mouse studies, mice with endothelial-specific *Picalm* deletions crossed with the *5xFAD* model[39,54] of β-amyloidosis carrying 5 familial AD mutations in APP and PSEN1 (*Picalm*$^{lox/lox}$; *Cdh5-Cre; 5xFAD* mice[39]) exhibited increased Aβ pathology, and increased CAA in arteriolar vessels compared to littermate controls (*Picalm*$^{lox/lox}$; *5xFAD* mice) (**Fig. 3**).

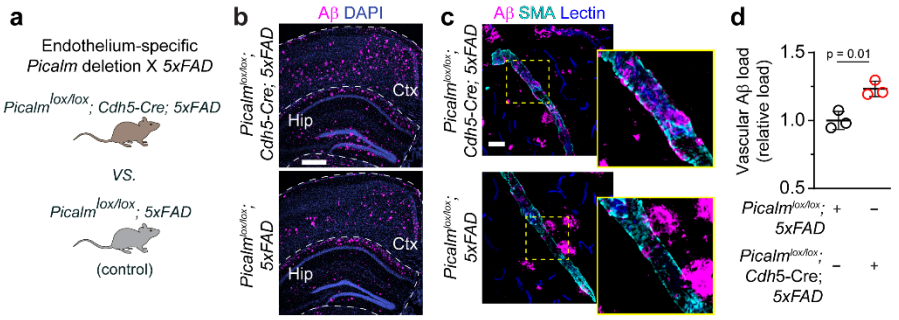

Fig. 3. Endothelial-specific *Picalm* deletion increases Aβ load and CAA in *5xFAD* mice. a. *Picalm*$^{lox/lox}$; *Cdh5-Cre; 5xFAD* mice with endothelial *Picalm* deletion were compared to control (*Picalm*$^{lox/lox}$; *5xFAD*) littermates with endothelial PICALM expression. **b.** Representative immunostaining of brain Aβ load in the cortex (Ctx) and hippocampus (Hip) of *Picalm*$^{lox/lox}$; *Cdh5-Cre; 5xFAD* and littermate controls. **c-d**. Representative immunostaining of α-smooth muscle actin (SMA)-positive arteriolar vascular-associated amyloid (CAA) in cortex in lectin-positive vessels (**c**) and preliminary quantification of arteriolar vascular amyloid load in cortex (**d**) in *Picalm*$^{lox/lox}$; *Cdh5-Cre; 5xFAD* mice and littermate controls. Data bars are mean ± S.D. In **c**, dashed boxes mark inset locations; bars in **b**=500 μm, **c** =30 μm.

Together these data and literature detailed above imply connections between PICALM expression, *PICALM* SNPs, CAA accumulation and *APOE* carrier status in AD progression and pathology.

Methodology. We will study vascular-specific changes in PICALM expression relative to CAA accumulation in *APOE4* carriers vs non-carriers in postmortem occipital or frontal cortical tissue, where CAA pathology most frequently occurs[55–57], using immunostaining techniques[32,38,52,53]. We will also investigate vascular tight junction expression (e.g. claudin 5, occludin) and perivascular fibrinogen as markers of BBB integrity and leakage[9,15,53,58]. We will explore the impact of seven prominent *PICALM* SNP alleles (*rs3851179, rs1237999, rs510566, rs7480193, rs41458, rs561655, rs642949*) that occur in about 30-45% of the population, on PICALM expression and other measures. SNPs were chosen based on literature reporting correlative effects in cohorts ranging from cognitively normal to AD[1,37,41–51,59,60]. Specifically:

Tissue. Human postmortem occipital or frontal cortical tissues will be sourced from ~1100 banked brains available in the USC ADRC tissue bank (courtesy of Dr. Hiniker). Samples will be sourced from *APOE4* carriers and non-carriers at high and low Braak stages, in age-, sex-, PMI- and Braak stage-matched groups. We will evaluate four groups: high or low Braak stage in *APOE4* carriers, and high or low Braak stage in *APOE4* non-carriers.

Tissue availability and demographics: Dr. Hiniker (ADRC Neuropathology Core) and her team reviewed the ADRC brain bank (~1,100 cases) to estimate availability and demographics for the four proposed groups: Braak I/II *APOE4*−, Braak I/II *APOE4*+, Braak V/VI *APOE4*−, and Braak V/VI *APOE4*+ (target n=9–12/group). The following counts are conservative because the database is currently being updated, and additional eligible cases are expected. Nonetheless, we identified >20 cases in the rarest group (Braak I/II *APOE4*+; ~1:1 male:female; mean age 78 years, range 58–96 years), >40 cases in Braak I/II *APOE4*−, and >60 cases in each high-Braak stratum. Across groups, there is sufficient age and sex distribution to support close matching to the Braak I/II *APOE4*+ cases. Therefore, we anticipate no difficulty assembling a robust, well-matched cohort for these studies. Tissue availability in the proposed cortical regions (occipital cortex or frontal cortex) is sufficient to support the planned analyses.

Immunostaining. Tissues will be sectioned and immunostained similar to previous work by us and others (**Fig. 1**) [32,38,52,53]. Tissue will be stained for PICALM expression, amyloid, markers of BBB integrity (e.g. tight junction proteins and fibrinogen), and vascular endothelium marker (e.g. CD31 or lectin), and imaged on our Nikon A1R confocal microscope equipped with objectives ranging from 4-60x magnification and 6 laser lines. Vascular PICALM expression, amyloid/CAA, and BBB integrity markers will be quantified. PICALM expression will be evaluated relative to CAA extent and localization, *APOE4* carrier status, and Braak stage. Similar analyses will be performed for BBB integrity markers. Images will be quantified using ImageJ/Fiji using our established protocols (e.g.[32,39,53,61,62]). Other image analysis software tools including Imaris and Nikon Elements are also available.

PICALM SNP genotyping. Genomic DNA will be isolated from the tissue samples using our well-established protocol with Zymo miniprep kits (Cat# D3025) and genotyped for 7 PICALM SNPs using validated commercially available TaqMan RT-PCR SNP genotyping assays (ThermoFisher Cat# 4351379) on our Bio-Rad CFX Connect RT-PCR system. PICALM SNP information will be correlated with PICALM vascular expression, *APOE* genotype, CAA accumulation, and BBB integrity markers.

Statistical analysis. Currently no data exists to our knowledge that compares immunostaining for vascular PICALM, CAA, and *APOE* genotype. This proposal will establish preliminary data on these comparisons that will inform future sample size estimates. Therefore, sample size estimations were calculated using means and SD from previous studies that explored key proposed comparisons via immunostaining[32,37,38,50,53,63,64], using GraphPad Prism assuming a two-sided α=0.05, 80% power, and homogeneous variances for 2 or 4 conditions to be compared. Group size estimates varied from approx. 3-12 per group, depending on marker analyzed, with many estimates falling between 7-12 per group. Therefore, to err on the conservative side, we estimate a sample size for human tissue immunostaining studies of 9-12 per group (e.g. high/low Braak scores, *APOE4* carrier/non-carrier). Demographic information for human tissues will be de-identified by the tissue repository. We will use the Kolmogorov-Smirnov test to evaluate the normality of the data, use t-test or ANOVA for normally-distributed data or Mann-Whitney U test or Kruskal-Wallis test as nonparametric alternatives for descriptive statistics, and linear regression models (or alternative non-parametric regression models) followed by false discovery rate correction to control for multiple tests. $p<0.05$ will be considered significant. Experiments will be blinded.

Expected outcomes. We expect vascular PICALM expression will inversely spatially coincide and correlate with CAA accumulation and vascular pathology in an *APOE* allele-dependent context. We expect that vascular PICALM expression will be reduced in *APOE4* carriers vs. non-carriers, and loss will be more severe with higher Braak stage and CAA pathology. We also anticipate that PICALM expression will correlate with specific PICALM SNP alleles. Although associations may not reach significance for all SNP alleles, the information generated will inform on which SNPs would be most relevant to pursue in future studies.

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64. Magaki, S. *et al.* The effects of cerebral amyloid angiopathy on integrity of the blood-brain barrier. *Neurobiol. Aging* **70**, 70–77 (2018).

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022
FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Kassandra		Elliott		PI	105,222.00	2.4			21,044.00	6,671.00	27,715.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name:

Total Senior/Key Person **27,715.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
1	Research Associate	1.2	0.0	0.0	7,210.00	2,286.00	9,496.00
1	Lab Tech	2.4	0.0	0.0	11,648.00	3,692.00	15,340.00
2	**Total Number Other Personnel**					**Total Other Personnel**	**24,836.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**52,551.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	17,449.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**17,449.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**70,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	70,000.00	46,200.00
		Total Indirect Costs	**46,200.00**

Cognizant Federal Agency DHHS Jeanette Lu 415-437-7820
(Agency Name, POC Name, and POC Phone Number)

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**116,200.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	116,200.00

L. Budget Justification* File Name: Budget justification Kisler.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Budget Justification

Title: ***Synergistic Impact of Alzheimer's Disease risk factors PICALM and APOE on cerebral amyloid angiopathy and vascular pathology***
Investigator: Kassandra Kisler

We request a total of **$70,000** for various supplies and salaries for the proposed study. Budget details and justification are listed below.

Key Personnel

Kisler Elliott, Kassandra, PhD, PI (**20% effort; 2.4 calendar months**)
Dr. Kisler will be the Principal Investigator on this project. She will be responsible for the overall supervision of the project, planning and execution of experiments and preparation of future extramural funding proposals and manuscripts.

Dr. Kisler is an Assistant Professor of Research in the Zilkha Neurogenetic Institute and Department of Physiology and Neuroscience at the Keck School of Medicine (KSOM) at USC, and is Assistant Director of the Optical Imaging Core and Operations in the Center for Neurodegeneration and Regeneration at the Zilkha Neurogenetic Institute. Dr. Kisler investigates vascular effects of genetic risk factors for Alzheimer's disease including *PICALM* and *APOE4*, and has publications in *Molecular Neurodegeneration* 2023, *Journal of Experimental Medicine* 2022, *Nature Neuroscience* 2015, and others detailing studies of these proteins. She has extensive experience working with mouse models of Alzheimer's disease and vascular dysfunction and has experience with the techniques proposed in this study. Dr. Kisler is also an expert in optical microscopy techniques for biological applications, and has published several papers using two-photon microscopy and other *in vivo* techniques to observe and quantify neurovascular function in mouse models, including publications in *Nature Neuroscience* 2017 and 2019, *Nature Protocols* 2018, and others. She has studied the role of PICALM in neurovascular physiology and PICALM loss in AD models for over a decade, and APOE4 models for over 5 years. Dr. Kisler developed a drug screen that identified and validated the drug artesunate from an FDA-approved drug library screen to upregulate PICALM. For this project, Dr. Kisler will perform *PICALM* SNP genotyping assays, and train and oversee the work of research associate Dr. Zhang, and technician Mr. Spillard in the lab. Dr. Kisler oversee all aspects concerned with experimental design, planning of experiments, data analysis and interpretation.

Other Personnel

Mingzi Zhang, PhD, Research Associate (10% effort; 1.2 calendar months) will advise and contribute to tissue processing, immunostaining, and image analysis of human tissues. She will help train Mr. Spillard in human tissue processing and immunostaining. Dr. Zhang has worked with Dr. Kisler's team for several years and has extensive expertise in immunostaining human and mouse tissues and high-resolution imaging techniques.

Gavin Spillard, MS, Lab Technician (20% effort; 2.4 calendar months) is a skilled technician in Dr. Kisler's team. He has experience with tissue processing, staining, and imaging techniques and will perform human tissue processing and immunostaining for this study under the guidance of Drs. Zhang and Kisler. Mr. Spillard also has experience with genomic DNA extraction and genotyping assays, and will be trained by Dr. Kisler to assist her with *PICALM* SNP RT-PCR genotyping assays.

Materials and supplies ($4949)
General lab supplies including consumables, glassware and chemicals.

Immunostaining supplies ($6000)
Histological supplies including immunofluorescent reagents and various primary and secondary antibodies for immunostaining analyses of PICALM, BBB integrity (e.g., fibrinogen, tight junction proteins), endothelial cells (e.g, CD31, lectin), amyloid pathology/CAA (e.g., Aβ antibodies).

SNP genotyping supplies and assays ($6000)
Genomic DNA extraction kits, SNP genotyping assay plates/kits, reagents for RT-PCR.

Research computing ($500)
Computer peripherals and data storage, and softwares for data organization and analysis.

A. COMPONENT COVER PAGE

Project Title: 7.3_APOE-Dependent Impact on α-Synuclein and Aβ Cross-Seeding: Behavioral and Molecular Signatures of Co-Pathology_Ahammad_new
Component Project Lead Information: Ahammad, Rijwan

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

SPECIFIC AIMS:
Aim 1. APOE dependent cognitive impact of α-syn/Aβ co-expression in vivo.
Rationale: Mixed α-syn/Aβ pathologies may synergistically impair cognition, with APOE4 amplifying vulnerability. Early functional deficits may precede overt pathology. Hypothesis: APOE4 mice will exhibit greater cognitive and behavioral impairments following α-syn/Aβ co-expression compared with APOE3 mice.

Approach: We will inject AAV-GFP-α-syn and AAV-mCherry-Aβ42 under the neuron-specific synapsin promoter into the hippocampus of APOE3 and APOE4 knock-in mice. Behavioral performance will be assessed three months post-injection using Morris Water Maze (MWM), Y-maze and Novel Object Recognition (NOR). Behavioral readouts will be correlated with subsequent molecular endpoints in Aim 2.

Expected Outcome: APOE4 mice will show impaired spatial memory, recognition memory deficits, and altered exploratory behavior relative to APOE3 mice, establishing functional consequences of mixed proteinopathy.

Aim 2. Molecular and cellular characterization of APOE-dependent α-syn/Aβ co-pathology
Rationale: Behavioral deficits in Aim 1 likely result from APOE-dependent changes in protein aggregation and microglial clearance.

Hypothesis: APOE4 enhances α-syn/Aβ aggregation, co-localization, and impairs microglial clearance relative to APOE3.

Approach: Brain tissue from Aim 1 mice will be analyzed for α-syn/Aβ co-localization using Proximity Ligation Assay (PLA) with super-resolution microscopy. Soluble and insoluble aggregates will be quantified by fractionation and immunoblotting. Microglial activation, phagocytic activity, lysosomal/autophagy function, and synaptic integrity will be assessed, linking aggregate burden to immune and synaptic dysfunction. These analyses will determine how APOE genotype influences cellular responses to α-syn/Aβ co-pathology and its impact on neuronal health. Expected Outcome: APOE4 mice will show increased α-syn/Aβ co-localization, higher aggregate burden, and reduced microglial clearance, linking molecular pathology to behavioral deficits observed in Aim 1.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

PROJECT EXECUTION PLAN AND TIMELINE:

This 12-month project integrates coordinated animal, behavioral, molecular, and imaging studies designed to meet the objectives of Aim 1 and Aim 2. Activities are organized into sequential but overlapping phases to ensure efficient use of time, with dedicated periods for animal incubation, assay preparation, and concurrent analytical work. A 3-month post-injection period is required for α-syn/Aβ propagation; during this time, substantial preparatory, optimization, and analytical tasks will be conducted (detailed below), consistent with NIH expectations for continuous productivity.

Months 0–3 | Cohort Establishment, Reagent Validation, and Viral Delivery
• Finalize experimental cohort (genotyping, randomization, baseline health checks).
• AAV purchase and QC (titer verification, purity, infectivity testing).
• Conduct stereotaxic injections across all groups.
• Initiate post-injection incubation period for pathology development; perform weekly welfare monitoring and maintain experimental logs.
Milestone: Successful delivery of viral constructs.

Months 3–6 | Post-Injection Incubation & Full Behavioral Assessment (Aim 1)
• Continue post-injection incubation to allow α-syn/Aβ co-pathology development.
• Complete final optimization and validation of Proximity ligation Assay (PLA) using tissue collected from small pilot subsets (not the main cohort).
• Conduct low-stress behavioral tests (Open Field Y-maze Novel Object).
• Conduct high-demand behavioral test last (Morris Water Maze).
• Perform interim data QC and refine behavioral data processing pipeline.
Milestone: Completed and quality-checked behavioral dataset and validated assay conditions for downstream analysis.

Months 6–10 | Molecular, Cellular, and Imaging Analyses (Aim 2)
• Harvest brains from experimental cohorts at defined endpoints.
• Perform PLA, immunofluorescence, Western blotting, and biochemical assays.
• Conduct confocal and widefield imaging for synaptic and pathological markers.
• Begin preliminary quantification and statistical analysis of molecular/imaging data.
Milestone: Completed molecular and imaging datasets with primary quantitative outputs.

Months 10–12 | Integrated Data Analysis, Interpretation, and Dissemination
• Integrate behavioral, biochemical, and imaging datasets.
• Conduct final statistical modeling and verification of reproducibility.
• Prepare figures, summary tables, and draft internal reports.
• Initiate post-award dissemination activities (abstract submission, preliminary manuscript outline).
Milestone: Fully analyzed dataset ready for dissemination and grant follow-up.

Title:

APOE-Dependent Impact on α-Synuclein and Aβ Cross-Seeding: Behavioral and Molecular Signatures of Co-Pathology

Principal Investigator (PI): Rijwan Ahammad, PhD

Mentor and Sponsor: Robert Rissman, PhD

Affiliation: Department of Physiology and Neuroscience,

Alzheimer's Therapeutic Research Institute (ATRI),

Keck School of Medicine, USC

Contact Information: Email: rahammad@usc.edu, Phone: +1-619-808-8569

Response to Reviewer Comments

We thank the reviewer for the encouraging and insightful comments regarding the significance of addressing mixed amyloid and synuclein proteinopathies, the strong rationale for focusing on APOE genotype, and the assessment of the investigator as productive and well positioned within an excellent research environment. We appreciate the recognition that this project will generate valuable preliminary data to support future mechanistic funding. Below, we address the specific concerns raised.

(1) Exploratory nature of the proposed studies

We agree with the reviewer that the proposed studies are primarily exploratory and descriptive, and this is an intentional feature of the project design. The goal of this USC ADRC Developmental Project is to establish rigorously controlled, APOE genotype dependent behavioral and molecular phenotypes in a mixed α-syn/Aβ model, which remain poorly defined despite the high prevalence of co-pathology in human disease. These foundational data are essential to guide subsequent hypothesis-driven mechanistic studies in a future R01 aimed at defining how APOE modulates α-syn/Aβ interactions, aggregation, and clearance pathways.

(2) Feasibility of AAV-based α-syn and Aβ expression to induce phenotypes

To address the reviewer's concern regarding AAV-mediated expression of α-synuclein and Aβ without prior validation, the revised proposal explicitly incorporates pilot validation experiments before initiating full behavioral and molecular analyses. Neuron-specific AAV9 vectors expressing human α-synuclein and Aβ42 (AAV9-hSyn-hSNCA-P2A-EGFP and AAV9-hSyn-APP(SP)-Aβ42-P2A-mCherry) will be used to ensure human-relevant neuronal expression. The human synapsin promoter restricts expression to neurons, while P2A-linked fluorescent reporters enable direct visualization, quantification of transduction efficiency, and regional expression mapping.

Pilot endpoints will assess transgene expression, aggregation, and early biochemical and histopathological changes, providing a clear go/no-go decision point prior to advancing to behavioral testing. This staged approach is supported by prior studies demonstrating that AAV-based delivery of α-synuclein and Aβ is sufficient to induce amyloid and synuclein pathology *in vivo*. If robust pathology is not observed within 3-6 months, the experimental strategy will pivot to a seeding-based approach using α-synuclein pre-formed fibrils and Aβ oligomers to ensure induction of biologically relevant pathology. This dual-path strategy minimizes risk while preserving rigor and translational relevance.

(3) Inclusion of sex as a biological variable

We agree with the reviewer that the initial focus on male mice alone was insufficiently justified. In response, the revised proposal now includes both male and female APOE3 and APOE4 knock-in mice. Sex will be incorporated as a biological variable in experimental design and statistical analyses, thereby strengthening rigor and enhancing translational relevance.

SPECIFIC AIMS:

Alzheimer's disease (AD) is characterized by amyloid-β (Aβ) plaques and tau neurofibrillary tangles, yet these classical hallmarks alone do not fully account for progressive cognitive decline[1,2]. Therapies targeting Aβ and tau have demonstrated only modest clinical efficacy and fail to halt disease progression[3,4], highlighting the urgent need to investigate additional mechanisms that drive neurodegeneration. Recent neuropathological studies reveal that up to 50% of AD brains exhibit α-synuclein (α-syn) positive Lewy bodies, while many Parkinson's disease (PD) and dementia with Lewy bodies (DLB) cases contain Aβ or tau deposits[5-7]. This overlap suggests that neurodegenerative diseases may represent a continuum of mixed proteinopathies rather than distinct entities. Experimental data indicate that Aβ and α-syn can physically interact and cross-seed, mutually enhancing aggregation, synaptic toxicity, and cognitive decline[6,8]. However, the molecular factors governing these interactions remain poorly understood. Critically, no studies have systematically examined how APOE genotype governs α-syn/Aβ interactions *in vivo* at the level of protein aggregation, clearance pathways, and cell-type specific neuroinflammatory responses.

One strong candidate is apolipoprotein E (APOE), the major genetic risk factor for late-onset AD[9]. Beyond its established roles in Aβ clearance and neuroinflammation, emerging studies demonstrate that the APOE4 isoform exacerbates α-syn accumulation and neurodegeneration independent of amyloid burden[10-12]. This positions APOE as a potential unifying modifier of mixed proteinopathies, rather than a risk factor restricted to classical amyloid-driven AD. Yet, how APOE genotype modulates α-syn/Aβ co-pathology at the cellular and molecular level remains unknown. Specifically, it is unclear whether APOE4 alters α-syn/Aβ interactions through impaired proteostatic clearance, enhanced glial reactivity, or direct effects on neuronal vulnerability. Addressing this gap is critical, as co-pathology involving α-syn is strongly associated with faster clinical decline, increased therapeutic resistance, and greater disease heterogeneity in AD and related dementias[13]. We hypothesize that, *APOE4 amplifies α-syn and Aβ co-pathology by impairing protein clearance and exacerbating glial-driven neuroinflammation, leading to accelerated synaptic and cognitive dysfunction.* The proposed studies will leverage transgenic mouse models and cell-type specific analyses to define APOE-dependent mechanisms driving α-synuclein and Aβ interactions. By uncovering how APOE genotype shapes cross-protein aggregation, clearance dynamics, and neuroinflammatory cascades, this work will advance our understanding of why certain individuals particularly APOE4 carriers exhibit more aggressive and treatment-resistant forms of dementia. Ultimately, elucidating these mechanisms will identify new molecular targets for therapeutic intervention in mixed Alzheimer's and Lewy body dementias and inform personalized strategies for patients at the highest genetic risk. The proposed work will *establish a foundational dataset for a future R01 application aimed at multi-omics characterization of APOE-dependent α-synuclein and Aβ co-pathology across mouse models and human brain tissues, enabling comparative analyses between pure AD and mixed-pathology cohorts*.

Aim 1. APOE dependent cognitive impact of α-syn/Aβ co-expression *in vivo.*
Rationale: Mixed α-syn/Aβ pathologies may synergistically impair cognition, with APOE4 amplifying vulnerability. Early functional deficits may precede overt pathology. *Hypothesis:* APOE4 mice will exhibit greater cognitive and behavioral impairments following α-syn/Aβ co-expression compared with APOE3 mice. *Approach:* We will inject AAV-GFP-α-syn and AAV-mCherry-Aβ42 under the neuron-specific synapsin promoter into the hippocampus of APOE3 and APOE4 knock-in mice. Behavioral performance will be assessed three months post-injection using Morris Water Maze (MWM), Y-maze and Novel Object Recognition (NOR). Behavioral readouts will be correlated with subsequent molecular endpoints in Aim 2. *Expected Outcome:* APOE4 mice will show impaired spatial memory, recognition memory deficits, and altered exploratory behavior relative to APOE3 mice, establishing functional consequences of mixed proteinopathy.

Aim 2. Molecular and cellular characterization of APOE-dependent α-syn/Aβ co-pathology
Rationale: Behavioral deficits in Aim 1 likely result from APOE-dependent changes in protein aggregation and microglial clearance. *Hypothesis:* APOE4 enhances α-syn/Aβ aggregation, co-localization, and impairs microglial clearance relative to APOE3. *Approach:* Brain tissue from Aim 1 mice will be analyzed for α-syn/Aβ co-localization using Proximity Ligation Assay (PLA) with super-resolution microscopy. Soluble and insoluble aggregates will be quantified by fractionation and immunoblotting. Microglial activation, phagocytic activity, lysosomal/autophagy function, and synaptic integrity will be assessed, linking aggregate burden to immune and synaptic dysfunction. These analyses will determine how APOE genotype influences cellular responses to α-syn/Aβ co-pathology and its impact on neuronal health. *Expected Outcome:* APOE4 mice will show increased α-syn/Aβ co-localization, higher aggregate burden, and reduced microglial clearance, linking molecular pathology to behavioral deficits observed in Aim 1.

BACKGROUND & SIGNIFICANCE:

Alzheimer's disease is the leading cause of dementia, yet its biological complexity has hindered development of effective disease-modifying therapies. Classical hallmarks, extracellular Aβ plaques and intracellular tau neurofibrillary tangles, do not fully account for interindividual differences in cognitive decline or treatment response[1,2]. Emerging evidence shows that α-syn pathology frequently co-occurs with AD, with Lewy bodies detected in up to 50% of AD brains, and conversely, Aβ and tau deposits found in Parkinson's disease and dementia with Lewy bodies[5]. α-Syn co-pathology correlates with accelerated cognitive decline, neuropsychiatric symptoms, and reduced therapeutic responsiveness, highlighting the clinical significance of mixed proteinopathies[6,14]. Experimental studies demonstrate that Aβ and α-syn can interact and cross-seed, promoting aggregation, synaptic dysfunction, and neuronal toxicity[15,16]. However, the factors determining when these interactions become pathogenic *in vivo* are poorly understood, contributing to the heterogeneity of clinical outcomes. Apolipoprotein E, particularly the E4 isoform, is the strongest genetic risk factor for late-onset AD and a candidate regulator of mixed proteinopathies. APOE4 exacerbates α-syn accumulation, impairs proteostatic clearance, and amplifies glial-driven neuroinflammation independent of amyloid burden[12,17]. Yet its role in modulating α-syn/Aβ co-pathology, including effects on protein clearance, glial responses, and synaptic vulnerability, has not been systematically studied *in vivo*. Addressing this knowledge gap is critical, as APOE4 carriers are disproportionately affected by aggressive, treatment-resistant dementias. Understanding how APOE genotype shapes α-syn/Aβ interactions will provide mechanistic insight into disease heterogeneity, reveal new therapeutic targets, and inform precision strategies for patients at the highest genetic risk.

The proposed studies carries significant in three key ways. **First,** they directly interrogate Aβ/α-syn interactions as a mechanistically distinct driver of cognitive decline and disease heterogeneity, moving beyond incidental co-pathology. **Second**, APOE is positioned as a unifying genetic modifier of mixed proteinopathies. Using humanized APOE3 and APOE4 knock-in mice with controlled neuronal co-expression of α-syn and Aβ, these studies enable causal dissection of how genetic risk shapes aggregation, clearance, and neurodegeneration, analyses not achievable through correlative human studies. **Third**, the project integrates behavioral, molecular, and cellular endpoints in the same animals, linking cognitive dysfunction to α-syn/Aβ co-aggregation, microglial clearance, lysosomal/autophagic integrity, and synaptic disruption. Proximity ligation assays combined with super-resolution imaging provide single-aggregate resolution of protein interactions *in situ*, representing a technical advance in studying mixed proteinopathies. Together, this work establishes a novel framework for understanding genetic modulation of interacting proteinopathies, generates foundational mechanistic data for future R01 scale multi-omics studies, and lays the groundwork for precision therapeutic strategies targeting APOE-dependent mixed Alzheimer's and Lewy body dementias.

APPROACH:

Aim 1. APOE dependent cognitive impact of α-syn/Aβ co-expression *in vivo.*

Rationale and Feasibility: APOE4 is the strongest genetic risk factor for late-onset Alzheimer's disease and is known to exacerbate α-synuclein and Aβ-induced neurotoxicity, whereas APOE3 is relatively protective[18,19]. Mixed α-syn/Aβ pathologies frequently co-occur in aging and dementia, contributing to faster cognitive decline and reduced therapeutic response[20]. However, how APOE genotype modulates the behavioral impact of α-syn/Aβ co-pathology remains unclear. Recent studies using a parabiosis model of blood-derived Aβ transfer (AD•APOE4→APOE4) in which Aβ-rich plasma from AD mice expressing APOE4 is introduced into APOE4 recipients, demonstrated greater spatial and recognition memory impairment than in APOE4→WT controls[21]. These findings suggest that APOE4 enhances Aβ-driven neurotoxicity and synaptic dysfunction.

Importantly, multiple prior studies have demonstrated that AAV-mediated neuronal expression of human α-syn or Aβ is sufficient to induce robust protein aggregation, synaptic dysfunction, and cognitive impairment in mice. AAV-based delivery of human α-synuclein to the hippocampus or cortex induces progressive α-syn aggregation, neuroinflammation, and deficits in spatial learning and memory[22-25]. Similarly, AAV-mediated expression of Aβ42 or APP-derived Aβ results in intracellular Aβ accumulation, plaque-like pathology, synaptic loss, and behavioral impairments[26,27]. Notably, combined or sequential induction of α-syn and Aβ pathology using viral expression and/or seeding approaches accelerates pathological burden and worsens cognitive outcomes compared with either pathology alone, supporting the feasibility of modeling mixed proteinopathies *in vivo*[28].

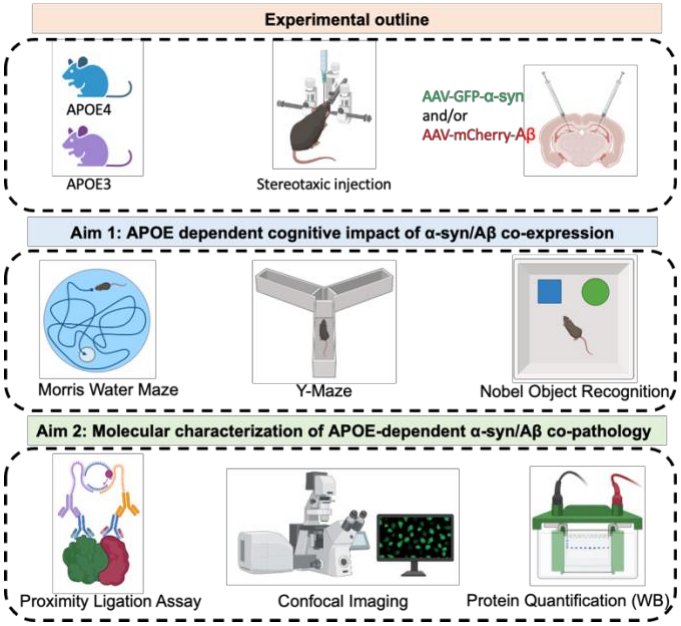
Our approach leverages well-characterized APOE3 and APOE4 knock-in mice maintained by the Dr.Rissman's animal Core. We will use stereotaxic delivery of neuron-specific AAV vectors encoding α-synuclein-GFP and Aβ-mCherry to induce controlled co-expression within the hippocampus. Behavioral assays

will be conducted in the Dr. Rissman's behavioral core using validated paradigms. I have extensive prior experience performing stereotaxic injections and behavioral assays (MWM, Y-Maze, NOR), ensuring technical feasibility.

Together, these published AAV-based and mixed-pathology models establish a strong precedent that targeted neuronal expression of α-syn and Aβ can drive both pathological and behavioral phenotypes, providing a solid foundation for the proposed APOE genotype-dependent analyses.

Experimental Approach: Humanized APOE3 (JAX Stock #029018) and APOE4 (JAX Stock #027894) knock-in mice will be used to model APOE genotype dependent effects on α-syn and Aβ co-pathology, consistent with previous APOE rodent studies[29]. Both male and female mice (10-12 weeks old; n = 8-10 per sex per group) will be included, and sex will be incorporated as a biological variable in experimental design and statistical analyses. Mice will be randomly assigned to six experimental groups: (1) APOE3 + AAV-GFP (control), (2) APOE3 + AAV-hSNCA-P2A-GFP, (3) APOE3 + AAV-hSNCA-P2A-GFP + AAV-APP(SP)-Aβ42-P2A-mCherry, (4) APOE4 + AAV-GFP (control), (5) APOE4 + AAV-hSNCA-P2A-GFP, and (6) APOE4 + AAV-hSNCA-P2A-GFP + AAV-APP(SP)-Aβ42-P2A-mCherry.

Neuron-specific AAV9 vectors driven by the human synapsin promoter were selected to express native human α-syn and a secreted form of human Aβ42, enabling assessment of APOE-dependent effects on aggregation, synaptic toxicity, and clearance under physiologically relevant conditions. Expression of native human α-syn allows interrogation of APOE genotype effects on α-syn handling and propagation without confounding over-aggressive mutations, while secretion of Aβ42 mimics extracellular Aβ exposure known to interact with APOE isoforms and influence synaptic and glial responses. P2A-linked fluorescent reporters (EGFP and mCherry) permit direct visualization of neuronal transduction efficiency, regional expression patterns, and co-expression within the hippocampus.



Bilateral stereotaxic injections (0.5 µL per site, 2 × 10^{12} GC/mL) will target the dorsal hippocampus (AP-2.0 mm, ML ±1.5 mm, DV-1.7 mm from bregma), a region highly relevant to APOE-dependent cognitive vulnerability. To mitigate risk and ensure feasibility, a dedicated pilot cohort (male and female APOE3 and APOE4 mice) will be analyzed prior to behavioral testing. Pilot endpoints will include confirmation of transgene expression, α-syn and Aβ aggregation, and early molecular and histopathological changes (e.g., synaptic markers, neuroinflammation). Advancement to behavioral testing will occur only upon demonstration of robust pathology and expected molecular signatures. This staged design ensures biological relevance, incorporates sex as a biological variable, and provides a rigorous validation framework prior to behavioral analyses.

Behavioral testing will begin 3 months post-injection, to correspond the onset of early α-syn/Aβ propagation. Spatial learning and memory will be assessed using the MWM across five days of acquisition (four trials/day, 60 s maximum per trial, 15-min inter-trial interval), followed by a probe trial on day 6 to measure memory retention. Working memory will be evaluated using the Y-Maze spontaneous alternation test, in which mice freely explore three identical arms (35 × 5 × 10 cm; 120° apart) for 8 minutes. To measure the primary outcome, spontaneous alternation percentage will be calculated as (Number of alternations / (Total arm entries − 2)) × 100. Recognition memory will be evaluated using the Novel Object Recognition (NOR) task (two days habituation, 10 min/day; 10-min training with identical objects; 1-hour test with one novel object) measured by discrimination index (DI) calculated as $(T_{novel}-T_{familiar})/(T_{novel}+T_{familiar})$. All behavioral testing and scoring will be performed by experimenters blinded to genotype and treatment. Behavioral data will be analyzed using a two-way ANOVA (APOE genotype × treatment) for between-group comparisons. For within-subject analyses, such as MWM acquisition across training days, repeated-measures ANOVA will be applied to evaluate learning curves over time. Appropriate post hoc multiple-comparison corrections will be performed where applicable. Following behavioral testing, mice will be euthanized. Plasma, CSF, and one brain hemisphere will be snap-frozen for biochemical analyses, while the contralateral hemisphere will be fixed for histology and imaging. Viral expression will be confirmed at the study endpoint via fluorescence imaging of

GFP and mCherry from the brain tissue. Selected brain tissue and CSF samples will be retained for pilot proteomic and phosphoproteomic analyses (planned for future R01 funding) to link APOE genotype, α-syn/Aβ co-pathology, and behavioral outcomes.

Expected Outcomes: APOE4 mice co-expressing α-syn and Aβ are expected to exhibit significantly delayed escape latencies and reduced time in the target quadrant in MWM, lower spontaneous alternation in Y-Maze, and decreased discrimination index in NOR, whereas APOE3 mice are expected to show milder deficits. These outcomes will directly test the hypothesis that APOE4 selectively increases vulnerability to mixed α-syn/Aβ pathology, rather than to either pathology alone. Importantly, even in the absence of APOE dependent differences, detection of behavioral impairment in co-expression groups would validate the pathogenic relevance of mixed proteinopathy and guide mechanistic analyses in Aim 2.

Potential Pitfalls and Alternatives: Variability in behavioral performance could obscure genotype-dependent effects; to minimize this, multiple complementary behavioral assays will be employed along with randomization and blinded assessment. If behavioral deficits are subtle, cohort size may be increased and/or testing extended to 4-6 months post-injection. Variability in AAV transduction or expression will be addressed by quantifying viral expression levels and applying blinded inclusion thresholds or covariate-based analyses. Potential locomotor or motivational differences will be monitored and incorporated as covariates in behavioral analyses.

To ensure biological relevance, pilot molecular and histopathological analyses will be conducted prior to behavioral testing. If robust α-syn and Aβ pathology is not observed within 3-6 months following AAV delivery, the experimental strategy will pivot to a seeding-based approach using α-syn pre-formed fibrils and Aβ oligomers to ensure induction of biologically relevant pathology. This alternative approach has been widely validated to promote aggregation, spreading, and synaptic dysfunction and will allow continued interrogation of APOE genotype dependent effects. This dual-path strategy minimizes experimental risk while preserving rigor, feasibility, and translational relevance.

If molecular-behavioral correlations are limited by detection sensitivity, immunoprecipitation-based enrichment or ultrasensitive digital ELISA (Simoa) will be employed to quantify α-synuclein and Aβ oligomeric species. Together, these mitigation strategies ensure rigorous assessment of both behavioral and molecular outcomes.

Aim 2. Molecular and cellular characterization of APOE-dependent α-syn/Aβ co-pathology

Rationale and Feasibility: Behavioral deficits observed in Aim 1 likely arise from underlying changes in protein aggregation, neuronal signaling, and microglial clearance, which may differ by APOE genotype. APOE4 has been shown to exacerbate α-synuclein pathology, gliosis, and synaptic dysfunction in mouse models and human Lewy body disease, even independent of Aβ[11,12]. This genotype also disrupts lysosomal and autophagy pathways, impairing microglial and astrocytic clearance of misfolded proteins[30,31]. Defining how APOE4 versus APOE3 regulates α-syn/Aβ co-aggregation, propagation, and glial responses will reveal mechanistic links between genotype and early neurodegenerative processes. I already have established expertise in confocal and super-resolution imaging, biochemical fractionation, and molecular analyses, and can integrate these approaches with the behavioral data from Aim 1, ensuring feasibility and robust mechanistic insight.

Experimental Approach: Brain tissue from Aim 1 will be used for confocal and super-resolution imaging to quantify α-syn and Aβ aggregation, co-localization, and regional propagation within hippocampal circuits (CA1, CA3, dentate gyrus). α-Syn/Aβ interactions will be detected using PLA, in which paired antibodies recognizing α-syn and Aβ are conjugated to complementary DNA oligonucleotides (Duolink® *In Situ* PLA kit, Sigma-Aldrich) that generate a fluorescent signal only when targets are within ~40 nm indicating direct molecular proximity. This provides single-aggregate level resolution of protein-protein interactions within neurons. Microglial activation and clearance will be characterized using IBA1 (activation), and CD68 (phagocytic activity), with quantitative analyses of puncta density, fluorescence intensity, and percent co-localization with α-syn/Aβ aggregates. Lysosomal and autophagy markers (LAMP1, LC3, cathepsin D) will be evaluated to determine APOE-dependent alterations in degradation pathways. Previously snap-frozen contralateral hemisphere from Aim 1 will be used for biochemical analyses. Sequential extraction will separate soluble and insoluble protein fractions, followed by Western blotting for total and phosphorylated α-synuclein (pS129), Aβ (monomeric and oligomeric), total tau, and phospho-tau (AT8, PHF1). Synaptic integrity will be assessed by quantifying pre- and post-synaptic proteins (synaptophysin, PSD95). These complementary imaging and biochemical analyses will define APOE-dependent differences in α-syn/Aβ aggregation, propagation, microglial response, lysosomal/autophagy function, and synaptic disruption, which will be correlated with behavioral outcomes from Aim 1 to identify molecular pathways linking APOE genotype to mixed proteinopathy and early neurodegeneration, establishing a foundation for future R01 studies.

Expected Outcomes: We anticipate that APOE4 mice will exhibit increased α-syn propagation, greater α-syn/Aβ co-localization, impaired microglial clearance, and pronounced lysosomal/autophagy dysfunction compared with APOE3 mice. PLA and super-resolution imaging are expected to provide quantitative, cell-type specific visualization of co-pathology, while biochemical fractionation and immunoblotting will reveal increased insoluble aggregates and synaptic loss in APOE4 mice. These data will establish mechanistic endpoints connecting APOE genotype to mixed-pathology progression and functional deficits observed in Aim 1, directly testing the hypothesis that APOE4 amplifies vulnerability to interacting proteinopathies.
Potential Pitfalls and Alternatives: Variability in viral expression or regional propagation of pathology could reduce signal consistency; this will be mitigated through optimized AAV titers, stereotaxic precision, and validation of transduction efficiency. If PLA signal is weak, sequential immunofluorescence or biochemical co-immunoprecipitation will be employed to confirm interactions. Should microglial or lysosomal readouts be inconclusive, additional markers of autophagy, lipid metabolism, or glial activation will be incorporated. If proteomic or biochemical yield is limited, tissue pooling or region-specific enrichment (hippocampus, cortex) will ensure sufficient material for analysis. These alternative strategies, coupled with extensive expertise in α-syn/Aβ pathology and access to advanced imaging and proteomics cores, ensure feasibility and high likelihood of generating informative and reproducible mechanistic data.

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PROJECT EXECUTION PLAN AND TIMELINE:
This 12-month project integrates coordinated animal, behavioral, molecular, and imaging studies designed to meet the objectives of Aim 1 and Aim 2. Activities are organized into sequential but overlapping phases to ensure efficient use of time, with dedicated periods for animal incubation, assay preparation, and concurrent analytical work. A 3-month post-injection period is required for α-syn/Aβ propagation; during this time, substantial preparatory, optimization, and analytical tasks will be conducted (detailed below), consistent with NIH expectations for continuous productivity.

Months 0–3 | Cohort Establishment, Reagent Validation, and Viral Delivery

- Finalize experimental cohort (genotyping, randomization, baseline health checks).
- AAV purchase and QC (titer verification, purity, infectivity testing).
- Conduct stereotaxic injections across all groups.
- Initiate post-injection incubation period for pathology development; perform weekly welfare monitoring and maintain experimental logs.

Milestone: Successful delivery of viral constructs.

Months 3–6 | Post-Injection Incubation & Full Behavioral Assessment (Aim 1)

- Continue post-injection incubation to allow α-syn/Aβ co-pathology development.
- Complete final optimization and validation of Proximity ligation Assay (PLA) using tissue collected from small pilot subsets (not the main cohort).
- Conduct low-stress behavioral tests (Open Field → Y-maze → Novel Object).
- Conduct high-demand behavioral test last (Morris Water Maze).
- Perform interim data QC and refine behavioral data processing pipeline.

Milestone: Completed and quality-checked behavioral dataset and validated assay conditions for downstream analysis.

Months 6–10 | Molecular, Cellular, and Imaging Analyses (Aim 2)

- Harvest brains from experimental cohorts at defined endpoints.
- Perform PLA, immunofluorescence, Western blotting, and biochemical assays.
- Conduct confocal and widefield imaging for synaptic and pathological markers.
- Begin preliminary quantification and statistical analysis of molecular/imaging data.

Milestone: Completed molecular and imaging datasets with primary quantitative outputs.

Months 10–12 | Integrated Data Analysis, Interpretation, and Dissemination

- Integrate behavioral, biochemical, and imaging datasets.
- Conduct final statistical modeling and verification of reproducibility.
- Prepare figures, summary tables, and draft internal reports.
- Initiate post-award dissemination activities (abstract submission, preliminary manuscript outline).

Milestone: Fully analyzed dataset ready for dissemination and grant follow-up.

Post-Award Period (Beyond 12 Months)

- Full manuscript preparation and journal submission.
- Presentation at national conferences and ADRC meetings.
- Development of R01 application incorporating feasibility data from this USC ADRC Development Project.

Plan for Future R01 Application Aligned with ADRC Goals

The proposed USC ADRC Development Project will generate critical preliminary data to support a future R01 investigating APOE-dependent mechanisms of α-synuclein-amyloid-β (Aβ) co-pathology in Alzheimer's disease (AD). My recently submitted K99/R00 proposes that α-syn regulates Aβ propagation, seeding, and downstream transcriptomic remodeling. The ADRC project extends this proposed framework into a translational *in vivo* model to address how APOE genotype influences α-syn-Aβ interactions, neuronal vulnerability, and early behavioral impairments. Data from this project will form the foundation for my future independent research program focused on AD co-pathology.

Key Question: *How does APOE genotype shape α-syn-Aβ interactions, neuronal vulnerability, and early behavioral impairments?*

Integration With Future R01:

1. Foundational *In Vivo* Feasibility: This ADRC Development Project will provide essential *in vivo* feasibility data using APOE3 and APOE4 knock-in mice with neuronal AAV co-expression of α-synuclein and Aβ. Behavioral phenotyping combined with PLA-based visualization of co-pathology will determine whether APOE genotype modulates α-syn-Aβ co-aggregation, define early behavioral phenotypes linked to genotype-dependent vulnerability, and generate effect-size estimates necessary for powering future R01 mechanistic aims. These experiments will establish the biological rationale for a genotype-focused investigation of α-syn-Aβ co-pathology and early functional impairments.

2. Hemibrain Tissue Banking for Multi-Omics: Half of each mouse hemibrain will be preserved as a tissue bank for multi-omics analyses during the R01 phase, including single-cell and single-nucleus RNA sequencing, proteomics, phosphoproteomics, and microglial subtype profiling. This repository will provide a ready-to-analyze biobank, reducing R01 startup time, enhancing feasibility, and enabling detailed mechanistic studies of APOE genotype-dependent effects on co-pathology.

3. Mechanistic Bridge to Future R01: While the K99/R00 proposes to investigate α-syn mediated Aβ propagation and transcriptomic remodeling in human iPSC neurons, the ADRC project adds a genotype-modifier dimension by assessing APOE3 versus APOE4 effects on co-seeding, plaque composition, synaptic injury, microglial states, and behavior. These data will directly support future R01 aims, including *(1) determining how APOE genotype controls α-syn-Aβ co-aggregation and synaptic injury, (2) identifying APOE4-amplified molecular pathways in AD co-pathology, such as endosomal dysfunction, lipid metabolism, and inflammatory signaling, and (3) testing strategies to rescue toxic co-pathology in human iPSC models.*

4. Methodological Development: During the ADRC project, key methodologies will be optimized to enhance rigor, reproducibility, and innovation. These include PLA for detecting α-syn-Aβ interactions, confocal and fractionation-based co-pathology imaging pipelines, behavioral workflows, viral delivery quality control, and early-stage adaptations of IP/RT-QuIC for α-syn/Aβ co-seeds. The resulting optimized assays will serve as core tools for the future R01, strengthening both feasibility and innovation while establishing robust technical platforms for mechanistic and translational studies.

Conclusion

This ADRC Development Project will generate feasibility data, optimize critical assays, and bank tissues for multi-omics, providing a cohesive foundation for a mechanistic R01. Together with insights from my K99/R00, these data enable an integrated research trajectory (**K99 → ADRC DP → R01 & R00**) focused on multi-protein co-pathology and genotype-dependent vulnerability. This trajectory aligns fully with ADRC goals in mechanism-driven discovery, early biomarker development, and therapeutic target identification.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Rijwan		Ahammad		PI	80,000.00	3.01			20,065.00	4,535.00	24,600.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **24,600.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
0	**Total Number Other Personnel**					**Total Other Personnel**	**0.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**24,600.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	1,500.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**1,500.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	8,400.00
2. Publication Costs	1,000.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. Transgenic Animal Breeding & Vivarium Costs	4,500.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**13,900.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**40,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	40,000.00	26,400.00
		Total Indirect Costs	**26,400.00**

Cognizant Federal Agency (Agency Name, POC Name, and POC Phone Number) DHHS Jeanette Lu 415-437-7820

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**66,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	66,400.00

L. Budget Justification*	File Name: R.Ahammad_Revised JUSTIFICATION.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

BUDGET JUSTIFICATION

Title: APOE-Dependent Impact on α-Synuclein and Aβ Cross-Seeding: Behavioral and Molecular Signatures of Co-Pathology

PI: Rijwan Ahammad, PhD | **Mentor and Sponsor:** Robert Rissman, PhD

A. KEY PERSONNEL:

Rijwan Ahammad, PhD, Principal Investigator, (3.0 Calendar Months)
Dr. Ahammad is a Postdoctoral researcher trained in biochemistry, molecular biology, and neuroscience. He will be responsible for designing, conducting, and analyzing all experiments associated with this proposal. He will also lead manuscript preparation, and presentation of findings.

Robert Rissman, PhD, Mentor (**No measurable effort requested)**
Dr. Rissman is Professor of Physiology and Neuroscience at the University of Southern California (USC) and Founding Director of the Neuroscience Translational Research Division (NTRD) at USC's Alzheimer's Therapeutic Research Institute (ATRI), located in San Diego, California, where Dr. Ahammad is currently a postdoctoral researcher. Acting as mentor, Dr. Rissman will continue to meet weekly with Dr. Ahammad to review data, troubleshoot experiments, and provide guidance on experimental design, interpretation, and career development. Alongside providing continued mentorship and professional guidance to Dr. Ahammad throughout of this grant, Dr. Rissman will provide the institutional resources to support Dr. Ahammad to successfully complete the aims of this proposal while facilitating integration with collaborators to ensure Dr. Ahammad's successful transition to independence.

C. EQUIPMENT: N/A

D. TRAVEL: (**$1,500)**
Requested as partial support for Dr. Ahammad to attend scientific conferences **AAIC**, to present data resulting from this project and to integrate new advances in Alzheimer's and synucleinopathy research.

E. PARTICIPANT/TRAINEE SUPPORT COSTS: N/A

F. OTHER DIRECT COSTS:

1. Materials and Supplies (12,500):

I. **Transgenic Animal Breeding & Vivarium Costs ($4,500):** Covers breeding and colony maintenance of APOE3 and APOE4 knock-in mice, including genotyping and monthly vivarium service charges. This budget supports approximately 50 mice needed for AAV injections and subsequent behavioral and biochemical assessments. Costs also include cage changes, bedding, chow, veterinary oversight, and use of behavioral core facilities for Morris Water Maze, Novel Object Recognition, Rotarod, and Open Field tests, including technician time, data acquisition, and maintenance of testing equipment.

II. **Animal Surgeries & Supplies ($2,200):** Includes expenses for AAV-Syn and AAV-Aβ42 viral vectors (Addgene or Vector Biolabs), as well as microinjectors, syringes, anesthesia, small surgical instruments, antiseptics, and recovery housing supplies. Funds also cover technician assistance and annual calibration of stereotaxic equipment.

III. **Histology and Immunostaining ($2,200):** Supports purchase of primary and secondary antibodies (α-syn pS129, Aβ, IBA1, CD68, TMEM119, APOE), fixatives, slides, mounting media, and reagents required for immunohistochemical and proximity ligation assays. These analyses will quantify α-syn/Aβ co-localization and microglial activation across brain regions.

IV. **Biochemical Reagents ($2,500):** Includes reagents required for RT-QuIC, Western blotting, and biochemical fractionation assays. Requested funds will support recombinant proteins, buffers, ECL reagents, precast gels, and other consumables necessary for quantifying α-syn and Aβ seeding activity and solubility profiles.

V. **General Laboratory Supplies ($1,500):** Includes general consumables such as pipette tips, Eppendorf tubes, gloves, buffers, and protease/phosphatase inhibitors required for routine experimental work, tissue homogenization, and sample processing**.**

2. Publication Costs ($1,000):
Partial funds requested for the costs of documenting, preparing, publishing, or otherwise making available to others, the findings and products of the work conducted under this award. Expenses also include costs for the production of conference/meeting related presentation materials.

A. COMPONENT COVER PAGE

Project Title: 7.4_APOE ε4-linked taste pathophysiology and therapeutic modulation by semaglutide_Roman-Ortiz_new
Component Project Lead Information: Roman, Ortiz Ciorana

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

Aim 1: Characterize the impact of APOE ε4 mutation on inflammatory, structural, and transcriptomic changes in taste buds. Preliminary data show that 3xTG Familial AD mice exhibit reduced taste bud size, inflammation and tau pathology in the taste fields in early adulthood. Here, we will extend these studies to investigate if these effects are recapitulated in APOE*4 AD mice at early (4-months) and later (12-months) age relative to age-matched wild type controls. In addition, we will assess if expressions of key taste and nutrient receptors, as well as their intermediaries, are altered in the taste buds at both timepoints.

Aim 2: Evaluate the effects of chronic SEMA treatment on APOE ε4-linked taste pathophysiology and metabolic behavior. SEMA can be used to mitigate early AD progression by amplifying insulin release and reducing inflammation. Preliminary work shows that chronic SEMA treatment improves sensitivity in the taste system in C57BL6/J mice. Given taste-related deficits are associated with AD, here we will examine if SEMA treatment ameliorates age-related APOE ε4 taste pathology and regulates bodyweight, blood glucose and nutritional balance in APOE*4 mice.Collectively, the proposed aims will elucidate how APOE ε4 pathophysiology presents in the taste system to disrupt metabolic health and how GLP1r signaling restores taste and nutrient signaling to reduce AD-related dietary habits. Additionally, this work will provide the first evidence of transcriptomic gustatory changes that are associated with the APOE4 allele, paving the way for future work aimed at understanding mechanistic links between metabolic disturbances and sensory deficits in the context of neurogenerative disease. Moreover, this work will reveal novel mechanisms through which incretin-based treatments exert neurometabolic benefits, including at the first site of nutrient sensing.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : CRO_ADRC DP_Proposal_clean copy.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Methodology
In this proposal, we will use B6(SJL)-Apoetm1.1(APOE*4)Adiuj/J mice, which express a humanized form of the ε4 allele of APOE ,

a variant that confers significant risk for developing AD. Age-matched
C57BL/6J (wild-type) mice will be used as controls. All animals will be maintained on regular chow. Male and female mice will be included in each experiment to determine if APOE4-related or SEMA-
driven effects on sensory function are dependent on biological sex.

Exp. 1.1: Examine if APOE ε4 induces taste tissue inflammation, reduces taste cell number, and/or impacts taste bud morphology. Fixed taste tissue will be collected post-mortem at 4-months or 12-
months of age. Using immunohistochemistry (IHC) and microscopy, we will assess tumor necrosis factor alpha (TNFα), phosphorylated tau protein (p-Tau231), and taste bud size and quantity in the
circumvallate papilla of APOE*4 and C57BL/6J mice. Quantification of fluorescence intensity, taste bud count and size will be performed in ImageJ by an experimenter blinded to group identity.

Exp. 1.2: Investigate if APOE ε4 alters the transcriptomic profile of the taste bud, including nutrient sensors. Fresh taste tissue will be collected post-mortem at 4-months and 12-months of age. UsingqPCR, we will measure gene expression of Glucokinase (GCK), Sodium glucose linked transporter 1 (SGLT1), Taste 1 Receptor Member 3 (TAS1R3), and Transient Receptor Potential Cation Channel Subfamily M Member 5 (TRPM5) in the circumvallate papilla of APOE*4 and C57BL/6J mice. For each gene, three technical replicates per animal sample will be used, with Actin-b serving as the reference control gene.

Exp. 2.1: Investigate GLP1 agonist treatment (SEMA) ameliorates age-related taste APOE ε4 pathology and markers related to energy homeostasis. Bodyweight, chow intake, blood glucose (using a Contour Next™ blood glucometer) and sugar intake will be measured to set the baseline; then animals will be divided into 3 groups counterbalancing for bodyweight, chow and sugar intake across groups. The groups will include SEMA-treated with ad Libitum access to chow, vehicle-treated with ad libitum access to chow and a vehicle-treated group that will receive a chow ration equal to the amount consumed by their experimental counterpart (Food-paired to SEMA-treated). The vehicle-injected food- paired group will control for effects related to changes in caloric intake, glycemic control, and/or weight loss, allowing us to ascertain causal mechanisms related to GLP1r agonism. Animals will receive daily injections of SEMA or vehicle according to their group. Semaglutide-injected animals will start with a lower dose of 5nmol/kg for 5 days and then increased to 10nmol/kg for the rest of the experiment; this mimics the dosage weaning schedule prescribed to humans. On day 10, we will re-assess blood glucose levels. Bodyweight and chow intake will be measured every day. Sugar intake (24h, home cage water and sugar bottles) will be measured after 11 days of treatment. Lastly, on day 14 animals taste tissue will be collected post-mortem. In one set of animals, we will measure TNFα levels, p-Tau231 presence, and taste bud size and number in the circumvallate papilla of APOE*4 and C57BL/6J mice, as in Exp. 1.1. In another set, as in Exp. 1.2, we will measure mRNA expression of GCK, SGLT1, TAS1R3, and TRPM5 in the circumvallate papilla of the age-matched APOE*4 and C57BL/6J mice.

Exp. 2.2: Assess how SEMA treatment affects meal patterns and sugar intake in APOE*4 and C57BL/6J mice. We will follow the same protocol as Exp. 2.1, however on treatment day 11, animals will be given a daily sugar bottle connected to a lickometer and automatic feeder (FED3) will be placed in their home cage to timestamp sugar and food intake for 10 days. Animals will remain under daily
injections (Vehicle vs 10nmol/kg SEMA) and we will measure bodyweight, sugar and chow intake daily. Bodyweight, food, water and sugar intake will be quantified daily and the intake timestamps (water, sugar and dispensed pellet) will be used to recreate individual meal patterns.

PI: Ciorana Roman Ortiz Mentor: Lindsey Schier ADRC Sponsor: Hussein Yassine

APOE ε4-linked taste pathophysiology and therapeutic modulation by semaglutide

Scientific Rationale

Alzheimer's disease (AD) profoundly disrupts neuronal metabolism, triggering oxidative stress, neuroinflammation, and ultimately, neuronal death. Type 2 diabetes mellitus (T2DM) further amplifies the risk of developing AD, particularly in individuals carrying the ε4 allele of the apolipoprotein E gene (APOE4). Notably, both T2DM and AD are frequently accompanied by nutritional shifts, including a heightened craving for sweetened, energy-dense foods, behaviors that may, in turn, accelerate disease progression. Yet, the biological drivers of these altered dietary patterns remain largely unknown. Taste cells, as the body's frontline nutrient sensors, play a pivotal role in shaping food preference and appetite. Their susceptibility to metabolism-driven cellular pathologies raises the distinct possibility that systemic metabolic dysfunction in AD may directly compromise the peripheral taste system. While some studies have examined these mechanisms in the context of T2DM, no research to date has explored how the APOE4 genotype specifically influences taste cell health and sensory function.

Interestingly, drugs used to treat T2DM have been found to slow AD progression by improving insulin signaling and reducing neuroinflammation. Glucagon-like peptide 1 receptor (GLP1r) agonists, such as semaglutide (SEMA), commercially known as Ozempic, Wegovy, and Rybelsus, have been shown to decrease Aβ deposition, tau hyperphosphorylation and neuroinflammation in AD models. GLP1 is expressed in taste cells that also co-express the sweet receptor (T1R3) responsible for sugar detection, while the GLP1 receptor (GLP1r) is expressed on nearby taste nerves, providing a potential peripheral target to rescue taste and nutrient-sensing deficits associated with AD pathology.

Understanding how APOE ε4-linked pathophysiology modifies the initial cephalic phase of ingestion will provide insight into how oral chemosensory deficits may serve as potential biomarkers of disease and exacerbate AD progression. Furthermore, elucidating how GLP1r-based therapies modulate AD-related inflammation and peripheral taste system function will advance our understanding of non-central GLP1 mechanisms that influence metabolic health and AD management.

Specific Aims

The APOE ε4 allele is the main genetic risk factor for late-onset sporadic Alzheimer's disease (AD), increasing the risk to develop the disease by 15-fold in homozygous carriers. It is currently estimated that 25% of the population carries at least one copy of APOE4 gene variant. Preventative interventions and cures are currently lacking, raising the urgent need to understand the key drivers of APOE-mediated AD pathophysiology and decline and to develop better pharmacotherapies. APOE ε4 is associated in cholesterol transport dysfunction, impaired insulin signaling, accumulation of abnormal neuritic plaques (β-amyloid) and neurofibrillary tangles (tau), along with immune response in the brain, which are all hallmarks of AD pathology. Due to the fact that cognitive decline is the most debilitating sign of AD, most of the research has focused on brain metabolism. Still, AD pathology has been found outside the brain, one example is in the olfactory mucosa where smell sensitivity deterioration correlates with AD progression and is considered the earliest sign of functional decline. Surprisingly, despite the fact that taste is a critical factor metabolic homeostasis and AD patients suffer from taste alterations, how AD pathophysiology manifests in the taste fields remains unknown. The gustatory system is highly plastic, shaped by internal conditions, such as insulin resistance, diet, and pharmacological treatments, highlighting the importance of understanding the mechanistic links between oral nutrient sensing and AD pathology.

Recent research has shown that semaglutide (SEMA), a glucagon-like peptide 1 receptor (GLP1r) agonists used for the treatment of diabetes and obesity, rescues glucose metabolism and anti-inflammatory mechanisms necessary for cognitive function and memory in AD models. However, the

specific effects of SEMA on nutrient detection and dietary choice have not been investigated in humans or in pre-clinical AD. Since GLP1rs are abundant along the entire gustatory sensory pathway, I hypothesize that GLP1 therapeutics may attenuate APOE ε4-related taste deficits help restore peripheral glucose homeostasis and maintain better nutritional habits.

Aim 1: Characterize the impact of APOE ε4 mutation on inflammatory, structural, and transcriptomic changes in taste buds. Preliminary data show that 3xTG Familial AD mice exhibit reduced taste bud size, inflammation and tau pathology in the taste fields in early adulthood. Here, we will extend these studies to investigate if these effects are recapitulated in APOE*4 AD mice at early (4-months) and later (12-months) age relative to age-matched wild type controls. In addition, we will assess if expressions of key taste and nutrient receptors, as well as their intermediaries, are altered in the taste buds at both timepoints.

Aim 2: Evaluate the effects of chronic SEMA treatment on APOE ε4-linked taste pathophysiology and metabolic behavior. SEMA can be used to mitigate early AD progression by amplifying insulin release and reducing inflammation. Preliminary work shows that chronic SEMA treatment improves sensitivity in the taste system in C57BL6/J mice. Given taste-related deficits are associated with AD, here we will examine if SEMA treatment ameliorates age-related APOE ε4 taste pathology and regulates bodyweight, blood glucose and nutritional balance in APOE*4 mice.

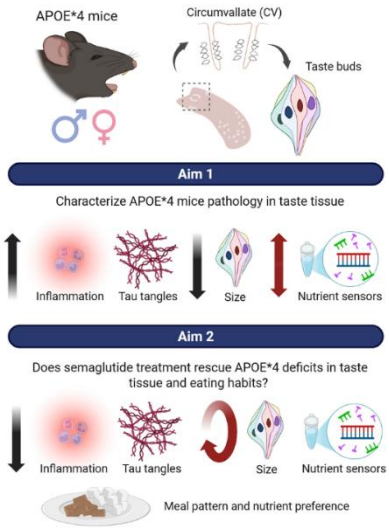

Figure 1 Graphical abstract of expected outcomes. In this proposal we will primary study APOE ε4 pathophysiology in taste tissue in APOE*4 AD mouse model. First in **Aim 1**, we hypothesize there will be an increase in inflammation and Tau tangles with a decrease in taste buds' sizes and altered transcriptomic of taste/nutrient sensors in taste buds in APOE*4 mice compared to controls. Then in **Aim 2**, we expect to see a rescue of APOE ε4-linked taste pathology deficits after 10 days of semaglutide treatment. Lastly, we will assess eating habits and preferences during SEMA treatment.

Collectively, the proposed aims will elucidate how APOE ε4 pathophysiology presents in the taste system to disrupt metabolic health and how GLP1r signaling restores taste and nutrient signaling to reduce AD-related dietary habits. Additionally, this work will provide the first evidence of transcriptomic gustatory changes that are associated with the APOE4 allele, paving the way for future work aimed at understanding mechanistic links between metabolic disturbances and sensory deficits in the context of neurogenerative disease. Moreover, this work will reveal novel mechanisms through which incretin-based treatments exert neurometabolic benefits, including at the first site of nutrient sensing.

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Exp. 2.2: Assess how SEMA treatment affects meal patterns and sugar intake in APOE*4 and C57BL/6J mice. We will follow the same protocol as **Exp. 2.1,** however on treatment day 11, animals will be given a daily sugar bottle connected to a lickometer and automatic feeder (FED3) will be placed in their home cage to timestamp sugar and food intake for 10 days. Animals will remain under daily injections (Vehicle vs 10nmol/kg SEMA) and we will measure bodyweight, sugar and chow intake daily. Bodyweight, food, water and sugar intake will be quantified daily and the intake timestamps (water, sugar and dispensed pellet) will be used to recreate individual meal patterns.

Study duration and potential follow up studies.

We estimate we can achieve our aims within 12 months as we keep active APOE*4 and C57BL/6J mouse colonies. For **Exp. 1.1** we previously collected male and females 4-months and 10-months old taste tissue that we plan to process the tissue for imaging at the end of this year. Cutting tissue, IHC staining, imagining and analysis will take 2 months. Currently, we are harvesting CV tissue for **Exp. 1.2**, We expect to complete tissue collection and start tissue processing during spring 2026. Transcriptomic analysis using qPCR for all the samples will take 3 weeks. Once we complete the tissue collection for **Exp. 1.2,** all new litters will be aged (4-months) to be used in **Exp. 2.1** which takes 1 month from SEMA treatment to tissue imaging analysis or qPCR transcriptomic profile. We expect to complete this experiment by early September. Lastly, **Exp. 2.2** will take a month between SEMA treatment and meal pattern analysis. Upon award notification, we will expand our mouse breeding colonies to ensure an adequate number of animals are available to achieve our aims within the proposed timeline. Time- and resource-pending, we can perform similar experiments at different ages to understand disease progression and identify the most effective therapeutic windows across the lifespan. As diet plays an important environmental factor in AD, we also plan to repeat these studies in APOE4 and age-matched C57BL6/J mice maintained on a westernized diet, high in fat and sugar.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.		Ciorana		Ortiz		PI	77,500.00	4.06			26,216.00	8,784.00	35,000.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **35,000.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
0	**Total Number Other Personnel**					**Total Other Personnel**	**0.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**35,000.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	5,000.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**5,000.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**40,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	40,000.00	26,400.00
		Total Indirect Costs	**26,400.00**

Cognizant Federal Agency (Agency Name, POC Name, and POC Phone Number) DHHS Jeanette Lu 415-437-7820

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**66,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	66,400.00

L. Budget Justification*	File Name: CRO_ADRC DP_BUDJUS FINAL.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

PI: Ciorana Roman Ortiz Mentor: Lindsey Schier ADRC Sponsor: Hussein Yassine

Budget Justification

Personnel

$35,000 are requested for postdoctoral stipend. Ciorana Roman Ortiz, PhD, Principal Investigator (34% annual effort). Dr. Roman Ortiz will perform and direct the project outlined in this application.

Supplies

$5,000 is requested in each year to defray the costs of materials used in the research project. Funds are requested for the purchase of materials and supplies, which will include, but are not limited to, qPCR probes and reagents, antibodies and semaglutide drug.

A. COMPONENT COVER PAGE

Project Title: 7.5_Platelet-derived growth factor receptor-β in Plasma and Cerebrospinal Fluid as a Marker of Pericyte Injury: Association Analysis with White Matter Changes in Alzheimer's Disease_ Fotouhi_new
Component Project Lead Information: Maryam, Fotouhi

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

Aim 1: Quantify molecular and imaging biomarkers of pericyte injury and glymphatic dysfunction in paired plasma, CSF, and MRI from 188 ADNI participants. Approach: We will quantify molecular and imaging biomarkers from 188 ADNI participants. First, we will measure soluble PDGFRβ concentrations in paired plasma and CSF samples using a validated electrochemiluminescence immunoassay on the Meso Scale Discovery platform. Then, we will quantify perivascular space (PVS) volume, count, and morphological measures in white matter, basal ganglia, and centrum semiovale of isotropic T1-weighted MRI with FLAIR white matter hyperintensity exclusion using a clinically validated automated PVS segmentation pipeline, followed by FreeSurfer region delineation and visual quality control.

Aim 2: Test cross-sectional and longitudinal associations between PDGFRβ levels and PVS burden and assess whether baseline PDGFRβ predicts PVS burden and cognitive decline. Approach: We will perform cross-sectional associations between baseline PDGFRβ levels (plasma, CSF) and quantitative PVS metrics (count, volume, density, diameter, length, tortuosity) in multivariable linear models adjusted for age, sex, BMI, vascular risk factors, and APOE ε4 status. Additionally, evaluate how baseline PDGFRβ measures predict longitudinal PVS and clinical disease progression over follow-up periods, using a mixed-effects model as appropriate. We hypothesize that elevated PDGFRβ levels will correlate positively with PVS volume and count at baseline. Longitudinally, we further hypothesize that higher baseline PDGFRβ concentrations will predict progressive PVS enlargement and cognitive decline, consistent with ongoing pericyte injury and worsening glymphatic dysfunction.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

File Uploaded : M.Fotouhi_ADRC DP_Proposal_2025.pdf

B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Methods
CSF and plasma PDGFRβ: We will quantify soluble PDGFRβ in baseline CSF (primary) and plasma (exploratory) from 188 ADNI participants using the Meso Scale Discovery (MSD) electrochemiluminescence sandwich immunoassaydescribed previously. The

assay spot-coats an MSD 96-well plate with a capture antibody against the extracellular domain of human PDGFRβ, followed by a biotinylated detection antibody and Sulfo-Tag readout on the MSD platform. Samples (CSF diluted 1:2) and a recombinant PDGFRβ standard series (100–6,400 pg/mL) are run in duplicate and quantified against a linear standard curve spanning a dynamic range of ~100–26,000 pg/mL ($r^2 \approx 0.996–0.9996$); intra- and inter-assay CVs were <5% in the original report. The method was validated for detection limits, dilutional linearity, parallelism, spike recovery, precision (repeatability/reproducibility), and robustness. The reference procedure measures CSF; plasma will be assessed
here as an exploratory extension using an optimized MSD assay for measuring plasma sPDGFRβ, noting the assay's large dynamic range, suggesting feasibility for blood, but requires confirmation.

PVS segmentation: T1-weighted images with isotropic resolution (voxel size ≤1.2 mm, slice thickness 1 mm) will undergo FreeSurfer recon-all processing for anatomical parcellation of basal ganglia (BG) and centrum semiovale (CSO) and intracranial volume estimation, based on ADNI quality control standards, excluding scans with motion artifacts, segmentation failures, or large infarcts. Denoised and bias-corrected T1w images will then be used for the 3D Frangi vesselness application using our clinically validated pipeline to produce regional PVS maps, with region-specific thresholds. Vendor-dependent contrast and signal variability across GE, Siemens, and Philips scanners will be harmonized via a diffusion-based multi-domain neuroimaging harmonization framework that preserves PVS-specific features while mitigating technical variance. White matter hyperintensity (WMH) masks will thereafter be generated from co-registered T1w and FLAIR images using the Sequence Adaptive Multimodal Segmentation tool 40 combined with the Lesion Growth Algorithm, (if only T1w is available 42), to exclude hyperintense voxels that confound PVS detection. The result is a 3D PVS mask. Quantitative features, including PVS count, total volume, and morphologic features (mean diameter, length, tortuosity), will be extracted within BG and CSO regions and normalized to regional white matter volume or intracranial volume. Regional PVS volumes for 34 bilateral Desikan-Killiany atlas regions will be computed, with CSO-PVS volume fraction derived by summing regional PVS volumes and normalizing by corresponding white matter volume. Failed segmentations are reprocessed once; persistent failures are excluded from primary analyses.

Platelet-derived growth factor receptor-β in Plasma and Cerebrospinal Fluid as a Marker of Pericyte Injury: Association Analysis with White Matter Changes in Alzheimer's Disease

Investigators

REC scholar: Maryam Fotouhi, MD, Department of Radiology; Faculty Sponsor: Jeiran Choupan, PhD, Department of Neurology; ADRC Sponsor: Hussein Yassine, MD, Keck Medicine of USC; Co-investigator: Abhay Sagare, PhD, Department of Physiology & Neuroscience

Scientific Rationale

Injury to pericytes directly disrupts the flow dynamics necessary for the brain clearance system, which involves the influx of cerebrospinal fluid (CSF) and the efflux of interstitial fluid [1–3]. Soluble Platelet-derived growth factor receptor-β (PDGFRβ) present in CSF serves as a quantitative biomarker of pericyte injury [4,5], specifically from exposure to hypoxia and amyloid-beta peptides[4], reflecting underlying blood-brain barrier (BBB) dysfunction and microvascular deterioration in neurodegenerative diseases [6]. Perivascular spaces (PVS), visualized by magnetic resonance imaging (MRI), are fluid-filled compartments surrounding cerebral microvasculature [7,8]; their pathological enlargement represents a radiological marker for impaired brain clearance and disrupted interstitial fluid homeostasis in aging and disease [9]. Investigating the link between the BBB marker of pericyte injury (PDGFRβ in plasma and CSF) and PVS morphology can help elucidate early vascular mechanisms driving Alzheimer's disease and cognitive decline.

Our *Hypothesis* is that higher soluble PDGFRβ is associated with greater quantitative PVS burden (volume, count) on MRI in the basal ganglia and centrum semiovale; secondarily, elevated baseline PDGFRβ predicts faster PVS enlargement and worse cognition, with potential modification by APOE ε4. This study aims to measure PDGFRβ concentrations in paired plasma and CSF together with MRI-derived PVS quantifications to assess their correlation as a marker of perivascular clearance efficiency and BBB integrity. The primary goal is to evaluate this correlation as a marker of Alzheimer's disease vascular pathology and the prognostic value of this biomarker. This project represents the first Alzheimer's Disease Neuroimaging Initiative (ADNI) assayed PDGFRβ linking to site-harmonized, automated 3D PVS quantification, directly testing a pericyte/BBB-perivascular clearance axis in vivo.

Significance. Despite decades of progress, Alzheimer's disease-relevant vascular biomarkers are lacking. The existing biomarkers exhibit inadequate specificity and suboptimal performance for this application [10–13]. This study directly fills that gap by testing PDGFRβ (plasma/CSF) against quantitative PVS as a vascular-clearance axis in disease. Pericyte degeneration and BBB dysfunction emerge early in Alzheimer's disease and related dementias and are quantifiable by PDGFRβ [14]; Increased PVS volume reflects impaired clearance [9]. Establishing a fluid-imaging association addresses a mechanistic gap in Alzheimer's disease and related dementias biomarkers, supports stratification beyond AT(N) [15]. In addition, demonstrating the prognostic value of PDGFRβ-PVS for future cognition would provide a noninvasive clearance phenotype for risk staging and trial enrichment.

Innovation. 1. Harmonized multi-site analytics with prespecified adjustments (age, sex, vascular risks, scanner/site, APOE ε4) and sensitivity analyses. 2. Assessing pericyte/BBB injury (PDGFRβ) with a quantitative regional PVS phenotype within individuals; 3. Automated 3D PVS segmentation with WMH exclusion, morphology metrics for reproducible effect sizes.

Impact. By forging a new research collaboration to assess novel PVS-MRI biomarkers, this application will decipher whether PDGFRβ level (plasma, CSF) can be a potential marker of the larger waste clearance system, which may shed new light on the potential mechanisms underlying neurodegeneration effects of pericyte degeneration in Alzheimer's Disease.

PVS enlargement reflects impaired brain clearance [16,17]. Recent longitudinal studies demonstrate that PVS burden accelerates in aging and is correlated with amyloid and tau positivity preceding clinical symptoms (Leone et al., 2025; Lynch et al., 2023; Menze et al., 2024; Sepehrband et al., 2021). Degeneration of pericytes results in diminished perivascular ensheathment by smooth muscle cells, impaired arterial pulsatility, and increased accumulation of vascular amyloid in arterioles [22–25], corresponding with the enlargement of perivascular spaces in affected areas. This pathological process indicates a disruption in the outward flow through arterioles, as the normal constriction and relaxation of pericytes contribute to the reflection wave that follows arterial pulsation, facilitating the clearance of interstitial fluid. The mechanistic relationship between pericyte injury and perivascular space enlargement represents a critical vascular-clearance axis in Alzheimer's disease pathogenesis. This application integrates cell biology with imaging to investigate a vascular-clearance

axis, surpassing single-modality investigations by connecting mechanistic cell injury to an in vivo clearance phenotype. These insights may position PDGFRβ as a potential biomarker for monitoring therapeutic interventions and an indication for perivascular-pathway targeted treatments, which aim to enhance protein clearance.

The PDGFRβ signaling pathway is essential for the recruitment, maturation, and survival of pericytes, contractile mural cells that stabilize the cerebral microvasculature. Through regulation of endothelial tight junctions, pericytes facilitate perivascular clearance and preserve BBB integrity, maintaining brain homeostasis [26–28]. Through α-adrenergic receptor signaling and smooth muscle actin-mediated processes, contractile pericytes change vessel diameter [29] by as much as 20%, making them the regulators of capillary diameter [30]. The pericyte-mediated spatial vascular tone variation allows for regional prioritization of blood flow under basal conditions, with pericytes maintaining capillary constriction at rest and exhibiting vasomotion patterns critical for cerebral perfusion regulation [31]. In *preclinical* mice and human studies of cognitive impairment and Alzheimer's disease, BBB-associated pericyte degeneration was shown to occur early in the pathological sequence, especially in susceptible areas, including the hippocampus [32,33]. Upon degeneration, pericytes release soluble PDGFRβ via ectodomain shedding facilitated by ADAM10, resulting in measurable indicators of pericyte injury that can be accurately quantified using developing immunoassays [34,35]. Increased levels of CSF PDGFRβ have been observed in individuals with mild cognitive impairment and Alzheimer's disease, with concentrations showing a positive correlation with indicators of blood-brain barrier dysfunction and neuroinflammation [5,36,37]. CSF PDGFRβ can serve as an early biomarker for cognitive dysfunction, independent of traditional amyloid and tau pathways.

This study aims to analyze the correlation of PDGFRβ in paired plasma and CSF from ADNI participants with quantitative measures of perivascular space. Outcomes include effect estimates that connect pericyte injury to clearance deficits, establishing a foundation for biomarker-guided trials focused on enhancing protein clearance in Alzheimer's disease.

Specific Aims

Aim 1: Quantify molecular and imaging biomarkers of pericyte injury and glymphatic dysfunction in paired plasma, CSF, and MRI from 188 ADNI participants. *Approach*: We will quantify molecular and imaging biomarkers from 188 ADNI participants. First, we will measure soluble PDGFRβ concentrations in paired plasma and CSF samples using a validated electrochemiluminescence immunoassay on the Meso Scale Discovery platform [28]. Then, we will quantify perivascular space (PVS) volume, count, and morphological measures in white matter, basal ganglia, and centrum semiovale of isotropic T1-weighted MRI with FLAIR white matter hyperintensity exclusion using a clinically validated automated PVS segmentation pipeline, followed by FreeSurfer region delineation and visual quality control [21].

Aim 2: Test cross-sectional and longitudinal associations between PDGFRβ levels and PVS burden and assess whether baseline PDGFRβ predicts PVS burden and cognitive decline. *Approach*: We will perform cross-sectional associations between baseline PDGFRβ levels (plasma, CSF) and quantitative PVS metrics (count, volume, density, diameter, length, tortuosity) in multivariable linear models adjusted for age, sex, BMI, vascular risk factors, and APOE ε4 status. Additionally, evaluate how baseline PDGFRβ measures predict longitudinal PVS and clinical disease progression over follow-up periods, using a mixed-effects model as appropriate. We *hypothesize* that elevated PDGFRβ levels will correlate positively with PVS volume and count at baseline. Longitudinally, we further *hypothesize* that higher baseline PDGFRβ concentrations will predict progressive PVS enlargement and cognitive decline, consistent with ongoing pericyte injury and worsening glymphatic dysfunction.

Methods

CSF and plasma PDGFRβ: We will quantify soluble PDGFRβ in baseline CSF (primary) and plasma (exploratory) from 188 ADNI participants using the Meso Scale Discovery (MSD) electrochemiluminescence sandwich immunoassay described previously [28]. The assay spot-coats an MSD 96-well plate with a capture antibody against the extracellular domain of human PDGFRβ, followed by a biotinylated detection antibody and Sulfo-Tag readout on the MSD platform. Samples (CSF diluted 1:2) and a recombinant PDGFRβ standard series (100–6,400 pg/mL) are run in duplicate and quantified against a linear standard curve spanning a dynamic range of ~100–26,000 pg/mL ($r^2 \approx 0.996$–0.9996); intra- and inter-assay CVs were <5% in the original report. The method was validated for detection limits, dilutional linearity, parallelism, spike recovery, precision (repeatability/reproducibility), and robustness. The reference procedure measures CSF; plasma will be assessed

here as an exploratory extension using an optimized MSD assay for measuring plasma sPDGFRβ, noting the assay's large dynamic range, suggesting feasibility for blood, but requires confirmation.

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Correlation statistical analysis: We will test adjusted associations between CSF and plasma PDGFRβ (pg/mL) and PVS burden cross-sectionally. For BG and CSO separately, we will fit multivariable linear models with PVS burden (primary outcome: volume, count; exploratory: morphology) as dependent variables and CSF PDGFRβ (pg/mL) as the main predictor. Covariates: age, sex, intracranial volume (for volume/count outcomes), vascular risk index (hypertension, diabetes, hyperlipidemia), APOE ε4 carrier status, and scanner/site terms to account for ADNI's multi-vendor, multi-field-strength design. We will inspect distributions and apply log/rank transforms as needed; check assumptions (linearity, homoscedasticity, residual normality, multicollinearity via VIF); and regression if indicated. Multiplicity across region×metric families is controlled with Benjamini-Hochberg FDR ($q = 0.05$) and will be considered exploratory. Secondary analyses mirror models with plasma PDGFRβ; effect-modification by APOE ε4 is tested via an interaction term (PDGFRβ×APOE ε4) with simple-slope estimates. Sensitivity analyses: (i) excluding high-influence observations, (ii) additionally adjusting for total WMH burden, (iii) repeating in the no-FLAIR subset using the T1-centered WMH fallback, and (iv) stratifying or adjusting for cognitive group (CN/MCI/AD).

Longitudinal analyses will test whether baseline PDGFRβ predicts trajectories of PVS burden and cognitive decline. Linear mixed-effects models with random intercepts and slopes will model PVS burden as a function of time (years from baseline), baseline PDGFRβ, and their interaction (PDGFRβ×time), adjusting for age, sex, vascular risk, APOE ε4, intracranial volume, and scanner/site. Parallel models will incorporate plasma PDGFRβ and PDGFRβ×APOE ε4×time interactions. Additional mixed-effects models will examine whether baseline PVS burden predicts longitudinal change in cognitive and clinical outcomes (Memory, Executive Function, CDR-SB, ADAS-Cog) under the same covariate structure. Power: with $N = 188$, two-sided $\alpha = 0.05$, we have ≈80% power to detect $r \approx 0.20$ (≈4% variance explained) between PDGFRβ and PVS volume in each primary region (BG, CSO). Secondary and exploratory analyses across region × metric families will be controlled for multiple comparisons using the Benjamini-Hochberg FDR method ($q = 0.05$).

Study Duration:

Year 1, Q1: **Aim 1**:Demographic and imaging repository setup and data cleaning; T1w/FLAIR imaging quality control; preprocessing (bias-field correction, skull-stripping, co-registration to T1w, spatial normalization); site/scanner metadata recording; PVS mask-generation pipeline setup, quantification, and mask generation; PDGFRβ assay plan for existing CSF/plasma samples (platform, calibrators/controls, duplicate placement).

Year 1, Q2-3: **Aim 1**: Run PDGFRβ assays; harmonize PVS measures across vendors/scanners; finalize regional parcellations and visual quality control; **Aim 2**: Primary statistical models for baseline and longitudinal cohorts; residual checking and model fitting.

Year 1, Q4: **Aim 2**: Complete analyses according to the statistical analysis plan; finalize figures and tables; manuscript preparation.

References:

1. Cai Y, Zhang Y, Leng S, et al. The relationship between inflammation, impaired glymphatic system, and neurodegenerative disorders: A vicious cycle. *Neurobiol Dis* 2024;192:106426.

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4. Sagare AP, Sweeney MD, Makshanoff J, et al. Shedding of soluble platelet-derived growth factor receptor-β from human brain pericytes. *Neurosci Lett* 2015;607:97–101.

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7. Kwee RM, Kwee TC. Virchow-Robin spaces at MR imaging. *Radiogr Rev Publ Radiol Soc N Am Inc* 2007;27:1071–86.

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C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

No Change

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

	Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.		Maryam		Fotouhi		PI	73,500.00	2.34			14,323.00	3,237.00	17,560.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **17,560.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
0	**Total Number Other Personnel**					**Total Other Personnel**	**0.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**17,560.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	2,000.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**2,000.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	15,040.00
2. Publication Costs	3,000.00
3. Consultant Services	0.00
4. ADP/Computer Services	2,400.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8.	0.00
9.	0.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**20,440.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**40,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	40,000.00	26,400.00
		Total Indirect Costs	**26,400.00**

Cognizant Federal Agency (Agency Name, POC Name, and POC Phone Number) DHHS Jeanette Lu 415-437-7820

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**66,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	66,400.00

L. Budget Justification*	File Name: M.Fotouhi_ADRC DP_BUDJUS FINAL.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Budget and Justification

Maryam Fotouhi, MD, Postdoctoral Scholar, 2.34 Calendar Months: Dr. Fotouhi is a postdoctoral scholar in Dr. Choupan's lab at USC's Mark and Mary Stevens Neuroimaging and Informatics Institute, where the mission is to advance understanding of brain clearance in health and disease and to translate these insights into clinical practice, diagnosis, and therapeutic development. Dr. Fotouhi has been working on the harmonization methodology for large-scale datasets on Alzheimer's disease and related dementias. She has extensive experience in computational neuroimaging analysis and dementia research. Dr. Fotouhi will perform the actual statistical analysis and interpretation under the guidance of Drs. Choupan and Co-Is. Dr. Fotouhi will be actively involved in all aspects of the project, including research planning, design, conduct, protocol development, data collection and analysis, interpretation of study results, communication, and dissemination of scientific findings, with a focus on the validation and analysis of brain clearance biomarker measures. She will have weekly meetings with Dr. Choupan to ensure integration and steady progress. Salary +Fringe = $14,323 + $3,237 = **$17,560**

Travel: Funds will be requested for project personnel to attend both domestic and international scientific conferences to present findings related to platelet-derived growth factor receptor-β in plasma and cerebrospinal fluid and perivascular space quantification. Travel funds will support attending one meeting such as the Alzheimer's Association International Conference (AAIC), American Academy of Neurology (AAN), and Alzheimer's and Parkinson's Disease (AD/PD). Annual travel budgets include **$2,000** for conferences in year 1. The travel costs include abstract fees, conference registration, airfare, hotel, and meals.

Biomarker Assay, Materials and Supplies for CSF and Plasma Measurement: We request **$15,040** for Soluble Platelet-derived growth factor receptor-β (PDGFRβ) testing kits and consumables to be performed by Dr. Abhay Sagare's lab at The Zilkha Neurogenetic Institute of USC.

Publication Costs: We are requesting **$3,000** per year to cover publication-related expenses for one manuscript in a peer-reviewed imaging or neurology journal.

Computer Services: We request **$2,400** for high-performance supercomputing time and maintenance services, which guarantee high throughput of the computational image dataset. The costs listed here are standard data analysis costs associated with imaging projects, which help defray the amortization costs for the equipment. The institute's data center provides 4,096 CPU cores/38 TB RAM plus a GPU server farm (~131,000 cores) for accelerated image analytics, ensuring timely execution of multi-stage pipelines. Storage exceeds 25 PB across Isilon and Ceph clusters, with 10 Gb links (and 40 GbE intra-Isilon) for high-throughput I/O during preprocessing and model inference. The environment is architected for high availability (Cisco UCS/Nexus, F5 load-balancing/firewall) with redundant power and cooling, supporting 24/7 secure processing of clinical MRI and linked data. The software maintenance provides our facility with immediate vendor software support, software licensing, and version upgrades for sophisticated clustering, visualization, development, and hierarchical file management software.

Fringe Benefits: The fringe benefits are calculated on salaries using the USC-DHHS Federal Rate Agreement from June 4, 2024, which sets Postdoc's fringe benefit rate at **22.60%.**

Facilities & Administrative (F&A) Costs: Facilities & Administrative costs were not included, as they are not allowed under the terms of the grant solicitation.

A. COMPONENT COVER PAGE

Project Title: 7.6_Iron Chelation as a Strategy to Mitigate Amyloid-Related Imaging Abnormalities During Anti-Amyloid Immunotherapy_Thorwald_new
Component Project Lead Information: Thorwald, Max

B. COMPONENT ACCOMPLISHMENTS

B.1 WHAT ARE THE MAJOR GOALS OF THE PROJECT?

Aim 1: Does a combination therapy with iron chelation attenuate Lecanemab-induced ARIA? Given the strong mechanistic link between amyloid removal, vascular destabilization, and iron-mediated
neurotoxicity, our first objective is to determine whether adjunctive iron chelation can reduce the occurrence and severity of Lecanemab-induced ARIA and associated ferroptotic injury. We hypothesize that DFO co-treatment with Lecanemab will preserve vascular integrity and attenuate ARIA-E/H by limiting iron-mediated oxidative stress. To test this, FAD mice (n=12/sex/group) at age 5-months are given DFO(10/mg/kg/day) by diet and/or Lecanemab (10mg/kg/week) by intraperitoneal injection (IP). They are assigned to the following groups: Control, DFO, Lecanemab, and DFO + Lecanemab. Mice not receiving Lecanemab are given a vehicle IP. After 2 weeks of treatment, mice undergo MRI to assess CMBs, BBB leakage, brain iron, and edema. Imaging will be conducted in two sex-balanced cohorts to ensure reproducibility, match timing of data acquisition, and maintain comparable treatment intervals. This aim will define quantitative MRI biomarkers of ARIA and determine whether DFO mitigates iron-driven injury during immunotherapy.

Aim 2: Does Lecanemab treatment result in a sex specific pro-ferroptotic phenotype? DFO showed promise in a clinical trial for AD with improved cognition12. Demonstrating that combination therapy
with DFO and Lecanemab confers an additive benefit would strengthen the translational value of metal chelation as both a preventive and therapeutic strategy. Aim 2 defines how Lecanemab and DFO, independently and jointly, influence ferroptotic signaling related to iron metabolism and amyloid in FAD mice. We hypothesize that the combination therapy will lower lipid peroxidation and amyloid load. Cortices will undergo bulk RNA- sequencing, soluble and insoluble amyloid, lipid peroxidation, and critical antioxidant and iron homeostatic proteins are measured in the cortex. Total iron will be assessed by inductively coupled plasma mass spectrometry (ICP-MS). By analyzing male and female brains, aim 2 will provide insight into the sex-linked molecular signatures of immunotherapy and chelation.

Overall Impact. As MoAbs become standard for AD, understanding their vascular and oxidative consequences is critical for improving safety and efficacy. This one-year project integrates high-field 7T MRI with transcriptomic and biochemical analyses to define how Lecanemab induces vascular injury and how DFO prevents it. (Fig. 1). By integrating 7T MRI, transcriptomics, and biochemical markers of ferroptosis, it addresses the urgent need for sex-informed therapeutic strategies that mitigate adverse outcomes while preserving cognitive benefit. Aims 1 and 2 run in parallel: Aim 1 quantifies vascular injury in vivo, while Aim 2 defines molecular correlates of ferroptotic stress ex vivo, providing a comprehensive evaluation of combination therapy outcomes. Results will guide future MoAb + DFO studies for adaptation to apolipoprotein E FAD (EFAD) models to assess genotype-dependent vascular and ferroptotic susceptibility to immunotherapy.

B.1.a Have the major goals changed since the initial competing award or previous report?

No

B.2 WHAT WAS ACCOMPLISHED UNDER THESE GOALS?

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B.3 COMPETITIVE REVISIONS/ADMINISTRATIVE SUPPLEMENTS

Not Applicable

B.4 WHAT OPPORTUNITIES FOR TRAINING AND PROFESSIONAL DEVELOPMENT HAS THE PROJECT PROVIDED?

NOTHING TO REPORT

B.5 HOW HAVE THE RESULTS BEEN DISSEMINATED TO COMMUNITIES OF INTEREST?

NOTHING TO REPORT

B.6 WHAT DO YOU PLAN TO DO DURING THE NEXT REPORTING PERIOD TO ACCOMPLISH THE GOALS?

Rigor and experimental design. Scientific rigor is addressed in each subaim under “Caveats.” All biochemical and imaging data will be collected and analyzed in a blinded manner. Power analysis for the primary contrast (LEC vs. LEC+DFO) within females, based on pilot MEMS-MRI data (d=0.8-1.0), indicated that with α=0.025 (co-primary MRI endpoints) and 80% power, 12 animals per group are required. Statistics by Two-way ANOVA with Tukey’s HSD or ANCOVA with sex as a covariate and Bonferroni correction for pairwise tests. Non-normal data will be log-transformed or analyzed by Kruskal-Wallis with Dunn’s post hoc. RNA-seq differential expression will be determined using DESeq2 with Benjamini-Hochberg FDR < 0.05

Summary. AD MoAbs that clear Aβ can trigger vascular injury known as ARIA. We propose that ARIA is not a random adverse event but a predictable iron-mobilization syndrome arising from Aβ plaque disaggregation, releasing plaque-bound iron, driving ferroptosis. Using FAD mice treated with LEC+DFO, we integrate multimodal MRI in Aim1 to assess brain structural changes complemented by transcriptomic and biochemical analyses in Aim 2, defining the interplay between iron load, vascular damage, and ferroptotic stress. This project reframes ARIA as an iron-driven, sex-modulated process and provides a translational framework for combining monoclonal antibody therapy with iron chelation to enhance safety in AD, FAD, and DSAD populations.

USC-ADRC Development Project Proposal

Title of Study: Iron Chelation as a Strategy to Mitigate Amyloid-Related Imaging Abnormalities During Anti-Amyloid Immunotherapy

Principal Investigator(s): Max Thorwald

ADRC Collaborator/Sponsor: Hussein Yassine

Project Period: April 01, 2026 - March 31, 2027

Specific Aims. Iron is implicated in neurodegenerative disorders, including Alzheimer's disease (AD), and is consistently found associated with amyloid plaques[1–3]. Excess ferrous iron (Fe^{2+}) is cytotoxic and drives lipid peroxidation, a well-established hallmark of AD pathology[4]. Increases in brain iron may occur from a weakened blood-brain barrier (BBB) or cerebral microbleed (CMBs), allowing blood-derived iron to enter the parenchyma. Blood contains hemoglobin and other iron-rich proteins with concentrations nearly 70-fold higher than those within brain tissue[5]. Incomplete breakdown of these iron-rich proteins by microglia results in hemosiderin deposition, a permanent redox-active reservoir that perpetuates cellular injury[6]. CMBs precede amyloid plaque formation and later co-localize with them, suggesting a mechanistic convergence between microvascular injury and amyloid deposition. CMBs accumulate throughout life but are markedly increased in AD[7,8] and remain largely undetected without Ultra High Field 7-Tesla magnetic resonance imaging (7T MRI), which can resolve CMBs well below the detection threshold of 3T MRI but remains unutilized in AD clinical trials.

The recent FDA approval of Lecanemab, an anti-amyloid monoclonal antibody (MoAb), represents a significant milestone in AD therapeutics. Lecanemab effectively reduces amyloid burden by targeting protofibrils, oligomers, and monomers, yet treatment produces amyloid-related imaging abnormalities with microhemorrhage (ARIA-H) and edema (ARIA-E) in nearly 20% of patients[9]. Plaque removal may destabilize amyloid-laden vessels, "unmasking" fragile vasculature and provoking leakage and oxidative stress. This influx of iron may result in ferroptosis, an iron-mediated form of cell death, a confirmed feature of AD[5,10]. When lethal levels of iron-mediated lipid peroxidation or a decrease in the antioxidant systems that detoxify lipid peroxidation occur, ferroptosis ensues. While definitive sex differences in ARIA severity have not been established, our pilot familial AD (FAD) results reveal pronounced female-specific ferroptosis-linked signatures; thus, sex will be rigorously tested as a biological variable. We hypothesize that the removal of amyloid plaques by Lecanemab permits vascular leakage and iron entry into the brain parenchyma, resulting in ARIA and ferroptotic injury. To mitigate these effects, *we* propose adjunctive treatment with deferoxamine (DFO), an FDA-approved iron chelator that crosses the BBB and lowers fibrillar amyloid and lipid peroxidation *peroxidation*[5,11]. Preliminary MRI data reveal treatment- and sex-specific changes in fluid diffusion and vascular permeability, while transcriptomic profiling shows broad divergence between males and females in response to Lecanemab alone or with DFO. These data underscore the need to define the molecular and vascular signatures underlying sex-specific responses to therapy relevant to ongoing clinical trials.

Aim 1: Does a combination therapy with iron chelation attenuate Lecanemab-induced ARIA?
Given the strong mechanistic link between amyloid removal, vascular destabilization, and iron-mediated neurotoxicity, our first objective is to determine whether adjunctive iron chelation can reduce the occurrence and severity of Lecanemab-induced ARIA and associated ferroptotic injury. We hypothesize that DFO cotreatment with Lecanemab will preserve vascular integrity and attenuate ARIA-E/H by limiting ironmediated oxidative stress. To test this, FAD mice (n=12/sex/group) at age 5-months are given DFO (10/mg/kg/day) by diet and/or Lecanemab (10mg/kg/week) by intraperitoneal injection (IP). They are assigned to the following groups: Control, DFO, Lecanemab, and DFO + Lecanemab. Mice not receiving Lecanemab are given a vehicle IP. After 2 weeks of treatment, mice undergo MRI to assess CMBs, BBB leakage, brain iron, and edema. Imaging will be conducted in two sex-balanced cohorts to ensure reproducibility, match timing of data acquisition,

and maintain comparable treatment intervals. This aim will define quantitative MRI biomarkers of ARIA and determine whether DFO mitigates iron-driven injury during immunotherapy.

Aim 2: Does Lecanemab treatment result in a sex specific pro-ferroptotic phenotype?
DFO showed promise in a clinical trial for AD with improved cognition[12]. Demonstrating that combination therapy with DFO and Lecanemab confers an additive benefit would strengthen the translational value of metal chelation as both a preventive and therapeutic strategy. Aim 2 defines how Lecanemab and DFO, independently and jointly, influence ferroptotic signaling related to iron metabolism and amyloid in FAD mice. We hypothesize that the combination therapy will lower lipid peroxidation and amyloid load. Cortices will undergo bulk RNAsequencing, soluble and insoluble amyloid, lipid peroxidation, and critical antioxidant and iron homeostatic proteins are measured in the cortex. Total iron will be assessed by inductively coupled plasma mass spectrometry (ICP-MS). By analyzing male and female brains, aim 2 will provide insight into the sex-linked molecular signatures of immunotherapy and chelation.

Overall Impact. As MoAbs become standard for AD, understanding their vascular and oxidative consequences is critical for improving safety and efficacy. This one-year project integrates high-field 7T MRI with transcriptomic and biochemical analyses to define how Lecanemab induces vascular injury and how DFO prevents it. (**Fig. 1**). By integrating 7T MRI, transcriptomics, and biochemical markers of ferroptosis, it addresses the urgent need for sex-informed therapeutic strategies that mitigate adverse outcomes while preserving cognitive benefit. Aims 1 and 2 run in parallel: Aim 1 quantifies vascular injury *in vivo*, while Aim 2 defines molecular correlates of ferroptotic stress *ex vivo*, providing a comprehensive evaluation of combination therapy outcomes. Results will guide future MoAb + DFO studies for adaptation to apolipoprotein E FAD (EFAD) models to assess genotype-dependent vascular and ferroptotic susceptibility to immunotherapy.

Significance. AD affects over 30 million people worldwide, and despite significant advances in amyloid-targeting therapies, there remains no curative treatment[1]. MoAbs, such as Lecanemab (LEC), effectively reduce β-amyloid (Aβ) plaque burden but frequently cause amyloid-related imaging abnormalities, including vasogenic edema (ARIA-E) and microhemorrhage (ARIA-H), in up to 20% of treated individuals[2]. While sex differences in ARIA incidence have not been firmly established, clinical observations suggest potential variation in vascular vulnerability between males and females, the biological basis of which remains poorly understood.

CMBs occur with aging and are markedly increased in all forms of AD[3–5]. CMBs allow iron-rich proteins, such as heme and hemoglobin, to enter the brain parenchyma. Microglia degradation of blood proteins results in a hemosiderin deposit, a complex aggregate of metals and metal-containing proteins[6–8]. These deposits undergo continuous redox cycling, damaging nearby structures such as cellular membranes. CMBs represent one of the earliest pathological events in AD[9], with Aβ plaques often forming around CMB sites[10], linking iron and Aβ deposition (**Fig. 1A**). FAD and Down syndrome with AD (DSAD) show an acceleration pattern, with early and prenatal CMBs for DSAD[11]. Averaging 50–300μm, CMBs are challenging to detect *in vivo*[10,12,13]. Clinical trials often implement 3T MRI capable of observing “CMBs” but with resolutions only reaching 1mm[14], underreporting the incidence of ARIA-H. Postmortem analysis estimates thousands of CMBs per AD brain, consistent with chronic, microscopic vascular injury[15].

Aβ plaques are coated with iron and heme[16,17], placing redox-active metals in direct contact with cellular membranes and initiating lipid peroxidation, a hallmark of AD[18–21]. This lipid peroxidation is predominantly irondependent, and when lethal levels are reached, a decrease in antioxidant systems that protect against lipid peroxidation, or both, iron-mediated cell death (ferroptosis) ensues[22,23]. Recently, we confirmed the presence of ferroptotic markers in sporadic AD human prefrontal cortex with concomitant increases in lipid peroxidation, reduced GSH-reliant antioxidant enzymes, and increased iron storage[20]. We then extended these findings to DSAD, showing that iron load and ferroptotic outcomes increased with APP gene dosage, which were normalized in rare variants of DS with a rare segmental deletion of the APP[21]. In both studies, females showed more pronounced ferroptotic markers than males. Iron therapies may benefit AD.

DFO, an FDA-approved iron chelator, improved cognition in an AD clinical trial[24] and in mice[25–27]. However, its potential as a neurodegenerative therapeutic was discounted due to the false assumption of poor BBB permeability[28,29], contrary to several reports[30–32]. We showed in female EFAD mice that acute DFO treatment reduced fibrillar Aβ, increased soluble Aβ monomers, and lowered lipid peroxidation, suggesting DFO may

disaggregate Aβ plaques by chelating plaque-bound iron (**Fig. 1B**)[20]. Our recent pilot data show that LECtreated FAD mice exhibit increased mortality and ferroptosis-related gene activation in females, and co-therapy with DFO improves survival and normalizes gene expression.

Sex differences have been observed in longitudinal studies and clinical trials, showing that females have a higher Aβ burden[33], faster progression of CMBs[34], and less cognitive benefits than males treated with LEC[35]. One case study of a female patient who died while receiving LEC showed increased CMBs on 7T MRI, not observable on 3T MRI during the study[36]. Mirroring human sex differences, female FAD mouse brains show more CMBs and nearly double the Aβ load compared to male brains at the same age[10,37,38]. We propose that LEC liberates plaque-bound iron while simultaneously allowing blood re-entry into CMB sites, thereby initiating ferroptosis and mortality (**Fig. 1C**). This proposal will integrate multimodal MRI with molecular ferroptosis profiling to determine whether iron chelation mitigates ARIA-like vascular injury and to define the sex-linked molecular signatures of iron dysregulation during Aβ immunotherapy.

Impact. This work reframes ARIA not as an unpredictable toxicity but as a predictable iron-mobilization syndrome during Aβ clearance. By integrating quantitative MRI biomarkers in Aim 1 with RNA, protein, and enzymatic activity assays in Aim 2, this study links vascular injury to downstream ferroptotic pathways central to AD neuropathology (**Fig. 1D**). Together, these complementary aims establish a mechanistic, sex-aware framework to improv MoAb treatment. This one-year project will directly inform future combination strategies coupling MoAb therapy with DFO to reduce ARIA risk in sporadic AD, FAD, and DSAD.

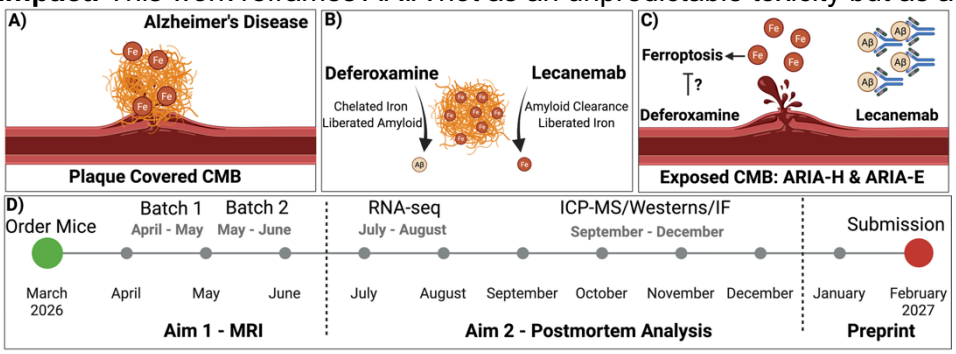


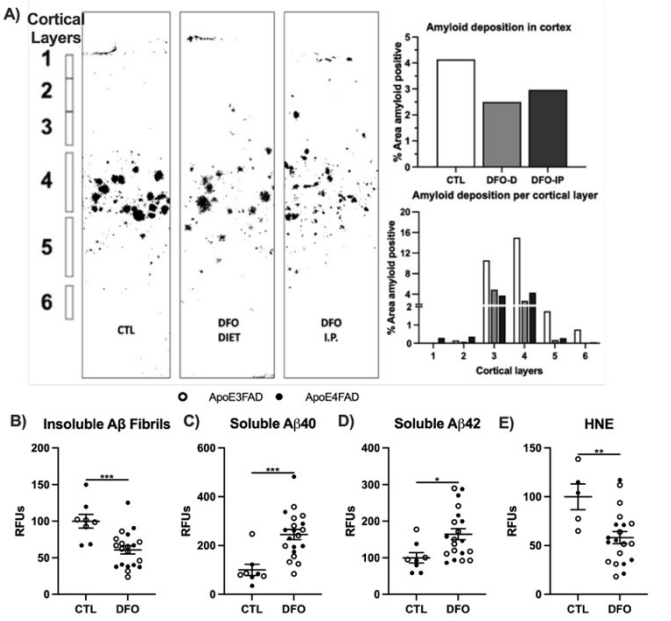
Figure 1: Linking Aβ, iron, and ARIA from MoAb's. **A)** Aβ plaques and CMBs co-localize, forming "plaque-covered CMBs" rich in redox -active iron. **B)** LEC removes Aβ but may liberate iron, whereas DFO chelates iron but may solubilize Aβ plaques, highlighting a potential therapeutic tradeoff. **C)** Combination therapy with LEC and DFO may simultaneously increase Aβ clearance and iron minimizing ferroptotic injury. **D)** Proposed timeline for the development project.

Preliminary Data. To determine whether iron chelation alters Aβ pathology, 6-month-old EFAD mice were treated with DFO (10mg/kg/day) either by diet for 2 weeks or IP injection for 1 week. Because both regimens produced comparable results, the data were combined. DFO reduced cortical fibrillary Aβ by 40% in as little as one week, as measured by IHC (**Fig. 2A**). The reduction in fibrillary Aβ was confirmed by dot blot using a fibril confirmation specific antibody (mOC87) through separation of soluble and insoluble proteins by formic acid extraction (**Fig. 2B**). Interestingly, soluble Aβ40 and Aβ42 levels increased after DFO (**Fig. 2C,D**), suggesting DFO disaggregates fibrillary Aβ by iron chelation, generating soluble Aβ peptides. The lipid peroxidation marker HNE was also reduced (**Fig. 2E**). Collectively, these findings demonstrate that DFO disaggregates Aβ plaques and mitigates lipid peroxidation, supporting the hypothesis that disruption of Aβ plaques liberates iron. More recently, I completed a second pilot study in 5-month-old FAD mice (n=5-6/sex/group), assigned to four groups: 1) FAD Control, 2) DFO, 3) LEC, and 4) LEC + DFO. LEC was administered by IP (10 mg/kg/week, 4 weeks) with ad libitum dietary DFO; controls and DFOonly mice received vehicle IPs. Marked mortality occurred primarily in the LEC group, prompting early termination of treatment at week 3. Survival analysis revealed significant differences between LEC-treated mice and both control and DFO groups, while combination therapy reduced mortality (**Fig. 3A**). This effect was sex-specific, with significant mortality in females but not males (**Fig. 3B,C**).

To assess the impact of these treatments on brain iron and BBB integrity, surviving animals underwent *in vivo* MRI before sacrifice at week 4. Multi-echo multi-slice (MEMS) MRI quantified brain iron, and dynamic contrastenhanced (DCE) MRI assessed BBB permeability. We previously validated MEMS MRI for iron in C57BL/6 mice with DFO[32]. Here, LEC increased total brain iron irrespective of sex, whereas the combination treatment normalized MEMS signal, consistent with chelation of **Figure 2:** DFO disaggregates Aβ fibrils and lowers lipid plaque- bound iron (**Fig. 3D**). DCE-MRI showed that peroxidation. **A)** IHC for Aβ plaques, dot blots for **B)** insoluble combination therapy reduced BBB permeability in males, Aβ, soluble **C)** Aβ40, **D)** Aβ42, and **E)** HNE adducts. t-test; suggesting vascular protection, while females showed a *P<0.05, **p<0.01, ***p<0.001. trend toward higher permeability (Fig. 3E). Due to the high mortality DCE was collected for one female.

After tissue collection, all brains, including those from premature deaths, underwent RNA sequencing. LEC treatment produced the most extensive transcriptomic divergence between sexes (~2,500 DEGs), whereas combination therapy yielded ~1,100 (**Fig. 4A,B**). In females, LEC increased expression of genes related to ferroptosis (GPx4, GPx1, Prdx5, Prdx2, SOD1), heme/iron metabolism genes (FTL, BLVRB), and the microglial activator Tyrobp (**Fig. 4C**). Males primarily upregulated transcripts linked to neuronal protection (FOSB, EGR1, NR4A1, SIK1) and BBB stability (Cldn12). GSEA reactome profiling indicated activation of the Nrf2 antioxidant pathway, heme metabolism, and HMOX1-mediated cytoprotection in females, consistent with a compensatory response to blood-derived iron.

With combination therapy, expression of ferroptosis regulators GPx4 and GPx1 was normalized, while genes supporting GSH synthesis (GSS), oxidative stress defense (Prdx5, Prdx3, Prdx2, SOD1), and selenium utilization (SELENOK, SELENOM) were upregulated. Distinct microglial patterns were observed: females showed enrichment of the TREM2-pathway



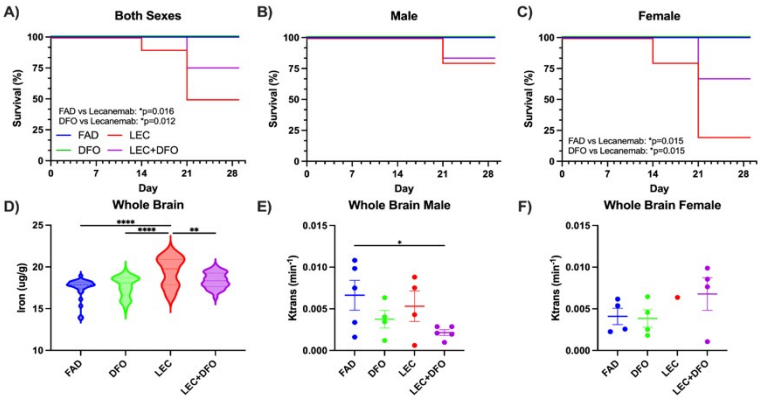


Figure 3: LEC causes increased mortality in female FAD mice. Survival curves for **A)** all animals, **B)** male, and **C)** females. **D)** Violin plot of MEMS slices for whole brain iron, Whole brain DCE-MRI for **E)** males, and **F)** females. Statistics: A -C; Mantel -Cox test, D ; ANCOVA with Bonferroni correction, sex as a covariate, E,F; One -way ANOVA with Tukey's HSD. *p<0.05, **p<0.01, ****p<0.0001.

(TYROBP, LGALS3, LYZ2) for debris clearance, whereas males upregulated hemoglobin-scavenging genes (CD163, CD93), indicating sex-specific strategies for iron detoxification and inflammation resolution. Males also maintained neuronal protective signatures (ARC, FOS, FOSB, NR4A1, SIK1). Together, these transcriptional profiles define an integrated cytoprotective network centered on ferroptosis resistance, iron sequestration, and vascular repair and suggest that females experience greater iron-driven oxidative stress with LEC. Combination therapy with DFO suppressed these stress signatures and diminished sex differences, indicating that iron chelation attenuates the oxidative and inflammatory sequelae of plaque-bound iron mobilization.

Specific Aim 1. Does a combination therapy with iron chelation attenuate LEC-induced ARIA? ARIA is a well-documented complication of MoAb treatments for AD and poses significant health risks that currently limit therapeutic use. The resulting CMBs and ARIA-H/E may exacerbate neuronal injury through the release of redox-active iron. Our preliminary MRI and survival data indicate that iron chelation mitigates these effects. Given the established link between Aβ plaques and brain iron, we hypothesize that co-administration of DFO will attenuate LEC-induced ARIA by limiting ironmediated vascular injury. To assess ARIA, we will reproduce the FAD pilot experiment in two independent batches to safeguard against treatment-related mortality and to accommodate our multimodal MRI (**Fig. 1D**). Five-month-old FAD mice (n=12/sex/group) will be divided into four groups identical to the pilot design. LEC will be administered for two weekly doses to minimize mortality while retaining ARIA pathology. One week after the second dose, animals will undergo a comprehensive multimodal MRI to assess iron load, vascular permeability, and tissue integrity. Imaging modalities include MEMS MRI for iron, susceptibility-weighted imaging (SWI) for CMB detection (ARIAH), fluid-attenuated inversion recovery (FLAIR) for edema (ARIA-E), diffusion tensor imaging (DTI) for whitematter integrity, and DCE-MRI for BBB permeability. These complementary modalities are routinely performed in tandem on the 7T

system, each adding only minutes to total scan time, and provide submillimeter resolution unavailable at clinical 3T field strengths. We expect that LEC will increase brain iron and BBB leakage, reproduce clinically observed ARIA features, and that combination therapy with DFO will normalize these parameters and improve survival. The integration of high-field MRI measures with molecular data from Aim 2 will establish the extent to which DFO mitigates the iron-driven vascular injury underlying ARIA. Caveats: This MRI workflow is fully established within USC's Functional Biological Imaging Core, and no technical barriers are anticipated. Splitting Aim 1 into two batches reduces imaging time and confirms reproducibility across sex balanced cohorts. The FAD model exhibits accelerated pathology compared with sporadic AD, potentially amplifying ARIA severity, but provides critical mechanistic insight relevant to high-Aβ conditions such as FAD and DSAD. Dietary DFO was selected to avoid repeated IP stress with no adverse effects in our pilot experiments. Prior studies using higher IP doses (100mg/kg/day) for up to 10 weeks showed no anemia or toxicity[39], and DFO's long clinical use supports its safety[40,41]. Other chelators are less established, reinforcing DFO as the preferred agent.

Specific Aim 2: Does Lecanemab treatment result in a sex specific pro-ferroptotic phenotype?

Ferroptosis, an iron-dependent form of lipid peroxidation-driven cell death, is increasingly recognized as a major contributor to AD neurodegeneration[20,21,42,43]. LEC treatment may inadvertently exacerbate ferroptotic stress by liberating redox-active iron from Aβ plaques, exposing neural and vascular membranes to oxidative injury. Sex differences may further modulate this vulnerability: female patients and FAD mice exhibit higher amyloid burden than age-matched males[33,37,38] and reduced therapeutic benefit from LEC[35]. Our pilot RNA-seq data support this mechanism: female mice treated with LEC upregulated ferroptosis-related genes, whereas males activated protective programs linked to neuronal stability and BBB integrity. Combination therapy with DFO normalized these responses and eliminated enrichment of ferroptotic and Nrf2-mediated stress pathways. We hypothesize that females exhibit heightened ferroptotic susceptibility due to the greater plaque abundance than males and that co-treatment with DFO will suppress these sex-specific ferroptotic effects. To test whether DFO suppresses LEC-induced ferroptosis, particularly in females, we will quantify key ferroptosis regulators (GPx4, GPx1, FSP1, SLC7A11, GCLC, GCLM, Prdx6, ACSL4) and iron-handling proteins (FTL, FTH1, TF, TfR, DMT1, SLC40A1, PRMT1) by Western blot. Antioxidant activities for reducing lipid peroxides are also measured. Cortex will be analyzed for soluble and insoluble Aβ (dot-blot)[44], total iron by ICPMS, tissue heme, and lipid peroxidation (HNE). RNA-seq and protein measures will evaluate ferroptotic regulators, heme-processing enzymes, and microglial activation markers. These endpoints will be correlated with MRI metrics from Aim 1 to define vascular-to-molecular convergence during ARIA. We anticipate that LEC will elevate ferroptotic gene expression, lipid peroxidation, and cortical iron, whereas combination therapy will normalize these indices and restore GSH-dependent antioxidant capacity. Females are expected to display stronger ferroptotic and inflammatory signatures than males, consistent with their higher amyloid burden and pilot data. Caveats: Although ferroptosis cannot be unequivocally proven in treatment, genes postmortem tissue without *in vitro* rescue experiments, our approach comprehensively profiles ferroptotic outcomes, including more than 20 validated regulators quantified in our prior studies[20,21,32].

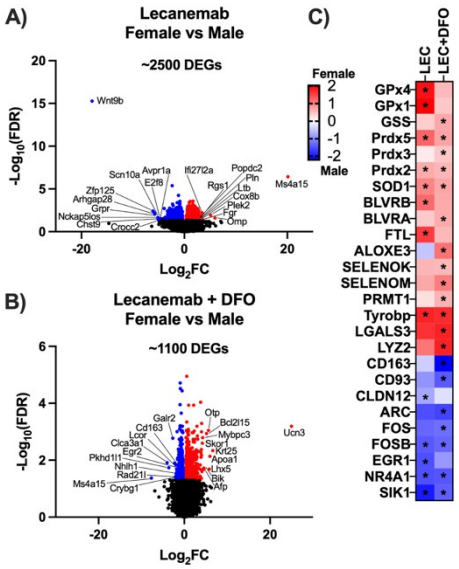


Figure 4: Female cortex has a stronger ferroptotic than males, consistent with their higher amyloid burden and pilot data. comparison volcano plots for **A)** LEC **B)** LEC+DFO, and **C)** genes postmortem tissue without *in vitro* rescue experiments, our approach comprehensively profiles ferroptotic outcomes, including more than 20 validated regulators quantified in our prior studies[20,21,32]. response to LEC. Sex related to ferroptosis, heme scavenging, microglial activity and neuronal repair. Red upregulated females, blue males.

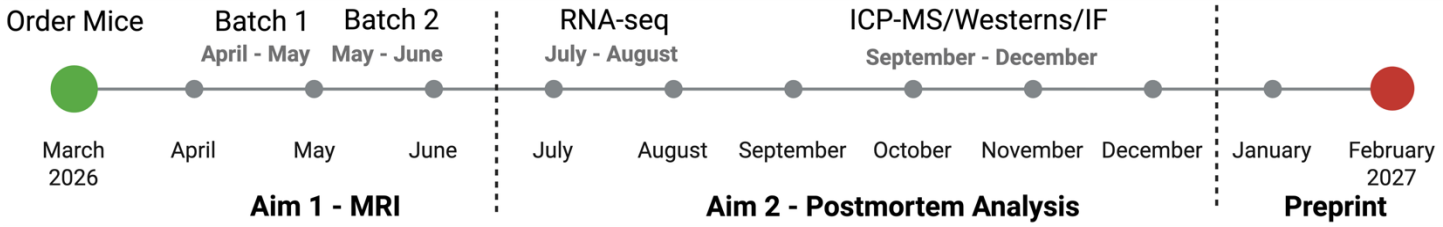
Rigor and experimental design. Scientific rigor is addressed in each subaim under "Caveats." All biochemical and imaging data will be collected and analyzed in a blinded manner. Power analysis for the primary contrast (LEC vs. LEC+DFO) within females, based on pilot MEMS-MRI data (d=0.8-1.0), indicated that with α=0.025 (co-primary MRI endpoints) and 80% power, 12 animals per group are required. Statistics by Two-way ANOVA with Tukey's HSD or ANCOVA with sex as a covariate and Bonferroni correction for pairwise tests. Non-normal data

will be log-transformed or analyzed by Kruskal-Wallis with Dunn's post hoc. RNA-seq differential expression will be determined using DESeq2 with Benjamini-Hochberg FDR < 0.05

Summary. AD MoAbs that clear Aβ can trigger vascular injury known as ARIA. We propose that ARIA is not a random adverse event but a predictable iron-mobilization syndrome arising from Aβ plaque disaggregation, releasing plaque-bound iron, driving ferroptosis. Using FAD mice treated with LEC±DFO, we integrate multimodal MRI in Aim1 to assess brain structural changes complemented by transcriptomic and biochemical analyses in Aim 2, defining the interplay between iron load, vascular damage, and ferroptotic stress. This project reframes ARIA as an iron-driven, sex-modulated process and provides a translational framework for combining monoclonal antibody therapy with iron chelation to enhance safety in AD, FAD, and DSAD populations.

ADRC Development Project - Iron Chelation as a Strategy to Mitigate Amyloid-Related Imaging Abnormalities During Anti-Amyloid Immunotherapy
PI: Thorwald

Timeline for the proposed project



This is the proposed timeline for the project. The study will be performed in two cohorts of 5xFAD mice to ensure ample animals for MRI to assess ARIA by June 2026. These months may shift by a few months if Jackson has insufficient animals, however we have more than enough time to complete all measurements between July-December.

Plan for K/R Grants
Below are grants pending or post IRG with me as PI related to ApoE, iron, and air pollution as they relate to AD neuropathology.

K22
1. Air Pollution Derived Metals as Drivers of Oxidative Stress and Alzheimer's Disease Neuropathology (Pending IRG Review)
2. More than Just a Lipid Transporter: Elucidating the Role of Nuclear ApoE in Alzheimer's Disease (Score 54, A1 submission pending).

R21
1. Iron and Lipid Peroxidation as Upstream Drivers of Amyloid production during Alzheimer's Disease (Pending IRG Review).
2. Role of Cerebral Microbleed-Derived Iron in Amyloid Plaque Initiation (Not Discussed, resubmission planned).

R01
The role of air pollution on DNA, RNA, and Protein in facilitating Amyloid precursor protein processing during Alzheimer's disease (Not Discussed, Letter of Appointment Required).

R35
Planned for June 2026 with a focus on ApoE. By combining protein language models with systematic *in silico* mutational and structural scanning, we reconstruct the isoform-specific fitness and structural landscapes of the

100+ documented isoforms for ApoE. This enables quantitative dissection of why specific ApoE variants persist in human populations and how evolutionary pressures trade off structure, disorder, and function.

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VERTEBRATE ANIMALS

1. Animal Numbers/Procedures:

This proposal uses the familial Alzheimer's disease (FAD) mouse and is approved under IACUC#20417. The FAD is a transgenic mouse model of familial Alzheimer's disease containing 5 mutations: APP KM670/671NL (Swedish), APP I716V, APP V717I, PSEN1 M146L (A>C), and PSEN1 L286V from a B6SJL background. These mice are readily available from Jackson labs. All mice will be housed until dissection.

All Aims (n=12/sex/group)
24: FAD
24: FAD + DFO
24: FAD + Lecanemab
24: FAD + Lecanemab + DFO

Iron chelation treatment: Pilot data shows equal efficacy of DFO treatment given through diet rather than intraperitoneal injection. DFO chow will be given to mice ad libitum starting at 20 weeks (5-months of age). Dosing (10mg/kg/day) is based on the approximate amount of food a mouse consumes daily.

Lecanemab treatment: Lecanemab is given to animals by intraperitoneal injection (10mg/kg/week) starting 20 weeks (5 months) until 22 months (5.5 months). Animals not receiving Lecanemab are given saline vehicle as a sham IP.

Tissue Collection: Mice are anesthetized by the inhalant isoflurane and blood collected through cardiac puncture using heparinized tubes. After cardiac-whole body saline perfusion, brains are rapidly removed and hemisected: one hemisphere for immunohistochemistry (IHC); the other hemisphere is microdissected (cortex and hippocampus) and flash frozen on dry ice.

2. Justifications:

In vitro cell studies or computer/mathematical models cannot satisfactorily reproduce the pathologic features and functional alterations in brain resulting from Alzheimer's pathology alone. Considering cost, ethical concerns, reproducibility, and our experience in the models being tested, inbred mice are optimum for these studies. We choose the FAD mouse model because they are the standard murine strains for studying AD pathology and are widely used in the AD field. Equal numbers of males and females will be included in all groups, consistent with NIH policy. Power analysis for the primary contrast (LEC vs. LEC+DFO) within females, based on pilot MEMS-MRI data (d=0.8- .0), α=0.025 0 , = 2 per group is sufficient. Group comparisons will use two-way ANOVA (TreatmentxSex) with Tukey's post hoc to assess main and interaction effects, or ANCOVA with sex as a covariate when sex is not a primary factor. Nonparametric tests (Kruskal-Wallis with Dunn's post hoc) will be used if data violate normality assumptions.

3. Minimization of Pain and Distress

Discomfort, distress, pain, and injury will be minimized in all phases of the experiment. All our staff and student are trained to carefully monitor behavioral changes, e.g., loss of appetite, grooming and activity. Moribund criteria include one or a combination of the following: >20% weight loss, sustained hunched posture, respiratory difficulty, hypo/hyperthermia, and inability of access food/water, inability to ambulate or make normal postural adjustments. Mice will be monitored for signs of stress or pain as well as general criteria outlined by IACUC (Body Condition Score 2, BC2) and are immediately euthanized.

4. Euthanasia As required. Euthanasia uses inhalation of CO_2 as approved by the American Veterinary Medical Association (AVMA) Guidelines for the Euthanasia of Animals.

C. COMPONENT PRODUCTS

C.1 PUBLICATIONS

Not Applicable

C.2 WEBSITE(S) OR OTHER INTERNET SITE(S)

Not Applicable

C.3 TECHNOLOGIES OR TECHNIQUES

NOTHING TO REPORT

C.4 INVENTIONS, PATENT APPLICATIONS, AND/OR LICENSES

Not Applicable

C.5 OTHER PRODUCTS AND RESOURCE SHARING

C.5.a Other Products

NOTHING TO REPORT

C.5.b Resource Sharing

Progress implementing the resource sharing plan:

NOTHING TO REPORT

C.5.c Data Management and Sharing

Not Applicable

D. COMPONENT PARTICIPANTS

Not applicable

E. COMPONENT IMPACT

E.1 WHAT IS THE IMPACT ON THE DEVELOPMENT OF HUMAN RESOURCES?

Not Applicable

E.2 WHAT IS THE IMPACT ON PHYSICAL, INSTITUTIONAL, OR INFORMATION RESOURCES THAT FORM INFRASTRUCTURE?

Not Applicable

E.3 WHAT IS THE IMPACT ON TECHNOLOGY TRANSFER?

NOTHING TO REPORT

E.4 WHAT DOLLAR AMOUNT OF THE AWARD'S BUDGET IS BEING SPENT IN FOREIGN COUNTRY(IES)?

Not Applicable

F. COMPONENT CHANGES

F.1 CHANGES IN APPROACH AND REASONS FOR CHANGE

Not Applicable

F.2 ACTUAL OR ANTICIPATED CHALLENGES OR DELAYS AND ACTIONS OR PLANS TO RESOLVE THEM

NOTHING TO REPORT

F.3 SIGNIFICANT CHANGES TO HUMAN SUBJECTS, VERTEBRATE ANIMALS, BIOHAZARDS, AND/OR SELECT AGENTS

F.3.a Human Subject

No Change

F.3.b Vertebrate Animals

File uploaded: M.Thorwal_VERTEBRATE ANIMALS RFA.pdf

F.3.c Biohazards

No Change

F.3.d Select Agents

No Change

VERTEBRATE ANIMALS

1. Animal Numbers/Procedures:

This proposal uses the familial Alzheimer's disease (FAD) mouse and is approved under IACUC#20417. The FAD is a transgenic mouse model of familial Alzheimer's disease containing 5 mutations: APP KM670/671NL (Swedish), APP I716V, APP V717I, PSEN1 M146L (A>C), and PSEN1 L286V from a B6SJL background. These mice are readily available from Jackson labs. All mice will be housed until dissection.

All Aims
10: FAD
10: FAD + DFO
10: FAD + Lecanemab
10: FAD + Lecanemab + DFO
6: FAD + mab158
6: FAD + human IgG1

Iron chelation treatment: Pilot data shows equal efficacy of DFO treatment given through diet rather than intraperitoneal injection. DFO chow will be given to mice ad libitum starting at 22 weeks (5.5-months of age). Dosing (25mg/kg/day) is based on the approximate amount of food a mouse consumes daily.

Lecanemab treatment: Lecanemab is given to animals by intraperitoneal injection (10mg/kg/week) starting 22 weeks (5.5 months) until 24 months (6 months). Animals not receiving Lecanemab are given saline vehicle as a sham IP or human IgG1.

Tissue Collection: Mice are anesthetized by the inhalant isoflurane and blood collected through cardiac puncture using heparinized tubes. After cardiac-whole body saline perfusion, brains are rapidly removed and hemisected: one hemisphere for immunohistochemistry (IHC); the other hemisphere is microdissected (cortex and hippocampus) and flash frozen on dry ice.

2 Justifications:

In vitro cell studies or computer/mathematical models cannot satisfactorily reproduce the pathologic features and functional alterations in brain resulting from Alzheimer's pathology alone. Considering cost, ethical concerns, reproducibility, and our experience in the models being tested, inbred mice are optimum for these studies. We choose the FAD mouse model because they are the standard murine strains for studying AD pathology and are widely used in the AD field. Equal numbers of males and females will be included in all groups, consistent with NIH policy. Power analysis for the primary contrast (LEC vs. LEC+DFO) within females, based on pilot MEMS-MRI data (d=0.8-1.0), indicated that with α=0.025 and 80% power, n=10 per group is sufficient. Group comparisons will use two-way ANOVA (TreatmentxSex) with Tukey's post hoc to assess main and interaction effects, or ANCOVA with sex as a covariate when sex is not a primary factor. Non-parametric tests (Kruskal-Wallis with Dunn's post hoc) will be used if data violate normality assumptions.

3. Minimization of Pain and Distress Discomfort, distress, pain, and injury will be minimized in all phases of the experiment. All our staff and student are trained to carefully monitor behavioral changes, e.g., loss of appetite, grooming and activity. Moribund criteria include one or a combination of the following: >20% weight loss, sustained hunched posture, respiratory difficulty, hypo/hyperthermia, and inability of access food/water, inability to ambulate or make normal postural adjustments. Mice will be monitored for signs of stress or pain as well as general criteria outlined by IACUC (Body Condition Score 2, BC2) and are immediately euthanized.

4. Euthanasia As required. Euthanasia uses inhalation of CO_2 as approved by the American Veterinary Medical Association (AVMA) Guidelines for the Euthanasia of Animals.

G. COMPONENT SPECIAL REPORTING REQUIREMENTS SPECIAL REPORTING REQUIREMENTS

G.1 SPECIAL NOTICE OF AWARD TERMS AND NOTICE OF FUNDING OPPORTUNITIES REPORTING REQUIREMENTS

Not Applicable

G.2 RESPONSIBLE CONDUCT OF RESEARCH

Not Applicable

G.3 MENTOR'S REPORT OR SPONSOR COMMENTS

Not Applicable

G.4 HUMAN SUBJECTS

Not Applicable

G.5 HUMAN SUBJECTS EDUCATION REQUIREMENT

NOT APPLICABLE

G.6 HUMAN EMBRYONIC STEM CELLS (HESCS)

Does this project involve human embryonic stem cells (only hESC lines listed as approved in the NIH Registry may be used in NIH funded research)?

No

G.7 VERTEBRATE ANIMALS

Not Applicable

G.8 PROJECT/PERFORMANCE SITES

Not Applicable

G.9 FOREIGN COMPONENT

Not Applicable

G.10 ESTIMATED UNOBLIGATED BALANCE

Not Applicable

G.11 PROGRAM INCOME

Not Applicable

G.12 F&A COSTS

Not Applicable

OMB Number: 4040-0010
Expiration Date: 12/31/2022

FINAL

RESEARCH & RELATED BUDGET - SECTION A & B, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

A. Senior/Key Person

Prefix	First Name*	Middle Name	Last Name*	Suffix	Project Role*	Base Salary ($)	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits ($)*	Funds Requested ($)*
1.	Max		Thorwald		PI	87,500.00	1.2			8,595.00	2,879.00	11,474.00

Total Funds Requested for all Senior Key Persons in the attached file

Additional Senior Key Persons: File Name: **Total Senior/Key Person** **11,474.00**

B. Other Personnel

Number of Personnel*	Project Role*	Calendar Months	Academic Months	Summer Months	Requested Salary ($)*	Fringe Benefits*	Funds Requested ($)*
	Post Doctoral Associates						
	Graduate Students						
	Undergraduate Students						
	Secretarial/Clerical						
0	**Total Number Other Personnel**					**Total Other Personnel**	**0.00**
						Total Salary, Wages and Fringe Benefits (A+B)	**11,474.00**

RESEARCH & RELATED Budget {A-B} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTION C, D, & E, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

C. Equipment Description	
List items and dollar amount for each item exceeding $5,000	
Equipment Item	**Funds Requested ($)***
Total funds requested for all equipment listed in the attached file	
Total Equipment	**0.00**

Additional Equipment: File Name:

D. Travel	**Funds Requested ($)***
1. Domestic Travel Costs (Incl. Canada, Mexico, and U.S. Possessions)	0.00
2. Foreign Travel Costs	0.00
Total Travel Cost	**0.00**

E. Participant/Trainee Support Costs		**Funds Requested ($)***
1. Tuition/Fees/Health Insurance		0.00
2. Stipends		0.00
3. Travel		0.00
4. Subsistence		0.00
5. Other:		
0 Number of Participants/Trainees	**Total Participant Trainee Support Costs**	**0.00**

RESEARCH & RELATED Budget {C-E} (Funds Requested)

RESEARCH & RELATED BUDGET - SECTIONS F-K, BUDGET PERIOD 1

UEI*: G88KLJR3KYT5

Budget Type*: ● Project ❍ Subaward/Consortium

Enter name of Organization: UNIVERSITY OF SOUTHERN CALIFORNIA

Start Date*: 04-01-2026 **End Date*:** 03-31-2027 **Budget Period: 1**

F. Other Direct Costs	Funds Requested ($)*
1. Materials and Supplies	6,060.00
2. Publication Costs	0.00
3. Consultant Services	0.00
4. ADP/Computer Services	0.00
5. Subawards/Consortium/Contractual Costs	0.00
6. Equipment or Facility Rental/User Fees	0.00
7. Alterations and Renovations	0.00
8. MRI	9,600.00
9. Vivarium and mouse cost	12,866.00
10.	0.00
11.	0.00
12.	0.00
13.	0.00
14.	0.00
15.	0.00
16.	0.00
17.	0.00
Total Other Direct Costs	**28,526.00**

G. Direct Costs	Funds Requested ($)*
Total Direct Costs (A thru F)	**40,000.00**

H. Indirect Costs

Indirect Cost Type	Indirect Cost Rate (%)	Indirect Cost Base ($)	Funds Requested ($)*
1. MTDC	66.0	40,000.00	26,400.00
		Total Indirect Costs	**26,400.00**

Cognizant Federal Agency (Agency Name, POC Name, and POC Phone Number) DHHS Jeanette Lu 415-437-7820

I. Total Direct and Indirect Costs	Funds Requested ($)*
Total Direct and Indirect Institutional Costs (G + H)	**66,400.00**

J. Fee	Funds Requested ($)*
	0.00

K. Total Costs and Fee	Funds Requested ($)*
	66,400.00

L. Budget Justification*	File Name: M.Thorwald_ADRC DP_BUDJUS FINAL.pdf

RESEARCH & RELATED Budget {F-K} (Funds Requested)

Budget Justification

Personnel

Max Thorwald, Ph.D., Principal Investigator (1.2 calendar months)
Max Thorwald is responsible for managing the project and coordinating between the Mack lab and MRI core for mouse transfer. Dr. Thorwald will analyze all data obtained data, prepare and submit manuscripts. He will also present the findings at scientific conferencs.

Materials and Supplies: (Total $28,526)
5xFAD mice are purchased from Jackson labs for this study. Dietary DFO will be given to mice daily via chow and Lecnaemab/IgG treatments once weekly by intraperitoneal injection. MRI will be performed the week following the final treatment for assessment of ARIA.

1. DFO Diet ($2,000): Deferoxamine $200 Sigma, Diet formulation 1,800 from Research Diets
2. Lecanemab ($3,000): at cost, single reagent.
3. Murine Lecnemab ($560): at cost, single reagent
4. Human IgG1 ($500): at cost, single reagent
5. MRI ($9,600): Scan time approximately 2 hours collectively for SWI, DTI, MEMS, FLAIR, and DCE MRI billed at $100/hr.
6. Vivarium ($12,272): Cost of purchasing 52 mice x $236 each. All mice and in vivo experimentation will occur in ZNI. Estimating 36 cages at 5 animals per cage for females and 1 animal for cage for males due to the aggressive nature of males at the per diem rate of $1.10 per cage per day for 2 weeks: $594